Meeting of the Board
17 – 20 October 2022
Incheon, Republic of Korea
Provisional agenda item 13

GCF/B.34/02/Add.01

29 September 2022

Consideration of funding proposals – Addendum I

Funding proposal package for FP191

Summary

This addendum contains the following seven parts:

a) A funding proposal titled "Enhancing Adaptation and Community Resilience by Improving Water Security in Vanuatu";
b) No-objection letter issued by the national designated authority(ies) or focal point(s);
c) Environmental and social report(s) disclosure;
d) Secretariat’s assessment;
e) Independent Technical Advisory Panel’s assessment;
f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
g) Gender documentation.

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Funding Proposal

Project/Programme title:	Enhancing Adaptation and Community Resilience by Improving Water Security in Vanuatu
Country(ies):	Vanuatu
Accredited Entity:	The Pacific Community (SPC)
Date of first submission:	2022/04/07
Date of current submission	2022/09/09
Version number	V.6

GREEN CLIMATE FUND

Contents

Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the GCF Information Disclosure Policy, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:
fundingproposal@gcfund.org
Please use the following name convention for the file name:
“FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]”

A

A. PROJECT/PROGRAMME SUMMARY

A.1. Project or programme	Project	**A.2. Public or private sector**	Public
A.3. Request for Proposals (RFP)	*If the funding proposal is being submitted in response to a specific GCF Request for Proposals, indicate which RFP it is targeted for. Please note that there is a separate template for the Simplified Approval Process and REDD+.* Not applicable		

A.4. Result area(s)

*Check the applicable GCF result area(s) that the overall proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of **GCF and Co-financers' contribution** devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively.*

	GCF contribution	Co-financers' contribution[1]
Mitigation total	Enter number %	Enter number %
☐ Energy generation and access	Enter number %	Enter number %
☐ Low-emission transport	Enter number %	Enter number %
☐ Buildings, cities, industries and appliances	Enter number %	Enter number %
☐ Forestry and land use	Enter number %	Enter number %
Adaptation total	100%	100% %
☒ Most vulnerable people and communities	34 %	34 %
☒ Health and well-being, and food and water security	33 %	33 %
☒ Infrastructure and built environment	33 %	33 %
☐ Ecosystems and ecosystem services	Enter number %	Enter number %

A.5. Expected mitigation outcome *(Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)*	*Not applicable.*	**A.6. Expected adaptation outcome** *(Core indicator 2: direct and indirect beneficiaries reached)*	*The number of direct and indirect beneficiaries for the project are presented below:*	
			Direct: 85,650 (including 42,825 women)	Indirect: 216,980
			~27% of the total population of Vanuatu[2]	~73% of the total population of Vanuatu
A.7. Total financing (GCF + co-finance[3])	28,290,014 USD	**A.9. Project size**	Small (Upto USD 50 million)	
A.8. Total GCF funding requested	23,325,033 USD			

[1] Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.
[2] This figure is based on the latest available population estimate of Vanuatu (307,000), according to the World Population Prospect, see: https://population.un.org/wpp/.
[3] Refer to the Policy of Co-financing of the GCF.

A.10. Financial instrument(s) requested for the GCF funding	*Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.* ☒ Grant 23,325,033 USD ☐ Loan Enter number ☐ Guarantee Enter number ☐ Equity Enter number ☐ Results-based payment Enter number		
A.11. Implementation period	5 years (2023 – 2028)	**A.12. Total lifespan**	2023 – 2045
A.13. Expected date of AE internal approval	*This is the date that the Accredited Entity obtained/will obtain its own approval to implement the project/ programme, if available.* Click or tap to enter a date.	**A.14. ESS category**	*Refer to the AE's safeguard policy and GCF ESS Standards to assess your FP category.* **B**
A.15. Has this FP been submitted as a CN before?	Yes[4] ☒ No ☐	**A.16. Has Readiness or PPF support been used to prepare this FP?**	Yes ☒ No ☐
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	**A.18. Is this FP included in the country programme?**	Yes ☒ No ☐
A.19. Complementarity and coherence	*Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.* Yes ☒ No ☐		
A.20. Executing Entity information	*If not the Accredited Entity, please indicate the full legal name of the Executing Entity(ies) and provide its country of registration and ownership type. Note that there can be more than one Executing Entity. Also indicate if an Executing Entity is the National Designated Authority. Refer to the definition of Executing Entity in the Accreditation Master Agreement.* As the Accredited Entity, SPC will also be the Executing Entity (EE). The Government of Vanuatu (GoV) will also be an Executing Entity for the Project, acting through the Ministry of Finance (MoF) and the Department of Water Resources (DoWR).		
A.21. Executive summary (max. 750 words, approximately 1.5 pages)			

[4] Concept Note for the project can be accessed here: https://www.greenclimate.fund/sites/default/files/document/21860-enhancing-adaptation-and-community-resilience-improving-water-security.pdf

1. The project, *Enhancing Adaptation and Community Resilience by Improving Water Security in Vanuatu*, is a climate adaptation investment, aiming to enhance climate resilience and increase water security of rural communities. The project will climate-proof water sanitation, and hygiene (WASH) infrastructure and enhance water management planning systems across Vanuatu. The approach will utilize and improve government-owned processes of water management. This will primarily be: the Drinking Water Safety and Security Planning (DWSSP) process, mandated through national policy frameworks such as the National Implementation Plan for Safe and Secure Water (NIP)[5] and the Capital Assistance Programme (CAP)[6].

2. This project is listed as a priority intervention in Vanuatu's Green Climate Fund Country Programme[7] and is being fully co-developed with the Department of Water Resources (DoWR) and other stakeholders for strong country-ownership. By addressing climate change-induced risks to water management and working directly with vulnerable rural communities (including community-based adaptation activities), the project is fully aligned with the Government of Vanuatu's climate change strategies and policies including the Climate Change and Disaster Risk Reduction Policy 2016–2030, National Adaptation Programme of Action, Nationally Determined Contribution, Vanuatu National Sustainable Development Plan 2016–2030 and the Vanuatu National Water Policy 2017–2030.

3. Observed changes in the climate of Vanuatu show there has been a gradual change in climatic conditions over the last 70 years. The temperature has increased at approximately 0.1°C per decade since 1970, and 30-year projections indicate this trend will continue. Despite no general upward trend in overall precipitation is observed or projected, extreme rainfall and tropical cyclone events are predicted to become more intense. These changes compound common water sanitation services and resource management challenges.

4. More intense extreme events will increase the incidence of flooding and induce greater damage to water utilities, pipes, and community-based water infrastructure in the baseline scenario. Increased climate-related damages can result in redundant infrastructure, cutting communities off to water supplies. Flooding events increasingly overwhelm drains, and latrine pits, resulting in stagnant, contaminated water that puts the health and wellbeing of the ni-Vanuatu population at risk. On the other hand, increased temperatures and increased intensity of major El Nino events are predicted to cause increased incidence of meteorological drought. Consequently, climate impacts directly threaten water security and sanitation across the country.

5. Compounding the negative impacts of climate change is restricted technical, institutional and financial capacity barriers related to water governance:

limited technical capacity, human resources, and financing at area council and provincial level to develop climate-sensitive water management plans;,
limited technical knowledge, human resources, and financing at provincial and national levels to establish and maintain climate-resilient infrastructure; and
insufficient awareness, knowledge, and skills on climate change issues and climate-resilient water management at provincial and national levels.

6. In 2020, the DoWR, in association with UNICEF, conducted an inventory of water systems, as part of a Water and Sanitation Sector Strengthening Programme. Only 79% of systems were identified as functional and only 37% of these systems were reported as providing water 24h/day daily, every day of the year.[8]

7. The project will address these barriers by implementing three distinct but mutually reinforcing Outcomes. They will target three climate change-related water impacts as described under GCF guidelines on water security projects: (i) water-related disasters; (ii) areas suffering from water stress; and (iii) poor water quality-induced fatalities.[9]

Outcome 1: Communities are empowered to plan and manage climate-resilient water resources
Activities will boost technical capacities across national extension structures to support climate-resilient water management planning at the community level. Enhancing capacities of rural water committees (RWC) to develop resilient and sustainable water management plans. Maintaining safe and secure water access in the long-term.

Outcome 2: Communities have enhanced climate-resilient rural water infrastructure
The project will enhance DoWR screening and approval processes for resilient infrastructure investments and increase financial flows to enhanced DWSSPs that require financial assistance to construct necessary resilient water infrastructure systems.

Outcome 3: Provincial and national institutions are strengthened to address climate risks associated with water security.

Activities will focus on capacity building across national and provincial levels to boost DoWR and WASH sector partners' technical capacity concerning climate-resilient water management. Further, the Outcome will enhance monitoring and evaluation practices across extension structures and facilitate improved knowledge management processes. This will enhance evidence-based knowledge transfer and raise awareness of climate-resilient water management practices.

8. Ultimately, the combined impact of these Outcomes will induce a paradigm shift in Vanuatu's water management systems. It will catalyse a shift towards safe, climate-resilient, and sustainable water utilization and improved water security at community levels.

[5] Drinking Water Safety and Security Planning (DWSSP) is a process of community engagement in identifying and discussing threats to safe and secure drinking water, and making plans to manage these threats. The resulting community DWSSP guides day-to-day water supply operation and maintenance, as well as improvements. These feed into the National Implementation Plan (NIP) for safe and secure community drinking water. The DWSSP and the NIP are the key processes to deliver safe and secure water to rural communities, directly addressing water security and improving community resilience. DWSSP are also required for school and healthcare facility water supply systems. The processes have been generally well received, supported through resources and training, and existing methodology to improve rural communities' water supply systems. See:https://www.nab.vu/sites/default/files/documents/Vanuatu%20National%20Implementation%20Plan%20for%20Safe%20and%20Secure%20Community%20Drinking%20Water.pdf

[6] The CAP will help communities to improve the safety and security of their drinking water supply schemes by providing financial assistance if they can demonstrate their need. An approved DWSSP is the essential prerequisite to making a CAP funding request.
See: https://mol.gov.vu/images/News-Photo/water/Vanuatu-CAP-Guide-annual-CAP-241018.pdf

[7] Vanuatu's GCF Country Programme is available here: https://www.greenclimate.fund/sites/default/files/document/country-programme-vanuatu.pdf

[8] UNICEF (2020). Rural water supplies in Vanuatu in need of significant improvements. (WASH Technical Paper/13/2020).

[9] GCF Water Project Design Guidelines (v 0.98 – 28 September 2021 – developed by Deltaeres).

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

9. The **climate context** is presented below. It is presented in the following sub-sections: adaptation context; vulnerability; climate data analysis (observed historical change and projected climate change); risks and impacts; and complementarity and coherence.

10. **Adaptation context:** Vanuatu is Small Island Developing State comprising 83 islands, 65 of which are inhabited, extending over 1,000 km in the Western Pacific between the equator and the tropic of Capricorn. The archipelago, which is of volcanic origin, is 1,750 km east of northern Australia, 540 km northeast of New Caledonia, and to the west of Fiji. Port Vila, on the island of Efate, is the capital of the country. Vanuatu is divided administratively into six Provinces and each province is administered by a provincial council (responsible for promoting regional autonomy). There are three municipal councils for the cities of Port Vila (Shefa), Luganville (Sanma), and Lenakel (Tafea). Each province is further divided into Area Council administrative units, with Area Administrators, who have devolved mandates from the DoWR.

11. The Vanuatu National Statistics Office (VNSO) estimated the 2021 population to be at 316,000.[10] The country is highly homogeneous – 99% of its population are the indigenous, Melanesian ni-Vanuatu peoples. Around three-quarters of the people live in rural areas, although Port Vila – and the surrounding capital region – account for about 21% of the total population. There are over 100 languages and dialects, of which approximately 80 are actively spoken, making it one of the one of the most linguistically diverse countries in the world. There are three official languages: Bislama, English and French.

12. Vanuatu's economy[11] is characteristic of small Pacific Island nations: small in scale, remote from major markets and internally dispersed. These factors combine to push up the costs of private production and public administration, lower the return on market activities and narrow the feasible set of economic opportunities.[12] The country's geographical location, undiversified economic structure, and high import dependence makes it vulnerable to external shocks, especially climate change-induced disasters and natural hazards.[13] Climate risks and shocks greatly impact hard-earned development gains and threaten food, energy and water security – the latter of which this project sets out to address directly.

13. **Water resources in Vanuatu:** The islands in the archipelago are highly heterogenous in geographic, topographic and climatic conditions. For example, some of the larger, more mountainous islands have good groundwater and surface water resources, whilst others have either ground or surface water or rely entirely on rainwater catchment. However, steep catchments and narrow coastal plains are ubiquitous in these islands and are vulnerable to flooding and sea-level rise. Water resources in the country, therefore, vary and are influenced by these climatic and geographic factors.

14. At the same time, **the island nation is prone to multivariable water-related climate risks** coupled with underlying social and economic vulnerabilities. Since Vanuatu's population is also concentrated along the coasts, the balance of freshwater and saltwater ecosystems also plays a vital role in the subsistence and commercial life of the population. The islands have uniquely fragile water resources due to its small scale, lack of storage and limited freshwater reserves – which are increasingly exposed to climate impacts. Climate impacts particularly destabilize natural resource-dependent livelihoods of rural communities (75% of the population)[14], who continue to rely on subsistence farming (with variable water access) in the different islands.

[10] Information on demographics available at: https://vnso.gov.vu/index.php/en/

[11] Vanuatu's economy is primarily based on subsistence or small-scale agriculture, providing a living for more than 70% of the population. Since the early 2000s, tourism, land sales and high commodity prices for copra and coffee, and donor funding have driven the economy. Economic impediments include: undiversified economic base, constrained transport infrastructure and a small domestic market. Migration and remittances have been a critical driver of increased living standards in Vanuatu and other Pacific Island Countries. However, while out-migration has increased household incomes, such processes are affected by external shocks (such as climate change). Due to COVID-19, seasonal worker schemes have been affected, causing broad-based and significant reductions in employment and earnings.

[12] World Bank Group. (2017) Systematic Country Diagnostic for the Eight Small Pacific Island Countries: Priorities for Ending Poverty and Boosting Shared Prosperity, available at: https://openknowledge.worldbank.org/handle/10986/23952

[13] World Bank (2017). Systematic Country Diagnostic: PIC8. Ibid.

[14] The Pacific Community (SPC) – see more details on https://sdd.spc.int/vu.

15. **Vulnerability analysis:** Vanuatu's climate profile is closely linked with, and is one of the important determinants of the performance of socioeconomic and political indicators in the country.

16. More than 10% of Vanuatu's population lives in extreme poverty, compared to the regional average of 3% for the Pacific. Vanuatu, the Federated States of Micronesia, Kiribati and the Marshall Islands collectively host over 90% of people in poverty in the southern Pacific. Annex 2 – the Feasibility Study and Annex 8 – the Gender Assessment and Gender Action Plan explore poverty, hardship and other indicators in greater detail.

17. Despite producing insignificant greenhouse gas emissions that cause climate change, SIDS populations (including in Vanuatu) are on the frontline of climate change impacts, including having a high burden of climate-sensitive diseases. Child mortality, for example, has fallen in Vanuatu, but remains high compared to other Pacific countries. Poor water security and WASH infrastructure is one of the leading contributors to enteric and other infectious disease burdens, which remains high in Vanuatu.

18. Vulnerabilities, in terms of limited climate resilience and water insecurity also encompass gendered aspects – explored in detail in Annex 8, and summarized below:
    - Different gender groups, particularly women and girls, have gendered needs, which remain underserved by existing WASH infrastructure and are particularly impacted during humanitarian emergencies;
    - Women continue to be underrepresented in water security and climate adaptation-related decision-making bodies, despite registering better performance when represented;
    - Insufficient, non-resilient WASH infrastructure continue to be compromised by climate risks, increasing time poverty among certain gender groups (women and children primarily collect water for domestic usage); and,
    - Since water is central to well-being, water insecurity indirectly and directly combines to exacerbate gender inequality in other sectors (such as limited land rights) and limit adoption of adaptation and resilience strategies.

19. This socioeconomic context informs the **primary challenge for water safety and security in Vanuatu**, which is: while access to a proximate source of drinking water is high (94% access to an improved drinking water source and 86% access on the premises)[15], the safety and reliability of the water consumed by most people in Vanuatu is not well known. This has short- and long-term health impacts on the indigenous ni-Vanuatu: unsafe drinking water can lead to diarrhoea and other water-borne sicknesses, while (in the longer-term) inhibiting the ability of the body to absorb nutrients and contributing to chronic undernutrition. Climate change is a threat multiplier in this context.

20. **In terms of sanitation**, the household living conditions analysis of the 2020 Census reveals that outside urban centres and provincial hubs (where flush toilets are available), people rely on shared or private pit latrines or have no toilets at all. This is particularly true in Penama where between 83–100% of Ni-Vanuatu living in Pentecost, Maewo and Ambae rely on these types of sanitation infrastructure.[16] Water rationing, both for household use and sanitation, and water shortages are common during the dry season within families. The DoWR Water Resources Inventory (WRI) indicates that many of the water sources are not available or have inadequate yields during the dry season. Meanwhile, rainfall patterns have been affected by climate change, and can often manifest as intense periods of extreme rainfall, leading to floods that will further exacerbate the WASH baseline context.

    The following sub-sections detail the observed and projected climate change in Vanuatu, and how these drive risks and impacts on the WASH sector.

21. **Observed historical trends and climate change**: Vanuatu's climate varies from wet tropical in the north to subtropical in the south. From May through September, south easterly winds support fine sunny days and cooler nights. November to April is the wet season with higher temperatures, heavy rain and occasional cyclones. The wettest months are from January through March.

[15] UNICEF (2020). Rural water supplies in Vanuatu in need of significant improvements. (WASH Technical Paper/13/2020).

[16] Vanuatu National Statistical Office; World Bank. 2014. Vanuatu: Socio-Economic Atlas, available at: https://openknowledge.worldbank.org/handle/10986/18669

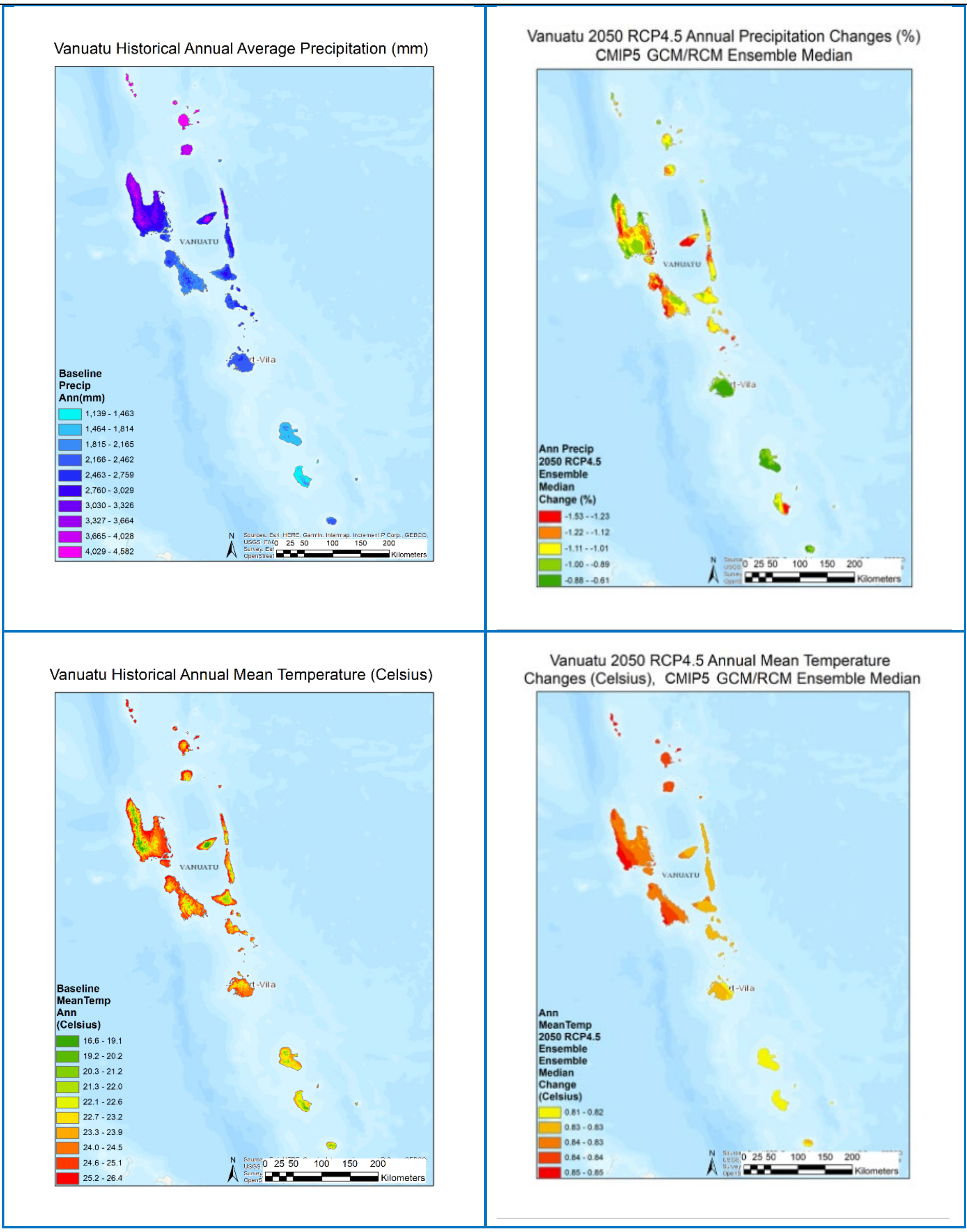


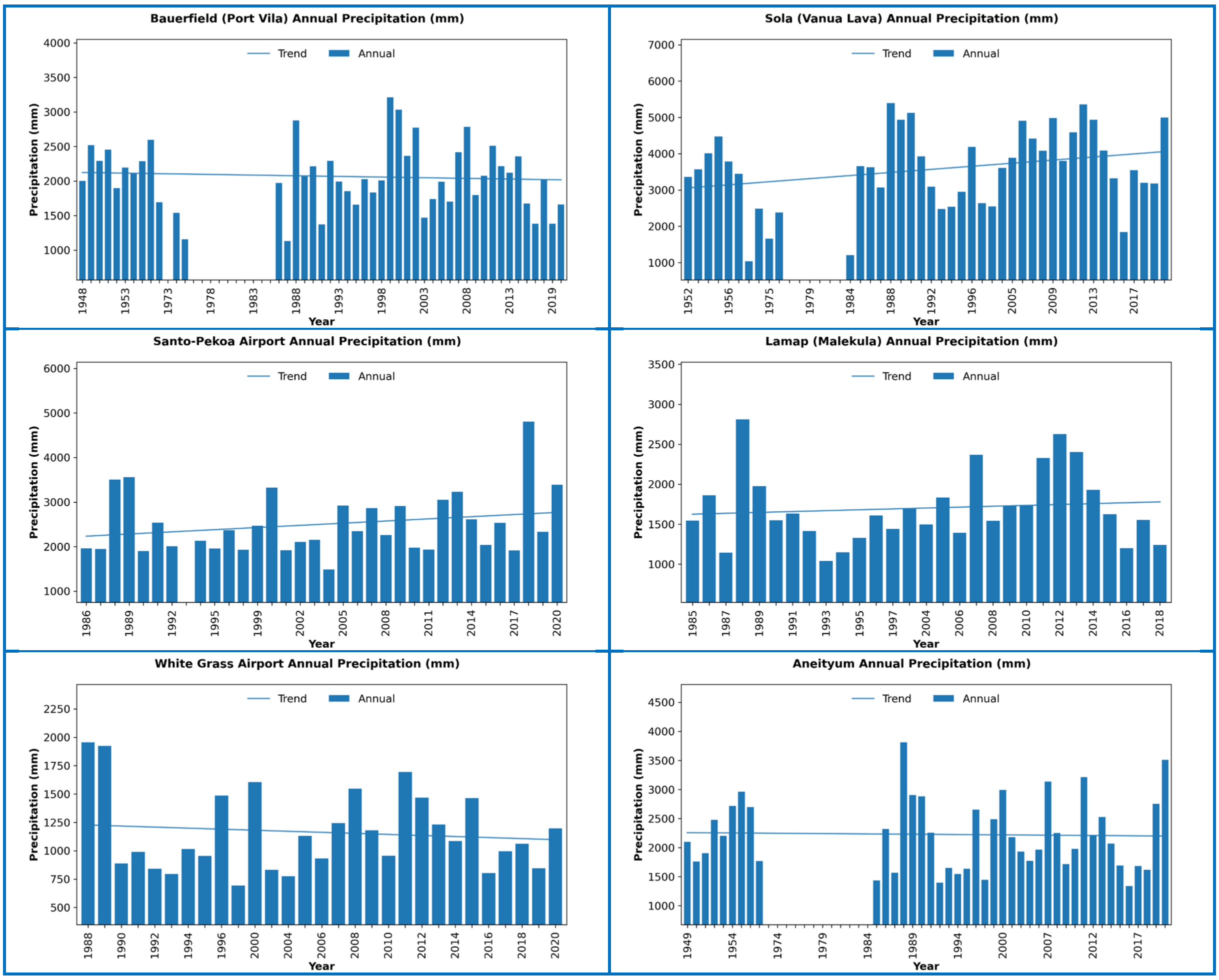
Figure 1: Sample maps for the historical (1981-2010 average) and RCP4.5 2050 annual precipitation and mean temperature for Vanuatu. The baseline annual precipitation and temperature were based on the adjusted WorldCLIM2 data, and RCP4.5 changes were based on the CMIP5 GCM/RCM ensemble median data. (Data and map source: authors)

Temperature – Average temperatures range in Port Vila from 27°C in February to 22°C in July. Temperatures have been rising in the at approximately 0.1°C per decade since the 1970s. Up to the 1990s, there was limited warming in the region, but from 1995 – warming noticeably accelerated, and temperatures between 2014–2018, were averaging around 0.5–0.6°C above the long-term average.

Table 1: Historical temperature trends by study site (Celsius)

Precipitation – Rainfall is also affected by latitude and altitude in the islands. The northern higher islands in the Banks and Torres groups receive an annual average of 4,000 mm rainfall, while the southern and lower islands may receive only half of such figures, showing regional disparity in the water sources available. The World Bank Climate Change Knowledge Portal states that a study has pointed to significant natural multi-decadal rainfall variability in the South Pacific Convergence Zone (which Vanuatu is situated within). Observing records over 400 years, it shows abrupt changes of ~1,800 mm can occur between wet seasons. However, no changes in rainfall patterns significantly outside the range of normal inter-annual variation have been documented and linked to human-induced climate changes.

Table 2: Historical annual precipitation changes (mm)



Extreme winds and Tropical Cyclones – November to April is also the tropical cyclone season, which impacts the water sector extensively. The geographical location of the archipelago in the southwest Pacific means that tropical cyclones (TCs) occasionally traverse the country with wind speeds of at least 62 km/hr or more. According to the Vanuatu Meteorological and Geo-Hazard Department statistics, the area of Vanuatu (land and sea) receives about 2-3 cyclones per season. The most significant frequency of these events is in January and February. On average, Vanuatu, and its marine economic zone experience 20–30 cyclones per decade, between categories three and five causing severe damage. TCs can affect any island of Vanuatu, with several impacts: extreme winds heavy rainfall, flash flooding, flooding of low-lying areas, coastal flooding, riverine flooding, storm surge, landslides, and very rough seas. These events regularly cause damage to life, infrastructure and public goods, as well as property in the islands – and also have direct and indirect impacts on water security and WASH infrastructure in the country. Since 2015 two category 5 (>250km/hr) Cyclones have directly passed over Vanuatu causing extensive damages:

- Cyclone Harold, 2020 160 000 people, more than half of the total population, required assistance. On certain islands this percentage raised as high as 90% of the population being impacted, e.g. Sanma province. Economic costs are still ongoing for the disaster with COVID19 pandemic contributing to slower response times and impacting response funding availability.
- Cyclone Pam 2015, displacing about 65 000 people and resulting in USD 600 million in economic losses, the equivalent of about 65 percent of the country's Gross Domestic Product

Drought – In Vanuatu, drought is frequently associated with El Niño Phases of the ENSO cycle, that causes a warm pool shift towards the eastern Pacific. The result is increased high pressure systems over the south Pacific, reducing rainfall levels and inducing meteorological drought. Whilst there is uncertainty on the predicted trends of El Nino occurrence, events have been stronger and more aggressive in the last 50 years. Four out of five extreme events since 1901 having occurred since 1970. This was most recently exemplified in the 2015/16 El Nino event that caused severe drought in Vanuatu which compounded negative impacts of the aftermath of Cyclone Pam that devastated infrastructure across the country[17]. The drought caused severe water security impacts resulting in increased soil aridity and crop loss, reducing food security. It was reported that the majority of stable crops (yams and taro) failed, forcing 90,000 people (about 30% of the total population) to rely on food aid[18].

Sea-level rise – According to the Pacific Climate Change Science Program[19], as ocean water warms, it expands, causing the sea level to rise. The melting of glaciers and ice sheets also contributes to sea-level rise. Instruments mounted on satellites and tide gauges are used to measure sea level. Satellite data indicate the sea level has risen near Vanuatu by about 6 mm per year since 1993. This is larger than the global average of 2.8–3.6 mm per year. This higher rate of rise may be partly related to natural fluctuations that take place year to year or decade to decade caused by phenomena such as the El Niño-Southern Oscillation.

22. **<u>Projected climate change</u>:** Vanuatu was ranked as being at the highest risk level in the 2019 World Risk Index for disaster exposure and has consistently featured among the top 10 most climate-impacted countries in the world.[20] For project preparation, therefore, it was crucial to conduct a climate modelling study that would identify these trends, and ensure that these are mainstreamed into the design and delivery of the project paradigm.

For this climate study (which is included in Annex 2 – Feasibility Study), historical data were analyzed based on six weather stations with reasonable data quality and quantity in Vanuatu. Climate change analysis was carried out for baselines and the year 2050, under RCP4.5 and RCP8.5 scenarios, applying CMIP5 22 CORDEX AUS Domain RCM and 40 GCMs ensemble approaches and other historical climate data sources, with outputs for the median, 5th and 95th percentiles. Due to the timing of the project's preparation phase, CMIP5 was the most current iteration of global modelling. All results are known to be consistent with CMIP6 scenarios and the findings of IPCC AR6. Ten populated locations in Vanuatu were selected for climate change scenario analysis: Port Vila, Luganville, Norsup, Port-Olry, Isangel, Sola, Lakatoro, Melsisi, Lamap, and Aneghowhat, indicated below:

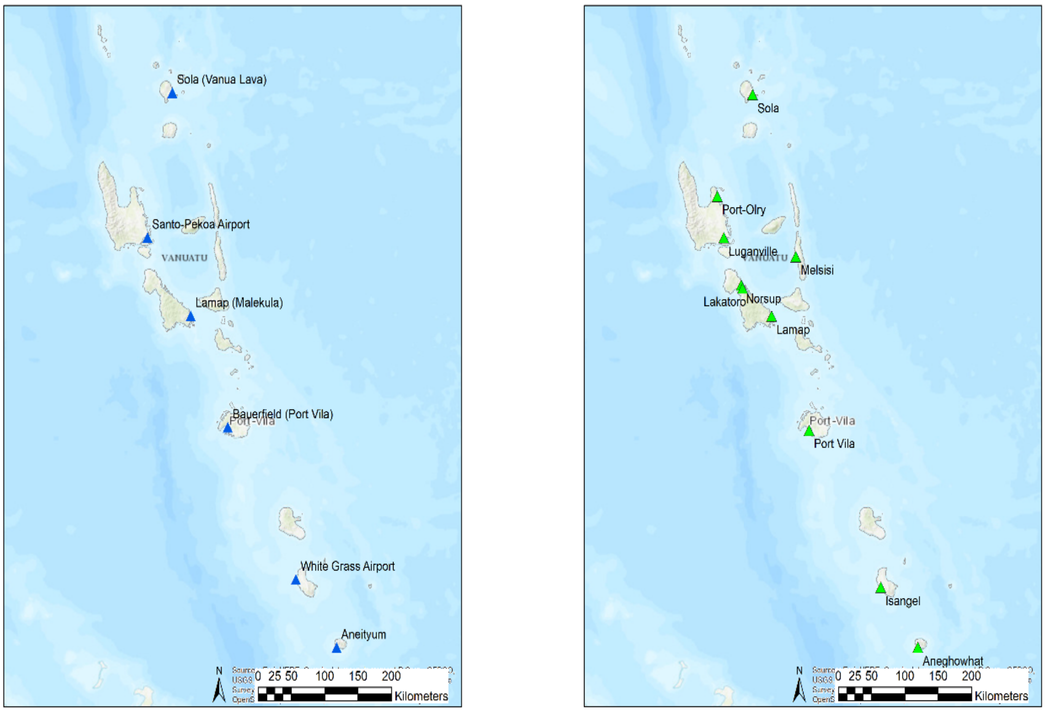
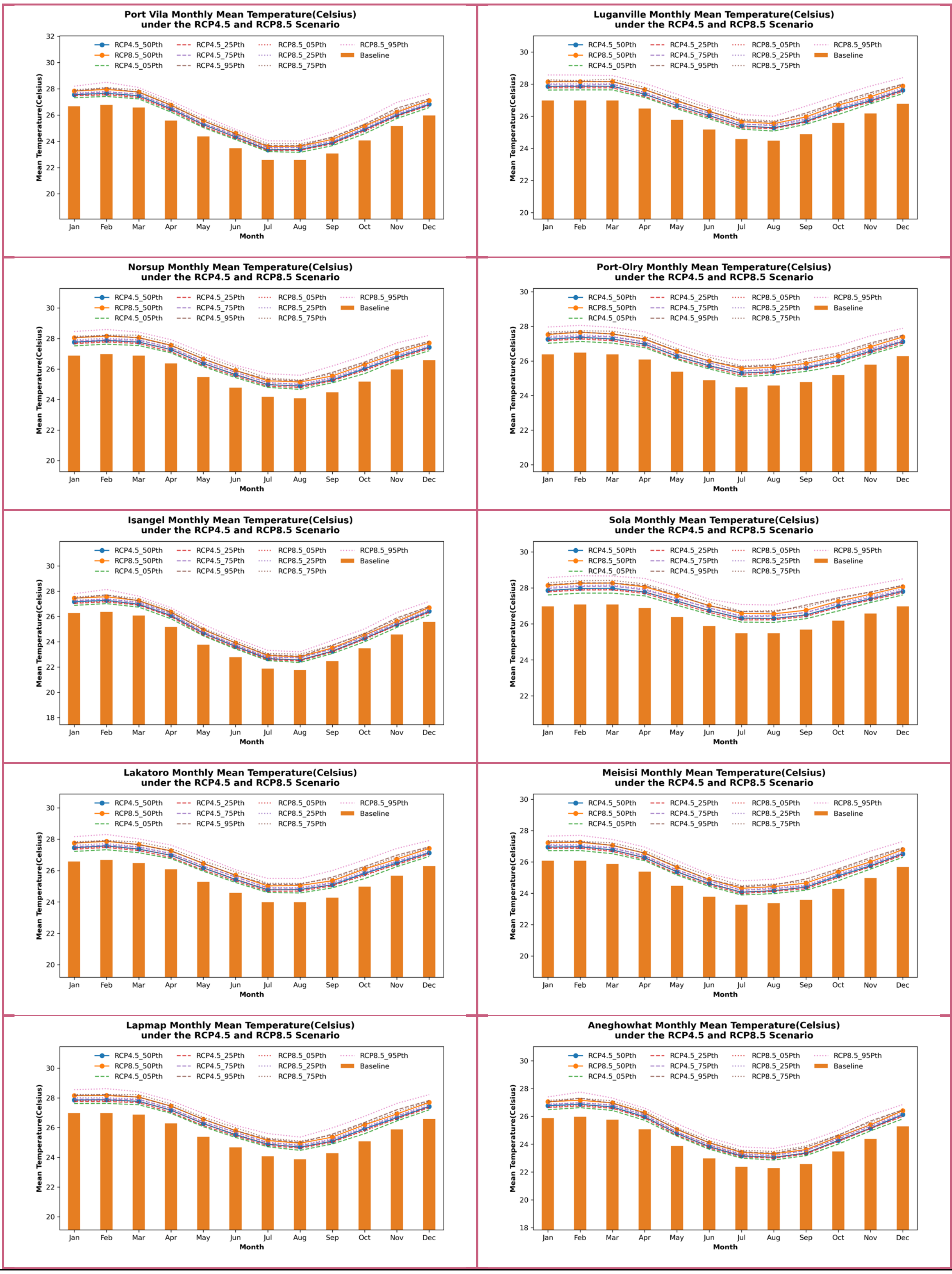


Figure 2: Selected sites for climate change scenario analysis in Vanuatu

[17] https://www.sciencedaily.com/releases/2019/10/191021153346.htm
[18] Savage A, Bambrick H, Gallegos D (2021) Climate extremes constrain agency and long-term health: a qualitative case study in a Pacific Small Island developing state. Weather Climate Extrem 31:100293. https://doi.org/10.1016/j.wace.2020.100293
[19] See: https://www.pacificclimatechangescience.org/wp-content/uploads/2013/06/15_PCCSP_Vanuatu_8pp.pdf
[20] World Risk Report 2021, see: https://reliefweb.int/sites/reliefweb.int/files/resources/2021-world-risk-report.pdf

23. **Temperature –** The climate modelling study revealed that climate change has manifested in the historical observational data, especially in the temperature data with an increasing trend. Projections for mean temperature showed increasing trends in all the selected stations. The projections indicate an approximate increase in temperature by 1°C across the locations under RCP 4.5 against the baseline.

Table 3: Baseline and future scenarios of monthly mean temperature by populated area (Celsius)



24. **Precipitation –** data shows no significant predicted change in the trend of annual rainfall. Percentage change in total annual rainfall is estimated at 0% with a variability of -12% to 14% under 2050 RCP4.5 projections, relative to a 1986-2005 baseline. Despite this, variability of individual rainfall events continues to be considerable across single seasons and years. Consequently, extreme precipitation events are expected to increase. Daily extreme precipitation was calculated based on the six meteorological observation stations data and average recurrence intervals (ARI) and projected for 2, 5, 10, 25, 50, 100, 200, 500 years. The extreme daily precipitation for all stations is predicted to increase at a steady rate and is deemed to be significant against the baseline. On average, across the country, daily extreme rainfall events are expected to produce an average of 6.5% more precipitation by 2050 under RCP4.5 projections. However, 95th percentile estimates indicate precipitation on any given event could be as high as 77.5% higher than baseline observations, indicating individual events over a season or year could cause dramatically higher precipitation events, causing major impacts as listed below. This is also correlated to predicted increase in Tropical Cyclone (TC) intensity and rainfall (see paragraph 26). For more detailed breakdown of the data please see Table 7 of Annex 2.

25. **Extreme wind –** wind speed in all the Vanuatu locations is very high, and under climate change scenarios, the median change indicates **increase in intensity**. Based on the GAR15 data, the 100-year return wind speed can range across the country of 165.5–175.8 km/h, and 1,000-year return is around 240–260 km/h, which are in the range of category five (252 km/h) of the Saffir-Simpson scale. Under climate change scenarios, the extreme wind speed is projected to increase between 1 and 5% in the ensemble median but reach more than 10% in the 95th percentile. This is also associated with a greater predicted intensity of TCs. Tables 16–18 in Annex 2 elaborates on the outputs regarding wind speeds from the climate data analysis.

26. **Tropical Cyclones –** the occurrence of and predictability of TCs is very hard to predict under current models due to their formation complexity and their influence by climatic cycles such as El Nino Southern Oscillation phases. TC tracking is also unpredictable, and it is not possible to accurately predict (at present) trends in paths of TC that will cross a given country. However, due to their size and speed of travel regional analysis is applicable to predicting overall trends in TCs. Analysis at the regional level indicates that there is:

- High confidence in the **increase in average TC intensity** (category 4-5). This correlates with the information presented in paragraph 25 that indicates that extreme wind speeds are predicted to increase (AR6 SPM B.2.4).
- Medium to high confidence in an increase in TC rainfall rates. This correlates with the information presented in paragraph 24 that indicates that extreme precipitation events are predicted to increase.
- High confidence that TC frequency will decrease over the coming century but that there is low confidence that Category 4 and 5 events will change. This correlates with Figure 3 that indicates that TC crossing rates for Vanuatu are decreasing.
- High confidence that sea level rise will increase TC-related storm surge events.
- In summary, although TC events are predicted to decline, it has been observed that the global proportion of major (Category 3-5) TC occurrence has increased over the last four decades (AR6 SPM A.3.4). TCs will likely generate greater windspeeds and precipitation causing greater risk of loss and damages and associated impacts as listed below[21] [22].

[21] https://doi.org/10.1007/978-3-030-32878-8_6
[22] IPCC, 2021: Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [Masson-Delmotte, V., P. Zhai, A. Pirani, S.L. Connors, C. Péan, S. Berger, N. Caud, Y. Chen, L. Goldfarb, M.I. Gomis, M. Huang, K. Leitzell, E. Lonnoy, J.B.R. Matthews, T.K. Maycock, T. Waterfield, O. Yelekçi, R. Yu, and B. Zhou (eds.)]. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, http://doi.org/10.1017/9781009157896

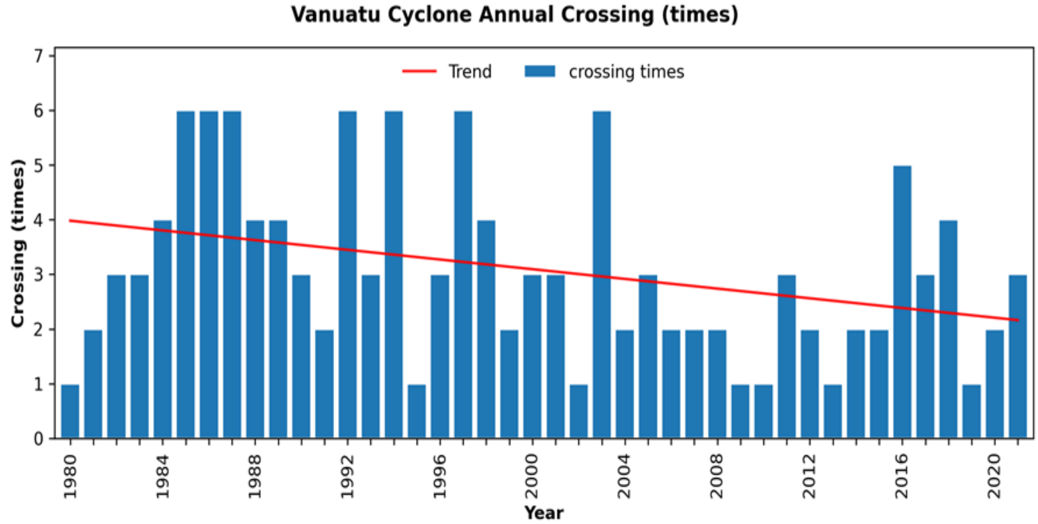


Figure 3: IBTrACS TC best track data for TC crossing Vanuatu

27. Drought – Drought severity and duration probability based on the Standardized Precipitation Evapotranspiration Index (SPEI) are projected to have a similar increasing trend for all the locations across the country under RCP4.5 and RCP8.5 2050. The increase of drought is caused by a slight decrease in precipitation and an increase of temperature. The results indicated that the probability of extreme drought increase in all the locations. Especially, when coinciding with El Nino phases of ENSO cycles. For detailed figures please see page 49 of Annex 2.

28. Mean sea-level rise – An increase in sea level rise over the next 30 years is projected. By 2050, an increase of 40–60 cm in mean sea level is projected under CMIP5 RCP4.5 and RCP8.5 scenarios with only slight variation from location to location[23]. In the coastal areas, where Vanuatu's population is concentrated, high-water levels in the historical period for 50-, 100-, 200-, 300-year return period are presented. The 100-year ARI extreme water level for the ten locations is around 0.7–1.0 m. By 2050, the extreme still high-water level could reach 1.4–1.8 m. Tables 29–30 in Annex 2 elaborates on the outputs about sea-level rise and coastal extreme water level analysis from climate data analysis.

Table 4: Projections of mean sea-level rise increase in Vanuatu by 2050 (cm)

Name	RCP4.5 5th percentile	RCP4.5 50th percentile	RCP4.5 95th percentile	RCP8.5 05th percentile	RCP8.5 50th percentile	RCP8.5 95th percentile
Port Vila	45.7	**54.2**	62.9	49.0	**57.9**	67.2
Luganville	40.6	**49.2**	57.9	43.9	**52.8**	62.2
Norsup	42.3	**50.8**	59.5	45.6	**54.5**	63.9
Port-Olry	39.3	**47.8**	56.5	42.6	**51.5**	60.8
Isangel	38.9	**47.4**	56.0	42.2	**51.0**	60.3
Sola	37.7	**46.2**	54.8	41.0	**49.8**	59.1
Lakatoro	42.4	**51.0**	59.7	45.7	**54.6**	64.0
Melsisi	42.5	**51.1**	59.8	45.8	**54.7**	64.1
Lamap	43.7	**52.1**	60.7	46.9	**55.8**	65.1
Aneghowhat	34.7	**43.2**	51.8	38.0	**46.8**	56.1

29. <u>Risks and impacts</u>: Water remains the main channel through which climate change will be felt in Vanuatu and remains the key sector for developing successful adaptation strategies. This is because climate change affects all aspects of the water cycle and is currently driving an ongoing intensification of the water cycle in the equatorial Pacific, which is already exposed to elevated occurrences of natural hazards. Climate change impacts on freshwater sources, for example, are evident in Vanuatu, and data analysis shows that climate variables, which can impact water access, supply and security, will continue to increase uncertainty over time.

[23] Earth Grid System Federation data portal. https://esgf-node.llnl.gov/search/cmip5/

30. Table 5 summarizes the findings of the climate modelling study and demonstrates the risks and impacts for the water sector (water sources, infrastructure and WASH) in Vanuatu.

Table 5: Climate and non-climate drivers impacting drinking water and WASH infrastructure in Vanuatu. Numbering in column three relates the individual impacts to the phenomena type and is cross referenced elsewhere in the document.

Phenomena	Type of driver	Risks and impacts for drinking water and WASH infrastructure
Increased air and water temperatures	Climate change driver	T.1. Reductions in catchment water sources due to increase*d* evapotranspiration – which will impact water points and sources that are accessible to communities; T.2. Increase in water treatment costs as quality of inflows decrease, due to lake stratification and fewer destratification (mixing) events; post-bushfire runoff of dissolved materials into receiving waters; changes in turbidity and chemistry of water – these will impact water quality as well as availability, particularly in remote areas; T.3. Increased occurrence of eutrophication and toxic algal blooms in rivers, reservoirs and lakes, due to increased water temperature – these occurrences reduce water quality and increase health burden among communities that cannot access alternative water sources or expensive options such as packaged drinking water; T.4. Reductions in groundwater levels due to accelerated conditions that increase biomass and higher levels evapotranspiration – which, in turn will affect water security and safety, as well as reduce options for water access; and T.5. Increase in operations, maintenance and repair costs for distribution networks as saline intrusion increases – which will particularly affect vulnerable communities that do not have access to disposable household income and/or cannot access government/donor funds to improve existing water systems and WASH infrastructure.
Increased intensity of extreme precipitation events	Climate change driver	P1. Damage caused by extreme winds or extreme flooding events, to WASH infrastructure, can lead to overflows in pit latrines or makeshift structures – which will immediately reduce access, as well as increase the health burden of exposed communities (and further affect marginalized groups within such communities); P2. Reduction in holding capacity of water sources due to increased overland flows, soil erosion and sedimentation as well as contamination of freshwater sources – this will impact water availability as well as quality, particularly in remote islands; P3. Inflows and recharge of groundwater as well as freshwater sources will have increasing levels of suspended solids and other contaminants – which will drive up the cost of purification to reach potability, as well as increase the costs for operations and maintenance of treatment plants (as well as create demand for establishing such systems); P4. Increase to damage costs to low-lying water infrastructure exposed to inundation hazards – which will drive up the costs for repair, operations and maintenance in the long-term, while having immediate effects such as reduced access and availability of potable water; P5. Increase in costs to maintain and repair distribution networks due to increasing erosion hazards caused by overland flows – which will drive up the costs for repair, operations and maintenance in the long-term, while having immediate effects such as reduced access and availability of potable water; and, P6. Decrease in groundwater recharge, as heavy precipitation can exceed soil infiltration capacity and, instead, increases surface runoff – which will necessitate the establishment of purification technologies that will be energy-intensive and difficult to scale up in rural contexts.
Extreme wind speed, tropical cyclones	Natural climatic driver (as Vanuatu is in the cyclone belt)	E1. Intense cyclones can bring torrential rains and have embedded severe thunderstorms that are accompanied by tornadoes – causing wind damage to limited WASH infrastructure and housing options; E2. TCs often damage precarious water points and infrastructure, particularly if these are not secured or underground – due to TC Harold

	Climate change driver (increasing the frequency and intensity of TCs)	in April 2020, WASH situation in Santo, Pentecost and Ambrym was degraded[24]; E3. Wind-borne debris often initiates or enhances damage to structures, which are then worsened by extreme precipitation – this can particularly affect rainwater harvesting infrastructure, particularly system components (gutters, downspouts, screens) that are usually exposed; E4. Heavy rainfall and subsequent inland flooding can also induce extensive damage and, in coastal areas, cause coastal floods and beach erosion – which can impact water sources nearer to the sea, cause increased salination and impact upon both water access and health burdens of dependent communities; and, E5. Even with climate-proofing, it is essential to note that some extreme events may overwhelm the best intentions in engineering for climate resilience – therefore, disaster preparedness must coincide with adaptation planning and infrastructure design[25], particularly in the context of a fragile WASH baseline in Vanuatu. This holds true for planning WASH infrastructure in emergency shelters as well – which often have to be relied on due to Vanuatu's increased exposure to TCs and other natural and climate-induced hazards.
Drought and decreased precipitation volumes, seasonal patterns changes	ENSO Oscillation – natural climatic driver Climate change driver	D1. Reduced streamflow, reduction in groundwater aquifers and increased pollution concentrations – which, in turn, impugns availability and quality of water supply for natural resource-dependent communities (primarily the rural population) of Vanuatu; D2. Increase in penetration of saline waters into estuaries and aquifers during drought season, causing imbalance between freshwater and saltwater – which may reduce water supply and quality available to communities as well as expose them to health hazards caused by overconsumption of sodium; D3. Increase in operations, maintenance, and repair costs for water infrastructure due to lower quality water – this will particularly affect systems that are vulnerable or require upgrade in rural and remote areas of the country, driving up the costs of water production for potable use. Water service providers may need to build greater numbers of capital-intensive rain-independent desalination and recycled water plants to provide adequate water *security for coastal populations.*
Sea-level rise	Climate change driver	S1. Saltwater intrusion and increased salinity of groundwater aquifers and distribution networks – this will lead to reduction in availability and quality of water supply (access to freshwater source points will reduce, and existing options may not remain potable for consumption), which will have direct impacts on water security as well as health burdens of communities; S2. Saltwater intrusion and increased salinity of groundwater will have direct impacts on drinking water and WASH infrastructure – this is by increasing costs and demand of desalination systems and distribution networks, which may be energy-inefficient and cumbersome in rural and remote communities (thus lowering overall water security); further, sea-level rise can be expected to physically damage both drinking and WASH infrastructure, and introduce yet another climate variable that would require mainstreaming in design processes. S3. Inundation of land due to raised seawater levels will also – increase the maintenance, operation, and repair costs of WASH infrastructure as exposure will increase. When this increased need cannot be met, communities will be exposed, and water insecurity as well as health burden from degraded WASH infrastructure will increase.
Deforestation	Anthropogenic driver	DF1. Deforestation can reduce communities' access to clean drinking water, by decreasing water yield.

[24] Government of Vanuatu – WASH Cluster (July 2020). See lessons learnt report here: https://reliefweb.int/report/vanuatu/tropical-cyclone-harold-covid-19-wash-cluster-lessons-learnt-july-2020
[25] The Water Supply and Sanitation Collaborative Council (WSSCC) demonstrates the need to incorporate disaster scenarios in planning of infrastructure and institutional elements of water supply and sanitation: https://www.preventionweb.net/files/8655_RefNoteWASHDisastersFebruary09.pdf

		DF2. Runoff and sediment deposition can increase the costs of water treatment, as well as pollute freshwater and groundwater sources, reducing water points and access.
Water pollution (direct) and other environmental pollution	Anthropogenic driver	WP1. Water pollution occurs when harmful substances—often chemicals or microorganisms—contaminate a stream, river, lake, ocean, aquifer, or other body of water, degrading water quality and rendering it toxic to humans or the environment. WP2. Water pollution can happen through point source pollution – wastewater (also called effluent) discharged legally or illegally by a manufacturer, oil refinery, or wastewater treatment facility, as well as contamination from leaking septic systems, chemical and oil spills, and illegal dumping. WP3. Other sources of environmental pollution can also find its way into water outlets, streams and water bodies. Particularly plastic and other anthropogenic waste, due to the lack of waste disposal and sustainable practices. Waste can often find its way into water sources, thereby reducing water safety and security in the long-term. WP4. This is because water is uniquely vulnerable to pollution. Known as a "universal solvent," water is able to dissolve more substances than any other liquid on earth.

31. The project will strongly contribute to addressing climate-related risks and impacts on drinking water and WASH infrastructure, while **supporting the implementation of Vanuatu's national priorities**. Indeed, climate-resilient WASH infrastructure has been identified as a priority intervention in Vanuatu's GCF Country Programme[26]. The Climate Change and Disaster Risk Reduction Policy 2016–2030, the National Adaptation Programme of Action (NAPA) and the Nationally Determined Contribution (NDC) also identified water resource management as a priority need for Vanuatu.

32. **Complementarity and coherence:** The table below summarises the main past, on-going and projects under development in Vanuatu which focus on water security and climate resilience. Synergies and complementarities are highlighted and more information on projects is provided in Section 11 of Annex 2 – Feasibility Study.

Table 6: Synergies and complementarities with other water-related projects and interventions

Project /Programme	Additionality / Complementarity with the proposal
Standardization of WASH in Emergency Information, Education and Communication (IEC) Material (UNICEF) 2015 - 2016[27] Emergency IEC materials were developed and delivered through training to Vanuatu stakeholders, to address critical life-saving safe water, sanitation and hygiene behaviors in an emergency context.	The GCF proposal will build on this previous experience to scale up the impact: IEC materials will be used in trainings where relevant to enhance adaptative capacity on WASH in case of emergency.
GCF – Climate Information Services for resilient development planning (SPREP) 2018 - 2022[28] The project addresses climate information gaps to develop and deliver climate information tools and resources, supporting coordination and dissemination of tailored information across five priority sectors.	The current GCF proposal will build on the climate information services to support decision-making processes on climate resilient WASH infrastructure throughout the project.
GCF – Community-based climate resilience project (Save the Children) 2022 - 2028[29] The project provides vulnerable remote and rural communities with access to climate information and early warning. It also integrates climate risks into community planning processes and build adaptive governance systems at the local level.	This proposal will expand on the initial results of the project, in particular on local adaptation plans which will identify key local issues that drive climate vulnerability in Vanuatu. It will also benefit from the implementation of nature-based solutions which will enhance the resilience of natural resources and improve water security.

[26] Vanuatu's GCF Country Programme is available here: https://www.greenclimate.fund/sites/default/files/document/country-programme-vanuatu.pdf
[27] https://www.nab.vu/project/standardization-wash-emergency-iec-material
[28] https://www.greenclimate.fund/project/fp035
[29] https://www.greenclimate.fund/sites/default/files/document/gcf-b32-02-add01.pdf

GEF – Enhancing water-food security and climate resilience in volcanic island countries of the Pacific (FAO) [*under development*][30] The objective of the proposal is to: - Expand and assess the role of groundwater resources - Introduce sound groundwater governance frameworks - Tackle hot-spots with on-the-ground-interventions - Reinforce institutional capacity in groundwater assessment, management and monitoring	Lessons learned on groundwater resources use and governance will be shared with FAO. Opportunities will be explored to amplify the benefits of the GCF proposal as a parallel investment for this GEF proposal under development.
GCF - Malekula Water Supply Project: Increase Resilience of Vulnerable and Marginalized Communities through Integrated Water Resource Management and Ecosystem-based Interventions (UNEP) *[under development]*[31] The proposal intends to develop climate-resilient water management plans for Malekula, provide climate-resilient water supply systems and catchment management in Malekula.	The UNEP project is specifically targeting Malekula island through ecosystem-based adaptation interventions to enhance ecological provisioning and regulatory services. These activities compliment the proposed project. Investments for water infrastructure made available by this project will enable communities to access greater water provision provided by UNEP's proposal. As such, GCF resources will not be invested in duplicated activities or impacts on the ground. This programming is seen as synergistic and complimentary to providing robust climate resilient water access on Malekula and in line with Vanuatu's wider water strategies. Lessons learned from this GCF proposal will be shared to ensure further complementarity with the UNEP proposal and aim to ensure maximum impact and effectiveness to induce a paradigm shift.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

33. This section presents the **barrier analysis** and the **project's proposed theory of change** to address these barriers.

Barrier analysis

34. Barriers for this project have been identified through a review of primary and secondary sources, as well as through a DoWR-led stakeholder engagement process. Primary sources include Vanuatu's draft GCF Country Programme, the Vanuatu Water Policy 2017–2030, and in-country consultations with water management experts at the national and provincial level. Consultations were conducted during the concept note stage, as well as during the full project preparation stage.

35. The secondary review included an extensive desktop study, which collated documents from different climate sector and development sector actors, focusing on climate change issues in the Pacific SIDS, water security and Climate-Resilient WASH (CR-WASH), and Vanuatu's socioeconomic and gender baseline. Additionally, post-disaster assessments on Cyclone Pam (2015) and Cyclone Harold (2020) were reviewed to assess climate impacts on Vanuatu's water infrastructure, with particular focus on available localised studies on Sanma, Torba and Tanna. The main barriers to climate change adaptation in the water sector in Vanuatu include:

36. **Barrier 1 – Community awareness and capacity – this is limited at community levels due to a lack of backstopping and outreach from extension agents to adequately design climate resilient DWSSPs and designs:** Resilience at the community level is key to ensuring beneficiaries receive sustainable water and sanitation services that can adapt to external shocks and processes of change, particularly climate change. The national/sub-national-level processes support coordination and service provision to the WASH sector, but it is the infrastructure and the capacity of communities to manage their system that can either deliver the needed health impacts and increased water security in Vanuatu or prove to be a hurdle in achieving the project goals.

Capacity barriers for climate-resilient WASH in Vanuatu, at local and community levels, have two features:

[30] Full project proposal under development, concept approved in 2020: https://www.thegef.org/projects-operations/projects/10712
[31] Concept note submitted in 2019: https://www.greenclimate.fund/sites/default/files/document/22710-malekula-water-supply-project-increase-resilience-vulnerable-and-marginalized-communities.pdf

- inadequate access to evidence-based planning among communities for climate-resilient water management; and,
- limited awareness, knowledge, and skills on climate change issues and climate-resilient water resource management at the community level.

Stakeholder consultations highlighted these observations: Red Cross Vanuatu (RCV) identified limited capacity to provide backstopping at provincial offices and community levels prior, during, and after the implementation of project activities, as a key gap. World Vision (WV), similarly, identifies that at the community and operator level, training and project intervention need to support design capacities, as well as be deployed quickly and efficiently.

Outcome 1 *will address this barrier 1 – section B.3 below provides relevant details.*

37. **Barrier 2 – Limited market signals and insufficient CR-WASH Infrastructure – Water is generally perceived as a public good in rural and remote settings in rural Vanuatu, and existing WASH infrastructure is lacking climate resilience in designs at local and community level:** In rural settings, water is a public good – although climatic and environmental conditions as well as resilience of infrastructure largely determine access to this essential utility. The DoWR recognizes that, particularly in these rural (often remote) areas, the absence of market signals (such as prices and permits), as well as inadequate planning and incentive structure can impact upon water security, leading to overextraction through the limited water source points currently available. This informed the establishment of the government-owned and community-facing water governance architecture in the form of the DWSSP, which the project will utilise to address climate-related challenges in the sector. Existing water challenges in the sector, particularly limited CR-WASH infrastructure, is being exacerbated by climate change, resulting in increasing temperatures globally and in Vanuatu, changes in precipitation patterns and extreme weather events, leading to excessive flooding and droughts and threatening water and food security, health and the environment.

One of the primary barriers in Vanuatu is limited WASH infrastructure and water supply sources. The DoWR conducted an inventory of water systems, as part of an MFAT-funded Water and Sanitation Sector Strengthening Programme – which found only 79% of systems functional.[32] Shefa Province, which hosts the capital city – Port Vila – had the best functionality at 98%.[33] Meanwhile, Torba, the most remote province in the north, had only 72% of systems functioning.[34] It is also important to note, that even operational systems, did not necessarily have water supply all day, every day. Further analysis revealed that only 37% of systems were reported as providing water 24h/day, every day of the year.[35]

Data collected from rapid assessments of 230 rural water supply systems, as well as inspection of 57 rainwater harvesting systems, following Tropical Cyclone Harold, revealed that –in the immediate aftermath, the only source of clean water for communities were coconuts. With extensive damage to water infrastructure, the WASH Cluster was quickly activated to send in contingency supplies, such as liquid chlorine and chlorine tablets, to stem chances of water-borne diseases. Given this baseline, infrastructure barriers have the following features in Vanuatu:

- insufficient or lack of WASH infrastructure, with sub-national variations;
- vulnerable WASH infrastructure (poorly maintained and not climate-resilient); and,
- no alternative water supplies in case of emergency (shelters are also ill-equipped).

Outcome 2 *will address this barrier – section B.3 provides relevant details.*

38. **Barrier 3 – Design standards – current designs do not account for full range of climate resilient approaches:** A sample of Drinking Water Safety and Security Planning (DWSSP) documents and their water supply designs were reviewed from an engineering perspective as part of the feasibility study,[36] whilst considering climate change and other vulnerabilities to secure water infrastructure.

[32] UNICEF (2020). Rural water supplies in Vanuatu in need of significant improvements. (WASH Technical Paper/13/2020).
[33] Ibid.
[34] Ibid.
[35] Ibid.
[36] During the project preparation stage, a technical review of the rural water infrastructure landscape as well as an analysis of the technical capacities of the DoWR was conducted. This exists as a standalone study and is also extensively incorporated into Annex 2 – the Feasibility Study.

Design standards, procedures, and choices of water systems and their components will provide the greatest amount of resilience against climate change and other vulnerabilities. Due to limitations of available data, finances and/or resources, and skills and experience, some climate-resilient technologies are not suitable for a given community/location. This is already recognised by system designers who focus on implementing simple yet effective solutions such as gravity-fed systems and community rainwater harvesting systems. The reviewed designs imply the need for greater consideration of water resource security hierarchy and threats such as saltwater intrusion.

Although the designs are to standard and typically optimised sufficiently, they do not extend beyond the minimum requirements of water supply for individuals. Nor do they provide backup or fail-safe mechanisms which could maintain water reliability during flood, drought or natural disasters. It was noticed that some designs are downscaled to minimum standards due to cost concerns; this should be avoided. The additional costs of a larger tank or pipe could greatly increase water security now and into the future.
Outcome 2 *will address this barrier – section B.3 provides relevant details.*

39. **Barrier 4 – Technical capacity - human resource capacity and knowledge of technical areas associated with climate-resilient WASH infrastructure is limited at the provincial and national levels: T**he National Sustainable Development Plan (2016–2030)[37], provisions for capacity and technical assessments to ensure a dynamic public sector in Vanuatu. The DoWR is the governing body for water resources, overseeing the provision of safe, secure and sustainable management of community water supplies, protection of water resources, catchment management, and WASH disaster programs. Their staff have been organised in three units: Monitoring and Evaluation, Technical Services, and Projects and Operations supported by an administrative department.

40. In 2016, the DOWR lacked human resource capacity, with a need to expand staff numbers, and only having two of three Section Heads in an acting capacity and with little management experience. Further, women held only administrative positions. This study identified that some staff worked without pay at the start of their employment. In 2018, only 5 of 16 planned recruitments were successfully completed. The DoWR have, demonstrably, experienced reduced staffing, including a period without in-house engineers, or dedicated Project Management Units. This, combined with competing urban water resource tasks, has burdened staff and hampered progress of rural water supply improvements. Civil engineers in NGOs were generally only available for collaborative projects. Indeed, the DoWR does currently have some civil engineers, however, personnel capacity to meet demand is severely lacking.

41. Stakeholder engagement, conducted during the preparation of this Feasibility Study, demonstrated that DoWR staff often undertake duties at a technical level below their terms of reference requirements. Training courses are frequently classroom-based, without direct in-field application to reinforce the absorption of skills, leading to ineffective outcomes. The training survey (self-assessments) conducted in February 2021, indicated the lack of (or under) confidence for a range of skills: basic water engineering skills, technical and project management, data processing and computer skills. The survey returned few responses on climate change, which could indicate a gap in understanding of climate risks interlinkages with water security and WASH infrastructure, generated likely from the well-established status quo of siloed action on these sectors. Water governance in Vanuatu is restricted by technical capacity barriers, amounting to the following features:

- limited technical capacity, human resources on climate-resilient water management at area council, provincial and national level; and,
- insufficient awareness, knowledge and skills on climate change issues and climate-resilient water management at the provincial and national levels.

Outcome 3 *will address this barrier – section B.3 provides relevant details.*

42. **Barrier 5 – Institutional constraints - the Department of Water Resources and associated water governance architecture lacks human resource and corporate/operations service capacity for effective implementation across devolved levels**
The DoWR underwent an institutional assessment primarily carried out through the Office of the Public

[37] https://www.gov.vu/index.php/resources/vanuatu-2030

Service Commission's Framework for Institutional Capacity Assessment and Development (available as Annex C of the Feasibility Study), which was designed to clarify the capacity needs, and assessment and development approach. The assessment has nine rubrics, with several indicators within each (I: Legal Framework, Purpose and Mandate; II: Strategic Plans and Budgets; III: Organisational Function and Design; IV: Operational Systems and Procedures; V: Human Resource Management; VI: Management Processes; VII: Performance; VIII: Workplace Culture and Leadership; IX: Infrastructure). Several indicators were pegged at basic or below moderate levels of capacity, such as: planning within the institution; operational procedures; and service delivery standards. Some indicators were highlighted as requiring immediate capacity development or attention, including staff retention; strengthened management and improved project implementation efficiencies and data accessibility.

The gap in institution building and capacity development, hence, is an important barrier that the project will contribute towards. Particularly:

- inadequate institutional capacities on climate-resilient water management corporate and operations services at the provincial and national level; and
- paucity of robust and accessible data for integrated and climate-resilient water management.

Outcome 3 *will address barrier – section B.3 provides relevant details.*

43. **Barrier 6 – Financial constraints – lack of financing restricts the ability for communities and local public institutions to meet the demand for CR-WASH infrastructure installations at provincial and community levels.**
The World Bank Systematic Country Diagnostic for Vanuatu (which was conducted in 2016 with 7 other Pacific Islands) emphasizes that while public expenditures tend to be high in relation to the size of the economies of these island nations, the absolute size of the public sector is still very small and often lacks the financial and human resources needed to provide adequate public services at national scales. This is supported by the high demand on CAP requests for investment in to DWSSPs for water infrastructure. Linked to Barrier 3 this has induced downscaling of infrastructure works or lack of funding for most appropriate climate resilience. This has led to increased vulnerability as exemplified by the aftermath of Tropical Cyclone Harold. Financial constraints for investment in water infrastructure therefore directly presents a barrier to climate-resilient water security and sanitation.
Outcome 2 *will address this barrier – section B.3 provides relevant details.*

Theory of Change

44. Based on this barrier analysis, a Theory of Change has been developed with outputs geared towards delivering improved water security and water governance with an overall focus on improving the climate-resilience of WASH infrastructure in Vanuatu.

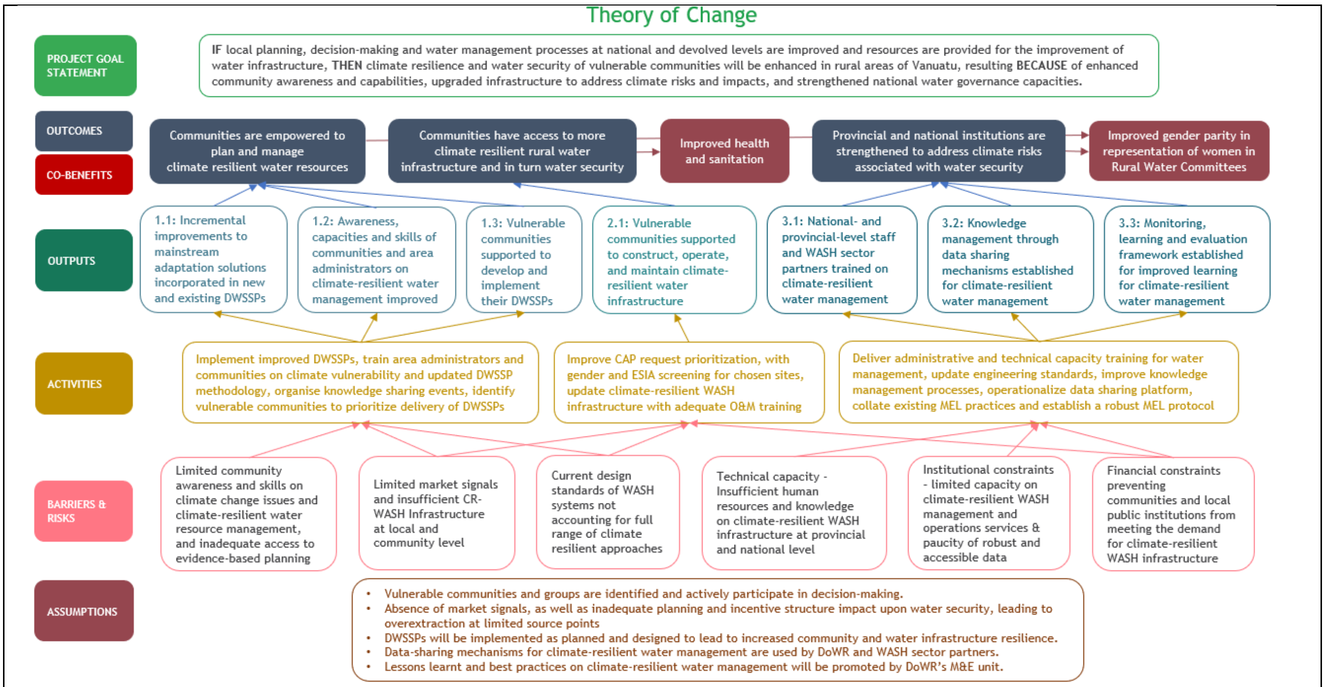


Figure 4: Theory of Change diagram

45. **Figure 4** presents the high-level, relevant GCF Integrated Results Framework indicators and how the project will contribute towards a paradigm shift for climate-transformational investments in the WASH sector of Vanuatu.

46. The project goal statement explains the causal linkages between the project outputs, outcomes and the goal that the project will contribute towards (in this case, increased water security and climate resilience of communities in Vanuatu). The statement states that **IF** local planning, decision-making and water management processes at national and devolved levels are improved and resources are provided for the improvement of water resources, **THEN** climate resilience and water security of vulnerable communities will be enhanced in the rural areas of Vanuatu, resulting **BECAUSE OF** enhanced community awareness and capabilities, upgraded infrastructure to address climate risks and impacts, and strengthened national water governance capacities.

47. The project will be delivered through three Outcomes to address the barriers above:

48. **Outcome 1: Communities are empowered to plan and manage climate resilient water resources**
This outcome will deliver on two aspects of community resilience towards climate change: firstly, through increasing the number of DWSSPs, while also introducing climate adaptation measures in these plans. It will also aid in retrofitting existing DWSSPs with a climate change perspective. Additionally, this Outcome will focus on building the institutional capacity for CR-WASH at national, provincial, area council and community levels.

This Outcome has been designed to create a paradigm-shift towards implementing CR-WASH in a scalable and sustainable way at the community level, building on UNICEF research that this can only be achieved by providing tailored and required assistance to communities. The DoWR Technical Assistance Programme (TAP) component, for example, which is implemented through the DWSSP process, provides a risk-based training method to communities in order to aid them in managing their systems and understanding required infrastructure to meet defined WASH targets. This will be support by the project to include greater focus on climate-resilience. Using these plans to both manage community systems and apply for capital assistance provides both the management capacity and infrastructure – and importantly, ownership – to rural communities. This Outcome will address Barrier 1, as described above.

49. **Outcome 2: Communities have enhanced rural water infrastructure**
This outcome represents the core investment envisioned for a paradigmatic shift towards CR-WASH infrastructure in rural contexts of Vanuatu. The objective of this Outcome is to strengthen water systems in prioritized rural communities (through the DWSSP process) and to address climate variability and

change risks and impacts (through the existing CAP). Investments in rural water supply infrastructure (whether new or improved), such as improved gravity fed systems, will be at once a private good (providing water supply to households), but when used correctly and consistently, also a wider, public good key to achieving SDG 6 targets.

At least 270 prioritized communities, schools or healthcare facilities are expected to be targeted as part of Outcome 2. GCF funding and co-financing will be mobilized to support climate-resilient infrastructures that will be developed, based on needs identified in the DWSSPs and through DoWR's expanded and improved capacity (delivered through Outcome 1). That every drop of water that has to be pumped, moved, or treated to meet health and food needs requires energy. The Oucome, therefore, will aim to deliver technology options that ensures effective co-management of water and energy, to ensure systems are reliable and climate resilient. This Outcome will address barriers 2, 3 and 6, as described above.

50. Outcome 3: Provincial and national institutions are strengthened to address climate risks associated with water security
A key barrier identified in the process of adapting to climate-related water risks in Vanuatu stem from constraints at the institutional level – both provincial and national. Water security is simultaneously impacted by and contributes to climate change – and institutional strengthening is a key aspect in addressing these multifaceted risks effectively. Outcome 3, therefore, focuses on improving provincial and national institutions in Vanuatu by increasing capacity of governance staff and WASH sector partners, while also provisioning for knowledge management, data sharing mechanisms and M&E framework.

The Government of Vanuatu recognizes that essential to the effective delivery of the National Implementation Plan are standardised support tools and processes, the foundation being community-level DWSSPs. Additionally, it also recognizes that essential to efficient delivery of the National Implementation Plan is reaching as many communities as possible by devolving responsibility and support to provincial government, coupled with national oversight and coordination of the many government agencies, implementing partners, and technical support agencies. Keeping this mandate as context, Outcome 3 will work towards gearing institutional capacity at provincial and national level to ensure the effective and efficient delivery of the National Implementation Plan and other related water management processes to manage the adverse effects of climate change. This Outcome will address barriers 4 and 5, as described above.

51. Six hundred communities will be direct beneficiaries as part of Outcome 1 (improved planning and community-based activities on climate-resilient water management), of which 220 will also benefit from improved climate-resilient water infrastructure as part of Outcome 2. 50 communities with existing DWSSPs will be targeted under Outcome 2 during the first year of the project. Further to this, 100 DWSSPs from the current 301 established to date will be updated to follow the new climate resilient methodology in Outcome 1. This makes a total of 750 communities directly benefiting from Outcomes 1 and 2. This will directly benefit 85,650 (including 42,825 women) beneficiaries, that is around 27% of the total population in Vanuatu. Indirect beneficiaries include the entire rural population in Vanuatu (216,980 individuals which is 73% of the total population), mostly through enhanced institutional capacities and processes leading to climate-resilient water security for rural communities (Outcome 3).

52. Table 7 below summarizes the identified climate risks and impacts, and how the Programme's Theory of Change addresses these climate change impacts through adaptation interventions in the three thematic areas.

Table 7: Climate impact mapping again adaptation needs and interventions to achieve desired adaptation benefits. The Impact numbering can be cross-referenced in Table 5.

Climate risk	Impact (table 5)	GCF results area	Adaptation need	Adaptation intervention	Adaptation benefit
Increasing temperature and drought occurrence	T2 and T3	ARA2	Improved awareness of communities on water quality under drought scenarios. Improved infrastructure and water treatment processes built into DWSSPs.	Community knowledge sharing and capacity building events. Training of trainers to support DWSSP designs. *(Activity 1.1.1)*	Climate-resilient DWSSP designs and greater access to finance through CAP

	T5, D3	ARA3	Assess CR-WASH infrastructure designs and update them to adapt to climate issues. Increased data management and knowledge generation support continual improvement of designs. Construction/upgrade of climate-resilient infrastructure.	Assessment of CR-WASH infrastructure. Data management systems enhanced, and data and knowledge disseminated. CAP request support to RWC. *(Activities 1.1.1, 1.2.1, 2.1.1, 2.1.2, 3.2.1 and 3.2.2)*	applications results in drought sensitive CR-WASH infrastructure construction and
	T1, T3 and D1	ARA 1	Improved understanding of drought on surface water quality and ground water tables	Assessment of drought sensitive CR-WASH intervention. Training of extension agents (DWSSP facilitators) on drought risk. Enhanced DWSSP designs *(Activity 1.1.1, 1.2.1 and 1.3.1)*	
Precipitation extremes	P2-6	ARA2	Increased awareness at community level of the dangers of overflows and contamination of water sources and buildup of sedimentation.	Training of trainers to support DWSSP design including watershed management to address soil degradation and sedimentation build up caused by extensive rainfall and increased contamination risk of water sources. Community knowledge sharing, and capacity building events *(Activity 1.1.1 and 3.1.1)*	Communities made aware of contamination risks and DWSSPs account for contamination risk with safe and sanitary CR-WASH infrastructure in place
	P1-5	ARA3	Assess and update CR-WASH infrastructure designs. Increased data management and knowledge generation. Construction/upgrade of climate-resilient infrastructure.	CR-WASH intervention account for flood related issues. DWSSP design processes strengthened CAP request support to RWC. *(Activities 1.1.1, 1.2.1, 2.1.1 and 2.1.2)*	
Extreme events	E4 and E5	ARA1	Extension agents identify communities most vulnerable to the risks of extreme events and include appropriate response measures into DWSSP designs.	Training of extension agents (DWSSP facilitators) to carryout technical assessments of community contexts to identify vulnerability to extreme events and tailor DWSSP designs to the context specific needs of communities. *(Activities 1.1.1 and 1.2.1)*	DWSSPs account for extreme event risk and CR-WASH infrastructure is built to withstand events and be maintained in the long-term.
	E1, E2, E3 and E5	ARA3	Assess and update CR-WASH infrastructure designs. Increased data management and knowledge generation. Construction/upgrade of climate-resilient infrastructure. Increased capacity to maintain and operate CR-WASH infrastructure in long-term. Update CAP risk ranking process with CR-WASH aspects	Assessment of robust CR-WASH interventions Data management systems and dissemination enhanced. Support RWCs to access CAP requests *(Activities 1.1.1, 1.2.1, 2.1.1, 2.1.2, 3.2.1 and 3.2.2)*	
Sea level Rise	S1-2	ARA2	Extension agents identify saltwater risks Support RWCs develop DWSSPs to manage impact.	Training of trainers identify saltwater risk and support DWSSP designs. Community level knowledge sharing, and capacity building *(Activity 1.1.1 and 3.1.1)*	DWSSPs account for saltwater intrusion and other water sources

			Improved community awareness on saltwater impact of se level rise		(desalination infrastructure as a last resort). Infrastructure is in place in appropriate contexts to overcome saltwater contamination.
	S2 and S3	ARA3	Assess and update CR-WASH infrastructure designs. Construction/upgrade of climate-resilient infrastructure	CR-WASH designs account for saltwater intrusion DWSSP design processes strengthened CAP request support to RWC. *(Activities 1.1.1, 1.2.1, 2.1.1 and 2.1.2)*	

53. Finally, the limited female representation within water governance and devolved utility authorities has been analysed in the Gender Assessment and Action plan (Annex 8). As a project co-benefit, increased gender representation is expected through 40% women participation in all registered RWCs by the end of the project. This will be primarily delivered through Outcome 1 (which will establish new DWSSPs and provide policy and climate training). The project will also ensure gender-balanced beneficiary distribution through the integration of gender-specific needs and priorities in the delivery of infrastructure and plumbers, community development and water committee management trainings (Outcome 2) and a gender balance in trainings on climate risks associated with water security (Outcome 3).

54. Improved health and sanitation have also been identified as a co-benefit for this project, particularly through Outcome 2. By strengthening rural water systems, the project will have a positive impact on sanitation and health of rural communities, reducing the risk of vector-, water-borne diseases and malnutrition.

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Outcome number	**GCF Mitigation Results Area (MRA 1-4)**				**GCF Adaptation Results Area (ARA 1-4)**			
	MRA 1 Energy generation and access	**MRA 2** Low-emission transport	**MRA 3** Building, cities, industries, appliances	**MRA 4** Forestry and land use	**ARA 1** Most vulnerable people and communities	**ARA 2** Health, well-being, food and water security	**ARA 3** Infrastructure and built environment	**ARA 4** Ecosystems and ecosystem services
Outcome 1	☐	☐	☐	☐	☒	☒	☐	☐
Outcome 2	☐	☐	☐	☐	☒	☒	☒	☐
Outcome 3	☐	☐	☐	☐	☒	☒	☐	☐

Co-benefit number	**Co-benefit**					
	Environmental	**Social**	**Economic**	**Gender**	**Adaptation**	**Mitigation**
Co-benefit 1	☐	☐	☐	☒	☐	☐
Co-benefit 2	☐	☒	☐	☐	☐	☐

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

55. The adaptation pathway proposed by this project will address the above three impacts through the following, interrelated outcomes:

Outcome 1: Communities are empowered to plan and manage climate-resilient water resources;
Outcome 2: Communities have enhanced climate-resilient rural water infrastructure; and,
Outcome 3: Provincial and national institutions are strengthened to address climate risks associated with water security.

56. This project will contribute to the second paradigm-shifting pathway in GCF's Water Security Sectoral Guide: *strengthening integrated water resources management and water management* and will deliver on all two sub-areas: (i) preservation of existing water resources; and (ii) development of new water supply sources.

57. This **objective** of the project, therefore, is to contribute to the management of water-related impacts from the above observed and projected climate change effects in Vanuatu. This project will address three water-related impacts from climate change in Vanuatu, which are increases in: (i) water-related disasters; (ii) areas suffering from water stress; and (iii) poor water quality-induced fatalities. [38]

58. The project will increase the adaptive capacity of rural communities to better cope with the additional burden of climate change on water security and safety by improving the current design criteria for DWSSPs (Outcome 1) and delivery of WASH infrastructure (through Outcome 2). This will be supported by Outcome 3 that will focus on capacity building of government institutions to enhance delivery of CR-WASH infrastructure. Presented below is a summary description of project Outcomes, outputs and activities. For detailed breakdown of each Outcome and their sub-levels, please refer to Annex 2 Feasibility Study.

59. **Outcome 1: Communities are empowered to plan and manage climate resilient water resources**
Rationale: *see Section B2 Outcome 1 for more detail on the Outcome's rationale.*
***US$***: *6.26 million (GCF US$ 5.64 million, co-finance US$ 617,400)*
Beneficiaries: *700 communities = 79,940 direct beneficiaries (including 39,970women), which is 25.2% of the total population of Vanuatu. These communities are selected at the provincial level based on prioritisation of most vulnerable communities to climate impacts. There are evaluated based on exposure to climate related impacts and current coping capacity of community systems to negative impacts.*
Complementarity with other Outcomes: *The Outcome will establish strengthened technical capacities across DoWR extension structures and within RWCs. This will strengthen integration of climate-resilience directly into DWSSP designs with a focus on adaption for resilience and long-term sustainability. The Outcome mutually reinforces Outcome 2, facilitating increased technical enhancement and knowledge transfer. Increased capacity in these areas enables greater consideration of climate considerations in water infrastructure investments.*
Outcome indicators: *contributes directly to GCF Core Indicators 5 and 6 – see section E for further detail.*

This Outcome has two outputs, one focused on improving the DWSSPs methodology, while the other focuses on improving the design, implementation and management capacities of communities and local governance mechanisms.

60. **Output 1.1: New and existing DWSSPs incorporate incremental improvements to mainstream adaptation solutions**
There is an identified shortfall of design and engineering standards experienced by the DoWR – which combines with the above water-related hazards of climate variability. This proposed output will address this gap by analysing conventional design protocols in the Department and increase climate impact resilience by implementing simple changes to the design criteria, providing staff exposure and experience to enhance climate-resilient designs. See Annex 2 'Feasibility Study' for detailed breakdown of technology design enhancements suggested through project formulation. These will be refined over project implementation.

The DWSSP methodology has been updated recently.[39] Through this project the methodology will be further updated with a specific focus on strengthening the mainstreaming of climate resilience. Annual incremental improvements will be applied to the methodology over implementation to account for experience-based learning in the enhancement process.

In the new DWSSPs developed with communities, the updated methodology will be used to ensure climate risks are mainstreamed as per Output 1.3. For the existing DWSSPs technical assistance will be provided to ensure these plans are retrofitted to better reflect climate risks and adaptation solutions. The importance of this Output stems from the reality that although Vanuatu has seen overseas development assistance (ODA) or ODA-driven investments in the WASH sector, these are often ad-hoc in nature and have not been made in a climate-resilient way or central process updated to have rigorous climate resilient factors incorporated. The updated, existing DWSSPs and the new DWSSPs, will therefore provide a sector leading framework that will encourage and leverage additional investments in the sector.

[38] GCF Water Project Design Guidelines (v 0.98 – 28 September 2021 – developed by Deltaeres).
[39] This was completed before the initiation of the project preparation stage in early 2021.

ACTIVITY	SUB-ACTIVITIES
1.1.1 - Implement DWSSPs processes that incorporate incremental improvements to mainstream adaptation solutions and update existing DWSSPs. Please refer to section 6.3 of Annex 2 Feasibility Study for details on DWSSP status across Vanuatu.	1.1.1.1: Review uptake and delivery of updated methodologies, making incremental improvements annually. Incremental improvements will be made by reviewing and updating the DWSSP template and process on a yearly basis to incorporate climate resilient design standards and design criteria in the risk assessment and improvement plan sections of the DWSSP. There will be small improvements made each year taking into consideration additional data and lessons learnt from implementation of DWSSPs and CR WASH infrastructure during the year. International and local consultants will be recruited to work with a National Project Engineer (NPE) to, carrying out a gap analysis on the current methodology and analysing lessons learned from ongoing DWSSPs. Following the above assessments, the consultants and the NPE will update the methodology to mainstream climate resilience factors into the documentation and ensure all DWSSPs moving forward account for climate risks and impacts. Updated methodology will be presented to the Project Steering Committee (PSC) for approval at regular intervals to enable incorporation of lessons learned over the projects 5 years of implementation. This will ensure that by completion of the project the DoWR and DWSSP methodologies are strengthened based on international best practice and tailored through evidence-based feedback to account for local contexts. 1.1.1.2: Integrate updated methodology into DWSSP processes. The NPE will provide training to DWSSP Facilitators at the national level. This will be conducted twice to accommodate facilitators covering all provinces. 1.1.1.3: Update existing DWSSPs, when appropriate. DoWR staff and provincial DWSSP facilitators will identify the 100 DWSSPs (of the 301 developed to date, please see Annex 2) most vulnerable to climate change through climate analysis and identification from the original documentation for updates to enhance the DWSSPs with new climate resilient factors. For cost efficiency this will be done through provincial level engagement with Provincial Area Councils at 3 intervals over the project. The remaining 201 DWSSPs will be updated by the DoWR through national processes post project. Upgrades will include enhancing the hazard risk assessments and improvement and preparedness plans as well as upgrading the climate change risk and improvement plan sections in alignment with the new DWSSP climate resilient methodologies.

61. **Output 1.2: Awareness, capacities and skills of communities and area administrators on climate-resilient water management improved**

Resilience at the community level is key to ensuring beneficiaries receive sustainable water and sanitation services that can adapt to shocks and processes of change. The national/sub-national-level processes support coordination and service provision to the WASH sector, but it is the capacity of communities to manage their system that delivers the needed sanitation and security impacts in Pacific communities. Awareness and empowerment, as well as capacity and skill-building will improve the climate-resilience and sustainability of community WASH services and infrastructure in rural Vanuatu. The Output's activities will be synergistic with, and build on, Output 1.1 activities by carrying out community level awareness and knowledge sharing campaigns to specifically enhance understanding of the importance of climate resilience and its incorporation into the DWSSP methodology. Further, climate vulnerability technical capacity training for area administrators within the government extension structure on climate resilient practices in the DWSSP processes will be held. This will enhance Area Council level technical understanding for development of new DWSSPs following the updated methodology in Output 1.3. Through this process the three outputs are seen as fully complimentary, providing in Output 1.1) relevant updates to methodologies, in 1.2) relevant awareness raising and technical capacity building on climate resilient, and in Output 1.3) actively engaging and providing support in the development of DWSSPs.

ACTIVITY	SUB-ACTIVITIES
1.2.1: Train area administrators and develop	1.2.1.1: Conduct training targeting area administrators on climate vulnerability and risks and updated DWSSP methodology. This will carried out at the

material for communities on climate vulnerability, risks and updated DWSSP methodology	provincial level by facilitators, who will provide tailored training on climate vulnerability and its relevance to building climate resilience through the updated climate resilience updates in the DWSSP methodologies. 1.2.1.2: Conduct training targeting communities on DWSSP processes, with interactive content and WASH advocacy materials. Technical knowledge on the impacts of climate change and solutions for climate resilience will be synthesised into appropriate learning material to be shared with communities by Area Councils, including easily comprehensible written documents (pamphlets and flyers for RWC to distribute) as well as through social media, and popular media (TV and radio) campaigns.
1.2.2: Organize knowledge sharing events	1.2.2.1: Organize community level knowledge sharing events to ensure widespread dissemination among communities. The Monitoring, Evaluation and Learning Officer (MEL) and Environmental and Social Safeguard (ESS) officers will establish a community of practice for knowledge sharing and dissemination. This will include provincial forums held at intervals over the project to enhance the profile of climate resilience and showcase success stories and lessons learned.

62. **Output 1.3: Vulnerable communities are supported to develop and implement their DWSSPs (600 DWSSPs by the end of the project cycle)**

The majority of WASH systems in the Pacific are isolated from government service provision and are managed by community committees. This is also the case for Vanuatu, establishing strong governance structures at grassroots levels that are accountable, with participatory decision-making mandates and are key to managing water resources safely in the country. The DWSSPs will provide a framework through which community-owned WASH infrastructure (new and/or improved) can be sustainably managed, localizing both priorities and interests. Through this output, the project will deliver the 600 targeted DWSSPs, identified through the country-owned (DoWR-mediated) NIP process[40]. For a full description of the process please refer to Annex 23. In summary, a community request (triggering) is received at the Area Council or Provincial Government. Support to communities to develop DWSSPs is provided with a priority to the most vulnerable communities climate threats impacting their water security. This takes into consideration drought maps and local knowledge from Area Councils, National Disaster Management Office, Ministry of Health and other Water, Sanitation and Hygiene (WASH) partners. Once a community is selected the support is provided to conduct a DWSSP with the Rural Water Committee and other community members. Following this a vulnerability analysis to quantify climate risks in relation to water quantity and safety is carried out by the Provincial Water Resources Advisory Committee (PWRAC) to prioritise the sites. After communities have completed low cost improvement plan developed during the DWSSP workshop they will be eligible for CAP funding. RWC's will be registered under the national and provincial RWC register (provincial RWC register to be established in this Output to filter into the National RWC register).

As presented in Annex 2 section 6.3, approximately 2000 DWSSPs must be developed by 2030. To reach this target a minimum of 200 DWSSPs are required to be developed per year up to 2030. This exceeds the DoWR current capacity. This project will therefore provide resources to enhance the capacity of the DoWR over the project cycle to target an additional 600 DWSSPs. This Output will be conducted in years 1-3 of the project. This will enable the 600 developed DWSSPs to access the resources as described in Outcome 2 to carry out the required infrastructure works over the remaining implementation period for communities most in need under the context of climate vulnerability.

The technical capacities provided by the project in terms of technical knowledge and human resources will be continued by the DoWR post project creating an enhanced enabling environment to carry out required operations at the scale needed. This in conjunction with institutional improvements also provided by the project (Outcome 3) will enhance efficiency in the process. This combined impact will enable the DoWR to have a long-term strategy and capacity to meet the 2030 target.

ACTIVITY	SUB-ACTIVITIES
1.3.1 Identify vulnerable communities through the	1.3.1.1: Recruitment of provincial engineers – these positions will be embedded in the DoWR and will provide centralised technical support at the

[40] https://mol.gov.vu/images/News-Photo/water/DoWR_File/Management_Plans/Vanuatu-NIP-Guide-annual-CAP-210818.pdf

NIP process to prioritize delivery of DWSSPs	provincial level to assist with monitoring contractors facilitating the community DWSSP workshops, assist with provincial level climate resilient trainings, assist with detailed assessments to create designs and bills of quantities for CR-WASH infrastructure improvements and complete construction monitoring of contractors implementing CR-WASH infrastructure improvements (refer to CAP, Outcome 2) 1.3.1.2: Refresher for community DWSSP facilitators halfway through the project to serve as a refresher training and to incorporate the various incremental improvements over the intervening implementation period incorporated under Output 1.1. 1.3.1.3: Development of 600 DWSSPs informed by best-available climate science, local information and traditional knowledge. This includes a five day workshop/training at the community level, conducted by DWSSP facilitators, to engage with RWCs and community members to develop DWSSPs in an inclusive and representative manner. This is in alignment with the requirements under the Water Resources Management Act and shows alignment with national legislation and policies. The ESS Officer and Provincial Engineers will monitor the community DWSSP facilitator contractors to ensure DWSSPs are to standard including the climate risk assessments, ESS assessments and improvement plans and that communities are well trained to improve community climate resilience. 1.3.1.4: Establish and register local water committees. Currently, there is no register at the provincial level. The project will establish an online register with the provincial governments to deliver this process. 1.3.1.5: Support provided by Provincial Engineers and Provincial DoWR officers to implement no and low-cost measures in communities.

63. **Outcome 2: Communities have enhanced rural water infrastructure**
Rationale: *see Section B2 Outcome 2 for more detail on the Outcome's rationale.*
US$: *20.0 million (GCF US$16.09 million, co-finance US$ 3.97 million)*
Beneficiaries: *270 communities (including 220 already targeted by Outcome 1, and 50 additional ones) = 30,834 direct beneficiaries (including 25,124 from Outcome 1 and 5,710 additional ones); which is 8% of the total population in Vanuatu. Applications to the CAP have set prioritisation criteria for approvals with a focus on the most vulnerable and at need communities. These will be applied to all submitted proposals, meaning all selected beneficiaries for funding under this Outcome will be driven by vulnerability assessment.*
Complimentary with other Outcomes: *As above, this Outcome is mutually reinforcing with Outcome 1. The technical capacities put in place in Outcome 1 will ensure that CAP requests incorporate greater emphasis on climate-resilience in infrastructure designs. Ultimately, the two Outcomes combine to ensure improved planning and increased technical abilities to ensure that climate-resilience is directly incorporated into the GoV NIP process, resulting in greater operationalisation of CR-WASH systems across the country.*
Outcome indicators: *contributes directly to GCF Core Indicators 6 – see section E for further detail.*

This Outcome forms the core investment required for a paradigmatic shift towards CR-WASH infrastructure in rural contexts of Vanuatu. There is one Output that will be delivered through two activities.

64. **Output 2.1: 270 vulnerable communities supported to construct, operate, and maintain climate-resilient water infrastructure**
This Output will focus on delivering climate-resilient water infrastructure to 270 communities in Vanuatu. Provision of this WASH infrastructure will include: rehabilitation, upgrading, and/or expansion of rural drinking water supply systems to serve at least 270 communities. Collaboration with provincial governments, drinking water providers, and target communities will ensure reliable and sustainable potable water services. To ensure improvements are sustained, the project will aim to achieve two mutually reinforcing intermediate results, through this Output:

i. Increase access to safely managed drinking water in rural Vanuatu through the rehabilitation or construction of small-scale infrastructure projects, identified through the country owned DWSSP and NIP processes

ii. Increase engagement of communities in management, oversight and accountability of drinking water service points and infrastructure (through, particularly, the RWCs).

In 2014-2016, the national Water Resources Inventory (WRI) under DoWR, and UNICEF conducted a water point mapping that identified 4,097 water systems across 44 islands of which:

- 15% were Indirect Gravity Fed systems (IDGF),
- 20% Direct Gravity Fed systems and
- 65% Community Rainwater Harvesting systems.

Only 79% of systems were identified as functional and only 37% of these systems were reported as providing water 24h/day daily, every day of the year.

The CAP facility under the NIP does not have capacity to upgrade or build new CR-WASH infrastructure for all of the required infrastructure needs at present. This project aims to enhance CAP and institutional capacities in the DoWR and to contribute an additional 270 CAP investments over the project life cycle. Please see Table 8 below and Annex 4's detailed budget notes for a detailed breakdown of planned infrastructure investments based on CAP and DoWR experiences to date. This is deemed the maximum level of support feasible considering the human resource and technical capacities conducting CAP operations at present plus the additional capacity provided by positions planned through this projects PMU. Noting the 270 is additional to the current CAP capacity which completed 100 investments in 2021 expected to increase each year with efficiencies. With the enhanced institutional processes in place from Outcome 3 it is envisioned that this project will also increase capacity of the DoWR to roll out greater than 100 CAP requests a year post project. Targeting the most vulnerable communities to climate change through the improved assessment process.

Table 8: average cost of investments through the CAP to date.

System Type	Average Cost (USD) - materials, shipping and labour
Rainwater Harvesting System	20,148
Direct Gravity Fed System	96,360
Desalination	91,980
Indirect Gravity Fed System	53,436

The 270 communities targeted for this Output will be selected from those DWSSPs developed under Outcome 1 and will be financed by GCF resources through the DoWR managed CAP. DWSSPs will be selected for CR WASH infrastructure projects through the method described in the 'Guide to the Capital Assistance Programme'. Please see Annex 22 sections 2.1 and 3.1 for pre-requisite conditions and list of eligibility criteria (these will be included in the Term Sheet and Annex 6 and will not be changed by through implementation) .
Once a DWSSP is deemed to meet the prerequisites and eligibility criteria a request for CAP funding will be developed by the RWC with assistance from the Provincial Engineers, DoWR Provincial Officers, Area Councils and PWRAC members. The project is then submitted through the review and approval process, please see Annex 22 for full details. In summary the process:

- Will check that each application contains all of the required documentation (application form and an approved DWSSP).
- Will calculate the overall risk scores (safety and security) for each application (see Section 4.1 Annex 22).
- Will use the overall risk scores to categorise each new application (see Section 4.2 Annex 22).
- Make adjustment to the category for community commitment (see Section 4.3 Annex 22).
- On a quarterly basis, PWARC will prepare a summary report of applications awaiting financial assistance and submit to DoWR all applications for Category 6 and 5 [and possibly Category 4] communities for detailed works design and costing.
- On an annual basis, complete the application form to the National Water Resources Advisory Committee (NWRAC), recommending the priorities for allocation of CAP funds
- NWRAC will, on an annual basis, review and make decisions on which community improvements will proceed, within the annual budget it has allocated to each province. NWRAC will notify provincial government of these decisions.

- DoWR will then administer the funds, contracting relevant suppliers as necessary to implement the constructions and requisite trainings as detailed in Activity 2.1.2 below.

To further support this process the project will assess the CAP criteria and upgrade the risk screening process to be more robust and provide a greater focus on climate vulnerability as described in sub activity 2.1.1.1 below. Further, the ESS Officer will provide technical capacity support to the Provincial Engineers and DoWR Provincial officers to provide technical support to communities in the development of ESS screening and impact assessments (for those projects deemed category B, estimated at <5% of applications based on CAP applications to date) for CAP proposals, as per Activity 2.1.1.2 described below.

Communities / water asset owners not covered by this project will be covered through updated CAP processes with alternative funding to meet the target National Sustainable Development Plan 2030 target of 'Ensur[ing] all people have reliable access to safe drinking water and sanitation infrastructure by 2030' (100% coverage of the population).

ACTIVITIES	SUB-ACTIVITIES
2.1.1: Improve CAP request prioritization, with gender and ESIA screening for chosen sites	2.1.1.1: Assess and update multi-criteria analysis for risk ranking process to prioritize CAP requests to prioritise sites for infrastructure planning. Additional criteria will be included to those risk criteria presented in Annex 22 to enhance focus on climate vulnerability. The new criteria to be incorporated includes, flood risk, sea level rise, degraded forest areas, groundwater resources and saline intrusion assessments. On approval of the changes by the NPSC, 2 trainings will be provided to provincial facilitators to highlight and explain these changes. 2.1.1.2: Conduct relevant surveys, gender, environment and social safeguards screening and impact assessments in chosen sites. The ESS Officer, Provincial Engineers, National Engineer and DoWR provincial officers will assist communities/RWCs to conduct these activities.
2.1.2: Upgrade CR WASH infrastructure, with adequate O&M training	2.1.2.1: Construct and upgrade infrastructure for climate resilient water sources, distribution, and storage. Please refer to Annex 2 Feasibility Study Annex C for indicative list (and pictures) of climate resilient Infrastructure works to be financed through the CAP. This includes technologies such as concrete protection on spring sources, well head aprons, cyclone screws and tie downs, appropriate foundations for tanks, robust river crossing infrastructure, housing around tap and pipe stands, well covers, deep ground water wells (not shallow), relocation of wells from brackish water sources. These are a few examples and each CAP proposal will be assessed on an individual basis against proved CR-WASH designs. 2.1.2.2: Train RWCs on operation and maintenance for the CR-WASH infrastructure. The Water Resources Management Act states that the Minister must not establish a Rural Water Committee unless he or she is satisfied that the members of the committee have undertaken a: 1. Community development training - technical training at community level, specifically focused on plumbers training and community mobilisation to understand, operate and maintain infrastructure. 2. Water management and financial training – provided to the RWC to ensure effective and sustainable management of the infrastructure in the long term. These trainings are detailed in the budget and reflect the national regulation under the Water Resources Management Act and subsequent lump costs the DoWR have defined as minimum requirement for adequate training.

65. **Outcome 3: Provincial and national institutions are strengthened to address climate risks associated with water security**
***Rationale**: see Section B2 Outcome 3 for more detail on the Outcome's rationale.*
***US$:** 0.893 (GCF US$ 0.718 million, 0.175 Co-financing).*
***Beneficiaries**: 2000 communities = Indirect beneficiaries: the entire rural population in Vanuatu (around 216,980 individuals, which is 73% of the total population). The Outcome will strengthen institutional*

capacities across Vanuatu's WASH sector with a focus on rural communities. This is complimented by enhanced M&E and knowledge management, that will increase lessons learned and aid refinement of processes, methodologies, and engineering designs in the future. Consequently, the total rural population of Vanuatu will benefit from the Outcome.

Complementarity with other Outcomes: *Institutional capacity building under Output 3.1 will directly facilitate outputs under Outcome 1 by enabling improved knowledge transfer across extension structures in the WASH sector. Output 3.2 and 3.3 will facilitate greater communication across WASH sector partners and generate key lessons learned from the project. Under Output 3.3, there will be continual monitoring and evaluation on an annual basis which will allow adaptation of processes, engineering designs or implementation modalities. All of these factors will generate essential knowledge that will build into Outcomes 1 and 2 to enhance their success and contribute to sustainability in the long-term.*

Outcome indicator*: Contributes directly to GCF Core Indicators 5 – see section E for further detail.*

Outcome 3 will enhance institutional capacity at provincial and national level to ensure the effective and efficient delivery of the DWSSP and other related water management processes to manage the adverse effects of climate change. The expected outputs, along with activities, are listed below:

66. Output 3.1: National and provincial-level staff and WASH sector partners trained on climate-resilient water management

This output will focus on providing training to different levels of staff within the water governance structure, as well as to WASH sector partners, to ensure climate change and management of climate risks are mainstreamed within existing processes of water safety and security. In doing so, it will deploy different types of training – from the provincial and key line ministry level, to WASH sector partners – ensuring stakeholders across the board (and in the WASH industry) are better able to deliver on Vanuatu's emerging and ongoing needs related to climate-resilient water management.

ACTIVITIES	SUB-ACTIVITIES
3.1.1: Deliver enhanced administrative and technical capacity training for water management	3.1.1.1: Train PWRAC and DoWR staff on updated DWSSP, climate risks and water management 3.1.1.2: One engineer recruited in in each of the six target provinces hosted by the DoWR and providing humanitarian engineering assistance by enhancing technical capacity of engineering related to water infrastructure at community level.

67. Output 3.2: Knowledge management through data sharing mechanism established for climate-resilient water management

This output will focus on knowledge management and will establish data sharing mechanisms available to the different stakeholders of the water and climate change sectors. Support, particularly, will be provided to WASH sector partners to be able to employ data for decision-making. One of GCF's paradigm shifting pathway for water security is creating and sharing knowledge to harmonise valuation methodologies with climate risks built into financial decisions for sustainable development – this Output will contribute towards this pathway and indeed pioneer a robust, climate-water- specific KM for Vanuatu.

ACTIVITIES	SUB-ACTIVITIES
3.2.1: Improve knowledge management processes	3.2.1.1: Review existing knowledge management and identify gaps to be strengthened 3.2.1.2: Establish knowledge management protocol through consultative process
3.2.2: Operationalize data sharing platform	3.2.2.1: Integrate data collected through DWSSPs into government knowledge management platforms, such as DoWR Information Management System. 3.2.2.1: Set and disburse relevant data management procedures to ensure data is compatible and aggregable within the system. 3.2.1.2: Engage relevant stakeholders to support effective utilization of data for decision-making by WASH sector partners

68. Output 3.3: Monitoring, learning and evaluation framework established for improved learning for climate-resilient water management

This output will establish a MEL system and contribute to the same paradigm shifting pathway for water security as the previous output (creating and sharing knowledge to harmonise valuation methodologies with climate risks built into financial decisions for sustainable development).
Having a MEL system, particularly for the sanitation and hygiene sub-sector will measure and ensure that inputs and activities lead to their intended results and outcomes; to adjust course where necessary (during the project lifetime); and to establish whether progress is being made; and lastly, to document lessons learnt and best practices. This will tie in with the DoWR's structure – which includes an M&E unit that keeps track of DWSSP progress and maintenance of system improvements.

ACTIVITIES	SUB-ACTIVITIES
3.3.1: Collate existing MEL practices within water governance structures in Vanuatu, and establish a robust MEL protocol	3.3.1.1: Stocktake existing MEL processes, and collate lessons learnt, case studies, and best practices available 3.3.1.2: Create an inventory of existing reports 3.3.1.3: Introduce robust MEL protocol, identified through a consultative process

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

69. The **implementation arrangements** are shown diagrammatically in Figure 5, which is followed by a narrative that provides the landscape of project oversight and supervision, elucidates the roles under each entity, and tracks how funds will flow if the proposed adaptation project is approved:

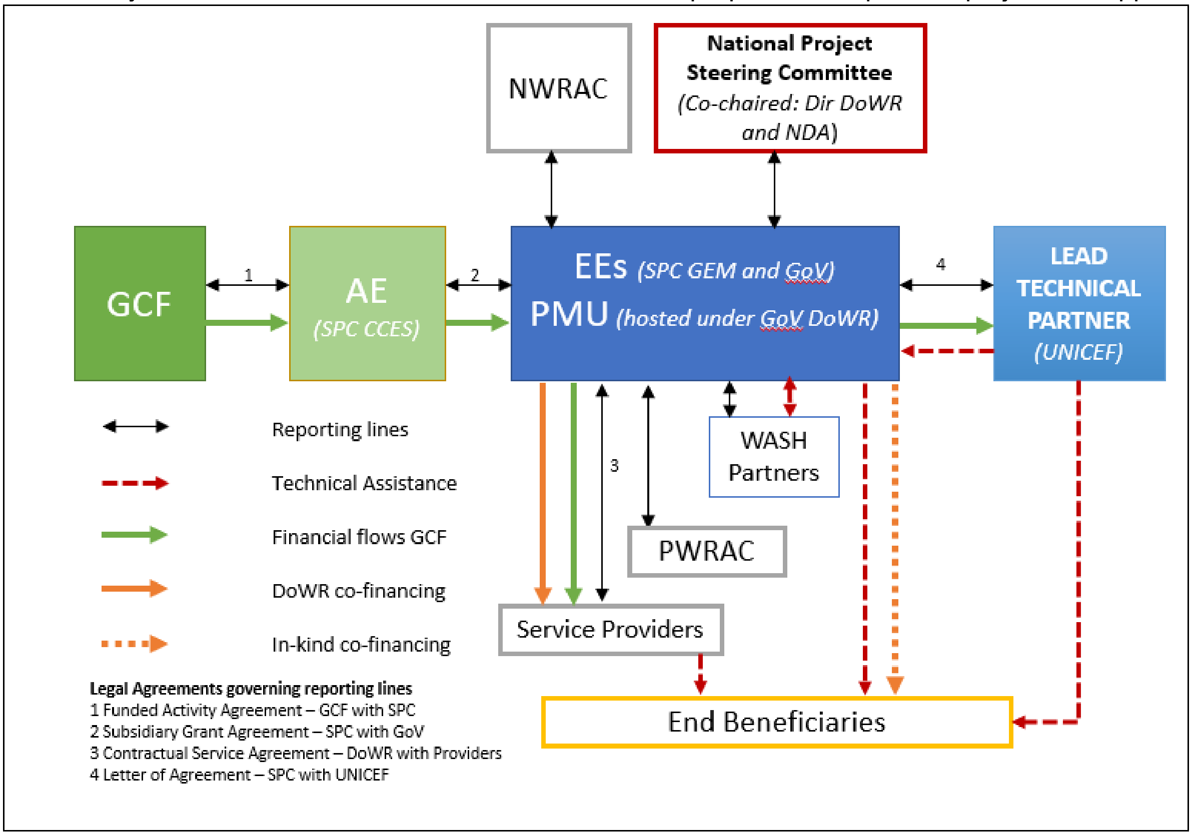


Figure 5 - Visual representation of the project implementation arrangements including financial flows, technical assistance roles and reporting lines.

70. **Accredited Entity:** SPC will be the Accredited Entity (AE) for this project. SPC was made a regional Direct Access Entity to the GCF in February 2019. SPC's comparative advantage, as the AE for this project, lies in its:

- extensive ties with Pacific Islands governments, administrations, agencies, and partners in all Pacific Island Countries.

presence of SPC's Melanesian Regional Office, which is hosted in Vanuatu, creating close linkages at country level that will enable greater communication between SPC and the PMU.
broad mandate on urgent development issues in the Pacific, including water, disaster-risk reduction, climate change and environmental sustainability, gender and human rights.
large funding base with multi-lateral and bilateral donors, allowing for co-financing options, cost sharing and combined programmatic funding.
extensive international partnerships which range from UN agencies to other IGOs, NGOs and civil society groups at grassroots level, including in Vanuatu.

Within SPC, the AE functions will be undertaken by the Climate Change and Environmental Sustainability programme, which hosts the Climate Finance Unit (CFU), SPC's focal point with the GCF. The CFU's role in the project will focus on oversight and quality assurance of project management and delivery in alignment with obligations agreed upon with the GCF through relevant financing and legal agreements. The primary roles of the AE include:

- managing and disbursing GCF funds to the GoV in its capacity as EE for execution;
- reviewing financial expenditures and progress reports submitted by the EE to ensure that GCF financial obligations are met in prior instalments to allow for subsequent disbursements;
- providing technical guidance as needed to ensure appropriate technical quality is applied to all Project activities as well as reporting outputs;
- submitting all relevant documentation to meet reporting obligations to the GCF (financial reporting, Annual Progress Reporting, Audits, Mid-term Report, Terminal Evaluation Report);
- ensuring the project complies with the terms agreed in the project's respective FAA as well as the AMA signed between SPC and the GCF;
- ensuring that the project is implemented in line with its environmental and social safeguards classification; and
- undertaking regular annual supervision missions according to the SPC's guidelines.

71. **Executing Entity: SPC** will also be an EE for project acting through the **Geoscience, Energy and Maritime (GEM) Division** for the project. Within GEM, the Disaster and Community Resilience Programme (DCRP) is a key technical pillar of the division and develops, implements and supports coordination of innovative applied science and technical action to respond to challenges faced by communities, as well as repeated disaster risks, in the Pacific region.

The DCRP will be essential in enabling community resilience through CR-WASH in Vanuatu given the breadth of its targeted work in disaster risk reduction initiatives alongside climate change adaptation, water security and sanitation, hazard and risk understanding, Ridge-to-Reef solutions and natural resource management. Further, the DCRP leads the SPC's efforts in operationalising the Framework for Resilient Development in the Pacific (FRDP) through the provision of coordinated technical support to our members. As such SPC through the DCRP is well placed to execute the project.

As EE, SPC will coordinate actions to ensure the project is adequately staffed and supported to efficiently implement the project. More specifically, the EE will oversee tasks within the overall project management structure consisting of:

- contracting PMU staff (Project Manager, National Project Engineer, Administrative Assistant and Environment and Social Safeguards Officer);
- providing guidance to support the PMU in alignment with the direction of the National Project Steering Committee (NPSC);
- supporting the PMU to ensure that all project-level reporting outputs are prepared in a timely manner and in alignment with GCF requirements;
- conclude and manage contracts of suppliers and services providers;
- quality assurance and enhancement of project-level reports submitted by the PMU;
- ensuring implementation of activities are in alignment with the project work plan; and,
- coordinating and facilitating project level audits in alignment with GCF obligations.
- In its role as EE SPC, will enter into a Letter of Agreement (LoA) with UNICEF, pursuant to which UNICEF will provide technical assistance for specific activities identified in the LoA. This agreement will detail and legally define UNICEF's roles and responsibilities as the Lead Technical Partner for the project.

SPC (GEM) as EE will execute all activities through the Project Manager, who will be contracted as a project position, and will provide technical support to DoWR on activities 1.1.1, 1.2.1, 1.2.2, .1.3.1, 2.1.1, 3.1.1, 3.2.1, 3.2.2 and 3.3.1

72. **Executing Entity: The Government of Vanuatu (GoV) will be the Executing Entity for the Project acting through the Ministry of Finance (MoF), the Department of Water Resources (DoWR) and the National Water Resources Advisory Committee (NWRAC)**

The Ministry of Finance – will receive GCF resources from the AE and allocated these into the CAP accounts to be administered by the DoWR**.**

The DoWR - housed within the Ministry of Lands and Natural Resources, is the focal government institution and responsible for activity execution for this project. It is. The DoWR oversees the provision of safe, secure and sustainable management of community water supplies, protection of water resources, catchment management, and WASH disaster programs.

The DoWR coordinates rural water supply scheme improvement plans, driven through the National Implementation Plan for Safe and Secure Community Drinking Water (NIP). The NIP process and integrated plans is the primary mechanism to which water systems must comply, and through which efficient and effective delivery of improvement services is mobilised. Through this, DWSSPs are planned for over 2,000 communities across Vanuatu. In many cases the implementation of these plans is beyond the financial capacities of the communities. The DoWR therefore administers the CAP under the decision-making authority of the NWRAC to support these communities in carrying out their DWSSPs through provision of financial support for infrastructure construction or upgrades.

The Department is also the lead coordinator of the Vanuatu WASH Cluster, with DoWR acting as secretariat and chair during emergencies. The Vanuatu WASH cluster is very active due to the high number of emergencies that the country experiences, and the DoWR is experienced in responding to water access and security issues during disaster and other fast-onset climate events.

Considering the above, the DoWR will have multiple roles in the implementation of this project: i) co-chairing the Project Steering Committee, ii) administering the funds of the CAP, and iii) housing the Project Management Unit (PMU) to ensure strong alignment of project implementation to national processes, strategies and regulation, whilst facilitating learning and ensuring strong national ownership. The DoWR will therefore carryout the following tasks related to project execution:

- contracting PMU staff (Provincial Project Engineers, Provincial Coordinator, Finance Officer, Procurement Officer and the Monitoring Evaluation and Learning Officer)
- manage funds within the CAP accounts (for GCF resources these will be under a separate ledger account to ensure traceability of resources);
- on approval of CAP funding the DoWR will administer the funds of the CAP, contracting and managing relevant services providers to carry out constructions at a local level and engaging with provincial engineers to ensure that the works are carried out to standard and in alignment with strengthened design specifications under Output 2;
- providing guidance to support the PMU in alignment with the direction of the National Project Steering Committee (NPSC);
- supporting the PMU to ensure that all project-level reporting outputs are prepared in a timely manner and in alignment with GCF requirements; and,
- ensuring implementation of activities are in alignment with the project work plan.

The GoV, acting through the DoWR, will execute all activities through the provincial engineers, provincial coordinators and other relevant PMU officers, who will be contracted as project positions. They will participate in the implementation of activities 1.1.1, 1.2.1, 1.2.2, .1.3.1, 2.1.1, 3.1.1, 3.2.1, 3.2.2 and 3.3.1.

The AE will enter into a subsidiary grant agreement with the GoV pursuant to which the AE will on-grant GCF Proceeds for the implementation of the activities by the GoV, acting through its DoWR, MoF and NWRAC.

73. **Project implementation:** This section will elaborate the different governance mechanisms for project execution, coordination and oversight.

74. **National Project Steering Committee:** A National Project Steering Committee (NPSC) will be formally established as a part of the inception workshop for the project and will be co-chaired by the **Director of the DoWR and the NDA** (or their nominated representatives). The NPSC will include the DoWR Director, the NDA, UNICEF Vanuatu Field Office, representative from the Vanuatu Association of NGO's, representative from Department of Women's Affairs, and a representative of the Ministry of Local Authorities of the Government of Vanuatu.

In addition, representatives from the AE, EE, the Project Manager, the New Zealand Ministry of Foreign Affairs and Trade (MFAT) Vanuatu WASH Coordinator[41], and selected technical partners (identified at inception) will be included in the NPSC as observers[42]. The PMU will act as the Secretariat for the NPSC. The NPSC will provide implementation guidance, strategic support and financial and procedural oversight to the project. Specifically, the NPSC will:

- provide strategic guidance and implementation oversight of the Project through review of progress and evaluation reports and provision of recommendations to the PMU for improved implementation;
- provide guidance and direction on cross-cutting issues which require consensus from the various stakeholders involved in the Project;
- ensure that institutional strengthening through the activities is consistent with the Project's overall objective as well as national policies and strategies;
- facilitate full cooperation of various stakeholders under their jurisdictions to provide access and support to the Project team in carrying out their tasks;
- represent the interests of civil society and communities derived through NDA and DoWR bilateral dialogues;
- approve the project's administrative, financial, accounting and operations manual;
- approve the project's Annual Work Program and Budgets (AWPB); and,
- ensure complementarity with other GCF-funded projects in the country.

The NPSC will act in accordance with best practices and standards for governing bodies and ensure that the PMU delivers expected results with best value for money, fairness, integrity, transparency and effectiveness. The NPSC will meet at least annually, as well as maintain regular contact with the AE, EE and PMU as needed on an *ad hoc* basis.

75. **National Water Resources Advisory Committee:** is an independent committee appointed by the Minister for Land and Natural Resources and is chaired by the DoWR and acts on the behalf of the GoV. As per Annex 22, the final decision for the approval of investments funded through the CAP will rest with the NWRAC. NWRAC will, on an annual basis, review and make decisions on which community improvements will proceed, within the annual budget it has allocated to each province. NWRAC will notify the DoWR and provincial government of these decisions. On approval, the DoWR will then administer the GCF resourced investments through the PMU.

76. **Lead Technical Partner: United Nations Children's Fund (UNICEF)** is one of the key players of the WASH sector in Vanuatu (and other Pacific islands), working with governments to achieve universal and equitable access to safe and affordable drinking water for all. UNICEF's approach is to strengthen national systems and capacity through a mix of community-based activities and collection of evidence. UNICEF Pacific's expertise and experience will be valuable to the project as the organization, in partnership with the government and civil society organisations, works primarily on the rights of children to survival and development through improved drinking-water, sanitation and hygiene behaviour.

Notably, several UNICEF Vanuatu-led assessments of the WASH sector, and WASH infrastructure have been used to inform the project design. Further, UNICEF has actively been engaged in the NIP processes and delivery of DWSSP and CAP capacity building activities at both facilitator and community level. SPC, as an EE, will enter into a Letter of Agreement with UNICEF, pursuant to which UNICEF, as Lead Technical Partner (LTP) will support the PMU in the delivery of technical assistance and capacity building activities. Specifically, in relation to sub activities 1.1.1.1, 1.1.1.2, 1.2.1.1,1.2.1.1, 1.3.1.1, 1.2.1.2, 1.2.2.1, 2.1.1.1, 3.1.1.1, 3.2.1.1, 3.2.2.1, and 3.2.1.2. In addition to this, UNICEF as LTP will provide technical recommendations to the PMU over implementation to ensure alignment (but not overlap) with ongoing

[41] MFAT is a long-standing partner of DoWR and funder of WASH sector initiatives in Vanuatu.
[42] At least one technical partner representative must be from a NGO/CSO focused on gender rights and empowerment.

UNICEF and WASH Partner operations in the sector. This will include, but is not limited to, providing advice on AWPBs for submission to the NPSC for approval. UNICEF's role in implementation of these activities as well as recipient of funds will be governed by a Letter of Agreement with SPC in its capacity as EE.

77. **Project Management Unit:** The PMU will be established under the Subsidiary Agreement between the SPC and the GoV. Staff will be contracted by DoWR or SPC, who will manage and support the PMU as described above. On implementation matters, the PMU will carry out AWPBs (in consultation with the LTP as a long-standing WASH sector support agency in the country) in alignment with NPSC approved documentation. Further, the PMU will be embedded within the DoWR to allow the Department to play a central role in guiding implementation of project activities in line with country regulations, policies, processes, and local contexts. Project implementation will be strongly aligned with national water safety and sanitation execution processes, with national and provincial DoWR staff providing guidance on the activity level. This will ensure more effective implementation through enhanced knowledge transfer between PMU staff and DoWR extension agents both formally and tacitly through day-to-day interactions.

The PMU will consist of various positions hired by both EEs in an open competitive process. The selection and recruitment PMU staff by SPC will follow SPC recruitment practices and be done in close consultation with the DoWR, who will provide no objection to Terms of Reference and will provide a country representative to participate in the interview panel. All positions will be advertised at local levels and preference will be given to local applicants if they meet the positions criteria. These positions include:

Project Manager– will be responsible for the day-to-day operations of the PMU, within the parameters laid down by the NPSC through the approved AWPBs and in alignment with their guidance and recommendations. They will also be responsible for monitoring project risks and maintaining the project risk log and will ensure regular communication with all selected service providers, CSO organisation and provincial governments in target locations. This will ensure smooth feedback across all project stakeholders and facilitate implementation as needed. With support from PMU members as required, the Project Manager will draft and finalise i) AWPBs and other relevant documentation for NPSC approval and ii) all reports required to meet GCF obligations under the FAA. Overall, it is the Project Manager's responsibility to ensure efficient and effective implementation of the project. This role will be recruited as a project position by SPC GEM.

Project Administrative Assistant – will support the PMU through carrying out all necessary administrative tasks required for the smooth operation of the project. They will also carry out secretariat roles for organisation for NPSC meetings and *ad hoc* communications as needed. This role will be recruited as a project position by SPC GEM.

National Project Engineer (NPE) – will review DWSSP and CAP proposals with a focus on the infrastructure design works to be conducted under the project. They will assess infrastructure designs recommended by the DoWR and provide technical updates to enhance climate resilience in design guidelines as necessary. Capacity building sessions will also be provided to support DoWR engineers and enhance their technical knowledge. In addition to technical assistance, the National Project Engineer will act as the focal point for provincial engineers, supporting them on highly technical designs and serving as quality assurance for DWSSP and CAP proposals. This role will be recruited as a project position by SPC GEM, in consultation with the LTP.

Provincial Engineers– six provisional engineers/water management specialists will be locally hired and will support the NPE in engaging with relevant RWCs to provide technical support on DWSSPs and CAP proposal designs. They will also work closely with provincial officials, selected service providers and relevant stakeholders to monitor construction and implementation of infrastructure and DWSSPs. Monitoring information and data will be coordinated in alignment with the MEL officer and the projects M&E plan. These roles will be recruited as a project positions by the Government of Vanuatu.

Provincial Coordinator – will be responsible for liaising with the Provincial Engineers and the National Project Engineer, identifying areas of support required for the provincial operations and ensuring equitable division of technical assistance between them. He/she will support the MEL Officer in coordinating data collection and aggregation for monitoring and evaluations and support the Finance Officer with liaising with relevant provincial staff for financial figures and reporting from the provincial level. Further, he/she will interact with the PWRACs and facilitate the submission of provincial submissions to the NWARC, acting as the focal point to ensure equitable spread in submissions targeting GCF resources. This role will be recruited as a project position by the Government of Vanuatu.

Project Finance Officer – will be responsible for supporting the Project Manager in managing the project budget and tracking finances in compliance with GCF obligations signed under the FAA and will contribute to the drafting of AWPBs. They will work in close collaboration with DoWR finance officers and provide technical advice and support on through capacity building to DoWR staff throughout project implementation. This role will be recruited as a project position by the Government of Vanuatu.

Procurement Officer – ensures that all contracts and goods and services procured through the use of project funds are in alignment with GCF procurement standards. He/she will work closely with the Project Engineer who will have the technical knowledge to advice on the procurement of technical equipment and materials as well as the experience and skill sets of service providers targeted for works. This role will be recruited as a project position by the Government of Vanuatu.

Project Monitoring, Evaluation and Learning Officer – will ensure that the M&E plan is followed and implemented on schedule, liaising with Provincial Engineers/Water Specialists on a regular basis. He/she will make sure all data collection and monitoring of IRMF and project-level targets in the logical framework and gender action plan is carried out in a timely and effective manner for reporting to GCF and the NPSC within appropriate timeframes. Further, the MEL officer will utilise this data to create learning products for dissemination and support the Project Manager in drafting any reports as needed. To facilitate this, he/she will also engage directly with PWRACs to gain feedback from local stakeholders and incorporate this into lessons learned for adaptive management. T This role will be recruited as a project position by the Government of Vanuatu.

Project Environment and Social Safeguards Officer – will support project implementation through providing technical assistance to enhance ESS (including gender related topics) into enhance DWSSP design processes, supporting training of extension agents and RWC proponents on ESS (including Gender) integration into DWSSPs and supporting the MEL Officer in conducting relevant monitoring and evaluation of project implementation against the ESMP and GAAP. This role will be recruited as a project position by SPC GEM as the EE.

78. The PMU may hire ad-hoc experts from local and international sources identified based on project needs, with the concurrence of the NPSC, to provide adaptive management support. Agencies and service providers specialized in WASH in Vanuatu, and other specialized entities will be linked to the PMU through this modality. The PMU will provide the NPSC with semi-annual progress reports and close its operations when the final project terminal evaluation report and other documentation required by the GCF and the SPC has been completed and submitted to the NPSC.

79. **Provincial Rural Water Advisory Committees (PWRACs):** At the local-level, PWRAC will coordinate closely with provincial entities (such as WASH officers, Area Administrators and service providers) and committees (such as existing Rural Water Committees – RWCs, and new RWCs established during the project cycle through the new DWSSPs) to select communities for DWWSSP development, carryout risk ranking to assess DWSSPs for consideration to CAP access, ensure smooth ground coordination in project implementation, provide ownership and ensure sustainability. The PWARCs will not have binding execution capacity but will be key in this context.

They will facilitate stakeholder understanding of the roles, functions, and responsibilities provided by the Project, including accessibility and accountability mechanisms to ensure conflict resolution as well as publishing the grievance redressal at local levels throughout the project implementation phase. Provincial stakeholders will be engaged throughout implementation and will provide feedback on the progress of the project to enable adaptive project management in response to the needs and priorities of the communities.

80. **Service Providers:** The PMU will recruit and work with Service Providers at the activity level. Service Providers will be selected through a procurement process, in alignment with policies and procedures compliant with SPC's policies, and will be managed by the Procurement and Finance Officer in the PMU. The Service Providers will receive funds, upon successful delivery and acceptance by the Project Manager that the conditions of their recruitment have been fulfilled. Service providers will be recruited by the Government of Vanuatu under subsidiary agreements in its capacity as EE.

81. **WASH Partners**: The purpose of Vanuatu WASH Sector Partners is to ensure the adequacy, coherence and effectiveness of overall humanitarian outcomes by mobilising all stakeholders in the WASH sector. The WASH Sector is led by the DoWR and co-led by UNICEF and is made up of Partners[43] deemed as crucial in the sector. These partners are expected to support the DoWR and UNICEF in facilitating the Sector objectives and activities, and to meet on a regular basis to share information and set priority activities for the Sector. Considering this remit and their ongoing work in the water sector in Vanuatu, the WASH Partners are deemed as an important group to provide technical guidance and enhance planning and coherence in the sector. They WASH Sector will meet regularly and coordinate with DoWR as the EE, providing feedback and advice on project operations (but will not have any implementation of decision-making authority related to the project). Similarly, the DoWR will provide guidance and updates to the WASH Sector Partners on the ongoing operations of the project.

82. **Flow of Funds:** On successful completion of conditions precedent to disbursement listed in the Project FAA, GCF will disburse funds to the AE. The SPC in its capacity as AE will then disburse funds to the DoWR as EE. Further, SPC, in its capacity as EE, will transfer funds to UNICEF as LTP (as a pre-selected technical partner) for provision of specific technical assistance activities detailed under the LoA, in alignment with the first AWPB approved by the NPSC at inception. Resources will be used to staff the PMU, secure service providers and goods/materials for implementation of activities.

All subsequent allocations and disbursements will be requested by the PMU based on AWPBs approved by the NPSC. The AE will make subsequent allocations and disbursements to the EE only in the case that previous instalments have been utilised in accordance with the GCF financial obligations and reported in Annual Progress Reports. Once there is consensus on the previous instalments financial and technical reporting, and on approval of the AWPB by the NPSC for the forthcoming disbursement request, the latter will be accepted and communicated to the EE. Following which disbursement will be made.

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

83. This section presents the financial barriers preventing the Government of Vanuatu from transitioning to CR-WASH in rural and remote contexts, and then presents conclusions from Annex 3 – Economic and Financial Analysis to demonstrate how the GCF funding will be mobilized for different Outcomes and the cost-benefit analysis of such adaptation investments.

84. **Financial barriers:** the DoWR has limited resources available for WASH in general, and is even more constrained for delivering climate-resilient water management and infrastructure. The latest GLAAS report[44] determined that the average national expenditure globally on WASH is $50 per capita or 1.27% of GDP. Vanuatu faces multiple constraints, leading an average expenditure of only $8 per capita or 0.26% of GDP, well below the global average. Over 90% of Vanuatu's WASH activities are funded by external agencies (particularly international donors), and less than 10% is through government budgets. The additional burden from climate change on the WASH sector exacerbates the limited public resources, and there are currently no dedicated financing options from the public and private sector to cope with increasing climate change impacts on water security and safety for rural communities in Vanuatu.

85. It is increasingly difficult for water security and safety programmes to make lasting and climate-resilient improvements because of the frequency and intensity of natural hazards. It is estimated that disasters caused by natural hazards cost Vanuatu's economy tens of millions of dollars a year. Funds are needed to repeatedly rebuild and recover after such hazards, directing resources away from long-term resilience-building activities. There are few sources of funding from the private sector for the proposed activities, as water is a public good under Vanuatu law and fully managed by rural communities.

86. Funding from GCF will allow the Government of Vanuatu to achieve a paradigm shift towards climate resilient water management for rural communities. To break the cycle of building and rebuilding, whether because of inadequate infrastructure or natural hazard impacts, this project proposes a strengthened approach that incorporates risk management and climate change resilience from the outset following government-owned and -led processes. By providing lasting, climate-resilient water security and safety solutions that communities can select and be trained to implement and maintain, this project will enable

[43] List of WASH Vanuatu WASH Cluster Partners can be found at: https://events.gov.vu/4w/index.php?action=partners&PHPSESSID=sghct5bh5cprfpfcgh14tlmnvm

[44] The GLAAS report can be accessed here: http://www.who.int/water_sanitation_health/publications/glaas-report-2017/en/

communities to better prepare for and recover from climate-induced disasters and will allow stretched government budgets to be directed to other important needs.

87. Funding from GCF will also ensure that government-owned and -led processes to deliver climate-innovative and climate-resilient WASH infrastructure increases the efficiency of the DWSSP methodology, which has been proven in the years of its implementation. For instance, communities that have already formulated their DWSSPs have spent almost USD 1 million of their own resources on no- and low-cost improvements.[45] Grant financing from the GCF is justified for this project because of the extreme vulnerability of Ni-Vanuatu communities to both climate change and natural disasters, and the lack of resources at all levels (local, provincial, national) to deal with projected climate change impacts.

88. Summary of economic and financial analysis:

A financial analysis for each of the four measures was carried out. The IRR and NPV was determined for two cases: 1) finance provided by commercial sources including a loan (80%) and equity (20%) from the developer and 2) finance provided through a grant from the GCF and "equity" from an investor (donor or government). The tariff was adjusted to produce, a) an Initial Rate of Return (IRR) equal to the Weighted Average Cost of Capital (WACC) , and, b) either a tariff set at the maximum ability to pay (665 VUV/month/household) or a tariff that would be sufficient to cover the OPEX for the technology in question. The results are summarized below:

Measure	Baseline			Project alternative		
	Tariff required to achieve IRR=WACC	% monthly income	% ability to pay	Minimum tariff (= ability to pay or OPEX coverage)	% monthly income	% ability to pay
M1 - Rainwater harvesters	$ 38.30	22%	657%	$ 5.83	3%	100%
M2 - Direct gravity fed system	$ 89.50	52%	1,536%	$ 6.00	4%	103%
M3 - Water desalination systems	$ 214.00	125%	3,674%	$ 15.00	9%	257%
M4 – IDGF systems	$ 104.50	61%	1794%	$ 12.00	7%	206%

In all cases the monthly household tariff that would be required for a commercial loan-based investment would be many orders of magnitude beyond what household can pay. In the project alternative, with GCF grant funding, the necessary tariffs (to, at a minimum, cover OPEX costs) are higher than the reported ability to pay for M2, M3 and M4. Since these investments are seen as a public good these additional costs will be covered by public funds.

An economic analysis of the project was performed to assess the incremental adaptation benefits to climate change. This analysis combines all three measures, scaled-up to the envisaged number of investments that could potentially be financed by the project. Additionally, the project-level analysis takes into account the entire proposed project budget including the costs of all the Outcomes (i.e. non-investment Outcomes as well) and project management costs and co-finance.

89. The following table presents the project-level cost-benefit analysis that consolidates all three previously elaborated adaptation measures and includes the non-investment part of the project budget. The discount rate of 12% was used throughout the entire analysis.

[45] Please see the RISCON database here: https://exchange.riscon.solutions/dwssp/

Table 9: Project-level cost-benefit analysis

Label	Unit	Source of information	Total
Year			
Costs			
M1 - CAPEX costs	USD	M1 - Rainwater harvesters	$ 2,981,904
M2 - CAPEX costs	USD	M2 - Direct gravity fed system	$ 6,648,840
M3 - CAPEX costs	USD	M3 - Water desalination systems	$ 367,920
M4 - CAPEX costs	USD	M4 - IDGF- systems	$ 2,618,364
Total	**USD**	**Calculated**	**$ 12,617,030**
Other project costs			
Total non-investment project costs	USD	Project proposal	$ 15,672,986
Percentage of funds utilisation - project lifetime distribution	%	Assumption	
Total non-investment project costs	**USD**	**Calculated**	$ 15,672,986
Total investment costs	**USD**	**Calculated**	$12,617,028
Total project costs	**USD**	**Calculated**	$28,209,014
Benefits			
M1 - benefits	USD	M1 - Rainwater harvesters	$ 12,153,815
M2 - benefits	USD	M2 - Direct gravity fed system	$ 58,917,056
M3 - benefits	USD	M3 - Water desalination systems	$ 526,844
M4 - benefits	USD	M4 - IDGF-systems	$ 19,310,424
Total benefits	**USD**	**Calculated**	$ 91,000,000

Net costs / benefits	**USD**	**Calculated**	$88,591,877
EIRR	**%**	**Calculated**	20%
ENPV	**USD**	**Calculated**	$37,150,065
Net costs / benefits per year	**USD / year**	**Calculated**	$3,543,675

90. The results clearly show that the project-level ENPV is positive, USD 37,150,065 and the programme-level EIRR is 20%. The conclusion is that the proposed programme is economically viable and can be justified on economic grounds, even with the non-investment budget costs and co-finance for which no direct benefits are envisaged. It is also noteworthy that the analysis included conservative assumptions and not all benefits have been included in the economic calculations since it was not possible to estimate their monetary values, but these benefits would nonetheless occur under the proposed interventions.

B.6. Exit strategy (max. 500 words, approximately 1 page)

91. The exit strategy for this project ensures that the barriers to climate resilience are overcome sustainably, so that project outcomes and processes continue beyond the planned project implementation period. Recognizing the importance of long-term sustainability, the project has been designed from the outset to align incentives between key stakeholders (through extensive consultation processes both during the conception and preparation phase – see Annex 7 for more details), strengthen and design delivery capacity and promote the long-term flow of financial resources in support of project objectives of improved climate resilience and water security in Vanuatu.

92. From the outset, the long-term sustainability of the project approach is ensured through the fact that it is building on and operating through existing government structures and programmes, rather than one-off project-specific processes. This includes the NIP framework, CAP investment programme and the DWSSP process, through which the project activities will be implemented. This prevents the creation of parallel structures and processes that only last for the duration of the project, instead operating through existing government structures and processes that have been in place before the project's conceptualisation and will continue to function after project conclusion, integrating lessons learned and best practices from the project. Moreover, the embedding of the PMU within the DoWR ensures that project activities are carried out in alignment with government activities. The strong government ownership of the NIP, CAP and DWSSP processes mean that the project approach will be continued beyond the project lifespan.

93. The project is envisaged to run for 5 years, and will reach all rural communities through improved institutional processes (Outcome 3), 600 communities with new, and 100 with updated, climate-resilient DWSSPs and low-costs adaptation activities (Outcome 1), and 270 communities with infrastructure upgrades (Outcome 2). These communities will then have long-term, climate-resilient water security and improved knowledge of climate risk management. The project will build technical capacity of the communities to undertake maintenance of the climate-resilient infrastructure (as part of Outcome 2). This means that after completion of the the GCF-funded activities, the communities will be empowered to operate and maintain their water security and safety infrastructure, and they will still be able to access government-funded support through the improved TAP and CAP.

94. The project will also provide training to community facilitators, water officers and other members of the water governance structures, who will be up-skilled and empowered to continue to work in the WASH sector and better address risks and impacts from climate change, thereby enhancing the overall capacity of the sector. In particular, this will improve the outcomes of the already existing NIP, CAP and DWSSP processes, ensuring that they will continue to deliver climate-resilient water security benefits beyond the lifespan of the project and outside of project-specific communities. The project also creates a positive feedback loop between adoption of climate-resilient water management and water governance structures by mobilising existing, government-owned processes. The project interventions are designed to reduce vulnerability to climate risks and burdens on water sector, which then provides a direct impetus to communities (who will be trained) to maintains system improvements to existing WASH infrastructure as well new WASH infrastructure. Additionally, an Operation and Maintenance Plan (Annex 21) has been designed as a part of the preparation phase, which has reviewed existing standards, and maintenance protocols. This will be mobilized in the implementation stage as well.

95. At the local level, the project has been designed through a participatory process involving all stakeholders, including water governance actors at the local level (municipal governments, departmental governments, chiefs and local governance bodies), local communities and beneficiaries. This ensure that the actions proposed in the project are socially viable, having built stakeholder ownership from the outset of the project design process. Likewise, the project contemplates effective local inclusion and participation in governance and decision-making during project implementation, through full recognition of the rights of women, men, and youth living in remote and rural communities. Annex 8 – Gender Assessment and Gender Action Plan provides more details on the gender-specific approaches and benefits that will be realised under the project.

96. By being fully integrated into existing national programmes, and in line with national policies and strategies of Vanuatu, as indicated in the table below, the project's outcomes and outputs will be sustained through central government funding that will continue to be provided, including for monitoring. At the same time, this project will be transformative by leveraging the very small amount of public funds used for water security and safety programmes. By addressing primarily the rural population, this project will ensure climate resilience is included in existing programs and allow government to include a focus on monitoring and maintenance rather than merely on service delivery.

Table 10: Legislation, policies and plans relating to the WASH delivery for the DoWR, MoET and MoH

	Sustainable Development Goals - SDG 6	National Sustainable Development Plan 2016 -2030	Decentralisation Act 2013
Key Agency	**DoWR**	**MoET**	**MoH**
Act(s)	Water Resources Management Act 2002 / 2016 Amendment	Education Act 2014	Public Health Act 1994 / 2018 Amendment
	Water Supply Act 1988 / 2016 Amendment		
	Vanuatu National Drinking Water Quality Standards 2016		
Strategies	Vanuatu National Water Strategy 2018-2030	Vanuatu Education and Training Sector Strategic Plan	

		National WinS Strategy 2019 - 2029	
Policies	Vanuatu National Water Policy 2017-2030		National Sanitation Hygiene Policy
			Vanuatu National Environment Policy and Implementation Plan 2016-2030
	Drinking Water Safety and Security Plan		
Safe Water Planning	National Implementation Plan for Safe and Secure Community Drinking Water	WASH in Schools Improvement Plan	WASH in Healthcare Facilities

Climate Change	Vanuatu Climate Change and Disaster Risk Reduction Policy 2016-2030	
	Nationally Determined Contributions – NDCs	• Target Wa1: By 2030, 100% of water-climate vulnerable rural communities in the six provinces have developed DWSSP and are able to address water needs in normal and (climate, disaster and environmentally) stressed times. • Target Wa2: By 2030, 6 climate-resilient water protection zones declared and sufficiently provides urban water supply needs in normal and (climate, disaster and environmentally) stressed times.

Standards Guidelines	Design and Construction Standards for Rural Water Supply	WinS Facilities Guide WinS 3-star indicators	Sanitation and Hygiene Guidelines

C. FINANCING INFORMATION

C.1. Total financing

(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount			Currency
	23.325033			million USD ($)
GCF financial instrument	**Amount**	**Tenor**	**Grace period**	**Pricing**
(i) Senior loans	Enter amount	Enter years	Enter years	Enter %
(ii) Subordinated loans	Enter amount	Enter years	Enter years	Enter %
(iii) Equity	Enter amount			Enter % equity return
(iv) Guarantees	Enter amount	Enter years		
(v) Reimbursable grants	Enter amount			
(vi) Grants	23.325033			
(vii) Results-based payments	Enter amount			

(b) Co-financing information	Total amount			Currency		
	4.940981			million USD ($)		
Name of institution	**Financial instrument**	**Amount**	**Currency**	**Tenor & grace**	**Pricing**	**Seniority**
Government of Vanuatu	Grant In kind	3,736,081 668,900	million USD ($)	Enter years Enter years	Enter%	Options
SPC	In kind	0.09	million USD ($)	Enter years Enter years	Enter%	Options
UNICEF	In kind	0.470	million USD ($)	Enter years Enter years	Enter%	Options
Click here to enter text.	Options	Enter amount	Options	Enter years Enter years	Enter%	Options

Total financing (c) = (a)+(b)	Amount	Currency
	28.290014	million USD ($)

(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	
	Leveraged finance includes operation and maintenance costs (OPEX) for the installed equipment post infrastructure works completion, amounting to $6,447,000 over the equipment lifetime. This financing is only possible post GCF investment through the CAP and completion of the infrastructure works. RWCs leverage the infrastructure works to generate the required OPEX finance for long-term operation and maintenance of the infrastructure post completion.

C.2. Financing by component

Component	Output	Indicative cost Options	GCF financing		Co-financing		
			Amount Options	Financial Instrument	Amount Options	Financial Instrument	Name of Institutions
Outcome 1: Communities are empowered to	Output 1.1 New and existing DWSSPs incorporate	0.315	0.256	Grants	0.059	In kind	Government of Vanuatu and UNICEF

plan and manage climate-resilient water resources	incremental improvements to mainstream adaptation solutions						
	Output 1.2 Awareness, capacities and skills of communities and area administrators on climate-resilient water management improved	0.777	0.672	Grants	0.105	In kind	UNICEF
	Output 1.3 Vulnerable communities are supported to develop and implement their DWSSPs (600 by the end of the project cycle)	5.165	4.712	Grants	0.453	In kind	Government of Vanuatu and UNICEF
Outcome 2: Communities have enhanced climate-resilient rural water infrastructure	Output 2.1 Vulnerable communities supported to construct, operate, and maintain climate-resilient water infrastructure (270)	20.063	16.093	Grants	3.970	Grants & In kind	Government of Vanuatu and UNICEF
Outcome 3: Provincial and national institutions are strengthened to address climate risks associated with water security	Output 3.1 National- and provincial-level staff and WASH sector partners trained on climate-resilient water management	0.269	0.144	Grants	0.125	In kind	UNICEF
	Output 3.2 Knowledge management through data sharing mechanism established for climate-resilient water management	0.100	0.100	Grants	-		
	Output 3.3 Monitoring, learning and evaluation framework established for improved learning for	0.499	0.474	Grants	0.025	In kind	UNICEF

	climate-resilient water management						
Project Management Component		1.076	0.874		0.203		
Indicative total cost (USD)		28.290	23.325		4.965		

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?	Yes ☒ No ☐
C.3.2. Does GCF funding finance technology development/transfer?	Yes ☐ No ☒

If the project/programme is expected to support capacity building and technology development/transfer, please provide a brief description of these activities and quantify the total requested GCF funding amount for these activities, to the extent possible.

Capacity building is a key element of this project and takes place under all three Outcomes. A summary of the training and GCF funding for each is given in the table below:

1.1.1 Two one-week training sessions of DWSSP facilitators in Port Vila including venue hire, travel and DSA costs	$32,100
1.2.2 Area administrator training, 3 times (one training+ two refreshers) for each of the 6 provinces	$138,100
1.2.2 Knowledge sharing events or activities with TA and training for water committees (3 in each province)	$260,700
1.3.1 Refresher training for DWSSP facilitators	$16,050
2.1.1 Training for facilitators based on updated CAP processes incl. plumber training, water management committee training	$11,100
2.1.2 Training workshop costs for each community prior to each project (plumber + water committee training), including community mobilisation and handover ceremony	$4,107,500
3.1.1 Two trainings for each of the 6 provinces that last one day each	$64,400
3.1.2 Two times trainings of one week each for WASH Sector partners on climate change risks, resilience and water safety	$16,500
3.3.1 Training sessions on M&E for WASH partners	$16,500
3.3.1 Workshop to discuss and disseminate best practice.	$8,600
TOTAL	**$4,671,550**

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

D.1. Impact potential (max. 500 words, approximately 1 page)

97. Water safety and security is vital for community resilience in the long-term as well as in the immediate aftermath of climate-induced disasters. GCF funding will positively impact and contribute to the management of water-related impacts from the observed and projected climate change effects in Vanuatu. This project will address three water-related impacts from climate change in Vanuatu, which are increases in: (i) water-related disasters; (ii) areas suffering from water stress; and (iii) loss of human life induced by poor water quality.[46] It will thus deliver on priority interventions identified in Vanuatu's GCF Country Programme[47] (pipeline project AA1).

98. The project will operate through and enhance the functioning of the government-owned DWSSP processes to deliver climate-resilient WASH infrastructure in tandem with training and capacity building. Since 2016, the DoWR has worked to consistently develop and standardise Vanuatu's planning for safe water. The NIP and DWSSP development have increasingly incorporated government agencies (at all levels), communities, NGO, and civil societies. The NIP process standardises the implementation of DWSSPs for communities, facilitators, government agencies, and donors. This process is a framework of logical steps required to develop, assess, fund and deliver DWSSP improvements, through which the project will operate to mobilise, utilise and finetune systems for delivering improved climate-resilient water supply safety and security.

99. The project's impact in reducing climate risks for water security is reinforced by recent revisions of the DWSSP processes. These significantly strengthen climate change and vulnerability considerations within DWSSPs, from training of trainers, community training and facilitation to data collection, improvement planning, operations and maintenance. The revised DWSSP also establishes a basis for parallel revisions of programmes such as the Ministry of Education's WASH in Schools and Ministry of Health's WASH in Healthcare Facilities. The DWSSPs have been generally well received and understood, with some resources committed to support rollout. By further improving these existing water management processes and community-based planning – particularly through enhanced integration of climate change into planning – the project will increase the adaptive capacity of rural communities to better cope with the additional burden of climate change on water security and safety, providing explicit capacity building and transitioning towards climate-resilient water management planning.

100. Through these actions, the project will provide climate-resilient water security benefits to 85,650 direct beneficiaries (including 42,825 women) and a further 216,980 indirect beneficiaries (see Table 7 for a more complete breakdown of direct and indirect beneficiaries across Outcomes). This contributes to GCF's fund-level impact A2.0 "Increased resilience of health and well-being, and food and water security" and indicator A2.3 (number of males and females with year-round access to reliable and safe water supply despite climate shocks and stresses). This will in turn contribute to achievement of GCF's fund-level outcome A7.0 "Strengthened adaptive capacity and reduced exposure to climate risks", to be measured through indicator A7.1 (use by vulnerable households, communities, businesses, and public-sector services of Fund-supported tools, instruments, strategies and activities to respond to climate change and variability).

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

101. This project will contribute to the second paradigm-shifting pathway identified in GCF's Water Security Sectoral Guide, namely "*strengthening integrated water resources management and water management",* and will deliver on both sub-areas: (i) preservation of existing water resources; and (ii) development of new water supply sources.

102. As outlined in Section D.1, climate mainstreaming into the government-owned DWSSP process is nascent. While climate considerations are increasingly being integrated in water security and safety management, there remain significant needs for improvements to deliver climate-resilient water security benefits for rural communities in Vanuatu. By scaling up the number of climate-resilient DWSSPs registered (Outcome 1) and then delivering climate-resilient infrastructure and system improvements through prioritized DWSSPs (Outcome 2), the project will strengthen a whole-of-government approach to achieving climate-resilient water safety and security through the DoWR with associated decision-making processes that are decentralized, transparent, participatory, community-focused, gender-mainstreamed and evidence-based.

103. In doing so, the project will transition the current precarious water security and safety situation experienced by Vanuatu's rural communities in the face of climate change to one of a reliable and predictable supply of safe

[46] GCF Water Project Design Guidelines (v 0.98 – 28 September 2021 – developed by Deltaeres).
[47] Vanuatu's GCF Country Programme is available here: https://www.greenclimate.fund/sites/default/files/document/country-programme-vanuatu.pdf

water. This will occur through climate-resilient, community-led planning and management practices (Outcome 1), and through the delivery of climate-resilient water infrastructure (Outcome 2). By building and operating climate-resilient water infrastructure, interruption to services during and after disasters will be reduced or avoided. Therefore, this GCF project will reduce the costs associated with continual repairing and rebuilding after disasters, allowing the limited public funds available to be leveraged for investment in new WASH infrastructure and sustainable exploration of water sources.

104. By empowering and supporting communities, with a focus on increased women's participation through Rural Water Committees (RWCs), to manage and operate their own water supplies in a climate-resilient menner (through dedicated activities under Outputs 1.4 and 2.1), this project will help to break the cycle of "*build-neglect-rebuild*" characterising many WASH programmes in Vanuatu in the past. Not only will the project change the knowledge and practices of WASH sector partners, but it will also improve the longevity of the current TAP and CAP's impacts through institutional strengthening (Outcome 3) at local, provincial and national levels.

105. The key pillars of the project's transformational impact will be the increased awareness, knowledge, skills and capacities at all levels (Outputs 1.2 and 3.1), the establishment of a community of practice at the local level (Output 1.2), and the mechanisms and systems strengthened for enhanced knowledge management, data sharing and MEL (Outputs 3.2 and 3.3).

106. The paradigm shift will also support transitioning towards gender-transformational WASH investments, as outlined in Section E where the gender co-benefit towards mandatory representation of women in rural water committees is described. The GoV, DoWR as well as SPC recognize that Vanuatu's currently limited WASH service delivery and infrastructure affect women and men differently. Traditionally, gender roles involve women and girls providing more labour and spending more time than men and boys in managing the household's water, sanitation and hygiene needs.[48] Increased time spent by women and girls to obtain water during dry seasons or after climate-induced emergencies increases their exposure to gender-based violence (GbV). Conversely, improving predictability and location of water sources - with better access points created through more resilient and securitised (such as improved lighting, locks) infrastructure - will reduce exposure to such risks, increasing personal safety and security in addition to the safety and security of water supply. Similarly, the participation of women in RWCs and in decision-making processes under the DWSSPs will ensure that gendered needs are mainstreamed into the design of new and maintenance of existing WASH infrastructure. This is in line with UNICEF research, which revealed that women-led as well as increased women's participation in water user committees result in better management such as improved fee collection and regular meetings of water supply points and sources. See Annex 8 – Gender Assessment and Action Plan for more detail on this issue.

D.3. Sustainable development (max. 500 words, approximately 1 page)

107. This project will generate significant environmental, social and economic benefits that will contribute to positive sustainable development outcomes for Vanuatu in the long term. For example, it will reduce the negative environmental impacts of poor wastewater management by improving water management behaviour and practices through the sanitation component of DWSSPs (Outputs 1.2 and 1.4). The project will also achieve better protection of water sources and forested water catchment areas (Output 1.4), through no- and low-cost community improvements, resulting in improved environmental quality.

108. Furthermore, the project will contribute to equitable access to water as a basic human right in Vanuatu, targeting especially the most vulnerable groups in rural communities. Significant social benefits resulting from project activities include positive health outcomes (as greater water safety and security lead to reduced stunting and diarrhoea) and improved social standing for women through their visible and meaningful roles in water committees (with 40% female members mandatory for registration of rural water committees). With the identified gender co-benefit, the project will catalyse gender mainstreaming for water governance through community-based, climate-resilient water management.

109. Similarly, the project will provide economic benefits through avoided health care costs (as improved water security and sanitation practices lead to improved health outcomes) and avoided damages in case of disasters (as climate-resilient infrastructure and training on maintenance thereof reduce the level of damage inflicted).

[48] Halcrow et al. (2010), Resource Guide: Working effectively with women and men in water, sanitation and hygiene programs, International Women's Development Agency and Institute for Sustainable Futures, University of Technology Sydney, Australia, available at: http://www.genderinpacificwash.info/system/resources/BAhbBlsHOgZmIj4yMDExLzAxLzI0LzE5LzA0LzI3LzIwMi9XQVNIX1JFU09VUkNFX0dVSURFX2ZpbmFsNHdlYi5wZGY/WASH%2520RESOURCE%2520GUIDE-final4web.pdf

110. The project responds directly to SDG13 – Climate Action, SDG6 – Clean Water and Sanitation, and SDG5 – Gender Equality through its objective of building climate-resilient and gender-equitable water safety and security in Vanuatu. Finally, by transitioning the water sector towards climate resilience and the management of climate risks, the project contributes towards the objectives of Vanuatu's National Sustainable Development Plan 2016–2030 and the Vanuatu National Water Policy 2017–2030.

D.4. Needs of recipient (max. 500 words, approximately 1 page)

111. This project will address vital needs relating to water security for rural communities in Vanuatu. Over 24,000 households across the country are reliant on either rainwater or surface water for their primary drinking water supply. Also, one third of households have no alternative source of drinking water. These challenges have historically been, and will continue to be, exacerbated by climate change.

112. The project addresses the fundamental human right of access to safe drinking water. Those communities that have elevated levels of vulnerability, low access to WASH services and are exposed to climate risks will be targeted through an existing, nationally-owned prioritisation process that the project will enhance. Besides those communities most in need of climate-resilient water management interventions to be directly targeted, the project will ultimately benefit all rural communities by strengthening the enabling environment for resilient interventions.

113. The project recognizes the primary challenge for water safety and security in Vanuatu, namely that while access to a proximate source of drinking water is high (94% access to an improved drinking water source and 86% access on the premises)[49], the safety and reliability of the water consumed by most people in Vanuatu is unknown. Particularly in rural areas, the absence of market signals (such as prices and permits) as well as inadequate planning and incentive structures destabilise water security.

114. Rural communities in Vanuatu use a combination of groundwater, surface water and rainwater that are increasingly precarious due to limited climate proofing of infrastructure and processes. Of these sources, rainwater systems are most common, with over 66% of all surveyed water supplies drawing from such infrastructure rising to over 75% of supplies in Malampa and Penama[50]. A significant portion of the population (36% of households overall and 44% in rural areas) relies on rainwater as either its primary or secondary water supply[51]. Groundwater-based water sources are less common, comprising only 13% of surveyed systems. Likewise, piped systems made up just 11% of surveyed water systems, of which 54% are fed by springs, 32% by surface water, and 14% by boreholes and wells. Only one third of households have access to water 24 hours per day, every day of the year. In more remote areas, using water from unprotected sources is common[52]; while the national average is around 12%, in area councils such as Erromango and Tanna, this rises to 70%.

115. Water rationing, both for household use and sanitation, and water shortages are common during the dry season. The DoWR WRI indicates that many of the water sources are not available or have inadequate yields during the dry season. Conversely, rainfall often manifests as intense periods of extreme precipitation, leading to floods that further exacerbate the WASH context. For sanitation purposes, households outside of urban centres and provincial hubs (where flush toilets are available) are reliant on shared or private pit latrines or have no toilets at all. This is particularly true in Penama where 83–100% of households rely on these types of sanitation infrastructure.[53]

116. The direct impacts of limited water supply and water security, contaminated drinking water and inadequate WASH infrastructure are increased morbidity (diarrhoea, stunting and other illnesses) and increased mortality, among both children and adults. Significant improvements in the management, operation and maintenance of rural water systems are needed to ensure water services are managed safely. The project will address these by introducing safely managed and climate-resilient WASH and drinking water services critical for preventing diseases and protecting human health during infectious outbreaks, including the current COVID-19 pandemic. Water insecurity also has secondary impacts on food security, as most Ni-Vanuatu rely on rain-fed agriculture.

[49] UNICEF (2020). Rural water supplies in Vanuatu in need of significant improvements. (WASH Technical Paper/13/2020).
[50] UNICEF (2020). Ibid.
[51] Ibid.
[52] Vanuatu National Statistical Office; World Bank. 2014. Vanuatu : Socio-Economic Atlas, available at: https://openknowledge.worldbank.org/handle/10986/18669
[53] Ibid.

117. The needs of recipients in Vanuatu are mapped in Annex 2 in terms of the variations and trends in drinking water sources as well as sanitation infrastructure[54], including:

- The national average of households using unprotected wells or surface water for drinking water is 11.7%. However, regional and intra-island variation is high: in West Santo (Sanma Province), South Erromango and Middle Bush Tanna (Tafea Province), such households range from 53.2–73.8%. This demonstrates that community vulnerabilities to climate-induced risks fall across a spectrum and must be addressed through localized interventions (see Figure 4, Annex 2).
- The national average of households primarily using household or shared tanks for drinking water is 34.3%. However, this is much lower in over 13 districts where such households form less than 12% (primarily Tafea Province, Maewo in Penama Province, parts of Malampa and Sanma Provinces). There is clear inequity in water access within and across islands, resulting in the climate burden causing water security risks being shared unequally in the country (see Figure 5, Annex 2).
- The national average of households using private or shared pit latrines or with no toilet facilities is 48%. However, this rises to 82.7–100% in most of Penama Province, while Sanma, Malampa and Tafea Provinces also have more limited access to sanitation infrastructure. Climate change poses risks to households using such limited sanitation infrastructure, with long-standing setbacks faced during hazards as experienced during recent Tropical Cyclones Harold and Pam (see Figure 6 – Annex 2).

118. The project will address these needs through a combination of institutional strengthening and capacity building at national and devolved levels, as well as through direct improvements to drinking water and WASH systems. The interventions to achieve these improvements are outlined in Section B.3.

D.5. Country ownership (max. 500 words, approximately 1 page)

119. This project is listed as the first priority project in the pipeline presented in Vanuatu's GCF Country Programme[55] and has been fully co-developed with the DoWR, along with other national stakeholders. By addressing climate change risks and impacts on water management, and by working directly with vulnerable rural communities on community-based adaptation activities, the project is fully aligned with the Government of Vanuatu's climate change strategies and policies, including the Climate Change and Disaster Risk Reduction Policy 2016–2030 (e.g. strategic priority 7.4.3) as well as the NAPA and the NDC, which both reference water resource management as a top priority. In addition, the project is fully in line with Vanuatu's National Sustainable Development Plan 2016–2030 (e.g. objective ECO2.2) and National Water Policy 2017–2030.

120. The project responds directly to the Government of Vanuatu's goal for every rural community to have a DWSSP by 2030, as outlined in the Water Resource Management Act[56] and the Water Supply Act[57]. The DoWR will be leading the implementation of the project by following the NIP and DWSSP processes, through which the project will scale up system improvements and WASH infrastructure in rural contexts. These processes are fully owned by DoWR, which is also the lead coordinator of the Vanuatu WASH Cluster, acting as the secretariat and chair during emergencies. This WASH cluster is very active due to the high number of emergencies that the country experiences, and the DoWR is experienced in responding to water access and security issues during disaster and other fast-onset climate events. There is thus already strong ownership of such processes and activities by DoWR, which this project will build on.

121. The Government of Vanuatu has also committed USD 3.9 million co-financing to support delivery of the project outcomes. This will flow primarily through the CAP process to support development of DWSSPs, further indicating the ownership and commitment of DoWR to the project activities and objectives.

D.6. Efficiency and effectiveness (max`. 500 words, approximately 1 page)

122. The overarching project goal is to ensure climate-resilient water security in vulnerable rural communities across Vanuatu. To do so, the interventions will address the social, gender, regulatory, financial, ecological, and institutional barriers that hinder communities from coping with the additional burden of climate change on water security management (see Section B1). The project thus serves to shift the water sector development pathway for rural communities in Vanuatu in a more climate resilient direction. A comprehensive description and illustration of the theory of change is presented in Section B2.

[54] World Bank (2014). Vanuatu: Socioeconomic Atlas, available at: https://documents1.worldbank.org/curated/en/827661468098051596/pdf/876750WP0P1336089B00PUBLIC00Vanuatu.pdf
[55] Vanuatu's GCF Country Programme is available here: https://www.greenclimate.fund/sites/default/files/document/country-programme-vanuatu.pdf
56 https://mol.gov.vu/images/docs/Water_Resources_Acts/DoWR_Files_2018/Water-Resource-Management-Ammendment-2016.pdf
57 https://mol.gov.vu/images/docs/Water_Resources_Acts/DoWR_Files_2018/Water-Supply-Act-1955--Revisions.pdf

123. The proposed project will build climate resilience and adaptation capacities in Vanuatu's WASH sector through the implementation of proven and efficient water management technologies. This will follow the nationally owned DWSSP processes that support the identification of locally led interventions and in doing so, lay the foundations for further scaling-up beyond the project lifetime.

124. The focus of the adaptation investments is on resilient and reliable water infrastructure. It will include:

- **Community Rainwater harvesting (RWH) systems.** These simple, modular systems consist of roof catchment, gutters, first flush and tank storage. The roof catchment could be household or community buildings. Storages can be linked and augmented as needed. Water can be extracted from a tap directly from the tank, a nearby tap stand, or several tap stands in a small distribution system.
- **Gravity-fed systems.** Gravity-fed systems deliver water through pipe and storage networks to demand centres. They typically collect water from reservoirs, springs, rivers and groundwater sources, and can vary in capacity and distance. Direct Gravity Fed (DGF) systems use only gravity to convey water from the source to demand centres, whilst Indirect Gravity Fed (IDGF) systems use pumping to elevate water before gravity conveys it to demand centre(s). DGF systems can provide large volumes of water at high pressure without on-going running costs. Their flexibility and scalability of distance and capacity is only hampered by source safe yield, and topology.
- **Desalination.** Solar-powered reverse osmosis desalination is increasingly being adopted around the world for its ability to reliably provide excellent quality water with low greenhouse gas emissions.
- **Groundwater systems.** Includes rehabilitation and construction of new groundwater wells systems.

Based on above, the project has the potential to generate a broad range of environmental, social, and economic benefits and co-benefits, ensuring that the project is holistic, efficient and effective:

- Increased capacity to identify, develop, and implement tailored and focused adaptation/water security measures and needs.
- Avoided losses associated with low water supply during extreme climate events.
- Avoided losses associated to water import which is costly due to Vanuatu's topography and remoteness.
- Avoided health care-related costs associated with water-borne diseases (WBDs).
- Avoided loss of human life associated with WBDs.
- Avoided loss related to labour for gathering water from remote, unsanitary, and non-adapted water sources.
- Avoided GHG emissions through renewable energy-based technologies.

125. Annex 3 – Economic and Financial Analysis demonstrates the viability of this project in greater detail. The financial analysis shows that without grants, the costs would be many orders of magnitude beyond what households can pay (ranging from 657% to 3,674% of ability to pay), and a substantial portion of average monthly incomes. With GCF grant funding, the financial analysis shows that for rainwater harvesters and direct gravity fed systems, an affordable tariff can be reached, whereas for water desalination and groundwater systems, additional public sector funding will be required. Concerning the economic analysis, the project-level ENPV is positive, USD 37,150,065 and the programme-level EIRR is 20%. This project is thus economically viable and can be justified on economic grounds, even though non-investment budget costs (including co-finance) have no envisaged direct benefits. This is the case even though the analysis included conservative assumptions and not all benefits were included in the economic calculations as it was not possible to estimate their monetary values. Nonetheless, such benefits will accrue to project beneficiaries in addition to those included in the analysis. Therefore, the adaptation project is expected to be highly effective.

126. The efficiency of the DWSSP process will also contribute to the overall efficiency and effectiveness of the project. This will be the primary modality through which the project will identify and prioritise communities in need of system improvements and the activities to bring about such improvements. Those communities that have already created DWSSPs have cumulatively spent almost USD 1 million of their own money on no- and low-cost improvements.[58] The project will benefit from the existing structures and policy frameworks to efficiently scale up investments by finetuning and mobilizing the DWSSP process through the DoWR.

[58] Please see the RISCON database here: https://exchange.riscon.solutions/dwssp/

E. LOGICAL FRAMEWORK

E.1. Project/Programme Focus

☐ Reduced emissions (mitigation)

☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

127. The project will increase the adaptive capacity of rural communities to better cope with the additional burden of climate change on water security and safety, by improving climate-resilient water management community- based planning, providing explicit capacity building, and fostering adaptation actions through improved local management practices and resilient infrastructures. Water safety and security is indeed particularly vital for community long-term resilience and in the immediate aftermath of climate-induced disasters.

128. This project will contribute to the second paradigm-shifting pathway that the GCF has identified in its Water Security Sectoral Guide: *strengthening integrated water resources management and water management* and will deliver on all two sub-areas: (i) preservation of existing water resources; and (ii) development of new water supply sources. Below, three assessment dimensions – scale, replicability and sustainability of this paradigm shift pathway are expounded.

129. 700 communities will be direct beneficiaries as part of Outcome 1 (improved planning and community-based activities on climate-resilient water management), of which 220 will also benefit from improved climate-resilient water infrastructure as part of Outcome 2. An additional 50 communities with existing DWSSPs will be targeted under Outcome 2 during the first year of the project. This makes a total of 750 communities directly benefiting from Outcomes 1 and 2. A preliminary estimate[36], of direct beneficiaries is 85,650 (including 42,825 women), that is around 27% of the total population in Vanuatu. Indirect beneficiaries include the entire rural population in Vanuatu (216,980 individuals which is 73% of the total population), mostly through enhanced institutional capacities and processes leading to climate-resilient water security for rural communities (Outcome 3).

Outcome	Targeted communities	Indirect / direct beneficiaries
1	700	79,940 direct beneficiaries (including 39,970 women), which is 25.2% of the total population of Vanuatu.
2	270 (including 220 already targeted by Outcome 1, and 50 additional ones)	30,834 direct beneficiaries (including 25,124 from Outcome 1 and 5,710 additional ones); which is 8% (2% additional) of the total population in Vanuatu.
3	2,000	Indirect beneficiaries: the entire rural population in Vanuatu (around 216,980 individuals, which is 73% of the total population)

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	Based on the data available, there are 2,447 registered water systems in Vanuatu. Of these, 11% (277) have DWSSPs registered and received system improvements. Nearly 90% of water supply systems therefore require DWSSPs to be developed and implemented. Current DWSSPs do not address climate-resilient WASH infrastructure. Most sources of drinking water cannot provide 24/7 service year-round due to climate related restrictions as detailed in section B2. .	<u>Low</u>	The 5-year project will reach 600 communities with new, climate-resilient DWSSPs and low-cost adaptation activities (Outcome 1), 100 communities with existing DWSSPs will also be updated in Outcome 1, and 270 communities (of which 220 communities will be those with new DWSSPs delivered through Outcome 1, with an additional 50 communities which have existing DWSSPs) with infrastructure upgrades (Outcome 2) and all ~2000 rural communities (216,980 individuals) through improved institutional processes (Outcome 3). The current total of DWSSPs delivered (277, according to DoWR data) will be scaled up by dispensing investments through the government-owned DWSSP and NIP processes. This should lay the groundwork for upscaling as government mechanisms will be utilized and improved, while attracting potential donors for later phases, with a similar goal of improving climate-resilient water security in rural Vanuatu.	Through Outcome 1 & 3 (see B2a), the project will address the institutional and regulatory barriers in Vanuatu – particularly, the limited financial and technical capacity that restricts the DoWR, provincial institutions and communities from scaling up existing technology as well as shift towards new technology. By project conclusion, Vanuatu will have 600 new DWSSPs and 100 updated DWSSPs which will focus on delivering climate-resilient WASH infrastructure, and improving water access and security.

Replicability	One of the primary barriers in Vanuatu is limited WASH infrastructure and water supply sources that include climate adaptation considerations. Although the project will benefit from established mechanisms for implementation (DWSSPs, NIP, TAP and CAP), investments in climate-resilient WASH infrastructure has been minimal, and the project will need to initiate this paradigm shift in Vanuatu.	Low	This project will aim to improve the ability of rural communities and water systems to anticipate, adapt to and recover from the negative effects of climate change through all three Outcomes. With the investments earmarked for infrastructure rehabilitation and development in rural Vanuatu through established mechanisms of identifying needs, prioritization and selection, and disbursement, the project will replicate interventions sub-nationally. Post project replication of this model is possible more widely across the country and in other SIDS in the region and globally.	Investments in rural water supply infrastructure (whether new or improved), will be at once a private good (providing water supply to households), but when used correctly and consistently, also a wider, public good key to replicating success and achieving climate resilience and SDG 6 targets in Vanuatu, and (more broadly) in the Pacific SIDS. Further, having relevant institutional documentation, systems and capacity building in place (under Outcome 3) will ensure that the project outputs are replicable post project, and that DoWR staff have capacity to continue to replicate project outputs across the country at scale beyond completion.
Sustainability	The DoWR – and other WASH sector partners – have been proactive in increasing sustainability of investments by establishing country-owned processes that this project will utilize during implementation (NIP, CAP and the DWSSP). Additionally, given the impacts of two recent Tropical Cyclones (Pam and Harold) – Vanuatu's National Water Policy identifies that sustaining infrastructure through climate-proofing as an important step towards water security.	Medium	The project will utilize existing institutional processes (NIP, CAP and DWSSP) to embed adaptation measures in both the design and ongoing maintenance of infrastructure. This will ensure communities in Vanuatu will develop climate resilience through improved water security, provided by infrastructure investments, as well as improved water management plans, which will help sustain these investments in the long term.	By being fully part of existing national programmes, and in line with national policies and strategies of Vanuatu, this project will be transformative by leveraging the limited amount of public funds used for water security and safety programmes. The project's outcomes will be co-financed through central government funding, including for monitoring, and sustained through the same mechanisms after the end of the project. Work in Outcome 3 will enhance the institutional processes and build

				capacity to contribute to sustainability. The project will build technical capacity of the communities to undertake maintenance of the climate- resilient infrastructure (as part of Outcome 2). The communities will also receive regular monitoring visits from the provincial and national government officials (as part of Outcome 1). Post project, communities will operate and maintain their water security and safety infrastructure, with government-funded support through the improved TAP and CAP processes. The project will also build technical capacity of DoWR staff which will improve the institutional capacity of the DoWR to continue to deliver CR-WASH infrastructure following the end of the project. Throughout the project training will be given to relevant extension agents, key institutional personnel and local communities on the impacts of climate change and what measures can be put in place to overcome these challenges. Training will be embedded in the institutional

				processes for ongoing sustainability. This includes training on: - climate vulnerability, risks and updated water management planning (Outcome 1) - operation and maintenance of water infrastructure installed in the project, to ensure long term operation of the investments (Outcome 2). - specific enhancement of engineering standards to ensure improved and more sustainable infrastructure is in place in the future (Outcome 3).

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

Select appropriate IRMF core and supplementary indicators to monitor project/programme progress. More than one IRMF (core and or supplementary) indicators may be selected as applicable for each GCF results area and project/programme outcome (as defined in the table in section B.2(b)). If IRMF indicators are unable to measure any given project/programme outcomes, project/programme-specific indicators should be developed under section E.5 (project/programme specific indicators).

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final[59]	
		Sources of information and methods used to collect and report data /information to measure	*The starting point or current value of the indicators before the implementation of the project*	*The estimated value of the indicator at the mid-point of the implementation*	*The estimated value of the indicator at the completion of the implementation*	*Externalities and factors outside project management's control that may impact the outcomes* *Data sources and methodologies applied for estimating baseline and targets*

[59] The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

		progress against targets				
TOTAL (Under ARA1, ARA2 and ARA3)	Core 2: Direct and indirect beneficiaries reached	Please refer to relevant rows below	Please refer to relevant rows below	**Direct beneficiaries:** 34,260 - 10.8% rural population (male: 17,130 female: 17,130) **Indirect beneficiaries:** 91,360 - 29.2% of the population (male: 45,680 female: 45,680)	**Direct beneficiaries:** 85,650 - 27% rural population (male: 42,825 female: 42,825) **Indirect beneficiaries:** 216,980 73% of rural population (male: 108,490 female: 108,490)	Noting that beneficiaries to the CAP overlap with those of the DWSSP.
ARA1 Most vulnerable people and communities	Core 2: Direct and indirect beneficiaries reached	Annual reporting based on monitoring and verification, national surveys, WHO/UNICEF's JMP survey, and independent interim and final evaluations carried through the project,	0	**Direct beneficiaries:** 34,260 - 10.8% rural population (male: 17,130 female: 17,130)	**Direct beneficiaries:** 85,650 - 27% rural population (male: 42,825 female: 42,825)	Total direct beneficiaries = Total beneficiaries from Outcome 1 (79,940 beneficiaries) and additional beneficiaries under Outcome 2 (5,710). Targeting criteria for project activities is to target the most vulnerable communities to climate change impacts, this is prioritised through

		completion reports of trainings and infrastructure investments. Quality assessed by the AE and EE.		**Indirect beneficiaries:** 91,360 - 29.2% of the population (male: 45,680 female: 45,680)	**Indirect beneficiaries:** 216,980 73% of rural population (male: 108,490 female: 108,490)	40% delivery by mid-term, and 100% by end-term is expected. The project's institutional strengthening in expected to **indirectly** impact 100% of the rural population = a total of 216,980 additional indirect beneficiaries to the 85,650 direct beneficiaries.. This represents 73% of the total population. All communities have, on average, an equal population. Sex ratio favours males – there are 105 males : 100 females as of 2020, and this may be reflected in the gender-disaggregation of the beneficiaries. For the purposes of estimation during preparation stage, 50-50 ratio has been used. Procurement is carried out on time and there are no political factors that slow down or interfere with project implementation. No natural disasters or extreme events cause catastrophic damages that impede the implementation of the project. Stakeholder engagement continues to be strong and there is a willingness at the beneficiary level to engage with the project.

ARA2 Health, well-being, food and water security	Supplementary 2.3: Beneficiaries (female/male) with more climate-resilient water security	Annual reporting based on monitoring and verification, national surveys, WHO/UNICEF's JMP survey, and independent interim and final evaluations carried through the project, completion reports of trainings and infrastructure investments. Quality assessed by the AE and EE.	0	**Direct beneficiaries:** 34,260 - 10.8% rural population (male: 17,130 female: 17,130) **Indirect beneficiaries:** 91,360 - 29.2% of the population (male: 45,680 female: 45,680)	**Direct beneficiaries:** 85,650 - 27% rural population- (male: 42,825 female: 42,825) **Indirect beneficiaries:** 216,980 73% of rural population (male: 108,490 female: 108,490)	Total direct beneficiaries = Total beneficiaries from Outcome 1 (79,940 beneficiaries) and additional beneficiaries under Outcome 2 (5,710). 40% delivery by mid-term, and 100% by end-term is expected. The project's institutional strengthening in expected to **indirectly** impact 100% of the rural population = a total of 216,980 additional indirect beneficiaries to the 85,650 direct beneficiaries. This represents 73% of the total population. All communities have, on average, an equal population. Sex ratio favours males – there are 105 males: 100 females as of 2020, and this may be reflected in the gender-disaggregation of the beneficiaries. For the purposes of estimation during preparation stage, 50-50 ratio has been used. Procurement is carried out on time and there are no political factors that slow down or interfere with project implementation. No natural disasters or extreme events cause catastrophic damages that impede the implementation of the project.

						Stakeholder engagement continues to be strong and there is a willingness at the beneficiary level to engage with the project.
ARA3 Intrastructure and built environment	Core 2: Direct and indirect beneficiaries reached	Monitoring surveys and reports provided by the PMU and quality assessed by the EE and AE to ensure alignment with the indicator and that no double counting or bias has been included. Engineering reports on completion of constructions provided by implementing partners submitted to PMU and EE's for quality assurance and verification against the indicator. Lessons learnt and best practice guide on	0 No individual has accessed CAP resources for enhance CR-WASH infrastructure as this will be brought in by the project.	14,618 individuals 7,308 female 7,308 male	36,544 individuals 18,272 female 18,722 male	40% beneficiaries reached by mid-term and 100% expected by project conclusion. These beneficiaries partially overlap with the beneficiaries of Outcome 1. This is because 220 DWSSPs selected will be from the 600 DWSSPs newly registered through Outcome 1 and an additional 50 existing DWSSPs in need of upgraded infrastructure will be targeted. This will total 270 DWSSPs, against which CR-WASH infrastructure is delivered. By increasing technical assistance and options to access to financial support, the delivery of functional, climate-resilient, water systems is made possible to communities. Members of communities are able to benefit from these investments through training and awareness activities, with the requisite behavioural needed to maintain these systems.

		technical interventions developed by the PMU and presented to EE's and AE for quality assessment and alignment to indicator.				40% communities reached by mid-term and 100% expected by project conclusion. Vanuatu is regularly exposed to natural hazards, which can be an impediment to the delivery of water infrastructure, particularly if the efforts and funds dedicated to water governance are redirected.
ARA3 Intrastructure and built environment	Core 3: Value of physical assets made more resilient to the effects of climate change and/or more able to reduce GHG emissions	Financial tracking and monitoring will generate reports carried out by the Finance Officer. Annual reporting based on monitoring and verification, national surveys, independent interim and final evaluations carried through the project, completion reports Reports for providers on trainings and infrastructure investments. Quality assessed by the AE and EE's.	The national Water Resources Inventory (WRI) conducted a water point mapping that identified 4,097 water systems across 44 islands of which: • 15% were Indirect Gravity Fed systems, • 20% Direct Gravity Fed systems and • 65% Community Rainwater Harvesting systems. Approximately 1/3 of these systems were not providing 24 hr supply, and half of water systems were not providing a year-round supply.	USD 3,818,379 Rain Water Harvesting USD 1,192,761 Direct Gravity Fed USD 2,767,536 Desalination USD 147,168 Indirect Gravity Fed USD 1,047,345	USD 9,545.947 Rain Water Harvesting USD 2,981,904 Direct Gravity Fed USD 6,918,840 Desalination USD 367,920 Indirect Gravity Fed USD 2,618,364	Infrastructure investments targeted will be conducted on a proposal basis through the CAP these are therefore considered as unidentified investmtnets in design. As such it is impossible through design to place a value on the underlying infrastructure in the country. However, the value of infrastructure to be upgraded or created is presented in the targets representative of the equipment and construction value of the new infrastructure estimated at this point. 40% delivery by mid-term, and 100% by end-term is expected. Procurement is carried out on time and there are no political factors that slow down or interfere with project implementation. No natural disasters or extreme events cause catastrophic

						damages that impede the implementation of the project. Please note that the baseline figure may change at inception with additional CAP investments over the approval and post approval processes period. This could marginally impact the percentages.

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)					
Core Indicator	**Baseline context (description)**	**Rating for current state (baseline)**	**Target scenario (description)**	**How the project will contribute**	**Coverage**
Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner	The Government of Vanuatu has put in place a water planning process based on a NIP and community-owned DWSSP process. At present this framework does not incorporate climate change considerations.	low	The project aims to achieve a paradigm shift towards climate resilient water security for rural communities across Vanuatu, by enhancing community-based planning and adaptation for climate-resilient water management (Outcome 1), developing climate-resilient rural water infrastructure (Outcome 2), and creating an enabling environment at provincial and national level to better address climate risks associated with water security (Outcome 3). Through activities under Outcome 1, the DWSSP design process will be enhanced to incorporate climate resilience directly into community level planning mechanisms. Further, institutional capacity building under Outcome 3 will result in increased technical and implementation capacity in the DoWR and other relevant ministries at both state and national levels. Through these institutional interventions,	The project will put in place a robust and replicable community owned process to incorporate climate change considerations into infrastructure improvement. The capacity of Vanuatu's institutions to deliver, maintain and increase these gains will grow.	National level (one country)

			the entire rural population of Vanuatu (216,980 individuals and 73% of the total population) will benefit indirectly from strengthened national and state institutions to enhance climate resilience.		
Core Indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation	Rural Vanuatu has a relatively high coverage (90%) of improved water supplies[60], although significant inequalities exist geographically.[61] [62] However, these "improved" systems do not incorporate climate resilience considerations and are susceptible to the negative impacts of climate change. Rural communities use a combination of groundwater, surface water, and rainwater, but rainwater systems are most common.[63] Despite having access, these systems often fail to provide 24-hour supply and year-long supply as well.[64]	low	Direct beneficiaries include those beneficiaries in communities targeted under Outcomes 1 and 2 that result in development and dissemination of new climate resilient DWSSPs and deployment of upgraded climate resilient infrastructure. Estimated direct beneficiaries is 85,650 (including 42,825 women), ~27% of the total population in Vanuatu. Indirect beneficiaries include the entire rural population of Vanuatu (216,980individuals and 73% of the total population), mostly through enhanced institutional capacities and processes toward climate-resilient water security for rural communities (Outcome 3 – provincial and national institutions are strengthened to address climate risks associated with water security). In the long-term these activities and outputs will result in greater technology deployment, dissemination and innovation transfer benefiting the whole rural population.	Community owned planning will take place and investments in infrastructure will be made – technology deployed and transferred.	National level (one country)

E.5. Project/programme specific indicators (project outcomes and outputs)					
			Baseline	Target	Assumptions / Note

[60] WHO/UNICEF define an improved drinking-water source as one that, by nature of its construction or through active intervention, is protected from outside contamination, in particular from contamination with faecal matter. Improved water sources include piped water into dwelling, plot or yard; piped water into neighbour's plot; public tap/standpipe; tube well/borehole; protected dug well; protected spring; and rainwater.
[61] Government of Vanuatu (2017 – 2030). Vanuatu National Water Policy. *Ibid.* found 93% "improved", which is broadly consistent with the 2015 WHO/UNICEF Joint Monitoring Programme's (JMP) estimate that 89% of the rural population of Vanuatu has access to an improved water source (one source of water).
[62] UNICEF (2020). *Ibid.*
[63] UNICEF (2020). *Ibid.*
[64] UNICEF (2020). *Ibid.*

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)		Mid-term	Final	
Outcome 1: Communities are empowered to plan and manage climate-resilient water resources						
Output 1.1 New and existing DWSSPs incorporate incremental improvements to mainstream adaptation solutions	• Percentage of new or updated DWSSPs that incorporate climate resilience	Monitoring by PMU submitted to the EE's for quality assessment/verification against the indicator. This will include summary information from the DWSSP approval processes under the DoWR.	0% 0 existing DWSSPs	60% new DWSSPs 60 existing DWSSPs	100% new DWSSPs 100 new DWSSPs	60% registration of DWSSPs by mid-term and 100% expected by project conclusion. In alignment with expectation that the DWSSPs will be carried out in fist 3 years of the project. Limited information on the DWSSPs process among communities may hinder this outcome. DWSSPs updated or developed using the methodology are expected to result in identification of climate responsive measures needed at community level, and aid in the prioritization of dispensing investments.
Output 1.2 Awareness, capacities and skills of communities and area administrators on climate-resilient water	• Number area administrators trained for greater technical knowledge on CR-WASH • Number of RWC members with increased CR-WASH awareness and	Monitoring reports including training evaluations and summary of trainings by PMU submitted to the EE's for quality assessment/verification against the indicator. Reports to included qualitative and anecdotal	0	Area Administrators = 38 RWC members = 180 (90 female)	Area Administrators = 96 RWC members = 450 (225 female)	40% beneficiaries reached by mid-term and 100% expected by project conclusion. 16 area administrators trained per province

management improved	capacity	information gathered through community engagements through implementation and supervisions. Training registration sheets collected and provided by trainers to the M&E Officer for collation into reports.				3 trainings per province with 25 RWC members attending each. 6 knowledge sharing events and activities with TA and training for rural water committees. The events would bring together a large group of stakeholders, and ensure that committees registered though DWSSPs can be brought together as a community of practice. Members of the communities are interested in climate-resilient water management and are available for training. Trained area administrators perform that role for a period long enough to allow for transmission and multiplication of knowledge. There is uptake among diverse members of the communities, due to active processes ensuring access and inclusion.
Output 1.3 Vulnerable communities are supported to	• Number of communities, who developed new DWSSPs using	Reports from DWSSP facilitators on completed DWSSPS included in interim and final progress	0	240	600	40% registration of DWSSPs by mid-term and 100% expected by project conclusion.

develop and implement their DWSSPs	updated methodology (though Output 1.1)	reports submitted to the EE's for quality assessment/verification against the indicator. Lessons learnt and best practice guide on technical interventions developed by the PMU and presented to EE's and AE for quality assessment and alignment to indicator.				Governance conditions in the target sites are favourable and communities are able to demonstrate no and low cost investments. There is uptake of the training provided and widespread understanding of the updated DWSSP processes. Trained community members and area administrators perform that role for a period long enough to allow for transmission and multiplication of knowledge. Please note at the time of project design 301 DWSSPs had been designed to date – therefore total DWSSPs at the time of completion in country is expected to be >901.
Outcome 2: Communities have enhanced climate-resilient rural water infrastructure						
Output 2.1 Vulnerable communities supported to construct, operate, and maintain climate-	• Number of communities supported to construct, operate, and maintain climate-resilient water infrastructure	Monitoring reports provided by the PMU and quality assessed by the EE's and AE to ensure alignment with the indicator and that no	0	108	270	Communities partially overlap with the beneficiaries of Outcome 1. This is because 220 DWSSPs selected will be from the 600 DWSSPs newly registered through

resilient water infrastructure		double counting or bias has been included. Travel log, meeting minutes and other reports for study tours presented in APR's and quality assessed by the PMU and EE and verified data against the indicator logged. Lessons learnt and best practice guide on technical interventions developed by the PMU and presented to EE and AE for quality assessment and alignment to indicator.				Outcome 1 and an additional 50 existing DWSSPs in need of upgraded infrastructure will be targeted. This will total 270 DWSSPs, against which CR-WASH infrastructure is delivered. By increasing technical assistance and options to access to financial support, the delivery of functional, climate-resilient, water systems is made possible to communities. Members of communities are able to benefit from these investments through training and awareness activities, with the requisite behavioural needed to maintain these systems. 40% communities reached by mid-term and 100% expected by project conclusion. Vanuatu is regularly exposed to natural hazards, which can be an impediment to the delivery of water infrastructure, particularly if the efforts and funds dedicated to

						water governance are redirected. Please note at the time of project design 47 CAP investments had been approved to date – therefore total CAP investments at the time of completion in country is expected to be >317.
Outcome 3: Provincial and national institutions are strengthened to address climate risks associated with water security						
Output 3.1 National- and provincial-level staff and WASH sector partners trained on climate-resilient water management	• Number of provincial and national institutions / offices strengthened through training of staff	Baseline, interim and final assessment reports provided by the PMU and submitted to the EE's for quality assessment. Pre- and post-training self assessments carried out by DoWR staff and assessed and collated by the M&E officer	0	7	35	Within these trainings 420 GoV (210 female) staff will be trained. This will be tracked an verified through attendance sheets. Alongside the DoWR, the: Ministry of Agriculture, Livestock, Forestry, Fisheries and Biosecurity; Ministry of Climate Change; Ministry of Education and Training, and the Ministry of Health will benefit from these trainings. By mid-term, trainings for DoWR will be prioritised and completed, and the remaining cross-sectoral trainings on CR-WASH will be provided by conclusion of the project.

						Therefore, 5 institutions will be strengthened nationally, alongside their provincial offices in each of the 6 provinces. Staff from the DoWR and other relevant governance institutions, at both national and provincial level, are available for training sessions. Staff are retained by these organizations for a period long enough to allow for transmission and multiplication of knowledge.
Output 3.2 Knowledge management through data sharing mechanism established for climate-resilient water management	• Data platform established, with robust knowledge management protocols	Monitoring reports provided by the PMU and quality assessed by the EE's and AE to ensure alignment with the indicator and that no double counting or bias has been included. Data platform established and published online and verified by the EE.	No mechanism and framework for robust knowledge management. DoWR has a data portal, that requires upkeep.	Knowledge management protocol established	Data portal improved and strengthened and operational online.	Stakeholders are interested in participating in coordination and consultative processes for knowledge management and data sharing mechanisms and can be trained on protocols established.
Output 3.3 Monitoring, learning and evaluation framework established for	• M&E process established within DoWR	Monitoring reports provided by the PMU and quality assessed by the EE's and AE to ensure alignment with the indicator and that no	Nascent processes exist but require strengthening.	Monitoring, learning and evaluation protocol established after	Monitoring, learning and evaluation operationalized with wide access	Stakeholders are interested in participating in coordination and consultative processes for the M&E framework established.

improved learning for climate-resilient water management		double counting or bias has been included.		consultation with DoWR and other WASH sector partners		
Project/programme co-benefit indicators						
Co-benefit 1	Improved gender parity in representation of women in Rural Water Committees (RWCs)	Registration of RWCs with members (indicating a minimum of 40% of women)	Women are not well-represented across water governance or devolved utility authorities across Vanuatu – exact figures are currently unavailable.	40% of RWCs registered, with 40% of women in each.	Remaining 60% of RWCs registered, with 40% women in each.	40% RWCs registered by mid-term and 100% expected by project conclusion. There is political will in involving women in different sub-national contexts and in water governance institutions. Women may not be, however, able to participate meaningfully participate due to sociocultural barriers, and demands on their time due to unpaid care work within the household.

E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
1.1.1: Implement DWSSPs processes that incorporate incremental improvements to mainstream adaptation solutions and update existing DWSSPs	The DWSSP methodology has been updated recently with significant overhaul towards climate mainstreaming. It is expected that the methodology will be further updated - with annual improvements. In the new DWSSPs developed with communities, the updated methodology will be used to ensure climate risks are mainstreamed. For the existing DWSSPs – workshops and technical assistance will be provided to ensure these plans are	1.1.1.1: Review uptake and delivery of updated methodologies, making incremental improvements annually 1.1.1.2: Integrate updated methodology into DWSSP processes triggered during the project 1.1.1.3: Update existing DWSSPs, when appropriate	1 annual review and stocktake of DWSSP process per implementation year, tabulating new and updated DWSSPs in specific communities achieved per year

	retrofitted to better reflect climate risks and adaptation solutions.		
1.2.1: Train area administrators and develop material for communities on climate vulnerability, risks and updated DWSSP methodology	For Activity 1.2.1, targeted trainings will be provided through technical assistance. Area administrators, if relevant, can be paired with DWSSP facilitators to ensure that trainings are not duplicated (this activity will tie in with 1.2.1).	1.2.1.1: Conduct training targeting area administrators on climate vulnerability and risks and updated DWSSP methodology 1.2.1.2: Conduct training targeting communities on DWSSP processes, with interactive content and WASH advocacy materials	Annual reporting, consolidating evaluations of training of area administrators and communities on climate-resilient WASH risk indicators: number of participants – gender disaggregated number and location of training courses delivered. 1 consolidated document of interactive training material and learning notes.
1.2.2: Organize knowledge sharing events	For Activity 1.2.2, interactive training material and learning notes will be developed. This could be done in partnership with UNICEF which already has tried and tested community-led efforts in Vanuatu.	1.2.2.1: Organize community-level knowledge sharing events to ensure widespread dissemination among communities	Annual event report will be produced.
1.3.1 Identify vulnerable communities through the NIP process to prioritize delivery of DWSSPs	Through this output, the project will deliver DWSSPs to 600 communities, through the NIP process. Activity 1.3.1 will involve PWRAC identifying vulnerable communities who request DWSSPs through the NIP process. For the new DWSSP facilitators – and existing staff – training will be provided on the DWSSP methodology. This will focus on climate risks and how these can be adapted to through the DWSSPs. Rural Water Committees (RWCs) will be established through the DWSSPs – with a mandatory quota of 40% women. Climate vulnerability assessments and community engagement are undertaken to ensure 600 DWSSPs developed during the project cycle are informed by the observed climate impacts.	1.3.1.1: Recruitment of provincial engineers – these positions will be embedded in the DoWR and will provide centralised technical support at the provincial level 1.3.1.2: Refresher for community DWSSP facilitators halfway through the project to serve as a refresher training and to incorporate the various incremental improvements over the intervening implementation period incorporated under Output 1.1. 1.3.1.3: Development of 600 DWSSPs informed by best-available climate science, local information and traditional knowledge 1.3.1.4: Establish and register local water committees. 1.3.1.5: Support provided by Provincial Engineers and Provincial DoWR officers to	1 annual review and stocktake of DWSSP process, tabulating new and updated DWSSPs in specific communities achieved per year [Part of reporting for 1.1, including following indicators, for reporting on 1.3: i. number of community facilitators recruited and trained, gender disaggregated ii. 600 local water committees established (at least 40% participation of women) iii. 600 climate vulnerability assessments Number of no and low cost measures delivered]

		implement no and low-cost measures in communities	
2.1.1: Improve CAP request prioritization, with gender and ESIA screening for chosen sites	These activities will establish climate-resilient drinking water infrastructure and build capacity among vulnerable communities to maintain and operate these, thus ensuring a paradigm shift from build-neglect-rebuild approach. Activity 2.1.1 will focus on improving the CAP risk matrix to ensure that vulnerable, remote communities are prioritized through the project.	2.1.1.1: Assess and update multi-criteria analysis for risk ranking process to prioritize CAP requests to identify sites for infrastructure planning 2.1.1.2: Conduct relevant surveys, gender, environment and social safeguards screening and impact assessments in chosen sites	1 report on communities supported and infrastructure delivered [merged with deliverable for 3.1] 1 consolidated, training reports on operation and maintenance capacity in communities [indicators: number of participants – gender disaggregated number and mapping (where in Vanuatu) of training courses delivered.]
2.1.2: Upgrade CR WASH infrastructure, with adequate O&M training	Activity 2.1.2 will deliver CR-WASH infrastructure to communities with selected DWSSPs.	2.1.2.1: Construct and upgrade infrastructure for climate resilient water sources, distribution, and storage 2.1.2.2: Train RWCs on operation and maintenance for the CR-WASH infrastructure 2.1.2.3: Finalise O&M for CR-WASH infrastructure for each community	
3.1.1: Deliver enhanced administrative and technical capacity training for water management	This activity will focus on enhancing the capacities of national and provincial level staff on climate-resilient water management and water governance. Activity 3.1.1. will attempt to expand the skill set available to the DoWR through external technical support.	3.1.1.1: Train PWRAC and DoWR staff on updated DWSSP, climate risks and water management 3.1.1.2: Host engineers (50% male 50% female) in each of the 6 provinces through humanitarian engineering assistance	1 training report on PWRAC & DoWR training [indicators: i. number of participants – gender disaggregated ii. number and mapping (where in Vanuatu) of training courses delivered.]
3.1.2: Update engineering standards and conduct workshops for WASH sector partners	Activity 3.1.2 will provide the necessary impetus to engage coordination between different WASH sector partners, as well as ensuring engineering standards are updated, approved with consensus to minimize or remove risks, are analysed through a robust safeguarding process and then, deployed.	3.1.2.1: Update engineering standards and deploy in provinces for new infrastructure 3.1.2.2: Workshops with WASH sector partners (incl. MoH, Med., and CSOs) on updated DWSSP, climate risks, and water management	1 written report on progress of updated engineering standards and how they have been rolled out presented at mid-term. Workshop reports on WASH sector partner events. [indicators:

			i. number of participants – gender disaggregated ii. number and mapping (where in Vanuatu) of workshops delivered.
3.2.1: Improve knowledge management processes	These activities will focus on knowledge management for the project, and improve data sharing portals in Vanuatu. Support will be provided to WASH sector partners to be able to employ data for decision-making.	3.2.1.1: Review existing knowledge management and identify gaps to be strengthened 3.2.1.2: Establish knowledge management protocol through consultative process	1 consolidated report on KM mechanism and data platform established and reported through the DoWR's M&E unit
3.2.2: Operationalize data sharing platform		3.2.2.1: Integrate data collected through DWSSPs into government knowledge management platforms, such as DoWR data portal 3.2.2.1: Set and disburse relevant data management procedures to ensure data is compatible and aggregable within the system. 3.2.1.2: Engage relevant stakeholders to support effective utilization of data for decision-making by WASH sector partners	
3.3.1: Collate existing MEL practices within water governance structures in Vanuatu, and establish a robust MEL protocol	This activity will take stock of existing MEL practices within government-owned process, and upgrade this system by introducting robust protocol using a consultative process, as well as through SPC's existing processes through the PEARL policy and *rebbilib.* See Section E.7 for details.	3.3.1.1: Stocktake existing MEL processes, and collate lessons learnt, case studies, and best practices available 3.3.1.2: Create an inventory of existing reports 3.3.1.3: Introduce robust MEL protocol, identified through a consultative process	1 consolidated report on MEL system established and reported through the DoWR's M&E unit

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

130. In its role as Accredited Entity, SPC CCES will oversee and supervise the implementation of this project, in accordance with the agreement signed between SPC and the GCF. SPC GEM, in capacity as EE, through the PMU will be responsible for project-level MEL and reporting in compliance with approved SPC policies and GCF requirements under relevant agreements. Further, SPC coordination between its CFU, Strategy, Performance and Learning (SPL) Team and the NPSC will provide supervision and technical assistance as needed to support the PMU implement tools and methods to monitor, evaluate and learn from the project activities.

131. In 2020, SPC along with the New Zealand Government Ministry of Foreign Affairs and Trade, and BetterEvaluation co-designed a participatory process to assess and understand the current MEL system and opportunities for capacity strengthening for projects in the region.[65] This MEL system is informed by a set of principles: Pacific ownership, a strengths-based approach to capacity development, adult learning, and supporting situationally appropriate choices of MEL methods and processes. The MEL for the project will derive from this *rebbilib* announced in conjunction with Pacific leaders and communities.

132. The logical framework contains performance indicators by Outcome and outputs, which will be monitored by the PMU and regular (semi-annual) updates provided to the NDA, NPSC and SPC CCES during program implementation. Additionally, the project will undertake rigorous KM and MEL exercises through Output 3.2 (knowledge management through data sharing mechanism established for climate-resilient water management) and Output 3.3 (monitoring, learning and evaluation framework established for improved learning for climate-resilient water management). Activities under these Outputs will improve the knowledge management platform (particularly the data portal) available in the country in line with GCF policies. Support will be provided to WASH sector partners to be able to employ data for decision-making.

133. A Monitoring and Evaluation and Learning Officer will be hired under the PMU to coordinate MEL across the project. This will include establishing M&E systems that are aligned with GCF, NDA and SPC policies and results framework. This MEL Officer will work together with the Project Manager under the guidance of the NPSC and the EE to develop a set of MEL tools, approaches and reporting arrangements for project activities. This will include annual performance reports and project closure reports. The training, coaching and support provided to beneficiaries of MEL capacity building will include capacity development in MEL, with a focus on how this can be used to maximize activity outcomes while building the evidence base for the results and impact of the initiative.

134. Monitoring will enable the PMU to make adjustments to respond to unexpected events during the implementation phase as well as to build trust and respond to stakeholders and affected communities. The scope, robustness, and frequency of project monitoring, and reporting will vary depending on the type of activities (as per section E above) and the significance of risks/impacts identified through the screening process. In addition, monitoring requirements will take into consideration the circumstances in which the project takes place and is implemented.

135. For the CR-WASH infrastructure to be installed through the DWSSP processes, Rural Water Committees (RWCs) set up as local water governance mechanisms will conduct participatory monitoring, reporting and evaluation, which will feed into the Activity Coordination Group, and scaled up to the provincial water governance bodies (consisting of Provincial Water Supervisor and Community Water Development Officers) for documentation and policy adjustments. One of the main units of the DoWR is the Monitoring and Evaluation Unit[66] – which will be responsible for ensuring the newly registered and existing DWSSPs are functional.

136. Finally, the CFU and GEM team will be jointly responsible for coordinating the independent interim and final evaluations for the GCF. The evaluations will be conducted using a question-driven approach, and will include assessments against the criteria of relevance, effectiveness and sustainability, among others. The evaluation criteria are further described in Appendix II of the GCF Evaluation Policy (2022)[67]. The Interim Evaluation will be instrumental in contributing (through operational and strategic recommendations) to improve implementation, setting out any necessary corrective and adaptive management measures for the remaining period of the project, and identifying relevant lessons learned for stakeholders in Vanuatu as well as the broader

[65] Pacific Monitoring, Evaluation and Learning Capacity strengthening Rebbilib / by Pacific MEL, Strategy Performance and Leaning (SPL) Unit, Office of the Director-General, SPC https://www.spc.int/DigitalLibrary/Doc/SPC/Publications/Pacific_Monitoring_Evaluation_and_Learning_Capacity_Strengthening_Rebbilib.pdf
[66] DoWR (Organizational Chart): https://mol.gov.vu/images/News-Photo/water/DoWR_File/Vanuatu_DoWR_Org_Chart.pdf
[67] https://www.greenclimate.fund/sites/default/files/document/evaluation-policy.pdf

Pacific region. The final evaluation will assess the relevance of the intervention, its overall performance, as well as sustainability, replicability and scalability of results, differential impacts and lessons learned.

137. The evaluation should also assess the extent to which the intervention has contributed to the Fund's higher-level goal of achieving a paradigm shift in adaptation to climate change in Vanuatu. Both evaluations will contribute to the evidence base for adaptation to climate change in Vanuatu and across the Pacific region and will be published on the SPC website and other relevant platforms.

RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Limited interest of engagement from local communities

Category	Probability	Impact
Technical and operational	Low	High

Description

Limited interest of engagement from local communities might lead to underdeveloped projects and reduced impacts given poor ongoing maintenance after installation of equipment.

Mitigation Measure(s)

The project will build on the NIP and the DWSSP which have been well received across the country and reinforced through consultations. These mechanisms are further supported through resources and training and are working mechanisms to improve rural communities' water supply systems. The process is triggered by communities and in most cases coordinated through area secretaries and Provincial Water Officers. By building on this process the project will be aided by and reinforce community ownership. Provincial authorities have been included in the development of the proposed project and have indicated considerable interest in the proposed activities. The project will target communities where supplies of reliable water is at risk so there is high interest in the proposed activities and an interest to maintain water infrastructure after installation. The project has been designed in such a way as to strengthen participatory governance, with focused outreach, engagement, and delivered benefits through trainings support to communities, to build leadership and ownership of the communities.

Geographic remoteness of project sites

Category	Probability	Impact
Technical and operational	High	High

Description

Some communities are located in remote islands and provinces. Transportation to, and communication and outreach/engagement with these remote communities is a significant challenge, that can result in project delays as materials required for infrastructure works are not obtained within planned timeframes, or travel of extension agents is limited, reducing interactions with communities and delaying training delivery.

Mitigation Measure(s)

Under the project, communities, and provincial authorities will be engaged through the DoWR and building on the established NIP and DWSSP processes to deliver project results – this dispersed approach, making use of existing delivery mechanisms helps to mitigate the risk, which will be strengthened within this project. This engagement will include communicating project needs to provincial authorities, and communities based on the relative geographic isolation of their community. SPC has well established and efficient procurement protocols, meaning greater flexibility in procurement to adjust to needs, and the SPC and DoWR teams will adjust to transportation and communication needs accordingly.

Interagency co-ordination

Category	Probability	Impact
Governance	Low	Medium

Description

Limited coordination between government ministries, SPC, communities, NGOs/CBOs, private sector and other stakeholders reduces the efficiency and effectiveness of implementation of project interventions. Further, this could result in project delays as decision-making processes are delayed by lack of inter-stakeholder coordination.

Mitigation Measure(s)

Strong institutional and implementation arrangements for the project's management framework will ensure effective coordination and collaboration between project partners. The PMU will facilitate constant dialogue between project partners and stakeholders and ensure that the main stakeholders communicate regularly and coordinate actions. The project will also build institutional capacities for coordination between national and regional government, executing entities and supporting organisations (outcome 3) covering data collection, communication, and knowledge

management processes and tools. Moreover, project activities focus specifically on building capacities in relevant institutions.[68]		
Insufficient capacity and resources to implement projects		
Category	Probability	Impact
Technical and operational	Medium	High
Description		
Communities have insufficient technical capacity and financial/material resources to implement priority projects which results in stranded or underperforming projects and limited results for local communities.		
Mitigation Measure(s)		
Existing technical capacity within communities, area committees and provincial authorities is quite limited which is why the project deliberately builds capacity within all these stakeholder groups on climate change, project development, and project management as well as provides sustained technical support, guidance, and oversight from project support bodies. In the event stakeholders need additional support beyond this in order to successfully implement a project the DoWR at the national level and the GEM division of the SPC, through the PMU will assist delivery, serve as the financial and procurement provider and directly support project implementation in collaboration with the relevant stakeholders.		
Localized data and information are insufficient to help support development of priority projects		
Category	Probability	Impact
Technical and operational	Medium	Medium
Description		
Localized data and information are insufficient to help support development of priority projects leading to underdeveloped projects, sub-optimal design, siting, and implementation of sub-grant activities, and ultimately limited climate adaptation outcomes.		
Mitigation Measure(s)		
The project works directly with national and provincial agencies to catalogue existing climate, vulnerability, and other data to help engage communities through the DWSSP process to prioritize investments, carry out necessary studies (for example related to groundwater) and identify gaps that need to be addressed. This data will be stored in a on a centralised and open access data repository to facilitate data use and learning and support local project development.		
Unfavourable exploitation of groundwater resources		
Category	Probability	Impact
Technical and operational	Low	High
Description		
In the case new boreholes would be needed as part of Outcome 2 activities (to be confirmed during the project preparation), there is a risk that groundwater abstraction might not be sustainable. All other activities will aim at improving the sustainable use of water resources.		
Mitigation Measure(s)		
For boreholes, if any, locations will be selected in advance and additional information (incl. water balance and sustainability of groundwater abstraction per aquifer via yield tests) will be provided when developing the funding proposal. A properly conducted yield test will ensure no over extraction. DoWR is currently developing national standards on borehole preparation, and design and installation of solar pump systems. The project design documentation will also clearly state monitoring and maintenance protocols that determine roles and responsibilities for water users that should be adhered to both during and past project implementation. The monitoring protocols will have strict threshold values determined from the yield tests to prevent over extraction beyond these thresholds, extraction does not exceed provisioning rates and ensure that water resources are sustainably managed.		
Difficulty in sourcing of technology and other inputs for project activities		
Category	Probability	Impact
Technical and operational	Medium	Medium

[68] No persons or entities listed on any UN Security Council sanctions list, including the UN Consolidated Sanctions list will be involved in any manner with the project or its activities, either as a counterparty, implementer, or beneficiary.

Description		
Procurement for the project in Vanuatu is complicated given the diverse, dispersed geographies of Vanuatu as well as constrained supply chains ongoing from the COVID-19 pandemic.		
Mitigation Measure(s)		
Existing capacity in communities, and within provincial authorities for procurement is limited. The project will support effective procurement through 1) centralised procurement by SPC following SPC's procurement policy to ensure that international standards of best practice are adhered to and costs are kept as low as possible, 2) strengthening DoWR procurement processes, and 3) technical support, guidance, and oversight provided by project support bodies. Beyond this the project will provide) training on climate change project development and project management to provincial authorities, including dedicated sessions of procurement practices. This should ensure more robust procurement practices beyond the project timeline.		
COVID-19 constraints for engagement and operations		
Category	Probability	Impact
Technical and operational	Medium	Medium
Description		
Depending on the timeline of the extended pandemic, stakeholder engagement could still be constrained by COVID-19 and the resulting shifts in staff and protocols. Further, there is a risk to execution officers and beneficiaries on increased exposure to the virus through project activities.		
Mitigation Measure(s)		
The project has been adaptive and proactive in finding pathways to engage with needed stakeholders at the local level during the COVID-19 pandemic by leveraging highly effective national consultants and local networks. This foundation will help the project to proactively plan for alternative pathways for stakeholder engagement. In addition, during cases of increased incidence of the virus, in alignment with government recommendations all necessary precautionary and protective measures will be in place for in person meetings (personal protective equipment use, hand sanitation, social distancing etc) and wherever feasible online virtual platforms will be used as appropriately.		
AML/CFT and Prohibited Practices		
Category	Probability	Impact
ML/FT	Low	High
Description		
Potential risks related to Anti-Money Laundering and Combating the Financing of Terrorism (AML/CFT).		
Mitigation Measure(s)		
The SPC has adopted Financial Policies for anti-money laundering and counter-terrorism financing to prescribe the principles and minimum safeguards to protect SPC from being misused for money laundering or terrorism financing. Although the risks of money laundering and terrorism financing are considered "low" for Vanuatu, SPC will take steps to ensure that its funds are not used to finance any illegal acts related to money laundering or terrorism financing. The Director-General will implement a continuous risk-based approach to identify, assess and understand SPC's money laundering and terrorism financing risks, and will take appropriate steps to mitigate those identified risks. The SPC will undertake due diligence in engagement with the development partner, supplier, implementation partner or any other entity involved in the activities of the SPC as part of Know Your Customer due diligence to ensure identity and safeguard the practices of the entity. Detailed due diligence will enable the SPC to assess and assess the risk of money laundering or terrorist financing. There are currently no international sanctions against Vanuatu. As for the prohibited practices, the SPC has procedures to ensure their avoidance including a whistle-blower policy. Manuals and emails for reporting of prohibited practices can be found at https://www.spc.int/accountability.		
Financial sustainability		
Category	**Probability**	**Impact**
Technical and operational	Medium	Medium
Description:		
Lack of financial capacity to ensure maintenance of infrastructure beyond project lifecycle		
Mitigation measures:		

DWSSPs and infrastructure upgrades will be financed through the DoWR managed CAP. Through Outcome 2 the project will ensure more robust selection criteria are put in place for CAP investments, including the enhancement of operation and maintenance planning. Through this process, maintenance planning will include indicative budgets for maintenance of infrastructure post project, to be funded by communities in alignment with Operations and Maintenance Plans submitted in proposals.

Major hazards

Category	**Probability**	**Impact**
Technical and operational	Low	Medium

Description:

Unexpected major natural hazards (major cyclones, earthquakes, volcanic eruptions, tsunamis etc) could damage project infrastructure or cause increased contamination of water sources.

Mitigation measures:

Through Outcome 1 the DWSSP methodologies will be upgraded to account for severe climate risks, such cyclones, ensuring that infrastructure is resilient to impacts of high winds and mitigates risk of contamination in flooding events. Design specifications to be provided through this project by DoWR are deemed adequate to withstand potential impact of climate related threats. This is based on enhanced designs standards refined by the NPE. Supporting this, operation and maintenance plans will be in place to implement regular assessment of infrastructure and ensure any damages are identified and resolved quickly.

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

138. In the development of this funding proposal and its accompanying feasibility study, an initial risk assessment for the intended adaptation activities has been conducted. All project activities are expected to be limited both in scale and risk given the size of investments and infrastructure designs planned. In line with GCF ESS requirements, an initial project screening was undertaken following SPC's SER Policy, which complies with GCFs standards on ESS. Following this, "Annex 6 – the Environment and Social Management Plan" has been developed for the project and presents a detailed environmental and social risk assessment for this project. Outcomes 1 and 2 are deemed to be in alignment with ESS category C standards as they focus on "soft" activities. However, Outcome 2 will finance the construction of water-related infrastructure works for 270 communities. It is envisioned that a low proportion (<5-10%) of these investments will fall into ESS Category B, causing potential or minimal localised impacts. Consequently, the overall categorisation of the project is category B.

139. For a full breakdown of assessed risks and mitigation approaches proposed for the project please refer to Annex 6, tables 3 and 4. A summary of potential environmental and social risks in the absence of mitigation measures implemented by the project are presented below.

<u>Environmental risks</u>

- Overexploitation of groundwater tables or surface water sources
- Pollution of the immediate environment from construction processes
- Salination of the immediate environment from desalination waste
- Shift in surface water resources
- Biodiversity lost in immediate surroundings
- Soil degradation in the immediate environment from construction processes
- Accidental contamination or pollution of water bodies from waste generated by construction processes

<u>Social risks</u>

- Community disputes resulting from lack of engagement in design
- Noise during construction processes
- Community disputes of access to water and land rights
- Lack of access to land in the case of fenced infrastructure
- Lack of gender equality in community decision-making processes

140. To ensure the compliance of investments under Outcome 2 with best practices in environmental and social protection and risk mitigation, the DoWR has put in place robust ESS processes for DWSSPs applying for infrastructure development assistance. This includes an initial ESS screening. On categorisation of any project to ESS Category B, the CAP proposal must institute an Environmental and Social Impact Assessment and develop an ESMP for consideration of the investment. Further, all project investments will comply with a set of exclusionary criteria (as detailed in Annex 6 section 7) to ensure all activities comply with GCF ESS Category B+C standards. The project ESS Officer will assess all proposals through the technical assistance role provided under Outcome 1 and will ensure that ESS assessments are in alignment with GCF standards to mitigate any potential risk and that no excluded criteria are included in proposals. They will also support the DoWR to strengthen its assessment processes.

Disclosure

141. In the case a investment is considered to meet ESS Category B criteria for environmental or social impact, the ESS Officer will make the report publicly available. In line with GCF disclosure policy and were not restricted by any partners or service providers confidentiality requirements, the ESS Officer will act accordingly to:

- publish the relevant report on SPC's website;
- ensure that copies of the report are accessible to potentially affected peoples at a local level in a local language if required; and
- provide the report to the partner or service providers in line with their requirements under any funding agreement.

Grievance Redress

142. SPC is committed to receiving any concerns or grievances from any affected persons or community, related to the environmental and social impacts of any SPC project. At any time, an affected member of a community may raise their concerns by lodging a complaint on SPC's website or to the PMU and NSPC in accordance with the procedures laid out in Annex 7 Stakeholder engagement plan. These mechanisms will be accessible and relevant to the affected person's context. Communities and stakeholders will be able to access the grievance complaint form. Such grievances will be raised to SPC as the Accredited Entity as well as to the NPSC.

Stakeholder Engagement

143. National level consultations on the project were held on 27 July -1 September 2021. This included a wide range of stakeholders. Supporting this, further provincial level consultations were held on the 20-29th of September 2021. For a full list of attendees and organisations consulted please refer to Annex 7 section 4.

144. Following this, the proposal has been designed in continuous collaboration with the NDA office, the DoWR and WASH sector partners to ensure it aligns with the needs of the target beneficiaries and the national policies and regulations. Including those related to Environmental and Social Safeguards.

145. During project development, key women's groups were also consulted to ensure appropriate gender mainstreaming and to validate the gender action plan. The Gender Analysis and Action Plan Annex 8 includes a summary of the outcomes of these consultations and how they were integrated into project design.

146. Lastly, a final validation workshop was held with project stakeholders on the 23rd of March 2022. Project documentation was submitted to all participants for no objection to the proposed project formulation. All comments were adequately addressed for the submission to the GCF secretariat.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

147. Annex 8 - the Gender Assessment and Action Plan – presents the gender mainstreaming exercise undertaken during the project preparation stage. The proposed project recognises the lack of a gender-responsive approach in Vanuatu that could limit the potential, inclusiveness and success of the project objective, stemming from:

- Poor or missing gender analysis or assumptions that climate services or adaptation actions are gender-neutral[69];
- Side-lining of gender needs or ethnic vulnerabilities in adaptation design, resilience capacity-building and mitigation services;[70] and
- Lack or scant allocation of financial means, gender budgets and dedicated resources towards mainstreaming gender action[71].

Annex 8, therefore, presents an assessment of the different roles, rights, needs and opportunities of women and men, boys and girls in the project context, and mobilises project resources to tackle gender barriers, and contribute towards improved gender equality in Vanuatu. Using gender as a lens will also help identify the social relationships between women and men, and other marginalized groups in Vanuatu. The adaptation pathway chosen by this project recognizes that access to WASH is fundamental to development, but it can be mobilized as critical climate adaptation strategy for poor and vulnerable communities, as well as marginalized groups within those communities, including women and girls.[72]

148. The project adheres to the GCF's Gender Policy and Action Plan (adopted in 2014, and updated in 2019); SPC's SER Policy (2018); and the Government of Vanuatu's National Gender Equality Policy 2020 – 2030 (NGEP),[73] prepared by the Department of Women's Affairs of the Ministry of Justice and Community Services. The NGEP prioritizes five strategic areas, including women's leadership and political participation and fostering gender-responsive and community-driven solutions to climate and disaster resilience.

[69] See Butterfield, R. (2018) 'Bringing rights into resilience: revealing complexities of climate risks and social conflict' in Disasters. Journal Article.

[70] Poor or missing gender analysis, or the lack of gender-responsive action, may lead to planners or personnel depending on women to assume a central role in their coping strategies, which may not be the practical reality for many vulnerable communities. Further, this also glosses over the existing burdens on women among such groups. See Nelson. V., Meadows. K., Cannon, T., Morton, J., & Martin, A. (2002) 'Uncertain predictions, invisible impacts and the need to mainstream gender in climate change adaptations' in Gender and Development. Journal Article.

[71] Department of Women's Affairs (2020). Beijing +25 – National Review Report. Available at: https://www.asiapacificgender.org/sites/default/files/documents/Vanuatu%20(English).pdf

[72] Winqvist, E. (ed.) (2020). Feminist Policies for Climate Justice. Stockholm: Concord. Available at: https://concord.se/wp-content/uploads/2020/06/fem-rapport-2020-final.pdf

[73] The NGEP prioritizes five strategic areas: 1) eliminating discrimination and violence against women and girls; 2) enhancing women's empowerment and skills development; 3) advancing women's leadership and political participation; 4) strengthening the foundation for gender mainstreaming; and, 5) fostering gender-responsive and community-driven solutions to climate and disaster resilience.

149. Gender barriers identified through stakeholder consultations with Department of Women's Affairs as well as among communities are:

- Different gender groups, particularly women and girls, have gendered needs, which remain underserved by existing WASH infrastructure and are impacted by humanitarian emergencies;
- Women continue to be underrepresented in water security and climate adaptation-related decision-making bodies, despite registering better performance when represented;
- Insufficient, non-resilient WASH infrastructure continue to be compromised by climate risks, increasing time poverty among certain gender groups; and,
- Water insecurity indirectly and directly combines to exacerbate gender inequality in other sectors (such as limited land rights) and limit adoption of adaptation and resilience strategies.

150. The project has identified the following co-benefit as a part of its Section E: Logical Framework.

Project/programme co-benefit indicators						
Co-benefit #	**Indicator**	**Means of Verification (MoV)**	**Baseline**	**Target**		**Assumptions/ Notes**
				Mid-term	**Final**	
Co-benefit 1	Improved gender parity in representation of women in Rural Water Committees (RWCs)	Registration of RWCs with members (indicating a minimum of 40% of women)	Women are not well-represented across water governance or devolved utility authorities across Vanuatu – exact figures are currently unavailable.	40% of RWCs registered, with 40% of women in each.	Remaining 60% of RWCs registered, with 40% women in each.	40% RWCs registered by mid-term and 100% expected by project conclusion. There is political will in involving women in different sub-national contexts and in water governance institutions. Women may not be, however, able to participate meaningfully participate due to sociocultural barriers, and demands on their time due to unpaid care work within the household.

151. This co-benefit will act on the finding[74] that women are underrepresented in water user committees in Vanuatu, and establish Rural Water Committees (RWCs) through the DWSSP processes that provision for a minimum of 40% representation.[75] Women's underrepresentation is detrimental, as UNICEF research has demonstrated that their meaningful participation can increase the effectiveness of community-based management models. This co-benefit feeds into the legal pre-requisite for the registration of RWCs (40% female representation is mandatory) in Vanuatu, under the Water Resources (Amendment) Act No. 32 of 2016.

152. This will be measured as an independent indicator and will be delivered in tandem with (primarily) Outcome 1, which will establish 600 new DWSSPs in Vanuatu, alongside providing policy and climate training. Further, the project will also ensure that infrastructure delivery (through Outcome 2), and trainings (as well as policy strengthening) provided through Outcome 1 and 3, incorporate findings from the gender consultations.[76] Participants, at the community-level stakeholder consultations, identified key gender groups and articulated their different needs. These were summarized by a breakout group (which reflected other stakeholder groups in the Sanma and Tafea provinces) in Penama province below.

Table 11: Climate resilient WASH needs of gender groups in Vanuatu

Gender Group	Needs
Elderly (60+)	▪ Easy access/tap stands to be in close proximity ▪ Taps fitted at a lower level for accessibility ▪ Ball taps for ease of use ▪ Solar lighting in the tap use area

[74] Mommen, B., Humphries-Waa, K., Gwavuya, S. (2017). "Does women's participation in water committees affect management and water system performance in rural Vanuatu?" in Waterlines, v. 2017.
[75] The Rural Water Committee usually takes overall responsibility for developing their supply-specific DWSSP. However, technical and facilitation support will be needed to develop a DWSSP. A community DWSSP team needs to be established, usually comprising members of the Rural Water Committee, women's, men's, youth and church group representation, local plumbers and carpenters, and the head person. This brings together a variety of perspectives from the people who know the water supply and how it is used. Further detail can be found here: Vanuatu's National Implementation Plan for Safe and Secure Community Plan, available at – https://mol.gov.vu/images/News-Photo/water/DoWR_File/Management_Plans/Vanuatu-NIP-Guide-annual-CAP-210818.pdf
[76] During implementation, these designs will be further defined and specified through further consultations and workshops at the community level. These will be dispensed through the process of developing 600 DWSSPs through Outcome 1.

Disability/Disabled	▪ Easy access ▪ Ramp for wheelchair/hand-rail ▪ Solar lighting in the tap use area
Women (menstruating/lactating)	▪ Separate shower facilities with dignity facilities ▪ Safe house for menstrual hygiene ▪ Separate individual tank with RWCs for menstrual hygiene / child-related water use ▪ Solar lighting for security / privacy
LGBTQI	▪ Separate shower facilities ▪ Solar lighting for security / privacy
6-18 years (school students)	▪ Separate water storage for: bathroom use and kitchen use (to reduce collection burdens)
0-5 years	▪ Safety valves to be fixed before taps are installed
NB. There is an urgent need to install gender-responsive signs to specifically assigned facilities at the community level. Alongside, working groups indicated the need for the inclusion of well-lit, demarcated WASH areas, with the provision of secure doors and locks – to ensure such facilities can be used by women safely.	

153. The proposed solution recognizes that bridging practical gender needs (e.g. access to water) with strategic gender interests (e.g. changes in power and roles) is critical to achieving transformational changes in gender equality in Vanuatu, where underrepresentation of diverse gender groups limits the efficacy of existing infrastructure and the potential of new investments. Through the planned trainings (where 50:50 gender balanced beneficiary distribution is targeted), alongside this co-benefit to ensure increased, meaningful participation of women in RWCs, the project will also be able to explore five WASH- and climate change-relevant elements of empowerment: participation, decision-making, information, capacity building and leadership.

154. In doing so, the assessment operationalises gender mainstreaming activities through the project's results framework by formulating a Gender Action Plan, based on the project's Logical Framework (Section E of the Funding Proposal).

155. This ensures that the proposed design[77], to deliver improved water security and increased climate resilience at community level, can have climate, environmental and social co-benefits, with identified indicators, timeline and means of verification. In addition, with the implementation of the gender action plan and the ESMP (that accounts for Gender Based Violence and Sexual Exploitation, Abuse and Harassment) it is envisioned that the project will have a low risk regarding gender issues. Consequently, the ESS categorisation as presented in paragraph 137 is deemed accurate from a gender perspective.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

156. SPC's Financial Management Information System (FMIS) is built on a Microsoft Dynamics NAV-based enterprise resource planning system that has been deployed to all of its regional offices, which provides SPC with financial management functionalities. Using FMIS improves the flow of financial information, supports financial monitoring and reporting, increases transparency and visibility, and strengthens internal control. SPC maintains organization-wide separation of income and expenditure by donor and project so that financial and other data can be recorded, classified and summarized to facilitate internal management and external reporting requirements.

157. As mentioned in Section C.4., SPC as the Accredited Entity to the GCF will have overall responsibility for quality assurance and oversight of project implementation in accordance with its policies and procedures. In addition to this, the AE will be responsible for the financial execution of GCF funds according to SPC rules and regulations mainly contained and detailed in the SPC Handbook (including those referred to financial monitoring, audit and procurement). Oversight and quality assurance may include monitoring missions, spot checks and participation in coordination and steering committee meetings. The project will be subject to SPC's audit regime will be conducted in conformity to accepted international standards on auditing as documented inGCF accreditation processes and approval from the GCF Board. This includes the external audit and internal audit functions with oversight provided by SPC's independent Audit and Review Committee. Audits are conducted on an annual basis or on more frequent basis at the request of the SPC governing body or as specified in any legal agreements with external donors. As such, audits for this project will all be carried out in compliance with SPC audit regime and the obligations detailed in SPC's Accreditation Master Agreement as well as the Funded Activity Agreement for this project.

[77] Please refer to Section E for the Logical Framework in the Funding Proposal (FP). The Gender Action Plan (GAP) (Section 8 of Annex 8) provides impact and outcome statement for each Outcome in relation to gender mainstreaming entry points and proposed actions for the project, to be evaluated against a set of indicators and targets as well as delineation of timeline, responsible parties and costing. The paradigm shift potential of the results framework is captured in the Theory of Change, available in the FP as Section B.2 and further described in Section D.

158. When under implementation, financial management will remain in compliance with SPC policies and aligned with Vanuatu' Public Financial Management Act to the greatest extent possible. This will support the ongoing sustainability of the project as it will already be operating within the framework of the rules and regulations that will govern the management of the project and implementation of activities after the GCF-funded period.

Procurement

159. Financial control and procurement processes of the project will be implemented as per SPC rules and regulations for Outcome 1 and 3, which were assessed by the GCF during accreditation. SPC has specific policies for financial management including "Finance Regulations", "Anti-money laundering and counter-terrorism financing", "Procurement" and "Grants and sub-delegation" as well as its "Staff Regulations and Manual of Staff Policies" that govern HR and other relevant processes and procedures. These and other policies are available on the SPC Intranet and are adhered to in the implementation of all SPC projects. All entities involved in the project's implementation will comply with these policies. For Outcome 2, resources on-granted through the CAP will follow the DoWR procurement and financial management processes. A capacity assessment of GoV financial management services has been conducted and their systems have been deemed to be in compliance with the SPC procurement policies. A separate ledger account for GCF resources will be established in the CAP to allow traceability of GCF funds. All projects funded under this ledger account will meet ESS standards of GCF and align to exclusion criteria described in Annex 6 as well as align with GCF prohibitive practices policy. The PMU ESS Officer and Finance and Procurement Officer will ensure compliance.

G.4. Disclosure of funding proposal

☒ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☐ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

☒	Annex 1	NDA no-objection letter(s) **(template provided)**
☒	Annex 2	Feasibility study - and a market study, if applicable
☒	Annex 3	Economic and/or financial analyses in spreadsheet format
☒	Annex 4	Detailed budget plan **(template provided)**
☒	Annex 5	Implementation timetable including key project/programme milestones **(template provided)**
☒	Annex 6	E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3): **(ESS disclosure form provided)** ☐ Environmental and Social Impact Assessment (ESIA) or ☒ Environmental and Social Management Plan (ESMP) or ☐ Environmental and Social Management System (ESMS) ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
☒	Annex 7	Summary of consultations and stakeholder engagement plan
☒	Annex 8	Gender assessment and project/programme-level action plan **(template provided)**
☒	Annex 9	Legal due diligence (regulation, taxation and insurance)
☒	Annex 10	Procurement plan **(template provided)**
☒	Annex 11	Monitoring and evaluation plan **(template provided)**
☒	Annex 12	AE fee request **(template provided)**
☒	Annex 13	Co-financing commitment letter, if applicable **(template provided)**
☒	Annex 14	Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

☒	Annex 15	Evidence of internal approval **(template provided)**
☒	Annex 16	Map(s) indicating the location of proposed interventions
☐	Annex 17	Multi-country project/programme information **(template provided)**
☐	Annex 18	Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
☐	Annex 19	Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
☐	Annex 20	First level AML/CFT (KYC) assessment
☒	Annex 21	Operations manual (Operations and maintenance)
☐	Annex 22	Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)[78]
☐	Annex X	Other references

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

[78] Annex 22 is mandatory for mitigation and cross-cutting projects.

No-objection letter issued by the national designated authority(ies) or focal point(s)

GOVERNMENT OF THE
REPUBLIC OF VANUATU MINISTRY
OF CLIMATE CHANGE
ADAPTATION, METEOROLOGY,
GEO-HAZARDS, ENVIRONMENT &
ENERGY & NDMO
PMB 9074, PORT VILA
VANUATU

LONG GOD YUMI STANAP

GOUVERNEMENT DE LA
RÉPUBLIQUE DE VANUATU
MINISTÈRE DE L'ADAPTATION AU
CHANGEMENT CLIMATIQUE, LA
MÉTÉOROLOGIE, LES RISQUES
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SPR 9074, PORT-VILA, VANUATU

TEL : (678) 22068

FAX : (678) 22068

Ref: PV/MoCC/DEPT-WATER/PROJ-GCF:4. B.1

To: The Green Climate Fund ("GCF")

Port Vila, Vanuatu

25 May 2022

Re: Proposal for submission to the GCF Secretariat by the Pacific Community (SPC) regarding "***Enhancing Adaptation and Community Resilience by Improving Water Security in Vanuatu***"

Dear Madam, Sir,

We refer to the project titled "***Enhancing Adaptation and Community Resilience by Improving Water Security in Vanuatu***" as included in the funding proposal submitted by the Pacific Community to us on the 25th of March 2022.

The undersigned "Ms. Esline Garaebiti Bule", is the duly authorized National Designated Authority (NDA) of Vanuatu.

Pursuant to GCF decisions B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Vanuatu has no objection to the project as included in the funding proposal;
(b) The project as included in the funding proposal is in conformity with the national priorities, strategies and plans of Vanuatu;
(c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been dully followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

REPUBLIC OF VANUATU
The Ministry of Climate Change Adaptation, Meteorolgy, Geohazards Environment, Energy & NDMO
Le Ministere Du Climat Change Adaptation, Météorologie Georiequs, Environement, Energie et NDMO
REPUBLIQUE DE VANUATU

Ms. Esline Garaebiti Bule
Director General
Ministry of Climate Change and GCF National Designated Authority
Vanuatu

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Enhancing Adaptation and Community Resilience by Improving Water Security in Vanuatu
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Public
Accredited entity	The Pacific Community (SPC)
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Vanuatu
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Wednesday, September 7, 2022
Language(s) of disclosure	English
Explanation on language	English is the official language of Vanuatu and a language understandable to the target population.
Link to disclosure	https://purl.org/spc/digilib/doc/zgdak
Other link(s)	https://www.spc.int/accountability/spcs-disclosure-of-relevant-ess-measures
Remarks	An ESIA consistent with the requirement for a Category B project is contained in the "Environmental and Social Impact Assessment and Management Plan".
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Wednesday, September 7, 2022
Language(s) of disclosure	English
Explanation on language	English is the official language of Vanuatu and a language understandable to the target population.
Link to disclosure	https://purl.org/spc/digilib/doc/zgdak
Other link(s)	https://www.spc.int/accountability/spcs-disclosure-of-relevant-ess-measures
Remarks	An ESMP consistent with the requirement for a Category B project is contained in the "Environmental and Social Impact Assessment and Management Plan".
Environmental and Social Management (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A

Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), IPP Framework (if applicable)	
Description of report/disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Thursday, September 15, 2022
Place	Government of Vanuatu's National Advisory Board (NAB) of Climate Change and Disaster Risk Reduction's disclosure page: https://nab.vu/node/27442 Address: National Advisory Board on Climate Change and Disaster Risk Reduction, Ministry of Climate Change, Namba 2 Area, Port Vila, Vanuatu As per the "Environmental and Social Impact Assessment and Management Plan", on sub project development those projects deemed Category B will be disclosed at physical locations convenient to the target communities in alignment with the GCF programming manual.
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, October 17, 2022

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP191

Proposal name:	Enhancing Adaptation and Community Resilience by Improving Water Security in Vanuatu
Accredited entity:	The Pacific Community (SPC)
Country/(ies):	Vanuatu
Project/programme size:	Small

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
The proposal targets vulnerable communities impacted by climate change through stronger floods and cyclones.	It is important that locally self-sufficient operation and maintenance structures are established to sustain the works.
It proposes an integrated approach to enhance water security by establishing new water management plans and developing essential water supply systems.	To establish and develop water supply and management systems including infrastructures in small islands, the cost per beneficiary is relatively expensive compared to the average.
Country ownership is high and well-aligned with national priorities. Government and other co-financing are available.	

2. The Board may wish to consider approving this funding proposal with the terms and conditions listed in the term sheet and in addendum XIII, titled "List of proposed conditions and recommendations", respectively.

II. Summary of the Secretariat's assessment

2.1 Project background

3. Vanuatu is a small island developing State comprising 83 islands located in the Western Pacific. The country's geographical location, undiversified economic structure, and high import dependence make it vulnerable to external shocks, especially climate change-induced disasters and natural hazards. Climate risks and shocks greatly impact hard-earned development gains and threaten food, energy, and water security. This project will focus on access to water resources, which is one of the fundamental human rights.

4. The objective of this project is to create safe, climate-resilient, and sustainable water utilization and improved water security at community levels in Vanuatu. This will be done by improving and scaling up the existing government-owned processes of water management, namely the Drinking Water Safety and Security Planning (DWSSP) process and the Capital Assistance Programme (CAP), both of which are mandated by the National Implementation Plan for Safe and Secure Water (NIP). The project will directly benefit 85,650 people (including

42,825 women) in rural communities, which account for about 27 per cent of the whole population of Vanuatu. These are the most vulnerable populations who are directly exposed to climate risks in Vanuatu and lack access to drinking water as well as water, sanitation, and hygiene (WASH) services. They will be targeted through an existing, nationally-owned prioritization process and benefit from the implementation of the project's interventions. The project will also indirectly benefit an additional 216,980 rural people (73 per cent of the population).

5. The total project financing is USD 28,290,014, with a request for GCF grants of USD 23,325,033. The Government of Vanuatu will provide co-finance of USD 3,736,081 in the form of grants and USD 668,900 in the form of in-kind contribution. The United Nations Children's Fund (UNICEF) and SPC will also provide in-kind contributions in the amount of USD 470,000 and USD 90,000, respectively. The project is submitted under environmental and social safeguards (ESS) category B.

2.2 Component-by-component analysis

Component 1: Communities are empowered to plan and manage climate-resilient water resources (total cost: USD 6.26 million; GCF cost: USD 5.64 million, or 90 per cent)

6. Component 1 aims to improve water security for rural people of Vanuatu by enhancing the existing DWSSP process and mainstreaming climate change adaptation solutions in these plans. Since 2006, the Government of Vanuatu, through the Department of Water Resources (DoWR) and the Ministry of Health's Environmental Health Unit, has been improving the DWSSP process under the NIP. This is one of the official processes by which the government ensures that all people have reliable access to safe drinking water and sanitation infrastructure. It is also a community engagement process through which communities identify and discuss threats to safe and secure drinking water and collectively make plans to manage these threats. The resulting DWSSPs guide day-to-day water supply operation and maintenance (O&M) as well as improvements to community water systems.

7. Drinking water safety planning is an internationally accepted approach to manage water resources. This was included in the 3rd edition of the World Health Organization Guidelines on Drinking Water Quality in 2004. Vanuatu adopted this approach in 2005 following the World Health Organization Workshop on Drinking Water Quality Standards and Monitoring in Pacific Island Countries. The addition of a second "S" (security) was done in response to the need to also plan for adequate supply of water especially in anticipation of and during times of drought. DWSSPs provide for a framework through which community-owned WASH systems (new and/or improved) can be sustainably managed. The process is triggered by communities, and through multiple steps including the formation and training of Rural Water Committees (RWCs), DWSSP related training, and simple no/low-cost improvements of the existing water facilities by the communities. The DWSSP process improves not only the water planning processes but also communities' ownership of their water resources and facilities.

8. The DWSSP methodology will be updated through this component with a specific focus on strengthening the mainstreaming of climate resilience. It will address the shortfalls in the existing design and engineering standards in the DoWR by analysing conventional design protocol, making changes in the design criteria, and providing the relevant staff with exposure and experience in climate resilient designs. The DWSSP methodology will be updated to mainstream climate resilience in the plans and will be further refined incrementally on an annual basis to account for the lessons learned during the five-year project implementation period. The upgrades will enhance the hazard risk assessments and develop improvement and preparedness plans as well as climate change risk and improvement plans. The updated methodology will be integrated into the DWSSP process, and training will be provided to all the DWSSP facilitators at the national level covering all the provinces.

9. Moreover, there are currently 301 completed DWSSPs and, as a result of this proposed project, 600 additional communities will benefit by developing their DWSSPs. These communities will be identified based on prioritization of vulnerability to climate impacts. The 600 additional DWSSPs will utilize the enhanced methodology that mainstreams climate resilience in planning while the existing DWSSPs will also be retrofitted to better reflect climate risks and adaptation solutions. Further to this, 100 DWSSPs from the current 301 established to date will be updated to follow the new climate-resilient methodology. In total, this component will benefit 700 communities or 79,940 direct beneficiaries (including 39,970 women), which is 25.2 per cent of the entire population of Vanuatu. Furthermore, various capacity development and knowledge sharing activities will be delivered to communities and area administrators within the government extension structure.

Component 2: Communities have enhanced rural water infrastructure (total cost: USD 20 million; GCF cost: USD 16.1 million, or 80 per cent)

10. Component 2 represents the core investment envisioned under this funding proposal in order to strengthen climate-resilient WASH facilities in rural communities of Vanuatu, accounting for about 70 per cent of the total project cost. The aim of this component is two-fold: (i) increase access to safely managed drinking water in rural Vanuatu through the rehabilitation or construction of small-scale infrastructure projects identified through the country-owned DWSSP and NIP processes; and (ii) increase engagement of communities in management, oversight and accountability of drinking water service points and infrastructure (through, particularly, the RWCs).

11. This will be achieved through scaling up the existing CAP that the Government of Vanuatu has successfully deployed to strengthen water facilities in Vanuatu. DWSSPs and the CAP are closely linked; an approved DWSSP is the essential prerequisite to making an application for CAP funding. Moreover, communities can apply for CAP funding only after they make an attempt to improve their own water systems (through simple no/low-cost enhancements). An approved DWSSP ensures that the requested financial assistance addresses priority threats to providing safe and secure drinking water and is for a sustainable upgrade or improvement option. Additionally, when communities go through the DWSSP process, it also means that they already have an RWC, which administers a plan for ongoing financial, operational, and managerial sustainability of water supply. All these prerequisites ensure that the facility improvements through the CAP funding do not end up being a one-off investment without proper maintenance afterwards. As such, this component is synergetic to component 1 in a sense that the technical capacities enhanced through activities under component 1 will ensure that CAP requests incorporate greater emphasis on climate resilience in the design of water facilities and ensure sustainability of the interventions.

12. Out of the 301 completed DWSSPs, 47 have had their water facilities upgraded through the CAP so far. With the activities under this component, this number will increase to 317 (270 additional DWSSPs will receive CAP funding, of which 220 are already targeted by component 1 and 50 are additional). This translates to approximately 30,834 beneficiaries (including 25,124 from component 1 and 5,710 additional beneficiaries).

13. This component will also assess the CAP criteria for funding and upgrade the screening process to be more robust and provide a greater focus on climate vulnerabilities. In addition, the Project ESS Officer will support the Provincial Engineers and DoWR Provincial Officers in providing technical support to communities that will need to develop ESS category B interventions (estimated at less than 5 per cent of applications, based on the trend of CAP applications to date). The upgraded CAP process, including the improved screening criteria, will also contribute to the wider rural communities of Vanuatu when the communities that would not be covered under this project will have their upgrades through other sources of funding.

Component 3: Provincial and national institutions are strengthened to address climate risks associated with water security (total cost: USD 0.893 million; GCF cost: USD 0.718 million, or 80 per cent)

14. The Government of Vanuatu recognizes that essential to the effective delivery of the NIP are standardized support tools and processes, the foundation being community-level DWSSPs. Additionally, it also recognizes that essential to efficient delivery of the NIP is reaching as many communities as possible by devolving responsibility and support to provincial governments, coupled with national oversight and coordination of the many government agencies, implementing partners, and technical support agencies. Keeping this mandate as context, component 3 will work towards strengthening institutional capacity at provincial and national levels to ensure the effective and efficient delivery of the NIP and other related water management processes to manage the adverse effects of climate change.

15. This component will focus on providing training to different levels of staff within the water governance structure, as well as to WASH sector partners, to ensure climate change and management of climate risks are mainstreamed within existing processes of water safety and security. Moreover, knowledge management and data sharing mechanisms will be established through this component, which would provide different stakeholders of the water and climate change sectors with better access to data and an opportunity to learn from each other. In addition, this component will establish a monitoring, evaluation, and learning (MEL) framework for improved learning for climate-resilient water management. Having an MEL system, particularly for the sanitation and hygiene subsector will: ensure that inputs and activities lead to their intended results and outcomes; adjust course where necessary (during the project lifetime); establish whether progress is being made; and lastly, document lessons learned and best practices.

Project management (total cost: USD 1.08 million; GCF cost: USD 0.87 million)

16. The GCF portion of this project management cost is less than 5 per cent of the total requested GCF funding and is compliant with the GCF policy on fees.

III. Assessment of performance against investment criteria

3.1 Impact potential *Scale: N/A*

17. Six hundred communities will be direct beneficiaries as part of component 1 (improved planning and community-based activities on climate-resilient water management), of which 220 will also benefit from improved climate-resilient water infrastructure as part of component 2. An additional 50 communities with existing DWSSPs will be targeted under component 2 during the first year of the project. Further to this, 100 DWSSPs from the current 301 established to date will be updated to follow the new climate-resilient methodology in component 1. This makes a total of 750 communities directly benefiting from components 1 and 2. This will directly benefit 85,650 beneficiaries (including 42,825 women), which is around 27 per cent of the total population of Vanuatu. Indirect beneficiaries include the rest of the rural population of Vanuatu (216,980 individuals or 73 per cent of the total population), mostly through enhanced institutional capacities and processes leading to climate-resilient water security for rural communities (component 3).

18. This project will address three water-related impacts of climate change in Vanuatu, which are increases in: (i) water-related disasters; (ii) areas suffering from water stress; and (iii) loss of human life induced by poor water quality. It will thus deliver on priority interventions identified in Vanuatu's GCF country programme (pipeline project AA1). Water security is vital for community resilience in the long term as well as in the immediate aftermath

of climate-induced disasters. GCF funding will positively impact and contribute to the management of water-related impacts of the observed and projected climate change effects in Vanuatu. The project's impact in reducing climate risks for water security is reinforced by recent revisions of the DWSSP processes. The revised DWSSP processes significantly strengthen climate change and vulnerability considerations within DWSSPs, from training of trainers, community training and facilitation to data collection, improvement planning, and O&M.

3.2 Paradigm shift potential *Scale: N/A*

19. The project is a climate adaptation investment aiming to enhance climate resilience and increase water security of rural communities. The project will climate-proof WASH infrastructure and enhance water management planning systems across Vanuatu. Given that only 79 per cent of systems were identified as functional and only 37 per cent of these systems were reported as providing water 24 hours per day in 2020, climate considerations are increasingly being integrated in water security and safety management. However, there remain significant needs for improvements to deliver climate-resilient water security benefits for rural communities in Vanuatu.

20. By scaling up the number of climate-resilient DWSSPs registered (component 1) and then delivering climate-resilient infrastructure and system improvements through the CAP (component 2), the project will strengthen a whole-of-government approach to achieving climate-resilient water safety and security through the DoWR with associated decision-making processes that are decentralized, transparent, participatory, community-focused, gender-mainstreamed and evidence-based. Also, by building and operating climate-resilient water systems, interruption to water services during and after disasters will be reduced or avoided. Therefore, the project will reduce the costs associated with continual repairing and rebuilding after disasters, allowing the limited public funds available to be leveraged for investment in new WASH infrastructure and sustainable exploration of water sources.

21. The project will also support transitioning towards gender-transformational WASH investments, including the gender co-benefit towards mandatory representation of women in RWCs. Traditionally, gender roles involve women and girls providing more labour and spending more time than men and boys in managing the household's WASH needs. Increased time spent by women and girls to obtain water during dry seasons or after climate-induced emergencies increases their exposure to gender-based violence (GBV). Conversely, improving predictability and location of water sources—with better access points created through more resilient and secured (such as improved lighting and locks) infrastructure—will reduce exposure to such risks, increasing personal safety and security in addition to the safety and security of water supply.

3.3 Sustainable development potential *Scale: N/A*

22. This project will generate significant environmental, social and economic benefits that will contribute to positive sustainable development outcomes for Vanuatu in the long term. It will reduce the negative environmental impacts of poor wastewater management by improving water management behaviour and practices through the sanitation component of DWSSPs. The project will also achieve better protection of water sources and forested water catchment areas, through no- and low-cost community improvements, resulting in improved environmental quality.

23. Moreover, the project will enhance the equitable access to water in Vanuatu, which is one of the basic human rights. It will target the most vulnerable groups in rural communities, which is expected to generate significant social benefits including positive health outcomes and

improved social standing for women through their visible and meaningful roles in water committees, with 40 per cent female members being mandatory for registration of RWCs.

24. Another benefit arising from the project is economic through avoided health care costs. Improved water security and sanitation practices will lead to improved health outcomes and therefore reduce health care costs. Moreover, because the project will enhance climate resiliency of water facilities, it will also lead to avoided damages in case of disaster, which is another economic benefit that the project is expected to generate.

25. In terms of its contribution to the achievement of the United Nations Sustainable Development Goals (SDGs), the project directly responds to SDG 13 (climate action), SDG 6 (clean water and sanitation), and SDG 5 (gender equality) through its objective of building climate-resilient and gender-equitable water safety and security in Vanuatu.

3.4 Needs of the recipient *Scale: N/A*

26. The project addresses the fundamental human right of access to safe drinking water. Communities that have elevated levels of vulnerability, low access to WASH services and are exposed to climate risks will be targeted. Besides these communities most in need of climate-resilient water management interventions that will be directly targeted, the project will ultimately benefit all rural communities by strengthening the enabling environment for resilient interventions. This project will address vital needs relating to water security for rural communities in Vanuatu. Over 24,000 households across the country are reliant on either rainwater or surface water for their primary drinking water supply. Also, one-third of households have no alternative source of drinking water. These challenges have historically been, and will continue to be, exacerbated by climate change.

27. Rural communities in Vanuatu use a combination of groundwater, surface water and rainwater that are increasingly precarious due to limited climate-proofing of infrastructure and processes. Only one-third of households have access to water 24 hours per day, every day of the year. The direct impacts of limited water supply and water security, contaminated drinking water and inadequate WASH infrastructure are increased morbidity (diarrhoea, stunting and other illnesses) and increased mortality among both children and adults. Significant improvements in the management and O&M of rural water systems are needed to ensure water services are managed safely. The project will address these by introducing safely managed and climate-resilient WASH and drinking water services critical for preventing diseases and protecting human health.

3.5 Country ownership *Scale: N/A*

28. The project is fully aligned with the Government of Vanuatu's climate change strategies and policies, including the Climate Change and Disaster Risk Reduction Policy 2016–2030 as well as the national adaptation programme of action and the NDC, both of which reference water resource management as a top priority. In addition, the project is fully aligned with Vanuatu's National Sustainable Development Plan 2016–2030 and National Water Policy 2017–2030.

29. The project directly responds to the Government of Vanuatu's goal for every rural community to have a DWSSP by 2030, as outlined in the Water Resource Management Act and the Water Supply Act. DoWR will be co-leading the implementation of the project by following the NIP and DWSSP processes, through which the project will scale up system improvements and WASH infrastructure in rural contexts. This project was fully co-developed with DoWR, along with other national stakeholders to address climate change risks and impacts on water

management, and to work directly with vulnerable rural communities on community-based adaptation activities.

3.6 Efficiency and effectiveness *Scale: N/A*

30. The project will build climate resilience and adaptation capacities in Vanuatu's WASH sector through the implementation of proven and efficient water management technologies. This will follow the nationally owned DWSSP processes that support the identification of locally led interventions and, in doing so, lay the foundations for further scaling up beyond the project lifetime.

31. The project-level economic net present value is positive at USD 37,150,065 and the programme-level economic internal rate of return is 20 per cent. This project is thus economically viable and can be justified on economic grounds, even though non-investment budget costs (including co-finance) have no envisaged direct benefits. This is the case even though the analysis included conservative assumptions and not all benefits were included in the economic calculations as it was not possible to estimate their monetary values. Nonetheless, such benefits will accrue to project beneficiaries in addition to those included in the analysis. Therefore, the adaptation project is expected to be highly effective.

32. The efficiency of the DWSSP process will also contribute to the overall efficiency and effectiveness of the project. This will be the primary modality through which the project will identify and prioritize communities in need of system improvements and the activities to bring about such improvements. The communities that have already created DWSSPs have cumulatively spent almost USD 1 million of their own money on low-cost improvements. The project will benefit from the existing structures and policy frameworks to efficiently scale up investments by finetuning and mobilizing the DWSSP process through DoWR.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

33. **Project background.** The project aims to improve water security for rural communities by targeting 270 communities, schools, and healthcare facilities throughout Vanuatu with rehabilitation, upgrading or expansion of water infrastructure such as concrete protection on spring sources, well head aprons, and rainwater harvesting. In addition, small solar-powered reverse osmosis desalination units will also be installed by the project. Other activities will focus on capacity-building of provincial and national WASH authorities as well as communities to plan and manage their water resources. Expected environmental co-benefits include avoiding GHG emissions owing to installation of solar power. Social co-benefits are expected from reduced health issues associated with waterborne diseases owing to improved sanitation and timesaving from collecting water from remote and unsanitary sources.

34. **Environmental and social risk category.** The confirmed and assigned environmental and social risk category of the project is equivalent to GCF's medium/category B. It is within the accredited entity's (AE) accreditation level. This is rationalized by rehabilitation or construction of small-scale community infrastructure for component 2 CAP activities that are expected to have risks and impacts that are generally reversible and site-specific, which can also be readily managed with mitigation measures.

35. **Safeguard instruments.** The AE provided an environmental and social impact assessment (ESIA) and management plan with the funding proposal. The document provides an overview of the Vanuatu context for environmental and social risk assessment, potential

environmental and social risks likely to be associated with the project activities and site-specific assessments and action plans that will be undertaken at the subproject level. An action plan is also included with proposed mitigation measures for risks that have been identified. The AE's procedure for conducting an ESIA is annexed in the document.

36. **Compliance with GCF's ESS standards.** The paragraphs below identify the key risks and/or impacts as regards the GCF ESS standards and how they are addressed in the proposal:

37. **ESS 1: Assessment and Management of Environmental and Social Risks and Impacts.** Activities included in the CAP component will be screened and categorized using the AE's ESS screening tool annexed in the document. Initial screening will be carried out by DWSSP community proponents. Less than 10 per cent of the subprojects are expected to be categorized as moderate environmental and social risk level and therefore require site-specific ESIAs and environmental and social management plans (ESMPs). The assessment of potential risks and impacts presented in the safeguards document is based on interventions proposed during consultations. Key potential risks are associated with labour and working conditions, pollution, community health and safety, and biodiversity. The description of baseline environmental and social conditions provided in the document is very minimal and will need to be elaborated at the subproject level.

38. **ESS 2: Labour and Working Conditions.** Construction related activities under component 2 have potential occupational health and safety risks for workers involved. This will be remediated by providing workers with personal protective equipment to wear during construction, safety training, and emergency plans in case of accidents. Materials to be used in the project will be sourced locally as much as possible and will avoid materials originating from harmful practices. The AE is recommended to establish a process for assessing risks related to labour in supply chains and mitigating them at the subproject level.

39. **ESS 3: Resource Efficiency and Pollution Prevention**. Installation of new and rehabilitation of existing wells and boreholes will be supported by the project where a water balance can illustrate the sustainability of abstraction of groundwater. In addition, long-term monitoring will be undertaken. Water pumps that will be installed by the project will be powered by renewable energy only to avoid pollution by combustion. The project seeks to phase out diesel-fuelled pumps to eliminate spillage of oils into community water sources. Measures to avoid contamination of water are centred around structural interventions that will be installed by the project to avoid pollution from latrines including preventing vectors, pests, and livestock from accessing community water sources. Waste is expected to be generated from construction activities. Additionally, the operational phase of the project will also generate waste from items such as plastic water tanks and solar components, including panels, after their lifespan is reached. The AE is recommended to ensure that it has a process for safe disposal of solar panels and batteries and handle any inconsistencies with best practices that are identified in the national guidelines that are yet to be finalized.

40. **ESS 4: Community Health, Safety and Security.** Construction, retrofitting, and installation activities may create temporary noise and air pollution for local communities. Noise will be limited by operating construction equipment during the day and other mitigation measures will be outlined at the subproject level. Construction may also pose a safety risk to communities. This will be mitigated by barricading sites during construction to prevent injuries. Water storage facilities will be enclosed to avoid breeding vectors that may cause diseases such as mosquitoes. The project's activities may fall under areas with longstanding land, communal or tribal disputes particularly where shared water resources are involved. Mitigation measures include assessment of conflicts, extensive community consultations, and obtaining community no-objection to selected sites.

41. **ESS 5: Land Acquisition and Involuntary Resettlement.** Construction-related activities will be widely consulted with communities and a memorandum of understanding will

be signed between DoWR, community leaders and landowners to secure land that may be required for water supply systems and to allow access for O&M purposes. The project excludes activities that will result in physical or economic displacement. However, there is a possibility that project activities may result in restrictions on communities in accessing natural resources or place constraints on some specific land uses temporarily or permanently such as cattle grazing. The AE is recommended to ensure that it has a process for conducting thorough due diligence where this risk is identified at the subproject level and apply necessary mitigation measures.

42. **ESS 6: Biodiversity Conservation and Sustainable Management of Living Natural Resources**. There is a potential negative risk on biodiversity as a result of brine disposal from desalination units. The units will be designed with a low recovery rate to minimize an increase in salt concentration where brine is discharged. Abstraction of surface water may also potentially impact marine life negatively.

43. **GCF Indigenous Peoples Policy and ESS 7: Indigenous Peoples.** The project is not expected to negatively impact indigenous peoples and takes place within a national context where around 95 per cent of the country is indigenous, mainly ni-Vanuatu, with other groups that include Wallisians, Futunans and i-Kiribati. Community consultations are planned to be conducted with the representation of all indigenous groups within a community. The AE notes that indigenous and local knowledge will be incorporated and ensured through consultations with communities. In addition, community engagement will ensure that construction does not occur on any indigenous lands or culturally important sites without free, prior and informed consent.

44. **ESS 8: Cultural Heritage**. The AE's due diligence indicates that project activities are unlikely to directly impact any areas of cultural heritage value. Nonetheless, a chance finds procedure has been outlined in the document in case tangible cultural heritage is found at the project sites.

45. **Sexual exploitation, sexual abuse, and sexual harassment**. The revised GCF Environmental and Social Policy adopted by decision B.BM-2021/18 requires safeguarding from sexual exploitation, abuse and harassment (SEAH) in GCF-financed activities. The project's ESMP has integrated various forms of SEAH as one of the social risks/impacts and identified safeguarding measures in its plan. The project will ensure that all project staff, consultants, facilitators, service providers and communities directly targeted are trained on prevention of SEAH principles and standards in alignment with GCF policies. In addition, the ESS Officer will lead the development of SEAH protocols into the DWSSP processes and provide training to the DWSSP facilitators and consultants. All these activities will contribute towards the prevention side of SEAH. With regard to SEAH response mechanisms (although the risks of SEAH are deemed to be low during the trainings), the project has also embedded SEAH responses in its grievance redress mechanisms (GRM). The project will post a multi-level GRM and include a specific SEAH protocol to ensure a survivor-centred approach is in place. This will allow for survivors to select multiple avenues to file a grievance. Varied options for grievance redress enhance confidence in the survivor to come forward and log a formal complaint as well as be assured of protection and confidentiality and that the perpetrator would not be involved in a specific GRM process. This will apply to all grievance mechanisms that will be controlled by SPC or the project management unit (PMU), ensuring that the rights, needs and wishes of the survivors are prioritized. This also ensures that their rights are upheld; privacy and confidentiality are maintained; non-discrimination measures are put in place; and translation opportunities are provided in the language the survivor feels more comfortable with. All the information related to GBV and/or SEAH cases will be kept confidential and identities protected. Moreover, the survivor must provide informed consent to progress with each stage of the complaints process. Should a case of SEAH or GBV be submitted, SPC will carry out duty of care to the survivors in line with its policies including, among others and where relevant, support for

the provision of medical services (including psychosocial support), legal counsel, community-driven protection measures, and reintegration of the survivor. These mechanisms are also in line with the national standard operating procedures for GBV service provides in Vanuatu. At the community level, the GRM will be addressed through the traditional governance structures and processes managed in individual islands, and the issues will range from utility access, conflicts among villagers, complaints from marginalized gender or vulnerable groups, issues related to water access points and GBV or SEAH. This level of the GRM will ensure that communities are able to resolve issues and conflicts with consensus as a first level, and then escalate to the project-level GRM only if deemed appropriate. The project is encouraged to ensure that the community-level GRM is also as inclusive and survivor-centred as possible in order to avoid reinforcement of gender stereotypes and/or any power imbalance that may lead to continuous victimization of GBV/SEAH survivors within the traditional complaints and grievance handling mechanisms.

46. **Implementation arrangements.** The PMU will recruit an ESS Officer who will be responsible for ensuring management of matters regarding safeguards—for example, by conducting quality assurance on screening done by RWCs; contracting consultants to prepare subproject level ESIAs and ESMPs and reviewing the same; and liaising with the Monitoring and Evaluation Officer on monitoring and reporting. Training and capacity-building will be provided to DWSSP community proponents to support effective environmental and social risk identification and management. A budget of related costs is also outlined in the document. The AE will provide oversight and supervise the management of environmental and social safeguards matters throughout the project's implementation.

47. **Stakeholder engagement and information disclosure.** Consultations were undertaken in 2021 at the national and provincial levels. The report on stakeholder engagement activities undertaken during the preparation of the project describes the main outcomes of the consultations, which have been integrated into the project's design. A stakeholder engagement plan is included in the document. It identifies the different types of stakeholders, their roles in the project, and when and how they will be engaged by the project. Component 1 activities allow for awareness and stakeholder engagement using community systems to develop plans for both management capacity and infrastructure and, importantly, ownership of water facilities by the rural communities. In addition, community members are expected to be actively involved in the implementation of component 2 activities. ESIAs and ESMPs of medium risk subprojects will be disclosed according to the GCF Information Disclosure Policy.

48. **Grievance redress mechanism.** The contact details of the redress mechanisms of both the AE and GCF are included in the document. Complainants may also contact the PMU with their grievances. Grievances received and registered by the PMU will be handled by the AE's legal experts and other relevant officers depending on the nature of the concern. In addition, grievance mechanisms administered by chiefs and community leaders will also be available to complainants. In line with the GCF Indigenous Peoples Policy, the GCF indigenous peoples focal point will be available for assistance at any stage, including before a claim has been made.

4.2 Gender policy

49. The AE has provided a gender assessment and gender action plan, and therefore complies with the requirements of the GCF Gender Policy and action plan.

50. The gender assessment conducted describes the gender issues at the country level and indicates the existence of national policy environment and supportive practices to promote gender equality and women's empowerment. Vanuatu ratified the Convention on the Elimination of all Forms of Discrimination Against Women in 1995 and has a National Gender Equality Policy that aims to engage in solutions to climate and disaster resilience that are

gender-responsive and community-driven. This project is expected to contribute towards that goal.

51. The gender assessment was undertaken through desk review and collection of inputs from various stakeholder consultations undertaken. The consultations and discussions were held with government (including with the Department of Women's Affairs), civil society organizations and community members.

52. The assessment indicated that the climate crisis affects women and girls in unique and disproportionate ways. The consultation pointed out the need to ensure the integration of the needs of women and other vulnerable groups in the WASH process. The finding indicates that lack of access to improved water supply places a disproportionate burden on women and girls who are the primary collectors of water for the family as in many countries. The assessment also points out the gaps in the existing infrastructure to serve the differentiated needs of women and men, and other vulnerable groups. Some of the specific needs of women that are not met are, for example, increased water needs during lactation and pregnancy and for the management of menstrual hygiene, child-related water use, and separate showers with dignity. These needs are not often considered in the design of the final infrastructure constructed. In addition, the elderly (for whom women are the primary caregivers) have increased need for water due to physiological requirements. Other issues presented related to the safety of infrastructure. The existing and newly constructed infrastructures require system improvements to increase safety for women. This is in terms of lighting, improved locks and doors as well as gated/protected areas. These are of paramount importance to ensure reduced incidents of GBV. In addition to safety and infrastructure design issues, women's representation in the same is limited. Women continue to be underrepresented in water security and climate adaptation-related decision-making bodies and in water user committees despite registering better performance when represented. The lack of representation, compounded with the other factors raised earlier, affects women's ability to access water with ease; therefore, they spend a lot of time in search of water. The limited year-round service of most water points, inadequate availability of water to meet daily needs, and dry seasons pose the greatest burden for communities, and women in particular as it is their job to fetch water (five hours per day). The water fetching role is in addition to feeding and caring for children, the elderly and people with disabilities as well as collecting firewood, washing clothes and gardening. Gender mainstreaming remains a need across all WASH stakeholders and requires ongoing improvement in a bid to be more responsive to the various needs of community members such as the elderly, female-headed households and people with disabilities, to name a few. These community members face inequalities in practicing and accessing the services and facilities and in a safe manner. Despite these challenges, the consultation held indicated that gender issues have been increasingly incorporated in WASH processes over the years.

53. Having noted the gaps, the gender action plan was developed with relevant activities, indicators, targets, timelines, gender expertise and budget. The findings from the gender analysis indicated that water committees where women were represented in the highest decision-making bodies registered better outcomes in terms of water provisions and services. In view of that, the gender action plan is geared towards ensuring increased meaningful representation of women in water committees with at least 40 per cent participation, while ensuring safe and efficient water supply through the engagement of women in the infrastructure design and implementation. The improved security and privacy features—such as adequate lighting, secure locks, and dignity areas with curtains/covers—are expected to address some of the challenges faced by women. The action plan includes activities towards increasing women's capacity in constructing, maintaining and operating water infrastructures while still assessing further, gender-based needs. Also included are activities towards effective knowledge management, collection of data, and documentation of lessons learned and best practices. The implementation of the gender action plan and mainstreaming of gender will be facilitated by various groups with different roles including the Gender Officer and the

Department of Women's Affairs. Meanwhile, technical assistance on gender responsive actions will be provided by UNICEF in addition to the Gender Protection Cluster, which will provide sensitization and advocacy support during trainings to ensure meaningful participation of women.

54. The AE is recommended to clearly articulate and demonstrate activities that speak to gender transformative and gender responsive actions. It is not clear if there is a distinction between the two and what results could be categorized as one or the other. The AE is recommended to use the planned consultations to identify transformative and responsive activities.

4.3 Risks

4.3.1. Overall project assessment (medium risk)

55. GCF is requested to provide a grant of USD 23 million to enhance climate resilience and increase water security of rural communities by investing in WASH infrastructure and water management planning systems. The Government of Vanuatu will provide a total of USD 4.4 million co-financing by way of grant and in-kind contribution. UNICEF and the AE will be providing in-kind contribution equivalent to USD 0.47 million and USD 0.09 million, respectively.

4.3.2. Accredited entity/executing entity capability to execute the current project (medium risk)

56. SPC, as a direct access AE, will serve as both AE and one of the EEs for this project. Headquartered in New Caledonia, it has been operating in the Pacific for over 70 years. Its Melanesian Regional Office in Vanuatu creates close linkages with the AE and the PMU in the country.

57. The Government of Vanuatu, through the Ministry of Finance (MoF) and the DoWR, will be a co-EE for the project. The capacity assessment of the DoWR has been carried out, and it identifies gaps and provides recommendations. However, the assessment acknowledges that the country lacks capacity in water supply engineering and in climate change and vulnerability assessment and impact mitigation throughout all levels and sectors.

4.3.3. Project-specific execution risks (medium risk)

58. **Project delays in remote locations:** The AE identified implementation delays in remote islands and provinces as a main challenge and high probable risk. This is also often associated with high cost of travel, communication and logistics. According to the AE, different projects being implemented in the country will be coordinated through relevant line ministries, including the DoWR, for logistics and procurement.

59. **O&M risk:** The project may face difficulties to maintain the new and upgraded equipment due to limited financing and/or expertise. A financing strategy for O&M is part of the O&M plan, and technical training (e.g., plumbers training, basic financial management training) on operating and maintaining the water system will be provided. Although direct revenue will not be generated from the programme, there will be a community household contribution that will depend on the financial arrangement agreed by the community through the DWSSP processes. However, as O&M responsibilities largely lie with the community, the project identified the limited technical and financial capacity of the community as one of the risks. If unexpected costs arise and/or migration of the trained community member happens, there will be constraints to ensure O&M beyond the project implementation period.

60. **Technical risks:** The installation of desalination systems or boreholes would require upgrades, repairs of existing distribution pipelines or new distribution systems. This will require well-designed construction plans and technical precision to be in line with the technical standards of the government. In addition, after drilling a borehole, water level check to prevent overextraction in the aquifer and water quality monitoring will be critical. The project will rely upon the technical capacity of the DoWR who will be responsible for supervision and monitoring.

61. **Project viability and concessionality:** Vanuatu is a small island developing State and funded activities for water are public in nature. The financial analysis shows that, without grants support, the cost of the investment will be beyond what households can afford (ranging from 657 per cent to 3,674 per cent of ability to pay). Therefore, full grant financing is requested. The economic analysis results in the economic internal rate of return of 20 per cent and the AE assessed that the adaptation project is economically viable and efficient.

4.3.4. Compliance risk (medium risk)

62. The Republic of Vanuatu is not currently subject to any United Nations Security Council Resolutions (i.e. sanctions).

63. SPC, as the AE for the project, confirmed that no persons or entities listed on any United Nations Security Council sanctions list, including the United Nations Consolidated Sanctions list, will be involved in any manner with the project or its activities, either as a counterparty, implementer or beneficiary.

64. SPC confirmed that it has undertaken an operational capacity assessment for the prospective grantee—the Government of the Republic of Vanuatu—and has determined that it has the requisite capacity to comply with SPC requirements to implement actions through grants awarded by SPC. However, the comparison of policies and procedures of the grantee with SPC policies reveal that there are some gaps that need to be taken into consideration when entering into a grant agreement with the grantee.

65. As part of the analysis, a review of the most recently issued audited financial statements of the grantee (issued on 28 October 2021) for the year ending 31 December 2017 was also carried out. This revealed that an adverse audit opinion was issued because of significant financial control weaknesses. The issues highlighted in the report may pose significant risk for any grant awarded to the grantee and requires action by the grantee. These issues will remain a risk to SPC when working with the grantee until they are addressed.

66. SPC informed GCF that, to mitigate these risks, any SPC division intending to award a grant to the grantee must provide training to the grantee to ensure the recommendations of this capacity assessment are implemented. The division must also carry out regular performance monitoring of the grant action, appropriate for the size and risk of the project in question.

67. An anti-money laundering and combating the financing of terrorism (AML/CFT) check on the grantee was carried out via Sanction Scanner and no red flags were raised.

68. SPC advised that it has conducted a risk assessment in relation to potential risks pertaining to AML/CFT as well as prohibited practices. SPC has assessed the probability of such risks to be low, whereas the impact would be high.

69. SPC has adopted financial policies for AML/CFT to prescribe the principles and minimum safeguards to protect SPC from being misused for money laundering or terrorism financing. Although the risks of money laundering and terrorism financing (ML/TF) are considered low for Vanuatu, SPC will take steps to ensure that its funds are not used to finance any illegal acts related to ML/TF.

70. SPC informed GCF that its Director-General will implement a continuous risk-based approach to identify, assess and understand SPC's ML/TF risks, and will take appropriate steps to mitigate those identified risks. SPC will undertake due diligence in engaging with the development partner, supplier, implementation partner or any other entity involved in the activities of SPC as part of Know Your Customer due diligence to ensure identity and safeguard the practices of the entity. Detailed due diligence will enable SPC to assess ML/TF risks.

71. SPC confirmed that it had assessed the project design procurement capacity of the executing entities (EEs), highlighting that their procurement standards are aligned with SPC standards. This was accepted by GCF at the accreditation stage. Furthermore, through subsidiary agreements, AML/CFT obligations will be passed down to the EEs, requiring them to monitor and impose SPC-endorsed standards throughout their operations.

72. During implementation, the SPC Climate Change and Environmental Sustainability (CCES) Programme will check that these standards are upheld through regular supervision missions and quality assurance reviews of annual performance reports and interim reports, including from SPC procurement specialists. In the case that any concerns are flagged in these processes, SPC will take appropriate measures to resolve issues and ensure no procurement occurs without appropriate amendment and checks in place.[1]

73. SPC confirmed that the project is not an on-granting project. As such, no cash, vouchers or other items of economic value will be distributed directly or indirectly to beneficiaries.

74. SPC informed GCF that it is committed to receiving any concerns or grievances from any affected persons or community related to the environmental and social impacts of any SPC project. At any time, an affected member of a community may raise their concerns by lodging a complaint via the SPC website or directly to the PMU and the national designated authority. Communities and stakeholders will be able to access the grievance complaint form. Such grievances will be raised to SPC as well as to the national designated authority. In further support, SPC also has a whistle-blower policy.

75. **Recommended risk rating**: The Office of Risk Management and Compliance (ORMC)/Compliance Team has conducted a review of the project in accordance with relevant GCF Board-approved policies and does not find any material issue or deviation with respect to compliance issues. Based on available information for this funding proposal, the ORMC/Compliance Team has determined a risk rating of medium and has no objection to this request proceeding to the next steps.

76. The ORMC/Compliance Team would like to remind SPC, as the AE, of its continuing obligations and responsibilities with regard to monitoring and reporting any risks for money laundering, terrorist financing, or prohibited practices among the intended counterparties, EEs, beneficiaries, persons involved, or any of the proposed activities.

4.3.5. GCF portfolio concentration risk (low risk)

77. In case of approval, the impact of this proposal on the GCF portfolio concentration in terms of result areas and single proposal is not material.

4.3.6. Recommendation

78. It is recommended that the Board consider the above factors in its decision.

Summary Risk Assessment		Rationale
Overall project	Medium	

[1] SPC manuals and contacts for reporting prohibited practices are available at https://www.spc.int/accountability.

AE/EE capability	Medium	The capacity of the AE/EE will be critical for successful implementation. The responsibility of O&M will mainly rely on communities, and may require additional support and supervision from the government for the sustainability of the project beyond its implementation period.
Project-specific execution	Medium	
GCF portfolio concentration	Low	
Compliance	Medium	

4.4 Fiduciary

79. SPC will be the AE for this project. As the AE, SPC will also be the EE. The Government of Vanuatu will also be an EE for the project, acting through MoF and DoWR.

80. Within SPC, the AE functions will be undertaken by the CCES programme, which hosts the Climate Finance Unit, SPC's focal point for GCF. The Climate Finance Unit's role in the project will focus on oversight and quality assurance of project management and delivery in alignment with obligations agreed upon with GCF through relevant financing and legal agreements.

81. SPC's Geoscience, Energy and Maritime (GEM) Division will be the EE for the project. Within GEM, the Disaster and Community Resilience Programme is a key technical pillar and develops, implements and supports coordination of innovative applied science and technical action to respond to challenges faced by communities, as well as repeated disaster risks, in the Pacific region. As EE, GEM will coordinate actions to ensure the project is adequately staffed and supported to efficiently implement the project. More specifically, GEM will oversee tasks within the overall project management structure. SPC GEM will execute all activities through the Project Manager, who will be contracted as a project position and will provide technical support to DoWR.

82. The Government of Vanuatu, acting through MoF and DoWR, will be the other EE for the project. MoF will receive GCF resources from the AE and allocate these into the CAP accounts to be administered by DoWR.

83. In its role as AE, SPC CCES will oversee and supervise the implementation of this project, in accordance with the agreement signed between SPC and GCF. SPC GEM, in its capacity as EE and through the PMU, will be responsible for project-level MEL and reporting on compliance with approved SPC policies and GCF requirements under relevant agreements.

84. SPC as the AE to GCF will have overall responsibility for quality assurance and oversight of project implementation in accordance with its policies and procedures. In addition to this, the AE will be responsible for the financial execution of GCF funds according to SPC rules and regulations mainly contained and detailed in the SPC Handbook (including financial monitoring, audit and procurement). Oversight and quality assurance may include monitoring missions, spot checks and participation in coordination and steering committee meetings. The project will be subject to SPC's audit regime, which will be conducted in conformity to accepted international standards on auditing as documented in GCF accreditation processes and approval from the GCF Board. This includes the external audit and internal audit functions with oversight provided by SPC's independent Audit and Review Committee. Audits are conducted on an annual basis or on a more frequent basis at the request of the SPC governing body or as specified in any legal agreements with external donors. As such, audits for this project will all be carried out in compliance with the SPC audit regime and the obligations detailed in SPC's accreditation master agreement as well as the funded activity agreement for this project.

4.5 Results monitoring and reporting

85. The goal statement of the theory of change captures the project's intervention logic. It included the most pertinent assumptions, risks and barriers that frame the proposed pathways of change to achieve increased climate resilience of rural water infrastructure and improved water security for vulnerable communities in Vanuatu. The Secretariat considers the barrier and related assumption on market signals and adequate planning and incentive structures for the water sector a critical assumption that could undermine the theory of change; it therefore included a relevant condition/covenant related to this.

86. The logframe captures the selected GCF Integrated Results Management Framework indicators, in particular (i) core indicator 2: direct and indirect beneficiaries reached; (ii) supplementary 2.3: beneficiaries (female/male) with more climate-resilient water security; and (iii) core indicator 3: value of physical assets made more resilient to the effects of climate change and/or more able to reduce GHG emissions. It also included core indicators 5 and 6 to capture enabling environment-related results.

87. At the output level, the AE has strengthened the means of verification for some indicators by specifying a combination of qualitative and quantitative data collection methods and primary and secondary sources. However, at the GCF outcome level, the means of verification remain highly reliant on quantitative sources, thus limiting the lesson learning potential from relevant monitoring and evaluation (M&E) activities.

88. The implementation timetable for the funding proposal has been completed appropriately. It shows all activities and key milestones associated with each phase of the project.

89. The M&E plan in annex 11 now covers the indicators comprehensively and includes an indicative budget for the data collection activities, which can be traced back to the project budget in annex 4. There is a good budget provision for project monitoring activities. The AE also clarified, through the budget notes in the project budget, the provisions for evaluative data collection activities that will inform the interim and final evaluations. The portion of the AE fees that will pay for the interim and final evaluations has been increased and, together with the budget that will be covered by the project budget, the evaluation budget is now at a reasonable level. Overall, the M&E budget is within the recommended 2–5 per cent range of the Evaluation Policy and the required budget lines have been clarified.

4.6 Legal assessment

90. The accreditation master agreement was signed with the AE on 8 November 2019, and it became effective on 10 April 2020.

91. The AE has provided a legal opinion signed by Georgia Ramsay, Principal Legal Officer of the AE, and dated 13 September 2022 confirming that it has obtained all internal approvals and it has the capacity and authority to implement the project.

92. The proposed project will be implemented in Vanuatu. GCF has signed a bilateral agreement on privileges and immunities with Vanuatu; however, such agreement is still pending its entry into force. This means that the privileges and immunities are only effectively accorded to GCF upon the agreement entering into effect. Therefore, until such moment, among other things, GCF is not protected against litigation or expropriation in this country, which risks need to be further assessed.

93. The Heads of the Independent Redress Mechanism and the Independent Integrity Unit have both expressed that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by GCF are made

only after GCF has obtained satisfactory protection against litigation and expropriation in the country, or has been provided with appropriate privileges and immunities.

4.7 List of proposed conditions (including legal)

94. In order to mitigate risk, it is recommended that any approval by the Board is made subject to the following conditions:

(a) Signature of the funded activity agreement, in a form and substance satisfactory to the GCF Secretariat, within 180 days from the date of Board approval; and

(b) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP191

Proposal name:	Enhancing Adaptation and Community Resilience by Improving Water Security in Vanuatu
Accredited entity:	The Pacific Community (SPC)
Country/(ies):	Vanuatu
Project/programme size:	Small

I. Assessment of the independent Technical Advisory Panel

1.1 Impact potential

1. This funding proposal for a small project in environmental and social safeguards category B is submitted by Vanuatu, a small island developing State[1] in the western Pacific, with 65 inhabited islands and a population last recorded at 272,000. This is an adaptation project, with The Pacific Community (SPC) as the accredited entity (AE) and also one of two executing entities (EEs), together with the Government of Vanuatu through the Ministry of Finance and the Department of Water Resources (DoWR). SPC is a regional direct access entity to the GCF, and is an international development organization owned and governed by its 27 country and territory members, to whom it provides scientific and technical advice and support. The total cost of the funding proposal is USD 28.29 million, as detailed in paragraph 38 below.

2. The overall objective of the project is to contribute to the management of water-related impacts from observed and projected climate change effects in Vanuatu, which are increases in: (i) water-related disasters; (ii) areas suffering from water stress; and (iii) poor water quality-induced fatalities. The project aims to enhance climate resilience and increase water security of 600 vulnerable rural communities across all six provinces of the country. Project activities will be undertaken to climate-proof water, sanitation and hygiene (WASH) infrastructure and enhance water management planning systems, enhancing community participation in the government-led Drinking Water Safety and Security Planning (DWSSP) process, mandated through national policy frameworks such as the National Implementation Plan for Safe and Secure Water (NIP) and the Capital Assistance Programme (CAP). The project will scale up the CAP programme to improve drinking water supply through investment in new rainwater harvesting tanks, gravity fed systems, boreholes and solar-powered desalination.

3. Observed changes in the climate of Vanuatu show that there has been a gradual change in climatic conditions over the last 70 years. The temperature has increased at approximately 0.1°C per decade since 1970, and 30-year projections indicate this trend will continue. Temperature increase will cause higher evapotranspiration which, when combined with increased seasonal variability of rainfall, will increase the likelihood of reduced soil moisture and agricultural drought. Despite no general upward trend in overall precipitation being observed or projected, extreme rainfall and tropical cyclone events are predicted to become more intense. Interannual rainfall variance is large, especially in the different El Niño–Southern Oscillation (ENSO) phases, where El Niño causes significant drought and La Niña significant heavy precipitation. These events are predicted to increase in intensity into the future, causing

[1] Vanuatu graduated out of least developed country status in 2020.

more severe drought and category 4 and 5 cyclones that can damage water infrastructure, as occurred with Cyclone Pam (2015) and Cyclone Harold (2020). Table 1 in the funding proposal, titled "Climate and non-climate drivers impacting drinking water and WASH infrastructure in Vanuatu", details the specific negative impacts of increased air and water temperatures, increased intensity of extreme precipitation events, extreme wind speed in tropical cyclones, increased length of dry spells, and sea level rise on the structure and functioning of water and sanitation infrastructure.

4. These changes are compounding existing development challenges related to water and sanitation service provision, and water resource management. More intense extreme events will increase the incidence of flooding and induce greater damage to water utilities, pipes, and community-based water infrastructure in the baseline scenario. Increased climate-related damages by extreme winds can result in redundant infrastructure, cutting communities off from water supplies. Soil erosion and siltation from heavy rains combined with human-caused deforestation can damage or pollute freshwater sources. Flooding events increasingly overwhelm drains and latrine pits, resulting in stagnant, contaminated water that puts the health and wellbeing of the ni-Vanuatu population at risk. With sea level rise, saltwater intrusion and increased salinity of groundwater will have direct impacts on drinking water and WASH infrastructure. On the other hand, increased temperatures and increased intensity of major El Niño events are predicted to cause increased incidence of meteorological drought. Consequently, climate impacts directly threaten water security and sanitation across the country.

5. Despite ongoing investments by government and its financial and technical partners in infrastructure for drinking water supply, supplementing inadequate freshwater resources across many of the country's 65 islands is a major challenge, intensified by climate change. A number of barriers exist to effective climate change adaptation in the water sector in Vanuatu. Awareness and capacity are limited at community levels due to a lack of backstopping and outreach from extension agents to adequately design climate-resilient DWSSPs and infrastructure. In remote settings in rural Vanuatu there is insufficient, poorly maintained or lack of WASH infrastructure (a development challenge), and current design standards do not account for a full range of climate-resilient approaches to withstand intense rain and wind, nor do they provide backup or fail-safe mechanisms that could maintain water reliability during flood, drought or natural disasters. A number of capacity constraints exist, including obvious financial constraints; limited technical capacity at area council, provincial and national government levels; inadequate institutional capacities; and lack of robust and accessible data for integrated and climate-resilient water management.

6. The project theory of change is that IF local planning, decision-making and water management processes at national and devolved levels are improved and resources are provided for the improvement of water resources, THEN climate resilience and water security of vulnerable communities will be enhanced in the rural areas of Vanuatu, resulting BECAUSE OF enhanced community awareness and capabilities, upgraded infrastructure to address climate risks and impacts, and strengthened national water governance capacities. Three components of the project aim at the following outcomes:

(a) Outcome 1: Communities are empowered to plan and manage climate-resilient water resources;

(b) Outcome 2: Communities have enhanced climate-resilient rural water infrastructure; and

(c) Outcome 3: Provincial and national institutions are strengthened to address climate risks associated with water security.

7. **Adaptation impact:** The project is expected to have significant adaptation impact by enhancing water safety and security and community resilience in the long term, in the face of

evolving climate risks as well as in the immediate aftermath of climate-induced disasters. By improving existing water management processes and community-based planning—particularly through enhanced integration of climate change into the DWSSP process—the project will increase the adaptive capacity of rural communities to better cope with the additional burden of climate change on water security and safety, providing explicit capacity-building and transitioning towards climate-resilient water management planning. Water systems will be made more resilient to the effects of climate change through updating design standards of infrastructure to improve the following: (i) increased storage capacity to cope with water shortages; (ii) more robust structures to cope with storms/floods; (iii) multiple water sources to provide alternatives and backups in times of emergency; and (iv) training and risk management approaches built into rural water committees (RWCs).

8. The project will provide climate-resilient water security benefits to 85,650 direct beneficiaries (including 42,825 women) and a further 216,980 indirect beneficiaries, the entire rural population, who are expected to benefit over time from enhanced institutional capacities and processes leading to climate-resilient water security for rural communities (outcome 3). As part of outcome 1 (improved planning and community-based activities on climate-resilient water management), 600 communities will be direct beneficiaries, of which 220 will also benefit from improved climate-resilient water infrastructure as part of outcome 2. An additional 50 communities with existing DWSSPs will be targeted under outcome 2 during the first year of the project, making a total of 650 communities directly benefiting from outcomes 1 and 2.

9. This contributes to GCF's fund-level impact A2.0 (increased resilience of health and well-being, and food and water security) and indicator A2.3 (number of males and females with year-round access to reliable and safe water supply despite climate shocks and stresses). This will in turn contribute to the achievement of GCF's fund-level outcome A7.0 (strengthened adaptive capacity and reduced exposure to climate risks), to be measured through indicator A7.1 (use by vulnerable households, communities, businesses, and public-sector services of Fund-supported tools, instruments, strategies and activities to respond to climate change and variability).

10. The independent Technical Advisory Panel (TAP) assesses the impact potential of the "Enhancing Adaptation and Community Resilience by Improving Water Security in Vanuatu" project as high.

1.2 Paradigm shift potential

11. As per the funding proposal, the project "will transition the current precarious water security and safety situation experienced by Vanuatu's rural communities in the face of climate change to one of a reliable and predictable supply of safe water". This will occur through climate-resilient, community-led planning and management practices (outcome 1), the delivery of climate-resilient water infrastructure (outcome 2), and strengthening of institutional capacity (outcome 3). By building and operating climate-resilient water infrastructure, interruption to services during and after disasters will be reduced or avoided. Therefore, this GCF project will reduce the costs associated with continual repairing and rebuilding after disasters, allowing the limited public funds available to be leveraged for investment in new WASH infrastructure and sustainable exploration of water sources.

12. Climate mainstreaming is still nascent in Vanuatu. While climate considerations are increasingly being integrated into the government-led DWSSP process, there remain significant needs for improvements to deliver climate-resilient water security benefits for rural communities. By scaling up the number of climate-resilient DWSSPs registered and then delivering climate-resilient infrastructure and system improvements through prioritized DWSSPs, the funding proposal argues that the project will "strengthen a whole-of-government approach to achieving climate-resilient water safety and security through the DoWR with

associated decision-making processes that are decentralized, transparent, participatory, community-focused, gender-mainstreamed and evidence-based".

13. In considering the extent to which these interventions might lead to a paradigm shift in water resource management in Vanuatu in the face of climate change, two issues were raised by the independent TAP. Firstly, enquiries were made as to the political feasibility of the shift from a public-funded model to a market-driven approach, as set out in the Vanuatu National Water Policy 2017–2030: Would communities be willing to contribute significantly to operation and maintenance (O&M) costs through payment of tariffs? In response, SPC indicated that tariffs are already being charged in over 50 per cent of communities, and pointed to a survey conducted by the Global Green Growth Institute, which found that 70 per cent of surveyed communities perceive that water quality is low and that vulnerability related to water access/security exacerbated by climate change is high. It also found that "93% of the targeted population is willing and able to pay for water (around 665 VUV/Month/HH)[2], and that populations agree with pre-paid services. As such, buy-in for the proposed systems is high and is exemplified in the oversubscription of CAP requests to date"—with communities showing a widespread willingness to make their own "low and no-cost contributions" to upgrading communal water infrastructure, before even receiving any government funding. In response to concerns about affordability, SPC stated that "Tariffs established by RWC's are heavily consulted with communities and proposals are posted locally for an objection period to allow communities to oppose any water tariff increases that are over their willingness to pay. This process is implemented to ensure that water remains within an affordable threshold for all community stakeholders." Overall, the new approach being promoted through the project is to strengthen a community-owned model of WASH infrastructure, which "can more easily transition into a market approach as socioeconomic development occurs".

14. This also partly addressed a second issue of concern raised by the independent TAP in relation to paradigm shift potential. This was the weakness of the Operations and Maintenance Plan included with the funding proposal, which contains only checklists and forms and no clear recommendations on whose responsibility specific elements of O&M will be, no itemized costings per technology type per year, and no indication of where the budget for such O&M will come from post-project. The independent TAP questioned whether the AE has considered that a solid O&M financing and organizational plan is vital to achieving post-project sustainability and ensuring that, as stated in the funding proposal, "this project will help to break the cycle of '*build-neglect-rebuild'* characterising many WASH programmes in Vanuatu in the past".

15. In response, the AE explained that O&M costs would be covered to a significant extent by tariffs, with explicit training of WRCs to set appropriate tariffs, collect revenue, and carry out O&M, both routinely and post-extreme events. The O&M Plan in annex 21 was described as indicative, presenting an appraisal of the government-owned process developed by Red Cross in consultation with DoWR and funded by the United Nations Children's Fund (UNICEF). "All communities will devise their own O&M plans through the DWSSP process and refine them further during the Water Committee Management Training that will be based on Annex 21, however a generic O&M plan cannot be developed that covers all RWCs as these are unidentified and will need to be operated and maintained based on-site specific circumstances (local *Kastom*[3] processes etc). A key aspect of CAP applications is the inclusion of a feasible (and financed) O&M plan in the DWSSPs. This will be cross checked by the project engineers and monitored through the Project. Post-project budget is expected from tariff aggregation done by RWCs and linked into the O&M plans. Again, this is a crucial part of the CAP approval criteria. Further to this, DoWR support and financing will be provided in exceptional cases, e.g. non climate disasters such as earthquakes."

[2] Abbreviations: VUV = Vanuatu vatu; HH = household
[3] *Kastom* is a Bislama word used to refer to traditional culture, including religion, economics, art and magic. It incorporates conventions, custom or customary law, social norms and traditions.

16. Overall, based on the above, the paradigm shift potential of the project is assessed as medium.

1.3 Sustainable development potential

17. The project contributes to three Sustainable Development Goals: SDG 13 (climate action), SDG 6 (clean water and sanitation), and SDG 5 (gender equality) through its objective of building climate-resilient and gender-equitable water safety and security in Vanuatu.

18. **Social benefits and social inclusion:** The project will contribute to equitable access to water as a basic human right in Vanuatu, targeting the most vulnerable groups in rural communities. Significant social benefits resulting from project activities include positive health outcomes. The Environmental and Social Management Plan ensures that designs of infrastructure for drinking water provision will include posting layout plans at the community level for comments and any objections. In this period, women, men, youths, and people with disabilities can object to the designs and plans if they feel that their needs have not been adequately addressed. In response to a question from the independent TAP on the involvement of civil society in the project, the AE clarified that representatives from the Vanuatu Association of NGOs are included in the Project Steering Committee. Additionally, the Vanuatu Society for People with Disability and others will be included as observers, including a technical partner from the NGO/CSO sector with a focus on gender rights and empowerment.

19. **Gender co-benefits:** As outlined in the Gender Assessment and Action Plan, Vanuatu's limited WASH service delivery and infrastructure affects women and men differently. Traditionally, gender roles typically involve women and girls putting in more labour and spending more time than men and boys in managing the household's WASH. Increased walking times to reach source water during dry seasons or climate-induced emergencies can increase instances where women and children are further exposed to gender-based violence. The project aims to reduce these burdens and risks by enhancing sustained access to good quality drinking water close to people's homes.

20. The project aims to achieve improved social standing for women through their visible and meaningful roles in RWCs, with 40 per cent female members mandatory for the committees' registration. In answer to a question from the independent TAP about whether this quota by itself would be sufficient to ensure meaningful participation and leadership roles for women, given the existence of strong sociocultural barriers in many island cultures and demands on women's time for unpaid care work within the household, the AE highlighted that UNICEF is currently assisting DoWR to hire a Gender Equality, Disability and Social Inclusion Officer and Consultant, who would also provide input into the GCF project. While SPC noted that RWCs apply their own decision-making practices based on local traditions and *Kastom,* which cannot be directly altered by the project, trainings provided in outcomes 1 and 2 will nonetheless "directly target RWCs on gender equality and aim to enhance the voice of women in community decisions". In response to a question from the independent TAP about the positive impact of women's leadership in RWCs, the AE indicated that a UNICEF study had found that RWCs with female chairs are more likely to perform better on the metrics of (i) having in place a form of revenue collection and (ii) water users themselves covering a large proportion of repairs to systems.

21. **Environmental co-benefits:** The project will reduce the negative environmental impacts of poor wastewater management by improving water management behaviour and practices through the sanitation component of DWSSPs. The project will also achieve better protection of water sources and forested water catchment areas through no- and low-cost community improvements, resulting in improved environmental quality.

22. In response to a query from the independent TAP about the risk of overextraction of groundwater, the AE responded that "potential negative groundwater impacts need to be checked and confirmed during the individual investment preparation. ESS screening and impact assessments will be completed at these sites (as they fall into ESS Cat B), including pump tests to mitigate the risk of overexploitation of groundwater resources. Monitoring of abstraction rates and depth of groundwater at regular intervals will be completed by RWC with training and equipment being provided to them during the water committee management training". Furthermore, the AE explained that "The DoWR, Provincial Government and Area Councils will assist the RWC to develop appropriate rules for their water systems to mitigate the risk of over extraction of groundwater resources, including the provision of equipment and training to measure groundwater and abstraction levels and advise on appropriate levels of abstraction and drops in groundwater levels".

23. **Economic and health benefits:** The direct impacts of limited water supply and water security, contaminated drinking water and inadequate WASH infrastructure are increased morbidity (diarrhoea, stunting and other illnesses) and increased mortality, among both children and adults. The project will provide economic benefits through avoided health care costs (as improved water security and sanitation practices lead to improved health outcomes), and enhanced human development capacity and productivity in the long term. Providing safe and secure water supply especially in the aftermath of cyclones will lead to avoidance of waterborne diseases. Loss and damage from disasters will also be reduced, as climate-resilient infrastructure and training on its maintenance will reduce the level of damage inflicted.

24. The independent TAP pointed out the importance of a well-planned programme to promote handwashing and hygiene in households, since this would be an essential accompaniment to water provision, in order to improve the health of the population. The AE acknowledged this, and noted that since the start of the COVID-19 pandemic there has been a continuous push by Vanuatu's Ministry of Health and the WASH sector to promote handwashing, supported through a number of current sanitation and hygiene pilot projects, including the hiring of area level sanitarians (funded by Asian Development Bank). This has resulted in all business houses requiring handwashing facilities and has significantly improved awareness around the importance of handwashing and the use of handwashing facilities. The AE also reported that UNICEF is scaling up a pilot WASH in Schools programme to all provinces, and is currently supporting the Ministry of Health to conduct a market study to assess the demand and supply chain for sanitation and hygiene, which will lead to a costed implementation plan.

25. **Traditional knowledge:** In response to a query from the independent TAP about the lack of detail in the funding proposal about how island communities' traditional knowledge would be incorporated into the project, the AE explained that utilizing traditional knowledge is essential to ensuring safety and security of the water supply, and that such knowledge will be captured as part of the DWSSPs and through the engineering assessments: "This knowledge includes finding out about the different traditional uses of different water sources and their reasoning for using them for these purposes, the reliability of different water sources i.e. if the existing community sources ever run dry, knowledge around potential land disputes (previous or current) and information on if the water source is in a taboo area that means it cannot be developed."

26. The independent TAP assesses sustainable development potential as high.

1.4 Needs of the recipient

27. Rural communities in Vanuatu use a combination of groundwater, surface water and rainwater that are increasingly precarious due to limited climate-proofing of infrastructure and processes. Over 24,000 households across the country are reliant on either rainwater or surface

water for their primary drinking water supply. Also, one-third of households have no alternative source of drinking water. The project addresses the primary challenge for water safety and security in Vanuatu that, while access to a proximate source of drinking water is high (94 per cent access to an improved drinking water source and 86 per cent access on the premises), the water quality is not always good. Moreover, water supply is increasingly unreliable on many islands, as a changing climate interacts with direct anthropogenic pressures on hydrological systems—for example, discharge of polluted wastewater into rivers or deforestation causing increased run-off, erosion and sedimentation, and less infiltration of rainwater.

28. Water rationing, both for household use and sanitation, and water shortages are common during the dry season. The DoWR Water Resources Inventory indicates that many of the water sources are not available or have inadequate yields during the dry season. Conversely, rainfall often manifests as intense periods of extreme precipitation, leading to floods that further exacerbate the WASH situation. In addition, safe and secure water supply is frequently compromised in the aftermath of cyclone events, which are expected to become more intense with climate change, posing serious public health risks. The project will address these needs through a combination of institutional strengthening and capacity-building at national and devolved levels, as well as through direct improvements to drinking water and WASH systems.

29. Vanuatu has the highest score in the world on the global disaster risk index (a score of 49.74), indicating a high degree of physical exposure, relative vulnerability and resultant risk of loss of life from natural hazards. This is intensified by the predicted impacts of climate change discussed above, combined with adaptation deficits (as well as development deficits) that need to be addressed to build community resilience to these impacts. Financing this adaptation gap has posed a significant challenge in Vanuatu, which remains highly aid-dependent despite its graduation in 2020 from least developed country to lower middle-income country status.

30. While steadily increasing economic and social indicators over recent decades are a positive sign for Vanuatu's climate resilience, gains remain unevenly distributed and the most vulnerable people and communities will require adaptation assistance for the foreseeable future, particularly given the country's significant exposure to extreme weather events and other hazards. The economic impact of the COVID-19 pandemic—with the collapse of the tourism sector and reduced tax receipts, combined with the loss and damage suffered from level 5 Tropical Cyclone Harold that hit the country in April 2020—pushed the domestic budget into deficit in 2020/21, and may have set Vanuatu back to gross domestic product levels typical of least developed countries.

31. The World Bank Systematic Country Diagnostic for Vanuatu (which was conducted in 2016 with seven other Pacific Islands) emphasizes that while public expenditures tend to be high in relation to the size of the economies of these island nations, the absolute size of the public sector is still very small and often lacks the financial and human resources needed to provide adequate public services at national scales. This is supported by the high demand on CAP requests for investment in DWSSPs for water infrastructure. This has induced downscaling of infrastructure works or lack of funding for most appropriate climate resilience, which has, in turn, led to increased vulnerability as exemplified by the aftermath of Tropical Cyclone Harold. Financial constraints for investment in water infrastructure therefore directly presents a barrier to climate-resilient water security and sanitation.

32. Overall, based on the above, the needs of the recipients are rated as high.

1.5 Country ownership

33. The project is listed as the first priority project in the pipeline presented in Vanuatu's GCF Country Programme and has been fully co-developed with DoWR, along with other national stakeholders. By addressing climate change risks and impacts on water management, and by

working directly with vulnerable rural communities on community-based adaptation activities, the project is fully aligned with the Government of Vanuatu's climate change strategies and policies, including the Climate Change and Disaster Risk Reduction Policy 2016–2030, as well as the National Adaptation Programme of Action and the Nationally Determined Contribution, which both reference water resource management as a top priority. In addition, the project is fully in line with Vanuatu's National Sustainable Development Plan 2016–2030, and National Water Policy 2017–2030.

34. As a further sign of country commitment, the Government of Vanuatu has committed USD 3.8 million in cash co-financing to support the project, especially component 2. This co-finance will flow mainly through the CAP, which is part of NIP, to support development of DWSSPs and concomitant infrastructure. The funding proposal states that "leveraged finance will include operation and maintenance costs for the installed equipment after completion of infrastructure works, amounting to an estimated $6,447,000 over the equipment lifetime. This financing is only possible post GCF investment through the CAP and completion of the infrastructure works. RWCs leverage the infrastructure works to generate the required finance for long-term operation and maintenance of the infrastructure post completion".

35. The project responds directly to the Government of Vanuatu's goal for every rural community to have a DWSSP by 2030, as outlined in the Water Resource Management Act and the Water Supply Act. DoWR will be leading the implementation of key elements of the project, following the NIP and DWSSP processes, through which the project will scale up system improvements and WASH infrastructure in rural contexts. These processes are fully owned by DoWR, which is also the lead coordinator of the Vanuatu WASH Cluster, acting as the secretariat and chair during emergencies. This WASH cluster is very active due to the high number of emergencies that the country experiences, and DoWR is experienced in responding to water access and security issues during disaster and other rapid-onset climate events. There is thus already strong ownership of such processes and activities by DoWR, which this project will build upon.

36. Country ownership of the project is thus seen to be high.

1.6 Efficiency and effectiveness

37. The results of the economic analysis conducted for the project clearly show that the project-level Economic Net Present Value is positive at USD 37,150,065 and the programme-level Economic Internal Rate of Return is 20 per cent. The conclusion is that the proposed programme is economically viable and can be justified on economic grounds, even with the non-investment budget costs and co-finance for which no direct benefits are envisaged. According to the AE, the analysis included conservative assumptions and not all benefits were included in the economic calculations, since it was not possible to estimate the monetary values of some of the co-benefits.

38. The total project cost is USD 28.29 million, 82.5 per cent of which is a USD 23.33 million GCF grant contribution. The remaining USD 4.94 million, or 17.5 per cent of the total project cost, is made up of grant and in-kind contributions from the Government of Vanuatu (USD 4.4 million), SPC (USD 90,000) and UNICEF (USD 470,000). Although Government of Vanuatu cash contributions to infrastructure in component 2, through the ongoing CAP, cover less than 20 per cent of the total costs, this contribution can be considered as significant given the country's financial position. The structure of the project as 100 per cent grant-financed represents an appropriate level of concessionality in the country's context, and given the fact that any revenues generated from sale of water to communities will be ploughed straight back into operating expenses (which revenues will likely not even be able to cover fully).

39. The independent TAP did, however, raise a query with the AE about why as much as 80 per cent of the cost of water infrastructure in component 2 was being covered by GCF funding, when section D.6 of the funding proposal on efficiency and effectiveness states that “the interventions will address the social, gender, regulatory, financial, ecological and institutional barriers that hinder communities from coping with *the additional burden of climate change* [the independent TAP’s emphasis] on water security management”. Elsewhere the funding proposal acknowledges that “one of the primary barriers in Vanuatu is limited WASH infrastructure and water supply sources", with some islands still lacking any improved systems and 21 per cent of existing systems being dysfunctional. The islands have fragile water resources as a result of factors unrelated to climate change, such as the “small scale, lack of storage and limited freshwater reserves”, which are now also “increasingly exposed to climate impacts”. Arguably, the key challenge here, i.e. lack of water safety and security, is primarily a development challenge, which should be tackled predominantly by government or Official Development Assistance resources, with GCF’s climate finance used only for the portion of the problem attributable to climate change.

40. In answer to this, SPC argued that “the fact that communities have fragile water systems is immensely climate relevant on its own accord, as it directly enhances their vulnerability to climate related impacts. Climate change is predicted to lead to enhanced drought periods and inundation events. As such, water security and safety are directly at risk from climate change in Vanuatu. This is especially the case in ENSO positive or negative years, which are predicted to have more severe impacts on water systems than ENSO neutral years. The fact that the water systems in place are currently fragile due to ‘small scales, lack of storage and limited freshwater reserves’ is a direct issue that enhances community risk and exposure of climate change impacts, regardless of the historical cause. Limited freshwater supplies will be greatly depleted over drought periods that are predicted to increase under projected climate trends or can be easily contaminated in inundation events or through enhanced saltwater intrusion from king tides, sea-level rise and storm surges. Communities are therefore sorely in need of diversified and more secure water sources to enable them to adapt and enhance their resilience to predicted climate change scenarios. Investment in these systems is therefore directly responsive to climate change adaptation needs, justifying the need for climate finance resources.”

41. Another area of cost-effectiveness into which the independent TAP enquired was why desalination is included in the project, given its relatively high cost compared with the other technology types. Using the figures provided in the economic analysis for the estimated average capacity of each of the four technology types (in litres per annum) and the upfront capital expenditure costs of each type, the independent TAP found that the cost per litre per annum of the planned desalination plants was USD 0.51, in comparison with USD 0.02 for gravity fed systems and USD 0.03 for groundwater systems (with rainwater harvesting systems falling in between at USD 0.22 per litre per annum). The AE explained that “some communities, in particular those located on low-lying reef islands, do not have reliable access to groundwater. Rainwater harvesting systems can also be unreliable during dry seasons or drought periods, especially within El Niño phases. In these particular contexts, solar powered desalination is deemed the most appropriate technology to provide these communities with safe and secure water sources in the face of climate change. This is not a common need, and the Project is only providing financing for 4 desalination units, to target the communities and locations that face significant water security challenges and have minimal alternative options for safe and sustainable sources of drinking water. This estimate is based on CAP requests to date. Despite the higher cost per litre of water, the need for these systems is highly context-specific and is justified in these rare cases, including considering the logistics and transport of fresh water to remote high risk areas when rainwater tanks run dry and hand dug wells become brackish”.

42. Desalination sites would also be required to have strong RWCs in place, given the more onerous O&M requirements. At the suggestion of the independent TAP, the AE also agreed that

recent advances in small-scale solar powered desalination technology[4] would be investigated, and the most affordable best practice technology would be utilized in project implementation.

43. The project's overall efficiency and effectiveness is rated as medium.

II. Overall remarks from the independent Technical Advisory Panel

44. The independent TAP endorses the funding proposal for the project "Enhancing Adaptation and Community Resilience by Improving Water Security in Vanuatu".

[4] The independent TAP highlighted emerging technologies that do not require battery storage or grid connection, but instead rely on sophisticated control systems to go into low/no-production mode depending on solar availability. See https://www.news24.com/news24/southafrica/news/revolutionary-solar-power-desalination-plant-could-provide-cheaper-option-but-it-needs-space-20180718 and https://solarmagazine.com/no-batteries-needed-low-cost-solar-desalination-system-green-namibia-desert-coast/

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP191)

Proposal name:	Enhancing Adaptation and Community Resilience by Improving Water Security in Vanuatu
Accredited entity:	The Pacific Community (SPC)
Country/(ies):	Vanuatu
Project/programme size:	Small

Impact potential
SPC is pleased to note that the independent Technical Advisory Panel (iTAP) concurs that the impact potential of the project is "high". Given Vanuatu's high vulnerability to climate change, the project will provide significant adaptation impacts in terms of water security and community resilience in the face of extreme climate risks following government-owned water management processes and community-based planning.
Paradigm shift potential
SPC notes the iTAP assessment of the paradigm shift potential of the project as "medium". By strengthening government processes through transparent, community-focused, gender-mainstreamed and evidence-based approaches, the project will achieve a paradigm shift away from the "build-neglect-rebuild" cycles characterising many such investments globally and towards a paradigm of water security that is sustainable, participatory, equitable and climate-resilient.
Sustainable development potential
SPC is pleased to note that the iTAP concurs that the sustainable development potential of the project is "high". Access to safe and secure water is fundamental to Article 25 of the Universal Declaration of Human Rights as well as the achievement of Sustainable Development Goals 6 and 3, amongst others. SPC is proud to be supporting Vanuatu in their progress towards achievement of national sustainable development outcomes in a climate-resilient, inclusive and community-focused manner.
Needs of the recipient
SPC is pleased to note that the iTAP concurs that the needs of the recipients are "high". Indeed, Vanuatu is one of the most vulnerable countries in the Pacific; notwithstanding underlying challenges related to water security, climate change will severely hamper the ability of ni-Vanuatu communities to achieve safe and secure wellbeing. This project will address many of these needs, providing opportunities for significantly improved living conditions for beneficiaries.
Country ownership

SPC is pleased to note that the iTAP concurs that the country ownership of the project is "high". As for all projects developed by SPC, it is predicated on a direct request to SPC from the government of Vanuatu for assistance in addressing national climate change priorities. The project has thus been government-owned from its inception and its design was government-led throughout.
Efficiency and effectiveness
SPC notes the iTAP assessment of the project's efficiency and effectiveness as "medium". While some interventions were assessed to have comparatively high financial cost, it is noteworthy that the costs to human lives, livelihoods and wellbeing of not taking such action or making such investments cannot be measured in financial terms. It is also significant that the costs associated with such investments in an island country such as Vanuatu are comparatively larger than similar investments elsewhere. In this context, such high costs are justified given the expected adaptation and sustainable development impacts vis-à-vis the needs of the recipients.
Overall remarks from the independent Technical Advisory Panel: SPC is pleased to note that the iTAP has endorsed the funding proposal for the project "Enhancing Adaptation and Community Resilience by Improving Water Security in Vanuatu".

Annex 8: Gender Assessment and Gender Action Plan

07 September 2022

This Gender Assessment and Gender Action Plan have been prepared for The Pacific Community (SPC), to inform the project design of the Green Climate Fund (GCF) Funding Proposal titled: *Enhancing Adaptation and Community Resilience by Improving Water Security in Vanuatu*. This project will deliver adaptation action for Vanuatu's water infrastructure and community users and will ensure gender mainstreaming in the paradigmatic shift being proposed.

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Contents

COVID-19 in Vanuatu

During the development of this report, the global community continues to grapple with the coronavirus pandemic, which is wreaking havoc on the lives of millions of people, both with regard to health and wellbeing, and socioeconomic indicators. COVID-19 is also having distinctive gendered impacts around the world.[1] The COVID-19 pandemic and major natural disasters hit the Vanuatu economy severely in 2020. Due to the authorities' decisive measures, Vanuatu primarily avoided domestic transmission of COVID-19, but the border closure, according to the International Monetary Fund, brought tourism (a key economic sector) to a virtual stop.

At the time of writing Annex 8 (March 2022), Vanuatu's health authorities announced community transmission of COVID-19, with restrictions on mobility and economy imposed on the main island of Efate (and its surrounding islands). A curfew running from 06:00 to 18:00 has severely constrained women's economic activity and incomes, notably female food sellers or *vatu mamas*.[2]

The coronavirus pandemic has exposed entrenched inequalities and stark disparities in power, clearly illustrating how a global crisis like this impacts people differently depending on where they live, their gender, their age and their socioeconomic status. The consequences of this crisis for gender equality are devastating, as has been anecdotally observed in Vanuatu during the stakeholder consultation processes for the project preparation phase. Worldwide, there has been increasing cases of gender-based violence, and a reversal of advances in sexual and reproductive health and rights. This is particularly concerning in the case of Vanuatu – which, along with the broader Pacific region, is endemic for heightened gender-based violence and violence against women.

The project has attempted to cover these nuances throughout this assessment, paying particular attention to how gender, WASH and climate change interact and identifying actionable interventions such as increasing women's representation in community-based water management groups and providing inclusive (and accessible) training on climate-resilient water management. Through the project's paradigm shift pathway, vulnerable (often remote) communities in Vanuatu (and marginalized groups such as women and girls within them) are expected to be able to benefit from overall increase in climate resilience and improved water safety and security.

[1] These links provide an overview of gendered impacts of COVID-19 globally:

https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(21)01651-2/fulltext

https://apps.who.int/iris/bitstream/handle/10665/332080/WHO-2019-nCoV-Advocacy_brief-Gender-2020.1-eng.pdf

https://www.who.int/news/item/03-05-2021-expanding-reach-addressing-gender-barriers-in-covid-19-vaccine-rollout

https://www.unwomen.org/en/digital-library/publications/2020/09/gender-equality-in-the-wake-of-covid-19

https://www.mckinsey.com/featured-insights/future-of-work/covid-19-and-gender-equality-countering-the-regressive-effects

[2] Latest figures on COVID-19 for Vanuatu are available on: https://covid19.who.int/region/wpro/country/vu

Definitions

To ensure a baseline understanding of climate issues, and the development of the analysis based on impact and exposure on women, men, girls and boys, the following definitions, adapted from the USAID Climate Change Adaptation Project Preparation Facility for Asia and the Pacific (Adapt Asia-Pacific)[3], are provided as reference for this Annex-8:

Term	Definition
Gender	A culturally-defined set of economic, social, and political roles, responsibilities, rights, entitlements obligations, associated with being female and male, as well as the power relations between and among women and men, boys and girls. The definition and expectations of what it means to be a woman or girl and a man or boy, and sanctions for not adhering to those expectations, vary across cultures and over time, and often intersect with other factors such as race, class, age and sexual orientation. Transgender individuals, whether they identify as men or women, are subject to the same set of expectations and sanctions. (OHA/PEPFAR, modified from the Interagency Gender Working Group – IGWG)
Gender Equity	The process of being fair to women and men, boys and girls. To ensure fairness, measures must be taken to compensate for cumulative economic, social, and political disadvantages that prevent women and men, boys and girls from operating on a level playing field. (IGWG training resources)
Gender Equality	The state or condition that affords women and men equal enjoyment of human rights, socially valued goods, opportunities, and resources. Genuine equality means more than parity in numbers or laws on the books; it means expanded freedoms and improved overall quality of life for all people. (IGWG training resources)
Gender-blind	are project designs and activities that ignore gender factors including roles and relations and can lead to reinforcement of gender-based discrimination and existing inequities. (USAID)
Gender-balance	requires that men and women be equally represented - either in equal numbers or in proportion to their presence - in particular settings. (USAID)
Gender Mainstreaming	Process of incorporating a gender perspective into organizational policies, strategies, and administrative functions, as well as into the institutional culture of an organization. This process at the organizational level ideally results in meaningful gender integration as outlined below. (adapted from IGWG training resources)
Gender and Socioeconomic Analysis	is the process of collecting information about gender, age and other social differences and analyzing the impacts of changing circumstances (i.e., climate change) on specific groups of people. This type of analysis provides the basis for identifying key gender considerations and designing a 'socially inclusive approach' that responds to the unique circumstances and needs of all project beneficiaries. (USAID)

[3] Gender and Social Inclusion Strategy of this project can be found here: https://www.nab.vu/sites/default/files/documents/Annex%208%20-%20Gender%20_social%20inclusion%20%20Strategy.pdf

Gender-based Violence	In the broadest terms, "gender-based violence" is violence that is directed at individuals based on their biological sex, gender identity, or perceived adherence to culturally- defined expectations of what it means to be a woman and man, girl and boy. It includes physical, sexual, and psychological abuse; threats; coercion; arbitrary deprivation of liberty; and economic deprivation, whether occurring in public or private. GBV is rooted in economic, social, and political inequalities between men and women. GBV can occur throughout the lifecycle, from infancy through childhood and adolescence, the reproductive years and into old age (Moreno 2005), and can affect women and girls, and men and boys, including transgender individuals. Specific types of GBV include (but are not limited to) female infanticide; early and forced marriage, "honor" killings, and female genital cutting/mutilation; child sexual abuse and exploitation; trafficking in persons; sexual coercion, harassment and abuse; neglect; domestic violence; economic deprivation, and elder abuse. (adapted from USG Strategy for the Prevention and Response to Gender-based Violence)
Sexual Exploitation, Abuse and Sexual Harassment	SEAH is the term used to refer to sexual exploitation, abuse and sexual harassment. Although sexual exploitation, abuse and sexual harassment can happen anywhere in society, when used as an umbrella term within the development and humanitarian sector, the term refers to SEAH perpetrated by those working in, or with, development and humanitarian organisations and within Peacekeeping Missions. (adapted from the Safeguarding Support Hub)

1. Introduction to the GCF – Annex 8

1.1 Objective of the study

This Gender Assessment and Gender Action Plan (GA-GAP) is a supporting document for the design of the proposed GCF project: *Enhancing adaptation and community resilience by improving water security in Vanuatu*. **It is the Annex 8 of the Funding Proposal**, which is being proposed as a full-scale project by the Accredited Entity (AE) - The Pacific Community (SPC) to the Green Climate Fund (GCF) and has been developed in close coordination with Vanuatu's Department of Water Resources (DoWR) and WASH sector partners.

SPC is currently implementing the FP169: *Climate change adaptation solutions for Local Authorities in the Federated States of Micronesia*[4] and has a track record in implementing gender-responsive adaptation solutions, within its broader portfolio of designing and implementing projects – as well as leading regional research – on cross-cutting issues such as climate change, disaster risk management, food security and human rights. The organization recognizes that:

- Climate adaptation and mitigation pathways are not gender-neutral[5];
- Gendered needs and vulnerabilities of marginal groups, such as Indigenous Peoples, need to be mainstreamed into adaptation design, resilience capacity-building and mitigation services;[6] and
- Gender-transformative impact can be driven through robust financial means, gender budgets and dedicated resources towards mainstreaming gender action in climate change and deforestation.[7]

Without gender and socioeconomic analysis, and a thorough Environmental and Social Management Plan (Annex 6 of the Funding Proposal), the benefits of increased support and access to climate-resilient WASH infrastructure, in tandem with awareness-raising and capacity-building, may accrue to better-off households or more mainstream groups that are able to capitalise on new opportunities and respond better to changes implemented through the project.

[4] More information on this project is available here: https://www.greenclimate.fund/project/fp169

[5] Current literature on climate change, and its effects and emergent risks, are predominantly produced in scientific circles. Yet, there is increasing evidence that adopting social science methods, and situating resilience and adaptation practice within a broader science-policy interface and right-based perspectives, can gear projects towards environmental and socioeconomic co-benefits. Particularly, this could better prepare communities to avoid resource strife and respond to the complexity of social arrangements, reducing far-reaching impacts of climate risks. See Butterfield, R. (2018) 'Bringing rights into resilience: revealing complexities of climate risks and social conflict' in Disasters. Journal Article.

[6] Poor or missing gender analysis, or the lack of gender-responsive action, may lead to planners or personnel depending on women to assume a central role in their coping strategies, which may not be the practical reality for many vulnerable communities. Further, this also glosses over the existing burdens on women among such groups. See Nelson. V., Meadows. K., Cannon, T., Morton, J., & Martin, A. (2002) 'Uncertain predictions, invisible impacts and the need to mainstream gender in climate change adaptations' in Gender and Development. Journal Article.

[7] OECD (2014). *Making climate finance work for women*. See: oecd.org/environment/making-climate-finance-work-for-women.

A 'gender lens'[8] is both necessary and relevant for the project to maximise its outcomes for the CR-WASH sub-sector, and ensuring resilience of water infrastructure against natural and climate-induced hazards, disasters and weather variations that cannot be avoided. This gender-responsive approach is also crucial for establishing institutional structures and broad-based political momentum and socioeconomic frameworks to mobilise medium- and long-term adaptation action in the country and derive socioeconomic and gender benefits from increased water security in the communities. Particularly, this project will aim to increase institutional representation and meaningful participation of women in community-based management of water sources, system improvements and new WASH infrastructure. Improved gender parity will help deliver the adaptation pathway chosen by the project, as well as sustain results to ensure that communities are able to transition to a water-secure and climate-resilient future.

The overall objective, therefore, of the gender assessment will be to provide a tool to promote gender mainstreaming in Vanuatu through the project paradigm – particularly, by identifying relevant entry points that the project can benefit from given its focus on water security and the climate-resilient water, sanitation, and hygiene (CR-WASH) sub-sector. Using gender as a lens will also help identify the social relationships between women and men, and other marginalized groups in Vanuatu.

Access to WASH is fundamental to development but it can be mobilized as critical climate adaptation strategy for poor and vulnerable communities, as well as marginalized groups within those communities, including women and girls.[9] The levels of global climate finance directed to the WASH sector are currently limited (according to WaterAid), which reflects that improved WASH is yet to be recognized as a key climate adaptation strategy, and for gender equality.
The focus of this study, therefore, is to present an assessment of the different roles, rights, needs and opportunities of women and men, boys and girls in the project context, and mobilise project resources to tackle gender barriers, and contribute towards improved gender equality in Vanuatu. In doing so, the assessment will seek to operationalise gender mainstreaming activities through the project's results framework by formulating a Gender Action Plan – Section 8, based on the project's Logical Framework (Section E of the Funding Proposal).

This will ensure that the proposed design[10] to deliver improved water security and increased climate resilience at community level in Vanuatu can have climate, environmental and social co-benefits in Vanuatu, with identified indicators, timeline and means of verification. The logical framework of the project has three Outcomes, two of which are focused on water infrastructure management and planning in tandem with a Outcome dedicated to institutional strengthening.
The outcome and outputs are detailed below:

[8] It is important to note that gender is socially constructed, and gender relations are contextually specific and often change in response to altering circumstances. MOSER, C. O.N. (1993): Gender Planning and Development: Theory, Practice and Training. New York: Routledge.

[9] Winqvist, E. (ed.) (2020). Feminist Policies for Climate Justice. Stockholm: Concord. Available at: https://concord.se/wp-content/uploads/2020/06/fem-rapport-2020-final.pdf

[10] Please refer to Section E for the Logical Framework in the Funding Proposal (FP). The Gender Action Plan (GAP) (Section 8) provides impact and outcome statement for each Outcome in relation to gender mainstreaming entry points and proposed actions for the project, to be evaluated against a set of indicators and targets as well as delineation of timeline, responsible parties and costing.

The paradigm shift potential of the results framework is captured in the Theory of Change, available in the FP as Section B.2.a.

Outcome 1:	**Communities are empowered to plan and manage climate-resilient water resources**
Output 1.1:	New and existing DWSSPs incorporate incremental improvements to mainstream adaptation solutions
Output 1.2:	Awareness, capacities and skills of communities and area administrators on climate-resilient water management improved
Output 1.3:	Vulnerable communities are supported to develop and implement their DWSSPs (600 by the end of the project cycle)
Outcome 2:	**Communities have enhanced climate-resilient rural water infrastructure**
Output 2.1:	270 vulnerable communities supported to construct, operate, and maintain climate-resilient water infrastructure
Outcome 3:	**Provincial and national institutions are strengthened to address climate risks associated with water**
Outcome 3.1:	National- and provincial-level staff and WASH sector partners trained on climate-resilient water management
Outcome 3.2:	Knowledge management through data sharing mechanism established for climate-resilient water management
Outcome 3.3:	Monitoring, learning and evaluation framework established for improved learning for climate-resilient water management

Through these Outcomes and outputs, the proposed project will mainstream gender by:

- Analysing and enhancing the baseline of gender and social inclusion in the National Implementation Plan for Safe and Secure Community Drinking Water (NIP); Capital Assistance Program (CAP) and more broadly, by equitably targeting beneficiaries through the Drinking Water Safety and Security Plans (DWSSPs);
- Establishing gender-transformative approaches for the design, delivery and maintenance plans of water infrastructure, whether existing or newly built or in the pipeline;
- Promoting gender parity, to the extent possible, through the technical and maintenance capacity-building outputs, alongside the institutional development targeted for increased water security in Vanuatu; and,
- Mainstreaming gender analyses and praxis for GCF investments in the CR-WASH subsector, both as a best practice in Vanuatu and for similar contexts in the Pacific Island Countries and Territories (PICTs).

1.2 Gender barriers in Vanuatu: a summary

A kaleidoscope of overlapping cultural, socioeconomic and political roles form gender norms, relations, uneven power dynamics and vulnerabilities in Vanuatu. For these reasons, the climate crisis affects women and girls in unique and disproportionate ways. The archipelago nation is spread out over 80 islands and has repeatedly been featured as one of the most at-risk and exposed country to climate change, notably in the 2019 World Risk Index.[11]

[11] Rurh Universität Bochum (2019). World Risk Report (focus on water supply). Available at: https://reliefweb.int/sites/reliefweb.int/files/resources/WorldRiskReport-2019_Online_english.pdf

Vanuatu is also classified as a Small Islands Developing States or SIDS[12], which are a distinct group of developing island countries facing specific social, economic and environmental vulnerabilities. As with other nations with SIDS status, repeated exposure to climate events and natural disaster phenomena presents a major obstacle to the sustained eradication of poverty and promoting shared prosperity in Vanuatu and undermines progress on gender equality indicators. This is compounded by SIDS characteristics such as: remoteness and deprivation from the benefits of scale, low income and assets, small domestic markets and heavy dependence on a few external markets and international support, high volatility of economic growth, fragile natural environments, and socioeconomic as well as gendered vulnerabilities.[13]

Additionally, until December 2020, Vanuatu was a Least Developed Country (LDCs).[14] During its graduation from the LDC list, United Nations Conference on Trade and Development recognized consistent strides made by the country to improve its social and economic development indicators but also that the exposure to environmental and economic vulnerabilities remain high, particularly in the face of climate change and other external shocks.[15]

In terms of gender characteristics, Vanuatu can be identified as a traditionally male-dominated and largely patriarchal society, with skewed sex ratio (105 males:100 females) in the islands. Overarching gender barriers include:

Limited participation of women in policy and decision-making spheres

Women have limited participation in decision-making spheres, with low representation in Parliament, as well as within devolved institutions of power. The national government's Beijing +25 review report, for example, highlights that no women were elected to parliament in the last two elections.[16] Further, in 5 out of 6 provincial councils, no women have been elected, with a sole female councillor elected in the sixth. Socio-anthropological research in Kurumambe and Purau villages in Tongoa, for example, highlighted how women must navigate strict rules governing their presence and speech in the *nakamal* (chief's meeting place) – effectively excluding them from holding titles or wielding decision-making power even at localised levels of governance.[17]

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[12] SIDS were recognized as having special status both for their environment and development at the Earth Summit, held in Rio de Janeiro, Brazil in 1992. Updated list of SIDS can be found on: https://www.un.org/ohrlls/content/list-sids.

[13] UN-OHRLLS – Office of the High Representative for the Least Developed Countries, Landlocked Developing Countries, and Small Island Developing States (2015). Small Island Developing States in Numbers: Climate Change Edition 2015. Report. Available at: https://sustainabledevelopment.un.org/content/documents/2189SIDS-IN-NUMBERS- CLIMATE-CHANGE-EDITION_2015.pdf

[14] See UNCTAD's LDC list here: https://unctad.org/news/vanuatu-graduates-least-developed-country-status

[15] See UNDESA's LDC profile for Vanuatu: https: //www.un.org/development/desa/dpad/wp-content/uploads/sites/45/LDC_Profile_Vanuatu.pdf

[16] Department of Women's Affairs (2020). Beijing +25 – National Review Report. Available at: https://www.asiapacificgender.org/sites/default/files/documents/Vanuatu%20(English).pdf

[17] Granderson, A. (2017) "The role of traditional knowledge in building adaptive capacity for climate change in Vanuatu" in Weather, Climate and Society.

https://www.researchgate.net/publication/317071599_The_Role_of_Traditional_Knowledge_in_Building_Adaptive_Capacity_for_Climate_Change_Perspectives_from_Vanuatu

Inequity in economic lives as well as gendered capacity and input gaps among men and women

Vanuatu is an agricultural society in which 75% of the total population lives in rural areas and depends largely on subsistence agriculture and fishing for daily subsistence and livelihoods, according to the Vanuatu National Statistics Office (2016). The 2016 mini-census found that 88% of household engage in some form of vegetable crop production, 57% in cash crop production, 69% in livestock production and 49% in fishing. A *Country Gender Assessment* led by the Food and Agriculture Organization (FAO) found that women and men participate in almost equal numbers in the agricultural sectors, but inequality manifests in the crops that are grown (often: men may grow cash crops) and in the technology employed (often: men have better access to agricultural inputs, owing to cash income from more lucrative crops).[18] Additionally, traditional and faith-based norms continue to operate and undermine gender equality in the essential inputs for productive and remunerative agricultural livelihoods, such as by: excluding women from matrimonial property, inheritance, and land rights).

Disproportionate share of unpaid care work continues to be undertaken by women

As in many other societies, the World Bank's *Gender and Investment Climate Reform Assessment* for Vanuatu finds: economically active women suffer from a double workday – combining responsibilities for home and family with their economic activities[19]. Although parity in enrolment rates has been achieved at primary and junior secondary school level, women remain less likely than men to have a completed secondary school education and to go on to tertiary education. This hints towards a gendered gap in educational achievements and skills, which translates into women being limited into low-skilled, low-paying jobs that are often found primarily in the informal sector.

In rural contexts, women face barriers in finding productive and remunerative work, and may be contributing to subsistence agriculture as well as various unpaid work in the household, thereby exhibiting high levels of time poverty.[20] Through stakeholder consultations for this project, it was revealed that women are primarily agents of domestic water provision, along with teenagers and young adults, which is corroborated by existing research by UNICEF and academics.[21]

High incidence of gender-based violence in Vanuatu

Vanuatu has one of the highest prevalence rates of violence against women and girls globally. Research conducted by the Vanuatu Women's Centre (in 2011) established that ~60% of women with an intimate partner had experienced physical violence, ~68% experienced emotional violence, and ~69% coercive behavioural control by men.[22] According to the World Bank's 2012

[18] FAO (2020). Country Gender Assessment of Agriculture and the Rural Sector in Vanuatu. Available at: https://www.fao.org/3/ca7427en/ca7427en.pdf

[19] AusAID & IFC (2010). Gender and Investment Climate Reform Assessment – Vanuatu. Available at: https://openknowledge.worldbank.org/bitstream/handle/10986/25925/111536-WP-IFC-GenderICReformAssessments-Vanuatu.pdf?sequence=1&isAllowed=y

[20] Time poverty is the burden of competing claims on an individual's time that reduce their ability to make unconstrained choices in how they allocate their time leading to increased work intensity and to trade-offs among various tasks. (A. Kes and H. Swaminathan. 2005. Gender and Time Poverty in Sub-Saharan Africa. In C. Blackden and Q. Wodon, eds. Gender, Time Use, and Poverty in Sub-Saharan Africa. Washington, DC: World Bank.)

[21] J. Willetts et al. 2010. Addressing Two Critical MDGs together: Gender in Water, Sanitation and Hygiene Initiatives. Pacific Economic Bulletin. 25 (1.

[22] World Vision (2018). Evaluation Report: Reducing GbV Project (Vanuatu Counselling Approach). Available at: https://reliefweb.int/sites/reliefweb.int/files/resources/Counselling%20Evaluation%202018%20.pdf

World Development Report on Gender Equality and Development, between 60 – 70% of women in Vanuatu (alongside Kiribati and Solomon Islands) experience some form of domestic violence.[23]

Male family members and boyfriends perpetrate most of the violence and it occurs in all provinces and islands, across age groups, education levels, socio-economic groups and religions. It is higher in rural (63%) than in urban (50%) areas. Social values held by both women and men reinforce the acceptability of violence towards women and girls and ~60% of women agree with at least one “reason” for men to beat their wives. In Vanuatu, gender-based violence, and particularly intimate-partner violence, is used as punishment and discipline and is accepted and condoned as ‘normal’ behaviour in many communities. This impacts both women and children.

Stakeholder consultations conducted during the project highlighted the need for introducing security and safety Outcomes for system improvements and new WASH infrastructure planned through the project – particularly demarcation of women and men’s toilets, locks and doors, as well as lighting to ensure safe passage and accessibility. Therefore, working to mainstream gender interests and ensuring dignity for women and girls (also around menstrual hygiene management) through planned project activities will overall increase community well-being, ensure social co-benefits and the success of the project adaptation pathway.

[23] See: https://www.worldbank.org/en/news/feature/2012/11/25/raising-awareness-of-violence-against-women-in-the-pacific

2. Methodology note for Annex 8

2.1 Primary data and information collected through key informant interviews and focused working groups

Due to pandemic-related obstacles, this gender assessment has been prepared using primary information and data collected by national stakeholder engagement experts between June – September 2021. An **Inception Workshop**, convened on 30th June 2021, commenced these consultations and engagement processes.[24] Following the Inception Workshop, one to one/face to face consultations were held with key stakeholders actively working or involved in the national water governance processes, humanitarian responses for water and utilities as well as the overall improvement of WASH service delivery in Vanuatu.

Using semi-structured interviewing methodology, ranging from one to two persons at a time and in accordance to COVID-19 guidelines laid down by the Government of Vanuatu, interviews were conducted with the **Department of Water Resources** (DoWR) and other key ministries; civil society organizations (CSOs); and, UN entities active in Vanuatu, particularly UNICEF (an important WASH sector partner for this project). The stakeholders ranged from employees of the DoWR to provincial government leaders and community members, who play a key role in ensuring water security in Vanuatu. Participants and stakeholders were identified using a combination of purposive and snowball sampling during the field missions led by the DoWR, and consulted according to their levels of knowledge, exposure and access on climate change, water infrastructure, and gender as well as socioeconomic issues to prioritise and vet information and data received.

The **Department of Women's Affairs** (DoWA) was among the key stakeholders consulted as a part of this process. A key finding from this consultation was that gender has been increasingly incorporated in WASH processes over the years. However, mainstreaming remains a need across all WASH stakeholders that requires ongoing improvement. WASH design interventions and processes within government sectors and NGOs need to be more responsive to the needs of children, the elderly and LGBTQIA+. Given the reach of the proposed project down to community-level interventions, it was determined consultation at the provincial level was also necessary to inform the project preparation stage. Accordingly, with guidance from DoWR, the provinces of Penama, Sanma and Torba[25] were selected. The consultations identified ongoing relevant water

[24] The workshop was conducted by SPC and involved the participation of key players including the Vanuatu Nationally Designated Authority (NDA) – Ministry of Climate Change, the Department of Water Resources, Ministry of Agriculture, Vanuatu Meteorology & Geo-hazards Department, Department of Strategic Planning, Policy & Aid Coordination. WASH sector partners included United Nation's Children Fund, Asian Development Bank, International Organization on Migration and the New Zealand High Commission.

[25] The Torba Province consists of the Torres and Banks group of islands in the north of Vanuatu. The Torres group is located in the extreme north of Vanuatu and comprises of 5 islands: Hiu, Metoma, Tegua, Loh and Toga. The Banks group include Mota Lava, Mota, Merik, Ureparapara, Vanua Lava and Gaua islands. The provincial centre or headquarters is located on the island of Vanua Lava. Penama encompasses the three islands of Pentecost, Ambae

security work that are being implemented by government and NGOs in the provinces, that provide a basis for future development opportunities in the sector. Participants also identified the key gender groups at the community level and articulated their different needs that should be incorporated into the design of new water security/WASH projects and programmes so that interventions are responsive to the needs and special circumstances of all beneficiaries.

The funding proposal package, including this Annex 8, was subsequently presented at a **Validation Workshop**, on 23rd March 2021. This ensured that national-level and provincial stakeholders were engaged, and their consensus was sought for, the gender mainstreaming goals identified by the project.

2.2 Secondary data and information collected through literature review

To collect secondary information and literature, the gender expert conducted an in-depth desktop review in tandem with the primary and formative remote research through national experts in the islands. The literature review focused on gender mainstreaming, water security and resilience as a broader topic, drawing from key players in the water industry such as the Global Water Partnership – GWP, the Global Facility for Disaster Reduction and Recovery – GFDRR, United Nations Development Programme – UNDP, United Nation's Children Fund – UNICEF and the World Bank, as well as donor agencies active in the Pacific region, such as Australian Aid (AusAID), Japanese International Cooperation Agency (JICA) and United States Agency for International Development (USAID).

Existing gender assessment and gender action plan for prior GCF investment in Vanuatu (FP035[26]), were also reviewed to understand the scope of gender-climate resilience work through prior GCF investment in the islands. Lessons learnt and best practices were reviewed from recently closed and ongoing projects such as: UNDP (V-CAP project) and Secretariat of the Pacific Regional Environment Programme – SPREP (VAN-KIRAP project). These studies and reports were supplemented by research from think tanks and research groups such as the International Institute on Environment and Development (IIED) and the World Resources Institute (WRI). Entries in peer-reviewed anthropology, geography- and climate-focused journals such as *Anthropology Today*, *Gender and Development*, *Disasters* (among others) were also reviewed to ensure that the assessment is able to capture latest studies and research in Vanuatu on the broader domains of climate resilience, WASH and gender.

and Maewo with the provincial headquarters located on Ambae, while Sanma covers the islands of Espiritu Santo and Malo where the provincial headquarter is located in the northern town of Luganville on Santo.

[26] The FP035: Climate Information Services for Resilient Development Planning in Vanuatu is a project being implemented by the Secretariat of the Pacific Regional Environment Programme to build technical capacity to harness and manage climate data and develop practical CIS tools in the country.
See https://www.greenclimate.fund/project/fp035 for more.

2.3 Concurrent triangulation of primary inputs and secondary data and information collection

The study employs a **concurrent triangulation methodology**, and deliberately sidesteps sequential research processes. In *sequential designs*, either the qualitative or quantitative data are collected in an initial stage, followed by the collection of the other data types during a succeeding, second stage.[27] In contrast, *concurrent designs* are characterized by the collection of both types of data during the same stage (which was the case for the development of this assessment).[28]

Within these two categories, there can be three specific designs based on: (a) the level of emphasis given to the qualitative and quantitative data (equal or unequal), (b) the process used to analyze and integrate the data, and (c) whether or not the theoretical basis underlying the study methodology is to bring about social change or advocacy.[29]

This approach has been deemed suited to this gender assessment, given that it will operationalize GCF and co-financing investments towards improved gender outcomes (such as: increased participation of women in rural water governance bodies, gender-transformative WASH infrastructure design and reinforced policy framework), through the paradigm of increased climate resilience and water security in Vanuatu.

Additionally, the stakeholder consultations undertaken in the islands, was formative in nature and rapid in mandate, necessitating a prompt methodology to establish an inventory of information and a gender baseline that can be operationalized to cover the complex aspects of CR-WASH in Vanuatu. Therefore, the concurrent approach is suited because of: the flexibility to use secondary quantitative and qualitative findings for the comprehension of the studied phenomenon in the primary research (mainly the stakeholder consultations), and the accuracy it can provide to samples of intermediate or small size (the stakeholders consulted during the fieldwork conducted by the national experts).

This also ensures that the assessment and action plan are able to propose measures that are relevant for the delivery of outputs identified in the results framework, particularly the identified gender-responsive co-benefit in Section E of the Funding Proposal: improved gender parity in the representation of women in Rural Water Committees (a minimum of 40%). Additionally, the project will be targeting a 50:50 ratio of beneficiaries at the community level in the 600 new DWSSPs, as well as among the chosen DWSSPs (270) that receive infrastructure support. The project will also upgrade the requirements for DWSSPs, mandating the inclusion of gender factors in the broader prioritization framework, as well as upgrading the methodology of the DWSSP.

[27] Creswell JW, Plano Clark VL, Gutmann ML, Hanson WE. Advances in mixed methods research designs. In: Tashakkori A, Teddlie C, editors. Handbook of mixed methods in social and behavioral research.

[28] Creswell JW, Plano Clark VL, Gutmann ML, Hanson WE. – Ibid.

[29] Ibid.

3. Policy environment for gender-transformative CR-WASH in Vanuatu

This section takes stock of the policy environment and legal frameworks available for gender-responsive climate change adaptation in the Vanuatu (including international conventions such as the Convention on the Elimination of all Forms of Discrimination against Women or CEDAW, national laws and policies, strategy documents on gender and climate change). It is important to note that one of the five strategic areas of the Government of Vanuatu's National Gender Equality Policy is *fostering gender-responsive and community-driven solutions to climate and disaster resilience*, which this project will contribute to by delivering gender-responsive, climate-resilient WASH infrastructure in rural and remote contexts of Vanuatu.

Identification of legal tools and enabling policies, particularly at the baseline, is crucial in ensuring that gender inequality can be addressed through government-owned and formal procedures. These tangible processes can also help drive ownership of the project, and its social impacts. Additionally, the inclusion of local and national gender partners develops capacity and technical knowledge towards future gender efforts while establishing ownership of the project and the change narrative being implemented.

Details	Year	Information
CEDAW	1995	Vanuatu ratified the CEDAW in 1995, and has received concluding observations on the combined fourth and fifth periodic reports of Vanuatu in 2016.[30] CEDAW has recognized Vanuatu's progressive reforms in Family Protection Act, the initiation of a new National Gender Equality Policy, Gender Equity in Education Policy, while reiterating the need for establishing effective remedies in both the formal and traditional justice systems. Awareness raising for women's rights, particularly through civil society organizations, is also key.
National Sustainable Development Plan (NSDP)	2016 – 2030	The NSDP 2016–2030, prepared by the Department of Strategic Policy, Planning and Aid Coordination, includes three main focus areas or 'pillars' that focus on 1) the environment; 2) the economy; and 3) society. The NSDP notes that Vanuatu aspires to "…an inclusive society which upholds human dignity and where the rights of all Ni-Vanuatu, including women, youth, the elderly and vulnerable groups are supported, protected and promoted in our legislation and institutions." The pillar on society includes commitments to 1) implement gender-responsive planning and budgeting processes; 2) prevent and eliminate all forms of violence and discrimination against women, children and vulnerable groups;

[30] United Nations CEDAW (2016). Concluding Observations – shared on 09 March 2016. Available at: https://documents-dds-ny.un.org/doc/UNDOC/GEN/N16/064/87/PDF/N1606487.pdf?OpenElement

		and 3) ensure all people, including people with disabilities, have access to governmental services (Government of Vanuatu, 2016).
National Gender Equality Policy (NGEP)	2020 – 2030	The NGEP 2020 - 2030, prepared by the Department of Women's Affairs of the Ministry of Justice and Community Services directs the government to implement gender mainstreaming nation-wide through all governmental departments. The NGEP prioritizes five strategic areas: 1) eliminating discrimination and violence against women and girls; 2) enhancing women's empowerment and skills development; 3) advancing women's leadership and political participation; 4) strengthening the foundation for gender mainstreaming; and, 5) fostering gender-responsive and community-driven solutions to climate and disaster resilience.
6 Gender Action Plans at provincial level	2020 - 2024	The purpose of this Gender Equality Action Plan is to provide a clear plan of action for government, civil society, private sector and development partners to coordinate actions to advance gender equality and the well-being of women and girls in all 6 Provinces in line with the National Gender Equality Policy (NGEP) 2020-2030.
WASH policies		The Water Resources Management (Amendment) Act No. 32 of 2016 and the Water Supply (Amendment) Act No. 31 of 2016, the Vanuatu National Drinking Water Quality Standards 2016, and the Vanuatu National Water Policy 2017–2030 all require community water supplies to have a Drinking Water Safety and Security Plan (DWSSP), an internationally-recognised approach for achieving safe drinking water. Both the current Vanuatu National Water Strategy 2008–2018 and the draft Vanuatu National Water Strategy 2018–2030 (NWS) include strategic directions and targets for introducing DWSSP i.e. the Drinking Water Safety and Security Planning process to rural community water supply schemes (i.e. departments, private, communities, schools, health facilities and households). The National Environmental Health Policy and Strategy 2012–2016 (NEHPS) also sets DWSSP targets. Drinking Water Safety and Security Planning (DWSSP) is a process of community engagement in identifying and discussing threats to safe and secure drinking water, and making plans to manage these threats. The resulting community DWSSP guides day-to-day water supply operation and maintenance, as well as improvements. Once the necessary community development training has been provided, the registration can be completed (Section 20F of the Water Resources (Amendment) Act No. 32 of 2016). The Act also requires at least 40% of the members of a RWC must be women – which has been added as a co-benefit to the project's Logical Framework (Section E – Funding Proposal).

4. Socioeconomic and gender baseline in Vanuatu

4.1 Methodology for the baseline

The contents of this assessment are derived from a combination of primary research (through national- and local-level stakeholder consultations) and secondary research (literature review, policy analysis and collation of secondary data). Site-specific, localised data has not been collected as the project will select sites during the implementation phase through the National Implementation Plan (NIP) process and mobilise those funds through the Capital Assistance Programme (CAP). In lieu, this GA-GAP uses available national aggregate statistics and composite indices. These data points have been chosen to nuance the primary and secondary research on climate resilience and the WASH baseline in the country. Drawn from primarily VNSO, and different UN entities, these speak to the issues identified by the GCF in its *Mainstreaming Gender* toolkit[31] as crucial socioeconomic indicators that can impact upon the performance of the project and therefore, are being mapped and mainstreamed in the project design stage.

4.2 National aggregate statistics with SDG indicators

The national-level data points are divided into four categories (poverty and hardship, education, labour and health), and presented alongside their corresponding Sustainable Development Goal (SDG). These indicators cover: **SDG 1**; **SDG 3**; **SDG 4** and **SDG 8**.

SDG 1	CHOSEN INDICATORS	VALUES
	• % of population below international poverty line • % of population below national poverty line • % of population below lower middle income poverty l • % of population in severe multidimensional poverty[32] • % of population vulnerable to multidimensional pover • % of male-headed households (HHs) • % of female-headed households	• 13.2% (2010) • 12.7% (2010) • 39.4% (2010) • n/a • n/a • 76% (2009) • 24% (2009)

[31] GCF (2017). Mainstreaming Gender in GCF projects. Publication Guide. Available at: https://www.greenclimate.fund/document/mainstreaming-gender-green-climate-fund-projects

[32] Multidimensional poverty: In the post-2015 SDG and Agenda 2030 Framework, SDG 1 targets poverty elimination – in all forms and dimensions. This mandate requires tools to enumerate (quantitatively) and assess (qualitatively) poverty levels in different countries. The Multidimensional Poverty Index, calculated by UNDP and Oxford Poverty and Human Development Initiative (OPHI), is an important tool as it complements the international poverty line statistics by showing the nature and extent of overlapping deprivations for each person. Currently, due to a paucity of data, the MPI is not calculated for Vanuatu.

Vanuatu's relatively high per-capita incomes combine with reasonably widespread land access for subsistence agriculture and informal, community-based social safety nets to keep the incidence of extreme poverty low. However, these high per-capita incomes overshadow the fact that Vanuatu (along with the Federated States of Micronesia, Kiribati and Marshall Islands) has higher than 10% extreme poverty (the regional average for the Pacific is around 3%). Vanuatu (along with the FSM, Kiribati and Marshall Islands) collectively hosts over 90% of people in poverty in the southern Pacific. Poverty is generally viewed as hardship culturally in this region.

- National and International Poverty Line:[33] National poverty lines are defined according to each country's specific economic and social circumstances, and the International Poverty Line is pegged at US$ 1.90 or 210 VATU. Since Vanuatu is a lower middle-income country, it is also important to measure poverty using the Lower Middle Income Class Poverty Line (US$ 3.20 or 354 VATU) – 39.4% of the population is below this line, according to the latest available statistics. With recent TCs Pam and Harold (strongest and second-strongest to ever affect Vanuatu), many households dropped back into hardship, reversing earlier gains.

- Multidimensional poverty: Currently, due to a paucity of data, the MPI is not calculated for Vanuatu. Like peer Melanesian economies, with large proportions of the population living semi-subsistence lifestyles in rural areas, consumption poverty is difficult to measure, however vulnerabilities are widespread – which are exacerbated by multivariable risks, including compounding climate hazards.[34]

- Gender-disaggregation of households:[35] In Vanuatu, due to the presence of multigenerational households, it is difficult to assess how household heads are designated and such roles adhered to. However, gender-disaggregation of private households, is available from a monograph published by the VNSO on the 2009 National Population and Housing Census – revealing that MHHs are the norm (76% compared to 24% of FHHs). The overriding factor influencing WASH in Vanuatu is rural versus urban location, according to the VNSO monograph. However, analysing the data by the sex of the household – lone MHHs have slightly better access to improved sanitation facilities compared to FHHs. FHHs also tend to have shared toilet facilities, possibly reflecting a lower standard of housing for women in particular cases and areas.[36]

[33]World Bank (2019). Poverty and Equity Brief (Vanuatu), available at: https://databank.worldbank.org/data/download/poverty/33EF03BB-9722-4AE2-ABC7-AA2972D68AFE/Archives-2019/Global_POVEQ_VUT.pdf

[34] Feeny, S. & McDonald, L. (2014). Vulnerability to Multidimensional Poverty: Findings from Households in Melanesia. Available at:https://www.tandfonline.com/doi/abs/10.1080/00220388.2015.1075974

[35] Micro data analyses, with consideration for macro, population and demographic factors, reveals the importance of introducing heterogeneity in household poverty figures as well as contextualizing how FHHS and MHHs function, conditional upon location, age, number of members, marital status, economic access, and other indicators. It is an imperfect metric but is currently one of the only viable option for exploring how gender impacts on household characteristics. See more: https://data.worldbank.org/indicator/SP.HOU.FEMA.ZS?locations=Z4-VU

[36] VNSO (2009). National Population and Housing Census. Gender Monograph: Women and Men in Vanuatu. See: https://mjcs.gov.vu/images/research_database/2009_Vanuatu_Census_Gender_Monograph.pdf

SDG 3	CHOSEN INDICATORS	VALUES
3 GOOD HEALTH AND WELL-BEING	• % exposure to gender-based violence (lifetime probability)	• 60% (2011)
	• # adolescent fertility rate, (modeled estimate, births per 1000 women)	• 48 births (2019)
	• # maternal mortality ratio (modeled estimate, per 100,000 live births)	• 72 deaths (2019)
	• # infant mortality rate (modeled estimate, per 1000 live births)	• 22 deaths (2019)
	• # children under five mortality (modeled estimate, per 1000 births)	• 26 deaths (2019)

Despite producing insignificant greenhouse gas emissions that cause climate change, SIDS populations (including the ni-Vanuatu peoples) are on the frontline of climate change impacts, including having a high burden of climate-sensitive diseases. Furthermore, the government highlights the threats climate change poses to human health in its Nationally Determined Contribution (NDC), which also recognizes the vulnerability of the country to climate change and the need to improve access to basic health services.[37] The importance of SDG 3 is underwritten by the scope of this project: investment in climate resilience and water security of communities can have diffuse and direct impacts on health outcomes of the population (particularly, in light of the ongoing COVID-19 pandemic).

- Maternal mortality ratio (MMR), infant mortality rate (IMR) and under-five mortality rate:[38] MMR, IMR and under-five mortality statistics are collected through the census to demonstrate the access and efficacy of health services and family planning in Vanuatu. Child mortality, for example, has fallen in Vanuatu, but remains high compared to other Pacific countries. Poor water security and WASH infrastructure is one of the leading contributors to enteric and other infectious disease burdens, which remains high in Vanuatu.

- Additionally, Vanuatu – like peer economies and SIDS, is vulnerable to tropical disease outbreaks. Climate change could affect the seasonality of such outbreaks, as well as the transmission of vector-borne diseases, underwriting the importance of robust WASH in the country.[39] For example, the seasonality and prevalence of dengue transmission may change in future climate change scenarios, and is modelled for different RCPs in Figure 1:

[37] https://cdn.who.int/media/docs/default-source/climate-change/who-unfccc-cch-country-profile-vanuatu.pdf?sfvrsn=451e1b3b_2&download=true

[38] The World Health Organisation (WHO) identifies MMR or complications during pregnancy and childbirth as a leading cause of death and disability among women of reproductive age in developing countries. This was an MDG and is now an SDG 3 indicator. Similarly, the IMR under-five mortality rates take stock of preventable child deaths at birth and below the age of 5.
https://data.unicef.org/country/vut

[39] https://cdn.who.int/media/docs/default-source/climate-change/who-unfccc-cch-country-profile-vanuatu.pdf?sfvrsn=451e1b3b_2&download=true

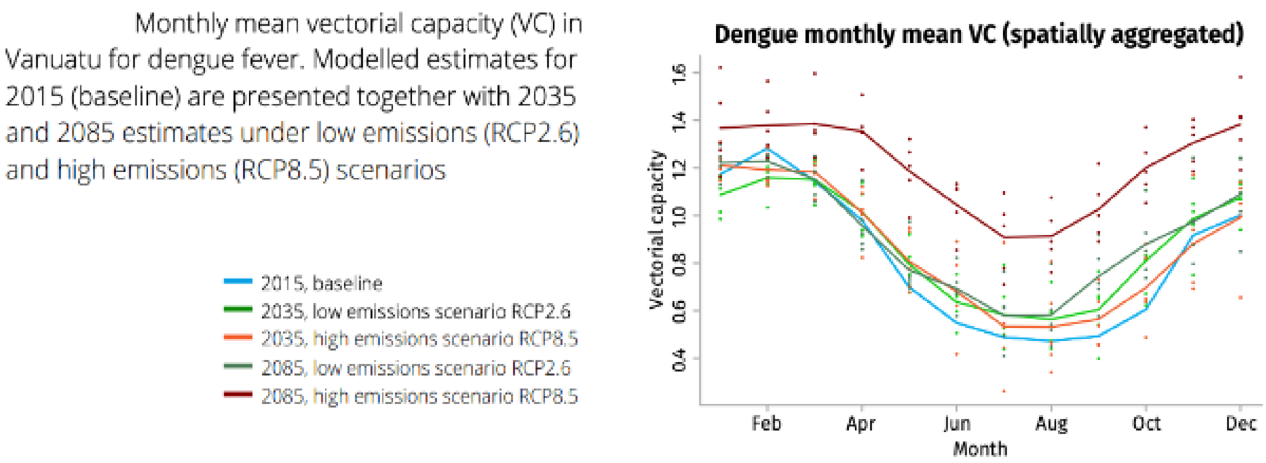

Figure 1: Dengue projections based on different RCPs in Vanuatu (WHO, 2020)

SDG 4	CHOSEN INDICATORS	VALUES
4 QUALITY EDUCATION	• % literacy rate, adult female	• 86.7% (2018)
	• % literacy rate, adult male	• 88.3% (2018)
	• % literacy rate, youth female	• 96.5% (2014)
	• % literacy rate, youth male	• 95.0% (2014)
	• # children out of school, primary, female	• 1503 (2015)
	• # children out of school, male	• 1797 (2015)
	• % progression to secondary, female	• 45.6% (2000)
	• % progression to secondary, male	• 44.2% (2000)

The education system of Vanuatu is atypical in that it represents an amalgamation of two disparate systems, the British and the French, that co-existed within the country prior to independence. Additionally, the church is an important education authority for primary and secondary schools. Government expenditure on education (as a percentage of total government expenditure) is 12.6% and the sector is the largest government service deliverer and employer.
Overall, there is no statistically significant difference between the performance of boys and girls, from available data. However, the Vanuatu Education and Training Sector Strategic Plan 2020 – 2030 identifies the importance of developing the capacities to identify further gender inequalities and address them through the Ministry of Education and Training (MoET). Particularly, a module on GbV has been developed by the Vanuatu Education Sector Program (VESP) in collaboration with the MoET as an awareness-raising exercise and for broader use in the education sector.

- WASH performance in Vanuatu's schools (per the UNICEF/WHO Joint Monitoring Project indicators) has been found to be limited. A baseline study of Penama province's 78 schools found that 63% had access to basic water, 26% had basic sanitation, and 27% with basic hygiene facilities.

- A Wash in Schools (WinS) program funded by the Australian Government (Western Pacific Sanitation Marketing and Innovation Program) in the broader region found that maximising girl children attendance can be achieve through the creation of 'girls' space' with dedicated bathrooms and reliable water supply.[40] Dignity kits are important to access in schools – with sensitization training for WASH in the national curriculum as well as for teachers' training syllabus. In Port Vila, through the said project, this is being encouraged through School Improvement Officers (SIO), who are now monitoring WASH as one of the fifteen school quality indicators.

SDG 8	CHOSEN INDICATORS	VALUES
8 DECENT WORK AND ECONOMIC GROWTH	• % labor force participation rate, female	• 43% (2019)
	• # labor force, total	• 129,602 (2020)
	• % vulnerable employment, female	• 70.6% (2019)
	• % vulnerable employment, male	• 65.6% (2019)
	• % wage and salaried workers, female	• 29.7% (2019)
	• % wage and salaried workers, male	• 33.8% (2019)
	• % employment in agriculture, female	• 56.7% (2019)
	• % employment in agriculture, male	• 56.7% (2019)
	• % of time, unpaid care work, female	• n/a
	• % of time, unpaid care work, male	• n/a

Vanuatu's economy is still primarily based on subsistence or small-scale agriculture, which provides a living for more than 70% of the population.[41] Since the early 2000s, tourism, land sales and high commodity prices for copra and coffee, and donor funding have driven the economy.[42] Major impediments to the economy include: undiversified economic base, constraints from poor transport infrastructure and a small domestic market.

- Although the gender differences are minor, it is telling that, according to World Bank statistics, about 70.6% of the active female labour force and 65.6% of the active male labour force are engaged in vulnerable employment. Roughly 20% of the workforce in Vanuatu is engaged in formal sector employment, and the formal economy only offers some hundreds of jobs each year. Consequently, the social protection system in Vanuatu is limited to formal sector employees – a miniscule minority of workers in the country.

- Given trends of low growth and limited employment opportunities, migration (including temporary migration for seasonal work in New Zealand and Australia) and remittances have been a critical driver of increased living standards in Vanuatu (and other Pacific Island Countries). However, although out-migration has significantly increased household incomes, such processes are affected by external shocks (such as climate change). Due to COVID-19, seasonal worker schemes have been affected in Australia and New Zealand, and the broader Pacific region, causing broad-based and significant reductions in employment and earnings.[43] About 8.1% of the ni-Vanuatu workforce rely on such programmes

[40] See: https://www.unicef.org/pacificislands/media/2186/file
[41] See: https://www.ilo.org/wcmsp5/groups/public/---asia/---ro-bangkok/---ilo-suva/documents/publication/wcms_366547.pdf
[42] Ibid.
[43] See: https://documents1.worldbank.org/curated/en/430961606712129708/pdf/Pacific-Labor-Mobility-Migration-and-Remittances-in-Times-of-COVID-19-Interim-Report.pdf

(Australia's Seasonal Worker Program – SWP and the Pacific Labour Scheme - PLS; New Zealands's Recognised Seasonal Employer Scheme – RSE), which indicates the precariousness of employment opportunities in the country. Both schemes actively promote the inclusion of women workers.

4.3 Composite indices (HDI, GDI, GII, GGI)

Designing a climate resilience-based WASH service delivery system requires situating the project's results framework on a thorough and context-driven baseline, which this feasibility study collates through various data sources. A collation exercise of scores and rankings from composite indices, especially due to paucity of updated, decentralized, nationally available data, has been included in this assessment to reflect Vanuatu's overall performance on different indicators. These indices have differing methodologies, and are being employed as indicative (and *not* conclusive) measures of current levels of development, gender equality, poverty, and labour force participation.[44]

This baseline, at the outset, uses scores of three different UNDP composite indices: Human Development Index (HDI), Gender Inequality Index (GII) and Gender Development Index (GDI) as points of departure. The latter two are currently unavailable for Vanuatu, but has been included to display the paucity of data characteristic of Pacific island contexts. Secondly, the baseline collates scores from the World Economic Forum (WEF)'s Global Gender Gap Index (GGGI), where Vanuatu was recently included.

INDEX (SCALE, ORGANIZATION)	RANK / SCORE
Human Development Index, out of 189 countries (UNDP)	**141**
Gender Inequality Index, out of 162 countries (UNDP)	**n/a**
Gender Development Index clustered with group (UNDP)	**n/a**
Global Gender Gap Index out of 153 countries (WEF)	**126**

Vanuatu's HDI value is 0.609 – which puts the country in the medium human development category – positioning it at 141 out of 189 countries and territories in the 2018 index.[45] This can primarily be linked to the SID-specific challenges that the World Bank identifies for the Vanuatuan economy:

[44] As Booysen's research shows, composite indices present both challenges and advantages. For example, numerous fallacies have been identified in the methodologies employed in composite indexing. These indices are mainly quantitative, and present empirical and aggregate measures of complex development phenomena, making values apparently objective, at the cost of subjective nuances. Yet, these also remain invaluable as useful supplements to income-based development indicators, understanding relative degrees of development, simplifying complex measurement constructs as well as providing access to non-technical audiences. See more in Booysen, F. (2002). "An Overview and Evaluation of Composite Indices of Development" in Social Indicators Research, (Vol. 59 No. 2). Journal Article.

[45] UNDP (2019). 'Briefing note for countries on the 2019 Human Development Report: Vanuatu' in the Human Development Report 2019. Research brief. Accessed online: http://hdr.undp.org/sites/all/themes/hdr_theme/country-notes/VUT.pdf

small size, remoteness from major markets and internal dispersion.[46] The country's geographical location, undiversified economic structure, and high import dependence makes it vulnerable to external shocks, especially climate change-induced disasters and natural hazards.[47] Such shocks greatly impact hard-earned development gains and threaten food, water and energy security – this project will address particularly water security issues.

Currently, the GII and GDI remain unavailable for Vanuatu due to the paucity of relevant and quality data.[48] The 2020 GGGI (WEF) includes four new countries, including Vanuatu for the first time. It is ranked 126 out of 153 countries and territories, on an aggregate score that combines performance in Economic Participation, Educational Attainment, Health and Survival and Political Empowerment.[49] Female representation in political spheres in Vanuatu displays a vast gender gap, there has never been a female head of state in the past 50 years, and there is currently no women in parliament or in ministerial positions, as evinced by the findings of the Beijing +25 report. Women's hardship trends, despite scant data, can be inferred from their dependence on lower skilled and remunerative activities such as subsistence agriculture as well as natural resource-based livelihoods.[50] On trend with the Pacific region, Vanuatu experiences a high incidence of gender-based violence and intimate partner violence.[51]

These indices are important to consider in the context of WASH and climate change, as they signal potential hurdles towards inclusive application of proposed alternatives towards resilience and water security in the country. However, this project attempts to mitigate these expected hurdles by locating the change paradigm on extensive stakeholder consultations presented in the Stakeholder Engagement Plan (Annex 7), Gender Action Plan (Section 8 of this Annex) as well as an Environment and Social Impact Assessment and Management Plan (Annex 6).

[46] World Bank (2017). Systematic Country Diagnostic: Republic of Kiribati, Republic of Marshall Islands, Federated States of Micronesia, Republic of Palau, Independent State of Samoa, Kingdom of Tonga, Tuvalu, Republic of Vanuatu (PIC8 – 8 Pacific Island Countries). Multi-country report. Accessed online: http://documents.worldbank.org/curated/en/313021467995103008/pdf/102803-REPLACEMENT-SecM2016-0025.pdf

[47] World Bank (2017). Systematic Country Diagnostic: PIC8. Ibid.
[48] UNDP (2019). 'Briefing note for countries on the 2019 Human Development Report: Vanuatu' in the Human Development Report 2019. Ibid.

[49] WEF (2020). The Global Gender Gap Report. Ibid.
[50] World Bank (2017). Systematic Country Diagnostic: PIC8. Ibid.
[51] The World Bank reports that based on the best-available data, women in the PIC8 suffer from either partner or non-partner violence at higher extents than any other part of the world. See API-GBV website for more.

5. Gender at the Green Climate Fund & the Pacific Community

The analytical prerogatives of this Gender Assessment and Action Plan are informed by both GCF's and SPC's respective gender and protection as well as social and environmental responsibility policies.

The **GCF** adopted a revised version of its 2014 Gender Policy and Action Plan on June 2018 in Korea. The revised Policy addresses pertinent issues on gender and climate change: the expansion of gender mainstreaming beyond the preserve of 'women's issues'; and the identification of synergies with the in-house Indigenous People (IP) Policy as well as the United Nations Framework Convention on Climate Change (UNFCCC)'s Gender Action Plan (GAP), Sustainable Development Goals (SDGs) and Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW).
Overall, the Policy and Action Plan reinforce the responsiveness of GCF to the multiple, heterogeneous, culturally diverse context of gender inequality to better address and account for the links between gender issues and climate change – a perspective that has been mainstreamed in the development of this full funding proposal for CR-WASH in Vanuatu.

The **SPC** adopted a set of general policies towards social and environmental responsibility in August 2020. The key objectives of this policy are to promote and drive continuous improvement of SPC's social and environmental performance, to embed a people-centred approach across its programmes, projects and activities, as well as to align with international recognized best practices for development.
The policy provisions for considerations of gender equality and social inclusion in SPC's mandate by coordination between key divisions within SPC that are responsible for mainstreaming human rights, gender, cultural heritage and socio-environmental issues. Through the adoption of this policy, the SPC will minimise social, cultural and environmental impact; contribute towards the economic empowerment of women, youth, persons with disabilities, and marginalised groups in all their diversities; promote local participation and ownership of development; and, contribute to the elimination of all forms of gender-based violence. These goals have informed the development of this full funding proposal for CR-WASH in Vanuatu.

The **GCF and SPC** are currently collaborating on shoring up gender-transformative adaptation solutions in the Pacific, through support at devolved levels: in the recently approved FP169[52], SPC developed a robust gender assessment[53] and gender action plan[54] to inform the disbursement of sub-grants to local authorities in the Federated States of Micronesia.

[52] See: https://www.greenclimate.fund/project/fp169

[53] See: https://www.greenclimate.fund/sites/default/files/document/fp169-gender-assessment.pdf

[54] See: greenclimate.fund/sites/default/files/document/fp169-gender-action-plan.pdf

6. Gender mainstreaming for CR-WASH in Vanuatu

6.1 Problem statement: WASH baseline in Vanuatu

Combined coverage of basic water and sanitation is lower in the Pacific than any other region of the world, with progress in the region described by the Joint Monitoring Programme (of UNICEF and WHO) as having stagnated between 2000 and 2015.[55] In Vanuatu, it is estimated that 44% of people have access to safely managed drinking water, 34% to basic sanitation, and 25% have a basic handwashing facility at home.[56] Climate change impacts such as sea-level rise, changing rainfall patterns and increasingly frequent extreme weather events makes Vanuatu particularly vulnerable to inadequate potable water, and insufficient water and infrastructure for hygiene and sanitation.[57]

Within Vanuatu, the report finds, the province of Sanma is reported to have the greatest burden of WASH-related diseases per 1,000 persons in the country.[58] Surveys completed by World Vision Vanuatu also show that WASH statistics in Sanma are significantly lower than national averages: 30% lower for access to clean water, and 21% lower for improved sanitation facilities.[59] Torba is the most geographically remote province in the country, with the highest number of islands – and despite having similar levels of cash and subsistence expenditure to all other provinces, Torba has a significantly lower monthly income. In both of these provinces and indeed throughout the nation, people with disabilities and women face many inequities in accessing safely managed WASH facilities, services and practices. These were reiterated in the stakeholder consultations conducted by national consultants during the project preparation stage.

In Vanuatu, the report further finds, there still is a stigma or 'tabu' around 'sikmun' or menstruation, which is believed to often lead to the isolation of menstruating women due to social and physical barriers, and to prevent girls from attending school. Preliminary research undertaken by World Vision Vanuatu and Care International in 2018 suggests that women and girls in Vanuatu access information on menstruation from parents, teachers, grandmothers, aunties, school and clinics. However, the report noted that mothers and family members did not

[55] World Vision (2020). Water, Women and Disability Study. Available at: https://www.wvi.org/publications/development-guide/vanuatu/water-women-and-disability-study-report

This is a recent study conducted by World Vision Australia in collaboration with the VNSO, Ministry of Justice and Community Services, Vanuatu Disability Promotion and Advocacy Association and Vanuatu Society for People with Disability. This study will be very relevant in the implementation stage for the delivery of Outcome 2.

[56] United Nations Children's Fund (UNICEF) & World Health Organization (WHO). (2019) Progress on household drinking water, sanitation and hygiene 2000-2017: Special focus on inequalities.

[57] World Vision (2020). Ibid.

[58] System, V. H. I. (2015) The burden of WASH-related diseases on health in Vanuatu.

[59] World Vision Vanuatu (2016). National Impact Assessment.

always have the confidence or information to discuss menstruation, and that consistent and specific information regarding menstrual hygiene management is lacking.[60]

In Vanuatu, current infrastructure is limited in serving gendered needs: women and girls often require increased water during lactation, pregnancy and for the management of menstrual hygiene, which is often not provisioned for. Additionally, the elderly (for whom women are the primary caregivers) have increased need for water due to physiological requirements. WASH infrastructure, current and new, also requires system improvements that increase safety for women: in terms of lighting, improved locks and doors as well as gated/protected areas, to ensure instances of gender-based violence are reduced. To ensure service delivery a climate-resilient, but also gender-transformative, for effective WASH, water security and accessibility in Vanuatu, therefore, will require the mainstreaming of these factors.

6.2 Rationale: gender and climate-resilient WASH

The UN Sustainable Development Goals have three dedicated goals for Gender Equality (SDG 5), Clean Water and Sanitation (SDG 6), and Climate Action (SDG 13). Integrated water management and improved WASH praxis connects these goals – although issues relating to gender-responsive climate adaptation and water governance are yet to be mainstreamed in programming and funding cycles of governments, donors and other stakeholders. Despite the centrality of WASH to climate resilience and adaptation, the links are not widely recognized – nor extensively explored – in policy, technical studies and programme development. Global climate finance flows are yet to mainstream CR-WASH extensively, particularly the nexus of gender inequality and its interlinkages with water insecurity and high levels of climate risk exposure. Water and sanitation sectors are systems of interconnected functions and continue to be seen as central to development aid (as opposed to being mainstreamed through climate aid).

The proposed GCF project deviates from this gender-blind approach and will lay the groundwork for gender-responsive and climate-resilient WASH service delivery (with greater water security for both female and male beneficiaries) in Vanuatu. Access to WASH, the project recognizes, is fundamental to development but it can be mobilised as a critical climate adaptation strategy for poor and vulnerable communities (as well as marginalized groups within those communities), including women and girls in Vanuatu. Therefore, this project will deliver on gender mainstreaming targets through the design and delivery of climate-resilient WASH infrastructure (Outcome 2), alongside improved technical capacity and water governance (Outcomes 1 and 3)

[60] The preliminary research, done by World Vision Vanuatu, also suggested that women from urban areas used a variety of options to manage menstrual bleeding including sanitary pads, double cloth, tampons, washable calico pads and in some cases silicone cups (although these have not been available for purchase in country until recently). Whereas women and girls in rural areas relied more on cloth and in some cases used natural materials like dry banana leaves. However, in both settings, products such as sanitary pads were not reported to be widely used due to lack of income, accessibility and availability. In some cases, there were reports that women would stay in an isolation hut for the duration of menstruation and used leaves. Additionally, it was noted that some women were not allowed to handle food our use shared plates and cups during menstruation as it is considered 'unhygienic' and sexual activities were postponed due to kastom (traditional) beliefs.
The report is available at: https://www.wvi.org/publications/development-guide/vanuatu/water-women-and-disability-study-report.

as well as the mobilization of government-owned programming (such as the DWSSP process) at the national, provincial and devolved levels.

Gender-transformative CR-WASH will aim to bridge the gap of practical, gendered WASH needs (of women and girls) as well as strategic gender interests (increased representation in water and climate-related decision-making bodies).

6.3 Barrier analysis: gender, water insecurity and climate-induced risks

Different gender groups, particularly women and girls, have gendered water needs, which remain underserved by existing WASH infrastructure and are impacted by humanitarian emergencies:

This is an infrastructure barrier, which combines with limited technical capacity. The absence of safe and secure water supplies, and decent sanitation facilities has disproportionate negative effects on the lives of marginalized and vulnerable groups. For women and girls, apart from daily water needs, specific hygiene needs (during menstruation, pregnancy, lactation, childbirth and child rearing), specific care roles (such as care work for the elderly and children) as well as age and different abilities (disability) increase daily water and sanitation requirements.[61] In underserved areas, such needs may not be mainstreamed, due to lack of safe and secure water supplies and infrastructure, discriminatory social norms as well as cultural taboos.[62] Exact estimates of water needs of various groups cannot be determined given the paucity of data, however, that gendered needs are not addressed, remains the status quo in Vanuatu – where investments in climate-resilient WASH infrastructure is currently limited.

Lack of hygienic conditions, a primary determinant of which is access to safe and potable water, also negatively impacts women's, girl's and newborn's lives and wellbeing by increasing their exposure to water- and vector-borne diseases. Diarrheal disease – which is the second leading cause of death in children under five years old, is directly linked to unsafe water and sanitation. According to WHO data, diarrheal diseases in Vanuatu reached 3.83% of total deaths in 2018.[63]

Being underserved by WASH infrastructure, whether it is due to lack of access and/or because of poor design, can also increase vulnerability to gender-based violence. WaterAid has observed that women and girls, often, adjust for insufficient WASH infrastructure by using available services primarily during evening hours or at night.[64] This increases their exposure risk to gender-based violence, particularly sexual assault and attacks, due to limited lighting and remoteness of these locations. Boys and men may also encounter violence in accessing water and sanitation, with local norms around masculinity inhibiting their ability to avoid or report on experiences of violence.[65]

[61] Winqvist, E. (ed.) Ibid.

[62] See: https://www.unicef.org/wash/menstrual-hygiene

[63] See: https://www.thelancet.com/pdfs/journals/laninf/PIIS1473-3099(19)30401-3.pdf

[64] Winqvist, E. (ed.) Ibid.

[65] https://journals.sagepub.com/doi/10.1177/0956247814564528

During humanitarian emergencies – which remains a cyclical and pervasive issue in Vanuatu due to repeated exposure to natural- and climate-induced hazards – accessibility is further reduced and exposure to WASH risks are exacerbated. As observed in the wake of Tropical Cyclone Pam – which caused widespread destruction across the eastern and south-eastern islands of the country – damage to water and sanitation assets were extensive.[66] This had immediate effects by producing spikes in WASH-related diseases, and by reducing services (and utilities) available to different population groups, and further to vulnerable groups within them, in Vanuatu. Women and girls are also exposed to post-disaster spikes in gender-based violence, while lacking access to essential services, and experiencing hindrances posed by emergency shelter design (with limited WASH services), and inadequate coordination on gender-specific issues by responders.

Women continue to be underrepresented in water security and climate adaptation-related decision-making bodies, despite registering better performance when represented:

This is a political barrier, which combines with limited policy capacity. Women in Vanuatu are underrepresented in decision-making and policy-formulating bodies, as evinced through figures of low (often no) representation in Parliament, as well as at the provincial, municipal and area council levels. According to the VNSO, women only represented 21% of senior officials, legislators and managers.[67] As stated previously, socio-anthropological research in Kurumambe and Purau villages in Tongoa, for example, highlighted how women must navigate strict rules governing their presence and speech in the *nakamal* (chief's meeting place) – effectively excluding them from holding titles or wielding decision-making power even at localised levels of governance.[68]

Consequently, women continue to be underrepresented in water user committees (WUCs), or community-based management bodies. This is despite women's leading role in household management of WASH needs, particularly through domestic water provision, which is importantly determined by access, affordability and reliability of water resources.

A national census of water points in Vanuatu was conducted between 2014 – 2016, which revealed around 8,000 water points in six provinces.[69] Most of these were not governed by communities – and among the 1,175 community-owned water systems, the majority (69% - 810) did not have a

[66] According to the Post-Disaster Needs Assessment for Tropical Cyclone Pam, conducted by the International Labour Organization (ILO), damage to water and sanitation assets in the health sector have been estimated at approximately VT 113 million (Chapter 3.2.2). Damage to water and sanitation assets in the education sector have been estimated at VT 270 million (Chapter 3.2.3). See: https://www.ilo.org/wcmsp5/groups/public/---ed_emp/documents/publication/wcms_397678.pdf

[67] Bowman, Chakriya; Cutura, Jozefina; Ellis, Amanda; Manuel, Clare. (2009). Women in Vanuatu : Analyzing Challenges to Economic Participation. Directions in Development ; private sector development. World Bank. © World Bank. https://openknowledge.worldbank.org/handle/10986/2624

[68] Granderson, A. (2017) "The role of traditional knowledge in building adaptive capacity for climate change in Vanuatu" in Weather, Climate and Society. Available at:

https://www.researchgate.net/publication/317071599_The_Role_of_Traditional_Knowledge_in_Building_Adaptive_Capacity_for_Climate_Change_Perspectives_from_Vanuatu

[69] Pilot data collection first took place in shefa province in 2014 and, based on the pilot findings, modifications were made in the questionnaire. Further data collection was done by dGMWr staff in Torba and sanma province, nGO staff in Tafea and Malampa and by students in penama province. all enumerators were trained by the dGMWr and UniceF with theoretical instruction on the tools and the definitions, followed by a practical data collection exercise in shefa province. The training was concluded with a review of the data collected during the field exercise, which provided a feedback loop. a verification of all collected data took place by displaying the data on a geographical map with habitations.
Mommen, B., Humphries-Waa, K., Gwavuya, S. (2017). "Does women's participation in water committees affect management and water system performance in rural Vanuatu?" in Waterlines, v. 2017.

committee in place, while the remaining (31% - 365) did. For these community-owned water systems with a committee, data was collected to understand community composition and the positions occupied by women.
For the 365 community water points with a WUC, there was an average of six members per committee and a total of 2,237 members across the six provinces. Of these WUC members, 16% (365) were female. Just over half of the WUcCs (51% - 186) reported having at least one key post held by a woman. Of these, approximately half the posts were as secretary (53%), 40% as treasurer, and only 7% as chair.

Despite overall low representation of women in WUCs, trends of improved water management were noted when women were better represented: WUCs with women in key posts met more regularly, where meetings were defined as meeting more than once per year. Regular user fee collection takes place more often when there is a female treasurer (73%), compared with a female chair (54%) and secretary (52%). Of the 186 WUCs with women in key positions, 43% of the systems were in good condition, 35% in fair condition, and about 22% in poor condition or not working at all. Similarly, of the 179 with only men in key posts, 35% were in good condition, 29% in fair condition, and 36% in poor condition or not working at all. Further analysis shows that WUCs with no women in key posts were also significantly more likely to function poorly or not at all compared to those that included women: 36% compared to 22%.

Insufficient, non-resilient WASH infrastructure continue to be compromised by climate risks, increasing time poverty among certain gender groups:

This is an infrastructure barrier, which is being compounded by climate impacts. WASH baseline in Vanuatu, as explored in Annex 2 – Feasibility Study – remains riddled with the lack of water safety (quality) and security (quantity), including: limited year-round service for most water points, inadequate output to cover daily water requirements. During the dry season and in drought conditions, including those associated with El Nino, some communities are forced to declare 'water emergencies.' Communities are also experiencing growing tensions over water access rights, necessitating the need to address climate threats as well as institutional (and infrastructure barriers).

In Vanuatu, women and girls (and often boys as well) usually bear the responsibility for collecting water, which is time-consuming and arduous. Lack of access to water supply systems has a significant impact on time spent transporting water, which can take up to five hours per day, according to UNICEF data.[70] This is reiterated in the government's 2004 report on the UN Convention on the Elimination of All Forms of Discrimination against Women (CEDAW), which describes the key role played by ni-Vanuatu women in the rural subsistence economy:

"Women play a critical role . . . devoting most of their working lives balancing their time between meeting family as well as community needs and cultural obligations. Women's work includes an array of work from . . . caring for children, old people, people with disabilities and the infirm to domestic tasks such as fetching of water and firewood, cleaning the house, washing clothes, cooking food and gardening. Women are involved in food production including animal husbandry and production of handicrafts such as mats, baskets for the home as well as weaving of mats, baskets and grass skirts for sale as well as for cultural purposes."

[70] UNICEF Pacific (2015) Gender equity and social inclusion in the wash sector in Vanuatu.

This inequitable gender-based allocation of unpaid domestic and care work, representing 'a double workday' for women who enter the workforce (whether formally or informally), often leaves women with little or no discretionary time. This is known as time poverty[71]: a predicament in which an individual is working long hours without choice because that individual's house is poor or would be at risk of falling into poverty if the individual reduced their working hours below a certain time-poverty line.

Time poverty has important repercussions for women's economic opportunities and health, and can actually limit them from accessing and benefiting from project interventions, trainings and adaptation strategies for increased climate resilience and improved water security.

Water insecurity indirectly and directly combines to exacerbate gender inequality in other sectors (such as limited land rights) and limit adoption of adaptation and resilience strategies:

This is an institutional barrier. Climate change represents the most complex challenge of our time – it is a threat multiplier[72] that has direct and indirect effects on existing sociocultural, economic, and political phenomena. This includes levels of gender equality and women and girls human rights, and how power, roles and responsibilities are organized among men, women, boys and girls within the household and in the economy.

Due to the impacts of climate change, agriculturists globally face decline in land productivity, greater land degradation, limited access to water, increase in pests, deforestation, destruction of crops and so on. A major issue among this, related to the project paradigm is water availability, which is impacting food production.[73] It is predicted that across the Pacific, island countries will face increasingly severe water shortages due to reduced precipitation. This anticipated intensification of water shortage will likely lead to land use change, possibly requiring farmers to move their crops away from drier areas and change to drought-resistant varieties and new propagation techniques.[74]

As a result of gendered norms and patriarchal structures, women may be disadvantaged and lack both the technical capacity as well as opportunity to adopt adaptation strategies. Women are often allocated less fertile land than their male counterparts, and further often face additional burdens of trouble accessing water, or using and implementing energy-saving solutions, or accessing extension and/or financial services. In Vanuatu, land laws do not explicitly discriminate against women, but 97% of the land in the country is held under customary tenure which generally means that women have very limited rights to control and manage land. Rigid social norms in rural areas, however, reinforce the decision-making power of men in Vanuatu, including regarding natural resources, land and other productive resources. Women's rights over land are increasingly threatened because the rights they do have—to use the land—cannot be registered. Further, women's participation in decision making on land use is limited to informal discussions

[71] Time poverty can also be as the lack of enough time for rest and leisure after accounting for the time that has to be spent working, whether in the labor market, doing domestic work, or performing other activities such as fetching water and wood. See: Bardasi, E., Wodon, Q. (2010) "Working long hours and having no choice: time poverty in Guinea" in Feminist Economics.

[72] Risk to stability in contexts vulnerable to climate change involves multiple drivers, many of which are pre-existing social, economic and political stresses with which climate and environmental change may interact and amplify. This is why climate change is often articulated as a threat multiplier.

[73] Mael, S.H. (2013) Climate Change and Agriculture in Vanuatu: A study of crops. FAO. Port Vila.

[74] Mael, S.H. (2013) Ibid.

within the household and in community gatherings.[75] Men control formal community decision making, and community chiefs are the final decision makers on the use of productive and natural resources.[76]
The National Council of Chiefs, or *Malvatumauri*, must be consulted by the Parliament on any matters related to land. Since women cannot be high chiefs, they are not part of the *Malvatumauri* and do not have a say on decisions related to tenure of customary land.[77]

Limited (and often lack of) land rights, despite being key stakeholders in land management and their responsibility for family food security, ensures that women have limited opportunities to expand their income, and their usufruct rights are directed towards (primarily) subsistence agriculture (as opposed to cash crops). These factors combine to directly and indirectly determine levels of food, energy and (importantly) water security, while impacting both readiness for and accessibility to climate resilience. Overall, land and water rights – in the context of mounting climatic and environmental change – function in tandem, and determine how vulnerable groups, and marginalized sections within them, are able to access opportunities as well as adopt adaptation strategies.

6.4 Co-benefit of the project: increased representation of women in climate-resilient water governance

The project has identified the following co-benefit as a part of its Section E – Logical Framework in the Funding Proposal.

Project/programme co-benefit indicators						
Co-benefit #	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions/ Notes
				Mid-term	Final	
Co-benefit 1	Improved gender parity in representation of women in Rural Water Committees (RWCs) -disagregated by women in executive level positions	Registration of RWCs with members (indicating a minimum of 40% of women)	Women are not well-represented across water governance or devolved utility authorities across Vanuatu – exact figures are currently unavailable.	40% of RWCs registered, with 40% of women in each.	Remaining 60% of RWCs registered, with 40% women in each.	40% RWCs registered by mid-term and 100% expected by project conclusion. There is political will in involving women in different sub-national contexts and in water governance institutions. Women may not be, however, able to participate meaningfully

[75] FAO. (2020). Country Gender Assessment of Agriculture and the Rural Sector in Vanuatu. Available at: https://www.fao.org/3/ca7427en/ca7427en.pdf

[76] FAO. 2020. Ibid.
[77] Ibid.

						participate due to sociocultural barriers, and demands on their time due to unpaid care work within the household.

This co-benefit will act on the finding[78] that women are underrepresented in water user committees in Vanuatu, and establish Rural Water Committees (RWCs) through the DWSSP processes that provision for a minimum of 40% representation.[79] This is also a legal pre-requisite for the registration of RWCs in Vanuatu, under the Water Resources (Amendment) Act No. 32 of 2016. This will be measured as an independent indicator, and will be delivered in tandem with (primarily) Outcome 1, which will establish 600 new DWSSPs in Vanuatu, alongside providing policy and climate training.

Further, the project will also ensure that infrastructure delivery (through Outcome 2), and trainings (as well as policy strengthening) provided through Outcome 1 and 3, incorporate findings from the gender consultations.[80] Participants, at the community-level stakeholder consultations, identified key gender groups and articulated their different needs. These gender groups and their specific gender needs were summarized well by a breakout group (which reflected other stakeholder groups in the Sanma and Tafea provinces) in Penama province below:

Gender Group	Needs
Elderly (60+)	▪ Easy access/tap stands to be in close proximity ▪ Taps fitted at a lower level for accessibility ▪ Ball taps for ease of use ▪ Solar lighting in the tap use area
Disability/Disabled	▪ Easy access ▪ Ramp for wheelchair/hand rail ▪ Solar lighting in the tap use area
Women (menstruating/lactating)	▪ Separate shower facilities with dignity facilities ▪ Safe house for menstrual hygiene ▪ Separate individual tank with RWCs for menstrual hygiene / child-related water use ▪ Solar lighting for security / privacy
LGBTQI	▪ Separate shower facilities

[78] Mommen, B., Humphries-Waa, K., Gwavuya, S. (2017). “Does women’s participation in water committees affect management and water system performance in rural Vanuatu?” in Waterlines, v. 2017.

[79] The Rural Water Committee usually takes overall responsibility for developing their supply-specific DWSSP. However, technical and facilitation support will be needed to develop a DWSSP. A community DWSSP team needs to be established, usually comprising members of the Rural Water Committee, women’s, men’s, youth and church group representation, local plumbers and carpenters, and the head person. This brings together a variety of perspectives from the people who know the water supply and how it is used. Further detail can be found here: Vanuatu’s National Implementation Plan for Safe and Secure Community Plan, available at - https://mol.gov.vu/images/News-Photo/water/DoWR_File/Management_Plans/Vanuatu-NIP-Guide-annual-CAP-210818.pdf

[80] During implementation, these designs will be further defined and specified through further consultations and workshops at the community level. These will be dispensed through the process of developing 600 DWSSPs through Outcome 1.

	▪ Solar lighting for security / privacy
6-18 years (school students)	▪ Separate water storage for: bathroom use and kitchen use (to reduce collection burdens)
0-5 years	▪ Safety valves to be fixed before taps are installed
NB. There is an urgent need to install gender-responsive signs to specifically assigned facilities at the community level. Alongside, working groups indicated the need for the inclusion of well-lit, demarcated WASH areas, with the provision of secure doors and locks – to ensure such facilities can be used by women safely.	

The proposed solution recognizes that bridging practical gender needs (e.g. access to water) with strategic gender interests (e.g. changes in power and roles) is critical to achieving transformational changes in gender equality in Vanuatu, where underrepresentation of diverse gender groups limits the efficacy of existing infrastructure and the potential of new investments. Through the planned trainings (where 50:50 gender balanced beneficiary distribution is targeted), alongside this co-benefit to ensure increased, meaningful participation of women in RWCs, the project will also be able to explore five WASH- and climate change-relevant elements of empowerment: participation, decision-making, information, capacity building and leadership.

The risk to this approach is that traditional gender roles in Vanuatu often prevent women from participating in such decision-making bodies.[81] Some communities are supportive of women's participation, whereas others refuse to let female members participate because of concerns over the erosion of community structures and traditional understandings about women's rolesThrough the trainings envisioned under Outcome 1, the AE will also ensure there is sufficient advocacy and sensitization preceding the increased participation of women in RWCs. .[82] The Gender and Protection Cluster of Vanuatu [83] as well as the Gender Officer hired during project implementation will demonstrate the expertise and capacity to address this issue of representation. They will be active as facilitating agents during the DWSSP development process, and provide sensitization and advocacy support during the trainings to ensure that women's participation in RWCs are meaningful.

6.5 Overall impact of the project: decreased time poverty for women and youth

Clean water is an essential resource for which there is no substitute. Each family needs a certain amount of water each day to survive, and the lack of access to improved water supply places a disproportionate burden on women and girls who tend to be the primary collectors of water for the family in many countries. The impact is felt the most by the poor who are more likely to lack access to improved water supply. Where there is no adequate water supply close by, people—mostly women and girls—have to travel, sometimes long distances, to fetch water.

This assessment has discussed that in Vanuatu, women and girls – as well as youth – are often responsible for domestic water provision. This has been corroborated by qualitative studies, as well as UNICEF experience. Further, during the stakeholder consultations held during the preparation stage – breakout groups confirmed this phenomenon.

[81] Bowman, Chakriya; et al. Women in Vanuatu - Ibid.
[82] Ibid.
[83] See: https://ndmo.gov.vu/resources/clusters/91-gender-protection

Recent research by ADB[84] has shown that: access to clean water supply closer to home could, therefore, have a significant impact on women's time poverty, especially for poor women and girls who otherwise tend to have the least access to clean water. Overwhelmingly, the evidence confirms time savings for women and girls when there is improved access to water supply. Time savings can have several spillover effects: significant and important impacts on girls' attendance at school, for example, although overall data on participation on market work is limited.

The quote below comes from a qualitative study looking at the relationship between access to water, sanitation, and hygiene, and gender equality in the Pacific, specifically Vanuatu.[85] This quote from a woman beneficiary of a water supply project in Vanuatu illustrates the choices that women make to manage time reallocation efficiently. The necessity of collecting water, and time poverty, can force women to make trade-offs in how they use their time, so that time saved gets reallocated to previously neglected household and care work rather than market work:

> "Before there was so much difficulty to fetch water so I used to shut my baby in the sleeping house so I could take the clothes down to wash and carry back the water. It used to be such hard and heavy work. When I came back the baby would be crying. Now there is less walking and work to get the water. I have more time at home to care properly for my children. I'm now teaching good hygiene practices and I am a good mother. Before I used to go to the hospital with my children all the time. But now they are healthy."

The Vanuatu study found that, in general, reduced labor in collecting water as a result of improved supply was highly valued especially by women, and also by men. Thus, the overall project paradigm – to deliver climate-resilient WASH infrastructure to enhance community resilience – will have a direct gender-transformative impact: it has the potential to reduce the time poverty that women and girls (primarily) face, by increasing access to water sources and improving overall water security.

[84] Asian Development Bank – ADB. (2015). Bala,cing the Burden – Desk Review of Women's Time Poverty and Infrastructure in Asia and the Pacific. Available at: https://www.adb.org/publications/balancing-burden-womens-time-poverty-and-infrastructure

[85] J. Willetts et al. (201). Addressing Two Critical MDGs together: Gender in Water, Sanitation and Hygiene Initiatives. Pacific Economic Bulletin. 25 (1).

7. Grievance Redress Mechanism, in the case of SEAH and GbV

In 1993, the Declaration on the Elimination of Violence against Women (A/RES/48/104) defines violence against women as “any act of gender-based violence that results in, or is likely to result in, physical, sexual or psychological harm or suffering to women, including threats of such acts, coercion or arbitrary deprivation of liberty, whether occurring in public or in private life.”

More recently, as both theoretical and practical work on GbV evolved and became enshrined in numerous conventions and declarations, article 3 of the Council of Europe’s Convention on Preventing and Combating Violence Against Women and Domestic Violence (Istanbul Convention) offered the following definition: “gender-based violence against women shall mean violence that is directed against a woman because she is a woman or that affects women disproportionately.” A critical issue identified in this assessment is Sexual Exploitation, Abuse and Sexual Harassment (SEAH)[86] – as part of the overall incidence of gender-based violence (GbV) – that is prevalent in Vanuatu.

Indeed, GbV statistics in Vanuatu are telling: Vanuatu has one of the highest prevalence rates of violence against women and girls globally. Research conducted by the Vanuatu Women’s Centre (in 2011) established that ~60% of women with an intimate partner had experienced physical violence, ~68% experienced emotional violence, and ~69% coercive behavioural control by men.[87] According to the World Bank’s 2012 *World Development Report on Gender Equality and Development*, between 60 – 70% of women in Vanuatu (alongside Kiribati and Solomon Islands) experience some form of domestic violence.[88] See Figure 2 below:

[86] Definition based on: https://safeguardingsupporthub.org/sites/default/files/2021-06/RSH_BiteSize_Understanding%20SEAH%20and%20GBV_final.pdf

[87] World Vision (2018). Evaluation Report: Reducing GbV Project (Vanuatu Counselling Approach). Available at: https://reliefweb.int/sites/reliefweb.int/files/resources/Counselling%20Evaluation%202018%20.pdf

[88] See: https://www.worldbank.org/en/news/feature/2012/11/25/raising-awareness-of-violence-against-women-in-thepacific

Therefore, SPC has designed a grievance redressal mechanism to ensure these issues are taken into consideration during project implementation. A grievance is a concern or complaint raised by beneficiaries of affected communities and stakeholders related to the perceived or actual impacts of the project activities. The objectives of setting up an appropriate grievance redress mechanism (GRM) are to:

- provide stakeholders with a clear process for providing comment and raising grievances and concerns in an anonymous manner;
- structure and manage the handling of comments, responses, and grievances in a timely manner; and,
- ensure that comments, responses, and grievances are handled in a fair and transparent manner and in line with local and national policies.

The GRM can serve as an effective tool for early identification, assessment and resolution of grievances and therefore for strengthening accountability to beneficiaries. The GRM is an important feedback mechanism that can improve project impact and respond to concerns and grievances of project-affected parties (e.g. related to the environmental and social performance of the project) in a timely manner. With restrictions on movement, it is important that, where possible, staff managing grievances can access systems remotely to enable GCFM processes to be conducted effectively. The SEP will keep the local communities and other stakeholders informed about the project's activities, to specifically address gender-based violence (GbV) and other cross-cutting issues.

All grievances will be closely monitored by the Accredited Entity to assess the number and type of grievances and evaluate any trends over time. This will be conducted by the relevant responsible parties as highlighted under SPC's policies for accountability[89]. All monitoring and reporting will be carried out conforming to confidentially and consent from aggrieved parties or survivors. This applied to all reporting obligations to the GCF as imposed through the Accreditation Master Agreement and Funded Activity Agreement.

7.1 GCF Grievance Redress Mechanism

Paragraph 69 of the Governing Instrument of the Green Climate Fund (GCF) requires the Board to establish an Independent Redress Mechanism (IRM) that will report to the Board. The Board established the IRM through the adoption of the Terms of Reference (TOR) of the IRM which sets out various matters, including the role and functions, governance and administrative arrangements of the IRM. In accordance with its TOR, the IRM is mandated to carry out the following functions:
(a) Review requests for reconsideration of a project or programme that has been denied funding by the Board and, as appropriate, make recommendations to the Board;
(b) Address grievances or complaints by a person, group of persons or community who/which have been or may be adversely impacted by a GCF funded project or programme through problem solving and/or compliance review, as appropriate;

[89] https://www.spc.int/accountability

(c) Initiate proceedings on its own to investigate grievances of a person, group of persons or community who/which have been or may be adversely impacted by a GCF funded project or programme;
(d) Monitor whether decisions taken by the Board based on recommendations made by the IRM, or agreements reached in connection with grievances or complaints through problem solving, have been implemented, and report on that monitoring to the Board;
(e) Recommend to the Board the reconsideration of existing policies, procedures, guidelines and systems of the GCF based on lessons learned or good international practices;
(f) Share best practices and give general guidance that can be helpful for the GCF's readiness activities and accreditation process and for supporting the strengthening of the capacities of accountability/redress mechanisms of the DAEs; and
(g) Provide education and outreach to GCF staff, relevant stakeholders and the public.
A request may be submitted to the IRM, by sending it to the mailing address or email address of the IRM as published on its website (https://irm.greenclimate.fund/case-register/file-complaint). A request may be submitted in any of the six official languages of the United Nations (UN), provided that where a request is in a language other than English, it must be accompanied by an English translation. The English version will prevail in the event of a conflict.

7.2 Grievance related to Sexual Exploitation, Abuse and/or harassment

In all situations involving complaints related to gender-based violence (GBV), sexual exploitation, abuse or harassment (SEAH), violence against children (VAC) and human trafficking (HT), the relevant grievance redress mechanism (8.3.3-3) will take on a "survivor-centred approach". This will apply to all grievance address mechanisms controlled by SPC or the PMU. In line with this approach, the following principles will be systemically applied through all steps and actions:

- The rights, needs, and wishes of the survivor (or victim) is the foremost priority of everyone involved with the project.
- The survivor has a right to:
  - be treated with dignity and respect instead of being exposed to victim-blaming attitudes.
  - choose the course of action in dealing with the violence instead of feeling powerless.
  - privacy and confidentiality instead of exposure.
  - non-discrimination instead of discrimination based on gender, age, race/ethnicity, ability, sexual orientation, HIV status or any other characteristic.
  - receive comprehensive information to help her or him make their own decision instead of being told what to do.
- The safety of the survivor shall always be ensured. Potential risks to the survivor will be identified and action take to ensure the survivor's safety and to prevent further harm including ensuring that the alleged perpetrator does not have contact with the survivor.

If the survivor is an employee of the Project, reasonable adjustments may be made to the survivor's work schedule and work environment to ensure their safety.

- All actions should reflect the choices of the survivor.
- All information related to the case must be kept confidential and identities protected. Only those who have a role in the response to an allegation should receive case-level information, and then only for a clearly stated purpose and with the survivor's consent. This applies to any documentation or reports related to the case. Identities will not be revealed unless explicit written consent is provided by the survivor.
- The survivor must provide informed consent to progress with each stage of the complaints process. Survivors may withdraw their consent at any time during the process.
- In the case that a case of SEAH or GBV is submitted. SPC as the Accredited Entity will carry out the duty of care to the survivor in line with its policies. This includes where relevant, support for the provision of medical services (including psychosocial support), legal counsel, community driven protection measures, and reintegration of the survivor. This will be conducted in a timely manner to ensure maximum safety and support is provided to the survivor.

7.3 SPC's Grievance Redress Mechanism

SPC has a Grievance and Redress Mechanism (GRM) in place to ensure that complaints are being promptly reviewed and addressed by the responsible units.[90] This process aims to address complaints from affected stakeholders, including communities, about the social and/or environmental performance of the project, and to take measures to redress the situation, where necessary. For the process to be efficient, project stakeholders have to be properly informed that SPC has such a mechanism established, and how they can access to it to settle their grievance.
The SPC GRM is operated through a web-hosted page on SPC site for the expression of concerns or complaints, which can be posted by email with the information in using the complaints' template.[91] Concerns expressed shall be received by the legal team who will reach out internally, primarily to the division in charge of the project or to relevant division. Grievances will be sorted out through a conflict resolution process. In case this process is not functional, other process will be used, such as a compliance system, the overall objective being to address and redress project stakeholders' grievances in a simple and efficient manner.

[90] https://www.spc.int/accountability

[91] (Please see Annex IV of SPC's GRM see SPC website: https://www.spc.int/sites/default/files/documents/Application%20SPC%20Social%20and%20Environmental%20Rresponsibility%20Grievance%20Mechanism.pdf).

7.4 Project-level Grievance Redress Mechanism

Through a project-level GRM, SPC will receive concerns or grievances from an affected community about the environmental and social plans or performance of the project. In that direction, communities and stakeholders will be sensitized about the existing grievance process and form. Both national level and provincial level government agencies will be responsible for supporting the communities with the information they need to properly submit a grievance letter. The national level and provincial level government agencies are taking part into the grievance and redress mechanism through documenting grievances and coordinating with SPC the process to settle the grievances. There are several processes to submit project related grievances:

1. Bring up the complaint during the meetings of the PWRAC or community awareness meetings. The complaint then must be directed to the project GCF focal point who will then forward to the SPC legal team.
2. Contact by email the Project Management Unit.
3. Contact by email the key project institution (DoWR), which will then forward to SPC.
4. Email SPC through the online process: https://www.spc.int/accountability.
   Email address complaint@spc.org

The Project Management Unit will receive and register grievances and will contact SPC legal team. He/she will provide an initial response within two business days to the person who submitted the grievance to acknowledge the grievance and explain that the grievance will be logged onto the SPC GRM. As a first timeframe, a response will be provided to the complainant within a two-month period, with indication of appropriate process to address the grievance. This duration should be sufficient to screen the complaint, outline how the grievance will be processed, screen for eligibility as well as assign organizational responsibility for proposing a response. This process will possibly involve engaging with other project stakeholders to resolve the issue.

SPC GRM is responsible to inform the complainant that he/she has the right to pursue other options to resolve the complaint if unsatisfied after the SPC GRM process, noting that the GRM may respond to questions from the complainant, but does not constitute an advisor or attorney for the complainant. All grievances will be recorded, and these records will be kept at a secure place for up to three years after the life of the project.

7.5 Community-level Grievance Redress Mechanism

At the community level in Vanuatu, concerns or grievances can be addressed through the traditional governance structures and processes managed by the chiefly systems of individual islands. The community-level GRM will mainly address issues related to utility access, conflicts among villagers, complaints from marginalized gender or vulnerable groups, issues related to water access points and gender-based violence. This level of the GRM will ensure that communities are able to resolve issues and conflicts with consensus, as a first level, and then escalate to the project-level GRM only if deemed appropriate. This will also ensure that, within the indigenous communities being targeted, the project benefits from active, traditional mechanisms of conflict resolution and decision-making structures.

The nakamal or Village Council is made up of chiefs and community leaders of a particular village. This authority is convened by the paramount chief or a designated customary leader and it deliberates and resolves matters at the specific village level which could include family matters, disputes/disagreements as well as land disputes.

The Ward Council of Chiefs sits above the Nakamal or Village Council and comprises chiefs and customary leaders from a number of different villages who all fall within a designated Ward Council. The Ward Council deals mostly with land ownership disputes.

Matters unresolved at the Ward Council are elevated to the Area Council of Chiefs or even higher to the Island Council of Chiefs if they are not resolved by the council below. In the event an individual or a group of individuals are aggrieved, their grievance can be raised for redress at the Nakamal or Village Council. If matters are not able to be resolved at this level, the paramount chief or head of the council may decide as follows:

1. elevate the grievance for redress at the Ward Council or with the Chief; or,
2. register the grievance directly with the representatives of the provincial authority for redress through the provincial institutional arrangements.

Matters raised with the representatives of the provincial authority are usually done through Area Administrators or Area Secretaries. These provincial officers then have the option to raise the issues for redress as follow;

- table the grievance for redress at the Provincial Area Council level through the Area-Technical Advisory Committee (Area-TAC);
- table the grievance for redress directly through the Provincial Technical Advisory Commission (PTAC); and,
- raise the grievance directly with the relevant national government representative present at the provincial level.

If and when the grievance is raised through the provincial institutional arrangements, the matter can then be elevated to the national government level for redress by the relevant government agency or ministry.

8. Key recommendations

Despite protracted gender issues, as identified in the gender assessment above, Vanuatu has made important gains in ensuring gender mainstreaming in climate change adaptation and disaster risk reduction. This is registered in the Government's communications on the implementation of the Beijing Declaration and Platform for Action covering 2014 – 2019.[92]
Additionally, the DoWA reviewed national policies (such as draft climate change/disaster risk reduction policies, the National Sustainable Development Plan 2016 – 2030) to find that programme directives are progressively reflecting gender issues. This also includes a 2017 Council of Ministers' Paper (decision 94/2017) advocating for robust gender-responsive planning and budgeting in five key ministries (Agriculture, Climate Change, Education, Internal Affairs, Lands) in the country – which is currently underway, being led by the DoWA[93].

Micro-studies focused on community research has also registered a positive trend of increased climate change and gender awareness.[94] Among the Nakanamanga-speaking peoples of Tongoa Island – indigenous concerns about self-reliance and livelihoods, cultural continuity and climate impacts, transition towards cash economy are oriented towards preservation of identity and autonomy rather than continuation of patrilineal customary laws. This includes safeguarding and reviving of traditional knowledge, while allowing for greater collective governance by including marginalized groups such as women and youth.

Gender equality, therefore, has gained progressive priority in the Government of Vanuatu's. This has also been the case with GCF's, SPC's, and other Pacific-focused agencies' portfolios – where gender is mainstreamed in an effort towards implementing a holistic social development mandate with inclusive engagement. With that policy background, this project partakes in the international conversation on gender mainstreaming in climate change adaptation efforts. If implemented effectively, this project has the potential to become a best practice gender mainstreaming guide for future interventions on climate-resilient WASH in Vanuatu, in the Pacific islands regionally as well as in SIDS and LDC/ex-LDC contexts globally.

Given mounting evidence that dynamic, long-term planning for climate change impacts and external shocks yield better results when inclusive, gender-responsive and stakeholder-friendly, this CR-WASH project has taken proactive steps to mainstream gender, where possible, in the project preparation stage. Key recommendations of the Gender Assessment are as follows (the Gender Action Plan in Section 8 reifies these into the project's Logical Framework – available as Section E in the Funding Proposal):

See next page.

[92] Department of Women's Affairs (2020). Beijing +25 – ibid.

[93] KII conducted with Department of Women's Affairs (DoWA).

[94] Granderson, A. (2017) – ibid.

- Ensure system improvements or installation of new infrastructure includes and reflects the needs of women, particularly in terms of safety and security (lights, secure locks, etc.);
- Include women and other marginalized groups in regular meetings, interactions and maintenance protocols of new and improved WASH infrastructure;
- Ensure mandatory representation of 40% women in Rural Water Committees for the registered DWSSPs;
- Conduct awareness-raising, gender advocacy and sensitization workshops among institution strengthening processes as and when required;
- Involve national-level (and where possible, province-level or local) gender officers and mechanisms to ensure gender mainstreaming is pursued through government-owned processes;
- Consult the Department of Women’s Affairs as well as the Gender and Protection Cluster on cross-cutting issues of gender, WASH and water security;
- Consult CSOs, with a track record of promoting gender interests and gender mainstreaming, when relevant, in the process of delivering system improvements and new infrastructure; and,
- Document and record success stories, best practices and lessons learnt to ensure there is feedback loop from gender-transformative activities, as well as to ensure a community of practice around gender and climate-resilient WASH is established.

9. Gender Action Plan

OUTCOME 1: Communities are empowered to plan and manage climate-resilient water resources								
IMPACT STATEMENT	Through Outcome 1, the project will address barriers of capacity and policy access at community levels in Vanuatu – particularly, the limited financial and technical capacity that restricts communities from scaling up existing technology as well as shift towards new technology and system improvements. This will result in Improved water governance and climate-resilient water management at community level, with all 600 DWSSPs registered. By project conclusion, Vanuatu will have 600 new DWSSPs which will focus on delivering climate-resilient WASH infrastructure to the communities, and improving water access and security through improved methodologies of needs assessment, prioritization and delivery.							
GENDER IMPACT STATEMENT	By ensuring improved water security, enhanced climate resilience and increased women's representation in water governance through the proposed intervention (Section E of the Funding Proposal presents the blueprint) – the project will reduce vulnerability and exposure to climate change, reduce water insecurity and climate-related WASH issues, and overall aid in reducing women's drudgery within the household as providers and managers of domestic water utilities, particularly in rural and remote settings.							
OUTPUTS	• Output 1.1: New and existing DWSSPs incorporate incremental improvements to mainstream adaptation solutions • Output 1.2: Awareness, capacities and skills of communities and area administrators on climate-resilient water management improved • Output 1.3: Vulnerable communities are supported to develop and implement their DWSSPs (600 by the end of the project cycle)							
ACTIVITY	SUB-ACTIVITIES	GENDER ENTRY POINT	GENDER-RESPONSIVE INDICATORS	BASELINE	TARGET	M&E TIMELINE	RESPONSIBLE PARTIES AND MEANS OF VERIFICATION	GENDER TECHNICAL ASSISTANCE AND BUDGET
1.1.1 Implement DWSSPs that incorporate incremental improvements to mainstream adaptation solutions and update	1.1.1.1: Review uptake and delivery of updated methodologies, making incremental improvements annually	The DWSSP methodology has been updated recently with significant overhaul towards climate and gender It is expected that the methodology will be further updated - with annual improvements.	Number of review reports of current DWSSP process.	0	5	Annual Annual	SPC DoWR Annual review and stocktake of DWSSP process per implementation year, tabulating new and updated DWSSPs in specific	USD 1,200 (10% of A2 international consultant to update the DWSSP methodology to mainstream gender issues more strongly) USD 3,250 (10% of the national

existing DWSSPs	1.1.1.2: Integrate updated methodology into DWSSP processes triggered during the project	Gender needs assessments (and templates for how to assess gender barriers, alongside socioeconomic baseline) will also be added to the DWSSP, where appropriate. In the new DWSSPs developed with communities, the updated methodology will be used to ensure climate risks are mainstreamed.	Number of DWSSP methodologies updated with project findings	0	5	Annual	communities achieved per year	project engineers time in training will provide support for integration of the new DWSSP methodology that has been enhanced to be more gender transformative)
	1.1.1.3: Update existing DWSSPs, when appropriate	For 100 existing DWSSPs - workshops and technical assistance will be provided to ensure these plans are retrofitted to better reflect climate risks, gender and adaptation solutions.	Number of DWSSPs updated with gender considerations.	0	100			
1.2.2 Organize knowledge sharing events	1.2.2.1: Organize community level knowledge sharing events to ensure widespread dissemination among communities	For Activity 1.2.2, interactive culturally appropriate training material and learning notes will be developed. This could be done in partnership with UNICEF which already has tried and tested community-led efforts in Vanuatu. Similar to Activity 1.2.1,	Number of community knowledge sharing events	0	10	Annual	SPC DoWR Annual event report including participant lists aggregated annually.	USD 27,477 (ESS and GES Officer time A1) USD 29,070 (10% of A3 knowledge sharing events have a gender focus)

		It will be important to include gender modules / sensitization modules (where relevant) to ensure that groundwork for increased women's participation is accepted and driven forward by communities. Further, the Gender and Protection Cluster will help in ensuring GbV issues are discussed in a culturally-sensitive manner in these trainings, as well as the need to introduce adequate safety measures, such as locks and lights, for the WASH infrastructure delivered.						
1.3.1 Identify vulnerable communities through the NIP process to prioritize delivery of DWSSPs	1.3.1.1: Recruit and train DWSSP facilitators 1.3.1.3: Development of 600 DWSSPs informed by best-available	Through this output, the project will deliver DWSSPs to 600 communities, through the NIP process. Activity 1.3.1 will involve PWRAC identifying vulnerable communities who request DWSSPs through the NIP process. For the new DWSSP facilitators – and existing staff – training will be provided on the DWSSP methodology.	Number of DWSSP facilitators trained on the new DWSSP processes gender considerations Number of DWSSPs with new gender mainstreamed DWSSPs	0 0	12 600	Annual	SPC DoWR Training reports and participant lists. Annual review and stocktake of DWSSP process, tabulating new and updated DWSSPs in specific communities achieved per year	USD 84,545 (A1 ESS and GESI officer) USD 6,000 (A2 local consultant for ESS and GESI)

	climate science, local information and traditional knowledge	This will focus on climate risks and how these can be adapted to through the DWSSPs. Rural Water Committees (RWCs) will be established through the DWSSPs – with a mandatory quota of 40% women. Climate vulnerability assessments and community engagement are undertaken to ensure 600 DWSSPs developed during the project cycle are informed by the observed climate impacts.						

OUTCOME 2: Communities have enhanced climate-resilient rural water infrastructure	
IMPACT STATEMENT	Through Outcome 2, the current total of DWSSPs delivered (277, according to latest DoWR data) will be scaled up by almost 100% (270 more – 220 new identified through Outcome 1 and 50 existing) by dispensing investments through the government-owned DWSSP and NIP processes. Communities will, therefore, receive climate-resilient system improvements along with training to maintain new WASH infrastructure.
GENDER IMPACT STATEMENT	Improved climate-resilient WASH infrastructure delivered to 270 communities through the DWSSP process is expected to have gender-transformative impact: DWSSPs will conduct gender assessments as part of the risk and need identification process which will directly address gendered needs; trainings imparted to community members will include 50:50 male and female participation to ensure both gender groups can operate and maintain infrastructure effectively and equitably; overall spill over effect of improved water supply and WASH infrastructure will address water insecurity among communities – and vulnerable sub-groups within them.
OUTPUTS	• Output 2.1: 270 vulnerable communities supported to construct, operate and maintain climate-resilient water infrastructure

ACTIVITY	SUB-ACTIVITIES	GENDER MAINSTREAMING ENTRY POINT	GENDER-RESPONSIVE INDICATORS	BASELINE	TARGET	M&E TIMELINE	RESPONSIBLE PARTIES AND MEANS OF VERIFICATION	GENDER TECHNICAL ASSISTANCE
2.1.1 Improve CAP request prioritization, with gender and ESIA screening for chosen sites	2.1.1.1: update multi-criteria risk screening to prioritize CAP requests to identify sites for infrastructure planning including gender components	These activities will establish climate-resilient drinking water infrastructure and build capacity among vulnerable communities to maintain and operate these, thus ensuring a paradigm shift from build-neglect-rebuild approach.	Updated multi-criteria risk screening tool.	0	1	Year 1	SPC DoWR Reports on communities supported and infrastructure delivered [merged with deliverable for 3.1]	USD 238,841 (B1 ESS and GESI costs)
	2.1.1.2: Conduct gender, environment and social safeguards screening and impact assessments in chosen sites	Activity 2.1.1 will focus on improving the CAP risk matrix to ensure that vulnerable, remote communities are prioritized through the project. Consultation processes will inform the design of the infrastructure delivered, with particular attention to gender groups, persons with disability and needs of elderly as well. This can include (this is not an exhaustive list but a suggestive list of action to address needs, when identified through the CAP risk matrix): - Improved security features such as adequate lighting,	Number of gender and ESS impacts conducted	0	270	Annually	Consolidated, training report on operation and maintenance capacity in communities	

		secure locks, dignity areas with curtains/covers - Ramps, different tap heights and other inclusive design features for persons with disability						
2.1.2 Upgrade CR WASH infrastructure with adequate O&M training	2.1.2.1: Construct and upgrade infrastructure for climate resilient water sources, distribution and storage, made more accessible and safer for use by both genders. 2.1.2.2: Train RWCs on operation and maintenance for the CR-WASH infrastructure inclusive of gender considerations	Activity 2.1.2 will deliver CR-WASH infrastructure to communities with selected DWSSPs. These infrastructures will ensure gender needs are met thanks to robust consultation processes that will happen before the disbursement of funds, using updated DWSSP methodology (Output 1.1). RWCs and other stakeholders in the rural water governance architecture will receive training to ensure gender-transformative change is generated through a government-owned and community-facing process. Gender equity will be ensured in training to ensure equal opportunity of skills development and	Number of Focus groups and key informant interviews identify positive change in accessibility and safety consideration of water access points at each CAP site Number of community level plumbers and management trainings, provided with a 50% female and 50% male participations[95]	0 0	270 270	Annually Annually	SPC DoWR Focus group and key informant interview reports Training reports and participant lists aggregated annually.	USD 466,860 (10% of costs in B6 focused on gender aspects)

[95] Number of trainings will be based on the number of CAP requests.

		capacity building (plumbers training and management training) is provided instilling a paradigm shift towards upskilling women at community level.						

OUTCOME 3: Provincial and national institutions are strengthened to address climate risks associated with water security								
IMPACT STATEMENT **Through Outcome 3**, institutional capacities and processes toward climate-resilient water security for rural communities, particularly at the national and provincial level will be enhanced. In the long-term activities and sub-activities under this Outcome will result in greater technology deployment, dissemination and innovation transfer benefiting the whole rural population.								
GENDER IMPACT STATEMENT	Institutional and regulatory barriers in Vanuatu will be addressed – particularly, the limited financial and technical capacity that restricts the DoWR and provincial institutions from scaling up existing technology as well as shift towards new, climate-resilient technology. This will ensure both women and men benefit from improved water security and safety, but also that national and provincial staff are sensitized to gender barriers, while MEL and KM outputs document important gender experience on the field for improving project delivery, and scale-up planning.							
OUTPUTS	• Output 3.1: National- and provincial-level staff and WASH sector partners trained on climate-resilient water management • Output 3.2: Knowledge management through data sharing mechanism established for climate-resilient water management • Output 3.3: Monitoring, learning and evaluation framework established for improved learning for climate-resilient water management							
ACTIVITY	SUB-ACTIVITIES	GENDER MAINSTREAMING ENTRY POINT	GENDER-RESPONSIVE INDICATORS	BASELINE	TARGET	M&E TIMELINE	RESPONSIBLE PARTIES AND MEANS OF VERIFICATION	GENDER TECHNICAL ASSISTANCE
3.1.1 Deliver enhanced administrative and technical capacity training for water management	3.1.1.1: Train PWRAC and DoWR staff on updated DWSSP methodology including on gender mainstreaming consideration and	This activity will focus on enhancing the capacities of national and provincial level staff on climate-resilient water management and water governance. Activity 3.1.1. will attempt to	Number of PWRAC and DoWR training sessions conducted with a 50% female and 50% male participation	0	35	Annually	SPC DoWR Training reports on PWRAC & DoWR trainings	USD 8,940 (10% of training in C1 will have a focus on gender mainstreaming)

	importance of accessibility and gender consideration to reduce time poverty and safety/security when locating and installing infrastructure.	expand the skill set available to the DoWR through external technical support. This will also include training on gender mainstreaming, which will be linked to the efforts made by the DoWA and the Gender and Protection Cluster. ‘ Alongside the DoWR, the: Ministry of Agriculture, Livestock, Forestry, Fisheries and Biosecurity; Ministry of Climate Change; Ministry of Education and Training, and the Ministry of Health will benefit from these trainings.					aggregated annually.	
3.1.2 Update engineering and conduct workshops for WASH sector partners	3.1.2.1: Update engineering standards to reflect gender concerns and deploy in provinces for new infrastructure	Activity 3.1.2 will provide the necessary impetus to engage coordination between different WASH sector partners, as well as ensuring engineering standards are updated, approved with consensus to minimize or remove risk to women and girls, are analysed through a robust safeguarding	Number of update engineering standards	0	1	Year 1	SPC DoWR Annual written report on progress of updated engineering standards and how they have been rolled out presented at mid-term. Workshop reports on	USD 38,045 (C3 ESS officer)

		process and then, deployed.					WASH sector partner events.	
	3.1.2.2: Workshops with WASH sector partners (incl. MoH, Med., and CSOs) on updated DWSSP, methodology with specific focus presented on gender considerations	Workshops with WASH sector partners will use gender modules developed for previous activities to ensure resilience approaches provide fair consideration to gendered needs, needs of persons with disabilities, needs of different age groups.	Number of workshops conducted with 50% male and 50% female participation.	0	5	Annual		
3.2.1 Improve knowledge management processes	3.2.1.1: Review existing knowledge management systems and identify gaps to strengthen mainstreaming	This activity will ensure that a gap analysis is undertaken that accounts for gender gaps and ensures that current KM systems are updated, and Gender mainstreaming issues are accounted for in the system	Gap analysis report completed	0	1	Year 1	SPC DoWR Finalised reports and documentation.	This will be incorporated into M&E activities as described and budgeted under Annex 11.
	3.2.1.2: Establish knowledge management protocol through consultative process that accounts for gender considerations to ensure that gender is mainstreamed in operations and	Further to this, the activity will specifically update KM protocols so that gender mainstreaming issues are factored into all knowledge products in an accessible and easy manner.	Updated protocol completed and operational	0	1	Year 2		

	standards are maintained.							
3.2.2 Operationalize data sharing platform	3.2.2.1: Integrate data collected through DWSSPs into government knowledge management platforms, such as DoWR data portal	The new DWSSP methodology will factor into to DWSSPs gender considerations. The activity will ensure that DoWR platforms and systems account for gender indicators	Number of updated DoWR KM and data platforms	0	1	Year 1	SPC DoWR Finalised reports and documentatio n.	
	3.2.1.2: Engage relevant stakeholders to support effective utilization of data for decision-making by WASH sector partners	Gender data and findings (collated from the project activities) will be shared on an annual basis	Number of gender focused report, on lived experiences, stories from the projects, qualitative improvements will be produced to document the gender-transformative impact of the project, best practices and lessons-learned for later phases and further interventions.	0	5	Annually		
3.3.1 Collate existing MEL practices within water governance structures in Vanuatu, and establish a	3.3.1.1: Stocktake existing MEL processes, and collate lessons learnt, case studies, and best practices available	This activity will take stock of existing MEL practices within government-owned process, and upgrade this system by introducing robust protocol using a consultative process, as well as through SPC's	Number of stocktake reports generated	0	1	Year 1	SPC DoWR Consolidated annual report on MEL system established incorporating gender indicators and	

robust MEL protocol		existing processes through the PEARL policy and *rebbilib*. The DoWA will be potentially consulted to ensure gender MEL processes are mainstreamed into water governance and climate resilience MEL. See Section E.7 – Funding Proposal for details.					reported to the EE's	
							Total	**USD 869,158**

TABULATION OF BENEFICIARIES		
Outcome	**Targeted Communities**	**Indirect / direct beneficiaries**
1	600	68,520 direct beneficiaries (including 34,260 women), which is 22.5% of the total population of Vanuatu.
2	270 (including 220 already targeted by Outcome 1, and 50 additional ones)	30,834 direct beneficiaries, with 50% of women (including 25,124 from Outcome 1 and 5,710 additional ones); which is 8% of the total population in Vanuatu.
3	2,000	Indirect beneficiaries: the entire rural population in Vanuatu (around 228,400 individuals, which is 75% of the total population, including around 114,200 women)

Annex A: Implementing roles and responsibilities for gender mainstreaming

Gender Officer in the Project Management Unit: An international Gender Officer will be hired on a part-time basis to increase the capacity of the PMU and provide technical assistance to deliver the planned activities of the Gender Action Plan. They will take the lead in supporting the DWSSP and CAP design processes (including ESS related topics), supporting training of extension agents and service providers and RWC proponents on Gender (including ESS) integration into DWSSPs in alignment with the project GAAP. They will support the development of gender and WASH training modules and protocols (including ESS) through Outcome 1 and 3, and ensure rapid gender assessments precede the system improvements and new infrastructure delivered through Outcome 2. For the latter, local gender experts may be hired on a need-by-need basis. They will also support the MEL Officer in conducting relevant monitoring and evaluation of project implementation against the ESMP and GAAP. The Gender Officer will review and assess all project activity implementation against the GAP and incorporate findings into relevant reports as obligated under the project Funded Activity Agreement. In the case that any issues are identified, the Gender Officer will draft technical recommendations to address these issues in implementation. These will be reviewed by the EE and where needed supported by technical assistance through supervision costs. If technical enhancements are required to address gender related issues through implementation, these will be incorporated into the AWPBs for approval by the NPSC.

Department of Water Resources: The DoWR will be the co-Executing Entity of the project. The Department will oversee the registration of DWSSPs – and ensure that 40% representation of women is ensured in that process. Since the DoWR owns the DWSSP process, it will take the lead role in delivering the co-benefit identified for the project.

Department of Women's Affairs: The DoWA, of the Vanuatu Ministry of Justice and Community Services, will play a facilitating role, as required, particularly in the awareness-raising, advocacy and trainings to ensure meaningful participation of women in community-based water management bodies as well as to ensure community well-being and acceptance of such planned activities of the project.

Gender and Protection Cluster: The formation of the Gender & Protection Cluster in Vanuatu on 13 March 2014 coincided with the Tropical Cyclone (TC) Lusi hitting Vanuatu. The response was the first time that gender and protection considered as part of post-disaster assessment and response phases. In the light of the experience gained during that emergency, and the frequency and severe impact of natural disasters in Vanuatu, it was decided that the work of the cluster should be ongoing. Thus, as part of the NDMO's coordination structure, the Gender & Protection Cluster contributes to improving preparedness for responding to natural disasters in a gender-and protection-sensitive manner, and ensuring that timly, effective and coordinated assistance is provided to persons affected by natural disasters in Vanuatu during emergency operations. The Gender & Protection Cluster comprises of representatives of National Ministries, UN Agencies, International and National NGO's, National Women's Organization, the International Federation of the Red Cross, the Vanuatu Red Cross Society, and other Organizations with a focus on protection. The DoWA is the designated Lead Agency of the Vanuatu Gender & Protection Cluster.

SPC: SPC is the Accredited Entity and will be the Executing Entity in the implementation phase. It will have overall responsibility to ensure the project co-benefit (increased participation of women in Rural Water Committees) is delivered during the lifespan of the project.

UNICEF: UNICEF is the lead among the different WASH sector partners that the project will consult during implementation. Due to its significant work on gender and WASH in the Pacific, the organization will provide crucial technical assistance as well as facilitation of gender-responsive actions of the project.

Annex B: Gender questionnaire for field missions

Introduction and background

The Green Climate Fund (GCF) has enlisted E Co. under the project preparation framework agreement to support the Pacific Community (SPC) to assist with the development of the full Funding Proposal (FP) package for the project titled: *Enhancing Adaptation and Community Resilience by Improving Water Security in Vanuatu*. The GCF result areas impacts are: increased resilience of – (i) most vulnerable people and communities; (ii) health and well-being, food and water security; and, (iii) infrastructure and built environment.

Your help is sought to inform the Gender Assessment and Action Plan of this project, and insights into Vanuatu's water sector and climate change adaptation actions. The delivery of the FP is expected by the end of this year, post which iterative interaction is expected from the GCF to finalise the project for the GCF approval process. The project is envisioned through three Outcomes (this framework is subject to updates during the preparation phase, and will be communicated to relevant stakeholders):

- Outcome 1: Evidence-based planning and decision-making for climate-resilient water management at the community level
    - Output 1.1: DWSSPs methodology improved to better address climate vulnerabilities and incorporate adaptation solutions
    - Output 1.2: Awareness, capacities, and skills of communities & area councils on climate-resilient water management increased
    - Output 1.3: Vulnerable communities have developed their DWSSPs
    - Output 1.4: Vulnerable communities have implemented no and low costs activities identified in their DWSSPs
- Outcome 2: Climate-resilient rural water infrastructure
    - Output 2.1: Climate change considerations incorporated into engineering standards and trainings
    - Output 2.2: Vulnerable communities supported to establish, operate and maintain climate-resilient drinking water infrastructure
- Outcome 3: Institutional strengthening at provincial and national level to better address climate risks associated with water security
    - Output 3.1: Improved DWSSPs prioritization process and CAP risk-ranking process at provincial level based on climate vulnerability indicators
    - Output 3.2: Increased capacities of PWRAC, DoWR staff, and WASH sector partners on integrated climate-resilient water management
    - Output 3.3: Robust knowledge management and data sharing mechanisms in place for climate-resilient water management
    - Output 3.4: Monitoring, Evaluation and Learning integrated in the National Implementation Plan for Safe and Secure Water to ensure continued climate resilience

We need your valuable insight to gauge how the project can be designed to be of relevance at the community and national levels; how to better improve the project's effectiveness; and create pathway for potential sustainability. Therefore, suggestions for improvements or other comments will be sought and may be incorporated into future work undertaken by this project, as well. If you have any additional comments or questions, please contact Debasmita (debasmita@ecoltdgroup.com).

Questions

1. Can you provide us an overview of gender mainstreaming for WASH at the policy level?

a) Has gender analysis been regularly undertaken to inform national policy responses to gender issues in the WASH sector at the national and provincial level?

b) Has an institutional audit been done to identify gaps in lead ministry capacity and practice (DoWR and other relevant institutions), in responding to gender issues?

b.i) When was the latest audit?
b.ii) How was the Department of Women's Affairs involved?
b.iii) In your opinion, what can be done more to address gender through these national institutions?
c) Do the lead water and sanitation sector ministries and its appointed agencies allocate resources for gender mainstreaming activities?
d) Are specific gender objectives articulated well within national water and sanitation policies and strategies?
e) What other gender and protection policies can be relevant for this project?

2. How familiar are you with this SPC- and DoWR-led GCF project?

a) This project is focused on delivering climate-resilient WASH infrastructure as well as introduce changes to the NIP processes. What are relevant gender issues that should be considered for this project?
b) The project is going to focus on northern and northwestern island groups – Torba, Sanma and Penama. What are your views on the selection of these specific sites?
c) How can the Department of Women's Affairs contribute to shaping the activities of the project under the project results framework? What are the synergies and what capacity can be demonstrated by the Department of Women's Affairs to potentially partner on this project?

3. Overall gender baseline and gender-based violence in Vanuatu

a) For similar projects, is there equal participation of men and women at all stages: initiation, design, site location, implementation, price setting, O&M and management?
b.i) What are the main gender issues of Vanuatu? (income inequality, access to decision-making institutions, high levels of gender-based violence?)
b.ii) Can you highlight and share documentation of ongoing and past interventions that have mainstreamed gender and/or chosen a gender-transformative project design?
c.i) In terms of gender-based violence, can you elaborate more on what happens during and in the immediate aftermath of a disaster?
c.ii) Can you provide studies / analyses conducted by your department of GbV levels in the post-disaster context?
c.iii) What are current and recently closed projects on GbV that have been successful, and in what areas of Vanuatu?

4. Gender-specific issues for WASH?

a) Is menstrual hygiene mainstreamed in WASH projects and interventions?
b) Can you provide examples of best practices / successful projects targeting menstrual hygiene in Vanuatu?
c) What can be interlinkages between climate impacted water infrastructure and menstrual hygiene? Are girls regularly limited from attending schools, for example, if toilets are not quickly reconstructed after cyclone damage?

5. Additional questions during provincial field missions

(please note interviewee name, with consent, location, date, time and their gender).

a) Are sector meetings at community level organized to overcome cultural barriers to women's participation, (cultural norms, seating arrangements, language and meeting times)?

b) Do operational agencies provide information for decision making on policies, strategies, plans and investments, in a format that is user-friendly and accessible to women, marginalized groups and the organizations that represent them?

c) Do policy makers and regulatory bodies make use of feedback mechanisms for complaints and challenges faced by citizens from their providers, including those on lower levels of service, such as for those relying on standpipes and kiosks?

d) Do service providers demonstrate commitment to the citizen voice by utilizing tools like citizens' charters, ICT, satisfaction surveys, toll free lines and effective complaint desks? Do you have access to grievance mechanisms?

e) Can you give examples of projects / groups that have, in your knowledge, intervened successfully and improved water infrastructure? How best can this be replicated in your commune?

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